

# US Interfacility Transport: Sourced Evidence Master v4.1

A complete, source-verified evidence base for the United States interfacility medical transport market: who moves, between which care settings, at what clinical acuity, paid by whom, at what price, served by which suppliers, and where the whitespace is measurable. v3 carries the entire verified v2.7 evidence base forward unchanged and adds the platform IFT database and the full government-connector layer. Built and verified 09-11 July 2026. Navigate with the Index tab: every tab, one click.

## What this workbook is, in one paragraph

Every number in these 335 tabs comes from a named government dataset, a peer-reviewed study, or a published primary document, or is computed by a visible Excel formula from numbers that do. Nothing is modeled, blended, or assumed. Where a figure the market usually wants does not exist in any public source, this workbook says so and names exactly what would produce it. Figures that failed that rule are quarantined on Excluded\_Not\_Sourced. PUBLIC-WEB company material is carried with its label and dates, never silently mixed with government data.

## How to read any number in three steps

Step 1: every tab tags its figures with Fact IDs (F01 to F621). Step 2: the Fact\_Ledger resolves each ID to a source, tier, and locator down to the table or page. Step 3: the Source\_Index identifies all 446 sources on one row each, with the full citation and URL on the Source\_Register. If a figure is labeled DERIVED, the ledger names its inputs and the cell shows the arithmetic. New in v3: Pull\_Manifest gives the exact endpoint, dataset UUID, filters, SHA-256 and timestamp of every live extraction, and V3\_Change\_Log records every edit made to carried-forward v2.7 content.

## The colour and label conventions

Blue font: a value hardcoded from a source document or dataset extraction. Black font: an Excel formula. Green font: a link to another tab. Tier A: verified against the primary document or extracted directly from the primary dataset by the committed pipeline. Tier B: published but secondary, or a figure that moves over time. SOURCED: carried from an upstream dataset and not independently reopened. DERIVED: computed here, with inputs named. PUBLIC-WEB: company/press material, labeled, dated, never blended with government figures. FRAMEWORK: definitions and scaffolds only - never numbers.

| The map: what each section answers |                                                                                                                          |                                                                                                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Section                            | Tabs                                                                                                                     | The question it answers                                                                                                                                                                            |
| Governance                         | Index, README .. Connector_Estate_Map, Engagement_Data_Map                                                               | Why every number is here, the findings, the complete audit trail, the v3 change log, the live-pull manifest, and the connector estate that feeds the data tabs.                                    |
| Demand                             | Demand_Drivers .. Growth_Evidence_Registry, InDepth_Q01-Q10                                                              | How many patients move, from which settings, for which conditions, driven by which measured forces - with the verbatim evidence registry behind each claim.                                        |
| Medicare claims core               | Medicare_PSPS .. MA_Geo_Variation                                                                                        | The public claims spine: PSPS 2010-2024 including denial rates, the MUP utilization/price series 2013-2024, dialysis/ESRD denominators, and the Medicare Advantage utilization bounds.             |
| Price and payment                  | Payment_Rules .. Commercial_Context_APCD                                                                                 | What Medicare pays, locality by locality (GPCI), a worked derived rate card, service-level economics, and published commercial context.                                                            |
| Supply                             | Supplier_Landscape .. Facility_Universe_State + the full registries (Hosp/SNF/Dialysis/IRF/LTCH/Hospice/HHA, PECOS, HSA) | Who provides transport and where: billing organizations, enrolled suppliers, workforce, certification vintages, ownership churn, and the facility universe that originates and receives transfers. |
| Market structure                   | State_Saturation_Raw .. Imbalance_Ledger + SP_Index and the 51 state profiles                                            | State and metro screens, county-level supply density and whitespace bands, and demand-over-supply imbalances.                                                                                      |
| Company & competitive              | MMT_NPI_Estate .. Contract_Benchmarks                                                                                    | The subject company NPI estate and dossier, named competitors, payment-integrity evidence, and contract benchmarks. PUBLIC-WEB material is labeled.                                                |
| Client alignment & costs           | Sizing_Playbook .. Cross_Source_Recon                                                                                    | How the team converges on a defensible number, production costs, and why four sources give four different counts.                                                                                  |
| Reference                          | IFT_Alt_Measures .. Dataset_Linkage_Map                                                                                  | Alternative counting routes, every code system that records a transfer, the KPI dictionary, and the regulatory register.                                                                           |

Integrity

Fault\_Audit .. Excluded\_Not\_Sourced

Every known threat to validity, and the quarantine of everything that failed the sourcing rule.

**What this workbook deliberately does not contain**

A single asserted total addressable market figure. A credible TAM requires a commercial transport volume, a Medicare Advantage transport volume, and an acuity-weighted price, and none of the three exists in any public source. The three-scenario TAM MODEL with its assumption register lives on TAM\_Model\_National; v3 adds the published bounds for its weakest input (MA\_Geo\_Variation) but does not convert the model into a fact. The quarantine list on Excluded\_Not\_Sourced grew in v3: the platform database sweep added 63 more modeled figures that are named and excluded rather than blended.

| Revision summary |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision         | Date         | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1                | 09 July 2026 | Initial evidence base: 20 tabs, facts F01 to F109, sources S01 to S38, findings 1 to 25, all verified against primary documents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2                | 10 July 2026 | The complete Medicare origin-to-destination dollar map rebuilt from 392,409 raw claim rows (reconciling to the cent), the air and dialysis channels, commercial rates, macro drivers, code crosswalks, and the dataset linkage map. Facts through F135; sources through S58.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3                | 10 July 2026 | Per-beneficiary normalization, acuity by channel, the supplier landscape, and the payment add-on cliff. Facts through F147; sources through S61.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4                | 10 July 2026 | The demand and supply stacks, the imbalance ledger with the whitespace register, an eleven-year supplier series, workforce, state saturation, metro landscape, alternative counting routes, and the consolidated fault audit. Facts through F163; sources through S72.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5                | 10 July 2026 | Final consolidation: the official state-file cross-check (bounding the supplier-file suppression tail at 1.0%), corrections from independent external review, and full internal-consistency verification. Facts through F165; 73 sources; zero formula errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6 (v3.0)         | 10 July 2026 | The platform integration: all v2.7 content carried forward unchanged (faithful-copy proof on Verification_Log Panel I) plus 209 new tabs built from the platform IFT database and 321 live government-API extractions - the MUP ambulance utilization/price series 2013-2024, the PSPS denial-rate series 2010-2024, CMS market saturation 2020-2025 with county whitespace bands, BLS QCEW EMS industry series 2014-2025, the PECOS supplier universe, the facility O/D universe, CY2025 GPCI localities with a derived rate card, certification-vintage supply series, the OMB county-to-CBSA crosswalk, and the sourced clinical/growth/company evidence registries. Facts through F430; 304 sources; 189 charts: ~6,970 printed pages (as shipped in v3.0 - this row records that revision. not the |
| 7 (v3.1)         | 11 July 2026 | The usability and closure pass: a hyperlinked Index tab covering every tab; findings 42-51 appended with LIVE formula references into the new evidence; the measured Census 65+/85+ state base 2020-2024 (State_Age_65plus - closes the state-grain half of P20); the OEWS May 2024 EMS occupation wage ladder (OEWS_EMS_Wages - closes P4); four gallery charts added to the Charts tab; state profiles gain 65+ joins; every source now carries a URL, a repo-file locator, or an explicit no-URL-captured statement; and two citation corrections applied at source (the PMID 25397857 attribution and a dead MedPAC link).                                                                                                                                                                          |
| 8 (v3.2)         | 11 July 2026 | The raw-granularity doubling: the provider-grain Medicare ambulance registry (every billing NPI by name, 2019 and 2024, ~90,000 rows - the most granular public Medicare ambulance record); hospital-to-ZIP transfer corridors (top-10 origin ZIPs per hospital); the county 65+/85+ age base 2020 vs 2024; county chronic-disease prevalence (CKD/CHD/stroke/diabetes, CDC PLACES, methodology stated); the quarterly QCEW industry series 2014-2025; the HCRIS hospital capacity panel FY2020-2022 (17,974 hospital-years); and the OIG ambulance-exclusion registry. All Tier A dataset extractions on Pull_Manifest.                                                                                                                                                                                |

|           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9 (v3.3)  | 11 July 2026 | <p>The presentation overhaul: every chart rebuilt through a single hardened house-style layer - bottom category axes with real year/state labels, explicit series colours and weights, legends only where they earn their space, subtle gridlines, zero chart-on-chart overlaps (19 collisions found and removed); the two dual-axis combo charts split into paired single-axis charts; the carried v2.7 charts re-parsed at full fidelity so every series regained its category labels and names; the Methodology tab rebuilt as a nine-section decision record covering the v3 pipeline, the verification gates and a sixty-second audit path (and repairing a v2.7 URL typo); and a workbook-wide format sweep - print setup normalized on every tab, truncated columns widened, header rows tightened.</p>                                                                                                                          |
| 10 (v3.4) | 11 July 2026 | <p>The specificity-and-analysis pass (worker handoff + eight-hour extension order): the facility-pay evidence layer (Facility_Pay_Layer - GADCS revenue-source census, the DocGo outside-per-trip decomposition, USAspending V225, all verified against the primary documents); the subject company's measured Medicare book (MMT_Medicare_Book - consolidated and per-NPI volumes, acuity mix with volume-weighted RVU, mileage economics, three-vintage trajectory); the handoff 5.1-5.3 repairs (committed fact-tag scan - zero stale tags found - and data-quality rows on all seven v3.2 raw tabs); findings 52 onward with live references; plus the analysis and cohort layers this revision row is extended by as they land.</p>                                                                                                                                                                                                |
| 11 (v3.5) | 12 July 2026 | <p>The completion pass: the market-wide 990 contractor sweep across the footprint nonprofit systems (Footprint_990_Sweep - the largest public window into facility-direct transport payments, with the top-five information floor stated); the Medicaid ambulance rate card extended toward the full ten-state footprint; and the retried parked pulls (SNF return-leg quality, Hospital-at-Home participants) landed where public data allowed and left as bordered PENDING with the named dataset where it did not. Findings continue past the v3.4 register; every new tab carries a read panel, a finding and a data-quality row.</p>                                                                                                                                                                                                                                                                                               |
| 12 (v3.6) | 12 July 2026 | <p>Return-leg structure pass: joins the SNF Quality Reporting Program claims measures to the CMS Nursing Home provider-info file on the CCN (all reporting SNFs, 1:1) and cross-tabs the bounce-back (potentially preventable readmission) and discharge-to-community rates by ownership, certified-bed scale and 5-star rating (SNF_ReturnLeg_Structure) - the structural cut of where return-leg transport demand concentrates. This revision also restores the verification gate script (verify.py), which the v3.5 package had picked up in a stale form.</p>                                                                                                                                                                                                                                                                                                                                                                       |
| 13 (v3.7) | 12 July 2026 | <p>CIM-grade presentation pass (Run 3, Block U): a committed format gate (format_gate.py) and a presentation pass (cim_format.py) now enforce the deck-ready standard on every tab - gridlines off, a section tab colour, freeze panes, a bold A1 title, and an explicit number format on every numeric cell (no raw floats, no unformatted counts, no Python artifacts), including the carried tabs the builder never touched. Adds the Style_Standard reference tab stating the rules. No evidence changed; every tab is now screenshot-worthy at 100% zoom.</p>                                                                                                                                                                                                                                                                                                                                                                      |
| 14 (v3.8) | 12 July 2026 | <p>Evidence-integrity pass (Run 4, outcomes 1-2). (1) The footprint-wide 990 sweep (Footprint_990_Sweep) now classifies every transport-keyword hit - GROUND TRANSPORT / AIR TRANSPORT / ED STAFFING / COURIER-LOGISTICS / OTHER-AMBIGUOUS - from the verbatim services text and the contractor name, with the rules printed on the tab; the read panels and finding now rest on the ground+air transport subset only, and the emergency-department staffing contractors are quarantined in a labelled exclusion panel rather than headlined as transport. (2) Every dollar cell now ships as a base-only / mileage-loaded PAIR: Derived_Rate_Card carries a two-way A0425 mileage-loading derivation (MMT book and national registry) and Scenario_Matrix Panel B2 states the base-only and mileage-loaded price per transport side by side, both live formulas, so the ~43-45% mileage share of allowed dollars is never dropped.</p> |

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|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15 (v3.9)  | 12 July 2026 | Usefulness-and-portability pass (Run 4, outcomes 4 and 6). (1) A new Study_Synthesis tab (up front, after the README) states the IFT thesis as nine measured pillars - demand, the measured TAM floor, price and its mileage load, input-cost risk, supply fragmentation, the subject-company book, the return-leg driver and the named risks - each headline a LIVE link to its home tab and each row carrying its guardrail, plus a plain-language firewall panel; it creates no evidence, only a map of what each block contributes. (2) The Index now prints each section's ROLE in the IFT study under its banner, so every tab inherits a stated usefulness, not just a title. (3) Contract-corpus portability: non-retrievable local file paths (pdfs/...) are stripped from every corpus citation and source locator - the retrievable public URL already rides in its own column - and the federal award ladder is widened to the top 25 Department of Veterans Affairs V225 awards by amount. A companion IFT_Deck_Feed_Extract workbook ships the three deck-facing tabs with values resolved. No existing fact/source/finding is renumbered. |
| 16 (v3.10) | 12 July 2026 | Editorial pass: every tab title and subtitle rewritten out of the conversational register into a declarative one. The "The question: ..." subtitle frame (313 tabs), the title flourishes (" - the decision record", ", measured:", ", ": the full certified registry"), and rhetorical asides ("stated plainly", "at a glance") are removed; titles are plain descriptive names and subtitles lead with what the exhibit shows and its source. Marquee narrative tabs (Methodology, Study_Synthesis) get a bespoke declarative subtitle. Presentation only: no data, finding, ledger or read-panel content changes, and title-block edits on carried v2.7 tabs are recorded and excluded from the carried-cell fidelity gate.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 17 (v3.11) | 13 July 2026 | Visual pass: house-style charts added to the load-bearing measured series that carried numbers but no chart - the Medicare interfacility transports and allowed dollars by year (Medicare_IFT_Series), the subject-company allowed dollars by vintage (MMT_Medicare_Book), the acute-to-acute ED-origin transfer series (Acute_IFT_Series), the NEMSIS type-of-service split (EMS_Transports), the payer revenue-per-NPI mix (Facility_Pay_Layer) and the RSNAT savings ladder (RSNAT_Series). Each is a live range reference and passes the chart-integrity gate. Plus a formatting tidy that collapses inconsistent double-spacing in body text on v3-authored tabs. No evidence changed.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 18 (v3.12) | 13 July 2026 | Presentation polish: RAG status cells (GREEN / AMBER / RED / PENDING) now carry a semantic fill so the exhibit register and quality gates read at a glance; the long reference tables (Source_Register, Source_Index, Findings, Slide_Feed) get zebra row banding matching the grey already on the Fact_Ledger and Verification_Log; and section-header cells are normalized to white bold. Pure formatting - no cell value, chart, finding or ledger entry changes, and the bordered PENDING markers keep their border.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 19 (v4.0)  | 13 July 2026 | Run 5 - the billing-flag pass. Five new evidence layers, all public and appended (facts from F610, sources from S436, findings from 108). Modifier_QM_QN_Series measures the hospital billing relationship directly from the QN/QM carrier claim modifier, 2010-2024, as a third measured leg on Insourcing_Bounds. Payment_Rules states the SNF consolidated-billing wedge from primary CMS sources. IFT_Software_Landscape maps the dispatch, ePCR, billing and transfer-center platforms from the public record with the request-cascade workflow. Hospital_Ambulance_Prevalence answers how many hospitals field their own ambulance (a strict ~14% HCRIS floor). MRF_Attempt_Log closes the commercial machine-readable-file attempt with three logged failures and names the interim anchor. The annual per-NPI book extension to twelve years remains the named next pull.                                                                                                                                                                                                                                                                        |
| 20 (v4.1)  | 13 July 2026 | Run 6 - the institutional wedge, the model's number-one open question. Institutional_Ambulance_Wedge sizes the Medicare ground ambulance volume billed on institutional (UB-04) claims rather than the carrier file, two free ways: top-down (MedPAC 11.4M all-claims 2023 minus the carrier file = a ~0.7M institutional residual) and bottom-up (the HCRIS line-95 hospital ambulance operating cost, with a GADCS-bounded implied volume). HCRIS_Institutional_Roster lists every hospital ambulance operator and its scale. The definitive per-claim route (ResDAC LDS, rev 0540-0549) ships as a bordered receiving schema with the application and commercial-extract specs ready to send. Appended F619 onward; no shipped ID renumbered.                                                                                                                                                                                                                                                                                                                                                                                                         |

Pending register: named enhancements, none assumed

Carried from v2.7 with v3 status: P1 HCUPnet condition-level transfer rates - OPEN (point estimates only without licensed microdata). P2 Ambulance-desert county tables - PARTIALLY CLOSED (county provider-count bands on Market\_Saturation\_Ambulance measure thin-supply counties; a population-weighted desert definition still needs a licensed drive-time layer). P3 ZCTA/county-to-CBSA crosswalk - CLOSED (CBSA\_Crosswalk\_Reference, OMB Bulletin 23-01). P4 BLS OEWS state wage files - CLOSED in v3.1 (OEWS\_EMS\_Wages: May 2024 state file, EMT/paramedic/ambulance-driver occupations, beside the QCEW industry series on QCEW\_EMS\_Employment). P5 USRDS prevalent dialysis counts - OPEN (Enrollment\_ESRD\_State carries the Medicare ESRD denominator instead). P14 locality price adjustment - CLOSED (GPCI\_Localities + Derived\_Rate\_Card). P20: state-grain 65+ base - CLOSED in v3.1 for the state grain (State\_Age\_65plus, Census Vintage 2024 measured estimates 2020-2024); the ACS API route stays OPEN for tract/county age detail (needs a free CENSUS\_API\_KEY). P21: NPPES monthly full-replacement file for the national supplier universe by taxonomy - OPEN (the live API caps at 1,200 rows per query; PECOS\_Suppliers\_State is the enrollment-based count today).

*Prepared with a deterministic pipeline committed to the repository (RCM\_MC/scripts/ift\_evidence\_v3): pull.py re-fetches every artifact on Pull\_Manifest, build.py reassembles this workbook byte-for-byte from the v2.7 master plus the caches, and verify.py re-proves the fidelity, recompute, and zero-error gates. Nothing in the build path is manual.*

Study synthesis: the IFT investment thesis in nine measured pillars

The interfacility-transport thesis as nine measured pillars, from demand through price and the mileage load to the subject company and the named risks. Every headline is a live green link to its home tab and every row carries its guardrail; like Investor\_QA and Slide\_Feed, this tab summarises linked cells and creates no evidence of its own.

HOW TO READ: a green number is the actual measured cell from the home tab, not a retyped figure, so it cannot drift from the evidence. A "->" link navigates to the home tab. The guardrail column states the one caveat that keeps each claim honest. The "what this study does NOT claim" panel below is the firewall.

| The thesis, nine measured pillars (green = live cell on the home tab) |                                                                                                                                                                                                                |                                   |          |                                                                                                                                                                                          |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thesis pillar                                                         | The measured claim                                                                                                                                                                                             | Headline (live)                   | Home tab | Interpretation guardrail                                                                                                                                                                 |
| 1. Demand is measured, not modeled                                    | Medicare fee-for-service billed this many hospital-to-hospital ground transports in 2024 - the hardest measured floor on IFT volume.                                                                           | 871,753 Medicare_IFT_Series       |          | Carrier claims only; hospital institutional billing and Medicare Advantage are excluded, so this is a FLOOR on volume, not the market.                                                   |
| 2. One TAM cell is fully measured; the rest name their gap            | Claims-visible volume times the measured Medicare allowed per transport equals 2024 Medicare FFS interfacility allowed dollars, by construction.                                                               | \$577,228,322 Scenario_Matrix     |          | This is the measured FLOOR, not the TAM. Every larger cell of the scenario grid is a bordered PENDING that names the dataset that closes it; mixing cells re-introduces double counting. |
| 3. Price carries mileage - and mileage is ~43-45% of it               | The measured Medicare allowed per transport INCLUDING A0425 mileage; base-only pricing silently drops this mileage share.                                                                                      | \$662.15 Derived_Rate_Card        |          | Every dollar cell in the book now ships as a base-only / mileage-loaded PAIR so the mileage share is never dropped (Scenario_Matrix Panel B2).                                           |
| 4. The mileage load is derived, not assumed                           | The loading factor (loaded allowed / base allowed per transport) is live and derived two ways - the MMT book and the national registry.                                                                        | 1.80x Derived_Rate_Card           |          | The derivation is printed on Derived_Rate_Card; the factor is a measured ratio of two allowed-dollar series, not a rule of thumb.                                                        |
| 5. Input costs are tracked, never blended                             | The statutory CY2026 ambulance payment update (AIF); diesel, wages and benefits are printed as separate measured series against it.                                                                            | 2.0% Input_Cost_Index             |          | No composite input index (house rule). The spread between this update and measured input inflation is the margin-risk watch item.                                                        |
| 6. Supply is fragmented, so competition is local                      | Distinct Medicare ground-ambulance billing organizations in 2024 - a large, slowly consolidating long tail.                                                                                                    | 8,721 Fragmentation_National      |          | National concentration is low; the real competitive set is per-state and per-corridor, so national shares understate local position.                                                     |
| 7. The subject company is measured, not asserted                      | MMT consolidated Medicare FFS ambulance allowed dollars, 2024 vintage (two NPIs) - the measured book behind the diligence.                                                                                     | \$22,004,764 MMT_Medicare_Book    |          | Medicare FFS only; the MA and facility-pay books are bounded on their own tabs, never summed into this figure.                                                                           |
| 8. Return-leg demand concentrates structurally                        | For-profit and lower-rated skilled-nursing facilities bounce patients back to hospitals more, and that is where the return transport originates - the structural cut of where demand lands.                    | see tab ->SNF_ReturnLeg_Structure |          | The join is on CMS claims quality measures (bounce-back / discharge-to-community rates), which are a demand PROXY, not transport counts.                                                 |
| 9. The risks are named and dated, not hand-waved                      | The forward picture rests on a dated statutory add-on cliff, the Medicare-Advantage invisibility of half the book, and local labor depth - each a bounded lever or a bordered PENDING, never a point forecast. | see tab ->Growth_Outlook_Shell    |          | Every forward lever sits on Scenario_Matrix Panel C (tornado) with public bounds, or stays a named PENDING; none is a typed guess.                                                       |

What this study does NOT claim (the firewall, in plain language)

|       |                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NOT 1 | A single "the TAM is \$X" number. The scenario grid is the answer; its one fully-measured cell is a floor and every larger cell names the public dataset that would close it.                                                   |
| NOT 2 | Any named roster of the subject company's customers, accounts or clients. The health-system material is a research cohort of representative multi-hospital health systems operating in the study footprint, selected for depth. |
| NOT 3 | Contract terms, performance metrics or survey statistics not tied to a public document. Unpriceable or unretrievable cells stay bordered PENDING naming their dataset.                                                          |
| NOT 4 | Commercial negotiated rates as fact. The MRF-forward price basis is a named PENDING awaiting Transparency-in-Coverage files; the blended basis awaits the Medicaid card and contract-rate extraction.                           |

Read panel

What this tab is: the one-page synthesis that ties every major evidence block to its role in the IFT study, so a reader knows not just what each tab carries but why it is here. The nine pillars walk demand (measured Medicare volume), the one fully measured TAM floor, price and its mileage load, input-cost risk against the statutory update, supply fragmentation, the measured subject-company book, the structural return-leg driver, and the named forward risks. Each headline is a live cell, not a restatement, so the synthesis moves with the evidence; each guardrail names the one caveat that keeps the claim honest; and the firewall panel states in plain language what the study refuses to assert. For the full three-sentence answer to any of the twelve investor questions, see Investor\_QA; for the exhibit-by-exhibit evidence status, see Slide\_Feed.

Methodology

The inclusion rule, why each evidence tab exists, the definitional choices that move the numbers, how each dataset was pulled so the workbook reproduces, and the rules for comparing figures. Sections 5 and 9 cover the pull pipeline, the verification gates and the audit path.

1. The question this workbook answers, and the one it refuses to answer

The question. How large is US interfacility transport, who moves, between which settings, at what acuity, paid by whom, and at what price. Every tab answers one slice of that and cites the document that produced it.

The refusal. This workbook does not size a total addressable market. A TAM requires a commercial-payer volume, a Medicare Advantage volume and an acuity-weighted price, none of which exists in any public source. The initial build carried three such assumptions (a 4.27x gross-up, a 2x to 4x commercial multiple, an acuity mix); all three were moved to Excluded\_Not\_Sourced and stayed there. revision 2 replaced two of them with published bounds rather than restoring the estimate.

The consequence you must accept. Some cells on this workbook say the number does not exist. That is the finding. An analyst who needs a TAM must buy data (Dataset\_Linkage\_Map lists which), not blend a public one.

2. The inclusion rule, applied without exception

| Rule                                                                                                                                                   | Why                                                                                                                               | What it prevents                                                                                                                  |                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| A number enters only if a named publisher document produced it, or an Excel formula computes it from numbers that did.                                 | Traceability must survive the analyst. Anyone opening a cell sees either a source-pinned constant or the arithmetic that made it. | Numbers with no parent. The single most common failure in diligence workbooks is a figure that no one can source six weeks later. |                                                                                                           |
| Blue font means hardcoded from a document. Black means a formula. Green means a link to another tab.                                                   | The colour is the audit trail. A reader can see the provenance of a cell without clicking it.                                     | Hardcoded values masquerading as calculations, and stale pastes of numbers that should have updated.                              |                                                                                                           |
| Every fact carries a Fact ID (F01-F165 in v2.7; v3 extends the ledger - see Fact_Ledger), a Tier, a Source ID and a locator down to the table or page. | Tier separates what was verified against the primary document from what was asserted by an upstream workbook.                     | Laundering. An assumption quoted twice does not become evidence.                                                                  |                                                                                                           |
| Anything labeled ILLUSTRATIVE, FRAMEWORK or INDUSTRY ESTIMATE is quarantined on Excluded_Not_Sourced.                                                  | These labels are how modeled numbers enter a deck and never leave.                                                                | A framework assumption being read off a chart axis as though it were measured.                                                    |                                                                                                           |
| Derived figures are labeled DERIVED and show their inputs.                                                                                             | A ratio of two published numbers is legitimate. A ratio of a published number and a guess is not.                                 | Same-source, same-year discipline on every multiple (see section 6).                                                              |                                                                                                           |
| The colour key, live:                                                                                                                                  | Blue - hardcoded straight from the cited document (e.g. 8,911)                                                                    | Black - a visible formula over sourced cells (e.g. =B12/C12 typed in the cell)                                                    | Green - a live link to another tab (e.g. =Fact_Ledger!D9)                                                 |
| Tier A                                                                                                                                                 | Verified against the primary document or dataset in this build; the locator points into that document.                            | Tier B                                                                                                                            | Published but secondary, or a figure that moves with revisions; carried with that caveat on the fact row. |

3. Why each evidence tab exists

| Tab                 | The question it settles                                                                    | Why it had to be built                                                                                                                                          | Primary sources |
|---------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Acute_IFT_Series    | How many acute-to-acute transfers happen, from the ED and from inpatient beds.             | HCUP is the only national all-payer census of hospital discharges. Its two origin channels are mutually exclusive and must be added, not averaged.              | S01, S04        |
| Receiving_Side      | Does the receiving side of the ledger agree with the sending side.                         | A transfer has two ends. NIS TRAN_IN counts arrivals independently of DISPUNIFORM departures. Agreement is the workbook internal validity test.                 | S04             |
| EMS_Transports      | How many ambulance legs exist, and what share are interfacility.                           | Hospital data counts patients moved; EMS data counts vehicles moving them. The two are different units and neither substitutes for the other.                   | S12, S25, S55   |
| Medicare_PSPS       | What does Medicare fee-for-service actually pay, by acuity and by origin-destination pair. | It is the only public file that carries dollars, acuity and endpoints on the same row.                                                                          | S13, S39        |
| Medicare_IFT_Series | Is the Medicare hospital-to-hospital book growing or shrinking.                            | No published table contains this series. It was built by pulling fifteen annual PSPS files and filtering the HH modifier.                                       | S32             |
| Medicare_OD_Matrix  | Where does the entire Medicare ambulance dollar go, pair by pair.                          | revision 1 carried the HH pair only. Building the full matrix revealed that the SNF interface is 2.75x the hospital pair, which reframes the market definition. | S39             |

|                                                                                                                                  |                                                                                                             |                                                                                                                                                                           |                                              |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Utilization_Normalized                                                                                                           | Is the Medicare decline demand or denominator.                                                              | FFS enrollment fell 15.5% over the same window. Without dividing by it, every trend on this workbook overstates decline by roughly a factor of two.                       | S39, S60                                     |
| Acuity_by_Channel                                                                                                                | Are the interfacility channels one clinical product or several.                                             | Staffing, unit economics and payer strategy all follow acuity. SNF-to-hospital is 94% emergency; hospital-to-SNF is 99% non-emergency. One name, opposite products.       | S39                                          |
| Air_Ambulance_IFT                                                                                                                | Where does high-acuity interfacility volume actually sit.                                                   | Rotary-wing is 65.6% hospital-to-hospital and grows per beneficiary while ground shrinks. Ground acuity mix cannot describe it.                                           | S40, S44                                     |
| Dialysis_ESRD_Channel                                                                                                            | What happens to a scheduled-transport book when a payer decides the volume is discretionary.                | The RSNAT rollout is a natural experiment with a date. The claims date-stamp the intervention, which makes it the template for Medicare Advantage behaviour elsewhere.    | S39, S41, S42                                |
| Payment_Rules                                                                                                                    | What governs the Medicare price, and when does it change.                                                   | RVUs are frozen since 2002, geographic adjustments since 2008, and the statutory add-ons expire 1 January 2028. The price is legislative, not market.                     | S14, S16, S28, S29, S61                      |
| Payer_Rates_Commercial                                                                                                           | What does everyone other than Medicare pay.                                                                 | The commercial multiple was the largest excluded assumption in v1. FAIR Health supplies published bounds; GAPB supplies the balance-billing regime.                       | S43, S44, S45, S46, S47, S48                 |
| Supplier_Landscape                                                                                                               | Who bills, and how concentrated is the market.                                                              | 8,721 NPIs with the largest under 1% of national volume. Any consolidation thesis starts here, and every count is a floor because of small-cell suppression.              | S59                                          |
| Cost_and_Capacity                                                                                                                | What does a transport cost to produce.                                                                      | GADCS is the only national cost collection, and MedPAC says it cannot yet support a margin. That warning travels with every cost figure.                                  | S22, S30, S31                                |
| Macro_Demand_Drivers                                                                                                             | What moves demand from outside the hospital.                                                                | Census 65+, Medicare Advantage share, rural closures, coordination infrastructure. Each is a measured series, not a narrative.                                            | S50, S51, S52, S53, S54, S55                 |
| Code_Crosswalks                                                                                                                  | How is one transfer written down in five different systems.                                                 | A transfer is an O-D modifier in claims, a discharge status on a UB-04, a DISP_ED value in HCUP, an eResponse.05 value in NEMSIS. Joining them requires knowing all four. | S56, S57, S58                                |
| Dataset_Linkage_Map                                                                                                              | What would it take to answer what this workbook cannot.                                                     | It names the join keys, the access terms and the two purchases that close the largest gaps.                                                                               | All                                          |
| Cross_Source_Recon                                                                                                               | Why do four sources give four different numbers.                                                            | They measure different units on different universes. The wedge is enumerated driver by driver so the spread is explained rather than averaged away.                       | S01, S04, S12, S13                           |
| <b>v3 evidence layers (live public-API pulls; one row per family; every pull is on Pull_Manifest with endpoint and SHA-256):</b> |                                                                                                             |                                                                                                                                                                           |                                              |
| Enrollment_ESRD_State + Enroll_State_*                                                                                           | The fee-for-service denominator, nationally and per state, 2013-2025.                                       | Every per-beneficiary rate in the workbook divides by these cells; the MA migration is the single biggest trend confounder.                                               | CMS Medicare Monthly Enrollment              |
| MUP_Ambulance_National/State, MUP_State_*, MUP_Providers_2013/19/24                                                              | Who bills ground ambulance, down to the individual NPI, and at what average submitted/allowed/paid amounts. | Supplier concentration and rate benchmarks need provider grain, not national sums.                                                                                        | CMS Medicare Physician & Other Practitioners |
| PSPS_Denial_Series + PSPS_Detail_*                                                                                               | Denial behaviour by code, modifier and level of service, 2010-2024.                                         | Denial rates are the revenue-cycle risk signal; no published table carries this series.                                                                                   | CMS Physician/Supplier Procedure Summary     |
| Market_Saturation_Ambulance + MS_County_*, MS_CtyWin_*                                                                           | County-level supplier density and utilization.                                                              | Saturation and dark-market analysis is a county question.                                                                                                                 | CMS Market Saturation & Utilization          |
| QCEW_EMS_Employment/State/County/Quarterly                                                                                       | The ambulance industry labor base: employers, jobs, wages, quarterly to 2025.                               | Crew labor is ~69% of ground ambulance cost; this is the cost-inflation instrument.                                                                                       | BLS QCEW NAICS 621910                        |
| Facility_Universe_State + Hosp/SNF/Dialysis/IRF/LTCH/Hospice/HHA registries, PECOS_Registry                                      | The universe of transfer endpoints and enrolled ambulance suppliers, row per facility.                      | Origin-destination analysis needs the actual endpoint rolls, not counts.                                                                                                  | CMS Provider Data Catalog; PECOS             |
| HSA_Hospital_Catchment + HSA_Corridors                                                                                           | Which ZIPs each hospital actually draws from - the top-15 corridors per hospital.                           | Transfer-lane planning is a corridor question; this is the only public flow file.                                                                                         | CMS Hospital Service Area file               |
| State_Age_65plus, County_Age_65plus, PLACES_County_Chronic                                                                       | The demand base: 65+/85+ population and chronic-disease prevalence by county.                               | Transport demand is driven by age and morbidity where the patient lives.                                                                                                  | Census Vintage 2024 estimates; CDC PLACES    |
| HCRIS_Hospital_Panel                                                                                                             | Hospital cost-report fundamentals, 17,974 hospital-years.                                                   | Occupancy and finance context for the sending side.                                                                                                                       | CMS HCRIS extract (vendored in repo)         |
| LEIE_Ambulance_Exclusions                                                                                                        | Ambulance-related program-integrity exclusions.                                                             | Counterparty screening; the OIG file is the reference list.                                                                                                               | HHS OIG LEIE                                 |



|                                          |                                                                                                                |                                                                    |                                                                  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|
| Metro_ * families + SP_ * state profiles | The same evidence re-cut per target metro and per state, formula-driven off the registry tabs.                 | Market entry decisions are made metro by metro and state by state. | Derived - whole-column INDEX/MATCH/SUMIFS over the registry tabs |
| Company/lnDepth/Reference tab families   | Connector-derived and repository evidence: unit economics, growth registries, question-by-question deep dives. | Carries the RCM_MC evidence base into the same sourcing regime.    | Repository modules + documents on Source_Index                   |

| 4. Definitional choices that change the answer, and the choice this workbook made |                                                                                                |                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Choice                                                                            | Options                                                                                        | What was chosen and why                                                                                                                                                                                             |
| What counts as interfacility                                                      | (a) Hospital-to-hospital only. (b) Any facility-to-facility pair. (c) Any non-scene transport. | (b), reported by pair. Restricting to (a) hides 2.4M SNF-interface transports; (c) sweeps in dialysis and discharge-home legs that behave nothing alike. Every tab reports pairs separately so the reader can pick. |
| Ground base codes                                                                 | Six codes (A0426-A0429, A0433, A0434) or seven including paramedic intercept A0432.            | Six. A0432 is a rural intercept payment, not a transport (3,312 services in 2024). Stated on Acuity_by_Channel so counts tie.                                                                                       |
| Service basis                                                                     | PSPS submitted services, or MUP-PHY final-action approved services.                            | PSPS for all volumes and dollars; MUP only for supplier counts. The two differ by 1.10x in 2024 and must never be mixed. Data_Quality_Register #34.                                                                 |
| FFS denominator                                                                   | Original Medicare any part (33.9M in 2024) or Part A and B both (27.7M).                       | A and B, the population that can generate a carrier ambulance claim. The choice moves every per-1,000 rate by about a fifth, so it is stated on the tab and in DQ #35.                                              |
| Fractional service counts                                                         | Truncate to integers or sum as floats.                                                         | Sum as floats. PSPS carries decimal service counts on shared services. The revision 1 truncation cost 39 services out of 10.6M. Correction #7.                                                                      |
| Rural closure count                                                               | Sheps 154 since 2010, or Chartis 182.                                                          | Sheps, with the Chartis number recorded beside it and a warning never to mix them. Different conventions on Rural Emergency Hospital conversions. DQ #31.                                                           |
| Air interfacility inference                                                       | Use O-D modifiers for both rotary and fixed-wing, or rotary only.                              | Rotary only. Fixed-wing is 94.6% II (airport to airport), so its modifiers identify runways. DQ #29.                                                                                                                |

5. The data pipeline, stated so it can be rerun

PSPS. GET [https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter\[HCPD\\_CD\]={code}](https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter[HCPD_CD]={code}), paginated at 5,000 rows, one call per HCPD per data year. Fifteen data years, twelve ambulance codes, 392,409 rows. Dataset UUIDs are printed on Verification\_Log Panel F. Two schema traps: the PSPS\_ field prefix begins with data year 2020 (not 2021), and service counts are decimal.

MUP-PHY. Same host, filter[HCPD\_CD]={code}, 2019 and 2024 datasets, eight base codes, 44,614 NPI-by-code rows. Rows with 10 or fewer beneficiaries are suppressed at source, so counts are floors.

Enrollment. Same host, Medicare Monthly Enrollment, filter[BENE\_GEO\_LVL]=National and filter[MONTH]=Year, 2013 to 2025.

Everything else was opened by hand: PDFs downloaded and parsed, HTML read, PubMed queried by PMID, NPPES and the Medicare Coverage Database queried live. Dates of access are on the Source\_Index.

Determinism. No sampling, no imputation, no model. Every figure is a sum, a ratio, or a value copied from a document. The workbook recalculates to zero formula errors across more than 50,000 formulas workbook-wide (verified in an independent LibreOffice recalculation; Verification\_Log Panel K carries the stamped counts).

v3 pipeline. Every API-backed tab is produced by a deterministic pull-assemble-verify pipeline shipped in the repository at RCM\_MC/scripts/ift\_evidence\_v3 (pull.py through pull7.py, assemble.py, verify.py). Each pull is cached to disk and recorded on Pull\_Manifest with the exact endpoint, dataset version UUID, filters, pages fetched, rows kept, SHA-256 of the canonical payload and the UTC retrieval time. Re-running the chain reproduces this workbook byte-for-byte up to retrieval dates.

Dataset versions. CMS datasets are pinned per data year by parsing the version UUID out of the public DCAT catalog, so a later CMS re-release cannot silently change a number: the UUID on Pull\_Manifest is the version that produced the cell.

Verification gates. V1: every carried v2.7 cell must recalculate to its original value (10,679 cells compared, zero differences allowed). V2: the whole workbook must recalculate with zero Excel error cells. V3: derived cells on pull-backed tabs are recomputed independently in python from the cached payloads. V7: scale gates on tabs, printed pages and file size. V8: fact and source IDs must be contiguous with no ghost references. The build runs twice so the verified numbers are stamped back into Verification\_Log Panel K and re-verified.

6. Rules for comparing numbers, which is where diligence workbooks usually break

Same source, same year, or say so. The 5.0x rotary air multiple divides FAIR Health Medicare allowed by FAIR Health Medicare allowed, both 2020, both from one white paper. The ground multiple of roughly 1.6x divides FAIR 2020 commercial by a 2024 claims-derived Medicare figure and is therefore labeled CROSS-YEAR AND CROSS-BASIS on its own tab. It is bounding evidence, not a rate card.

Never add a patient count to a vehicle count. HCUP counts discharges, NEMSIS counts activations, PSPS counts billed services. Cross\_Source\_Recon enumerates the wedge drivers instead of averaging four incompatible numbers into a fifth wrong one.

Deflate before you interpret. Any Medicare fee-for-service series read as demand must first be divided by fee-for-service enrollment, which fell 15.5% over 2019 to 2024 (Utilization\_Normalized).

A suppressed cell is a floor, not a zero. CMS masks cells under 11 services in PSPS and under 11 beneficiaries in MUP. Every count in this workbook derived from those files is a floor and says so.

A confidence interval that does not exist cannot be invented. HCUP weighted summary statistics carry no standard errors. HCUPnet does NOT supply them; they require licensed microdata and the AHRQ variance method (finding confirmed by independent external review, 10 July 2026). Until that route is taken, no HCUP trend in this workbook is called significant.

| 7. Corrections made to prior work, kept visible on purpose |                                                                                                            |                                                                                                                                                     |                          |  |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--|
| #                                                          | What was wrong                                                                                             | What is right                                                                                                                                       | Where                    |  |
| 1                                                          | CY2024 Ambulance Inflation Factor carried as 3.0%                                                          | 2.6%, per the CMS manual and MedPAC                                                                                                                 | Payment_Rules            |  |
| 2                                                          | Medicare FFS ambulance spend as \$4.0B in 2023                                                             | \$3.9B in 2022 program spending                                                                                                                     | Payment_Rules            |  |
| 3                                                          | GADCS for-profit cost per transport as \$1,778                                                             | \$1,788                                                                                                                                             | Cost_and_Capacity        |  |
| 4                                                          | GADCS mean of \$2,673 quoted without its median                                                            | Median is \$1,340. The mean is skewed and MedPAC says the data cannot support a margin                                                              | Cost_and_Capacity        |  |
| 5                                                          | Conversion factor assumed at \$278.00                                                                      | \$272.44, CY2024, published                                                                                                                         | Payment_Rules            |  |
| 6                                                          | 2007 NIS total omitted 16,542 records                                                                      | Restated                                                                                                                                            | Acute_IFT_Series         |  |
| 7                                                          | PSPS service counts truncated to integers                                                                  | Decimal counts summed as floats. 871,753 becomes 871,756 on the 2024 HH base set                                                                    | Medicare_OD_Matrix       |  |
| 8                                                          | PSPS_ field prefix said to begin in 2021                                                                   | It begins with data year 2020                                                                                                                       | Verification_Log         |  |
| 9                                                          | Two draft read-notes overstated their own findings                                                         | Corrected against computed values before release: HH per-capita decline is 8.0% (not flat) and the HH emergency-share shift is 2.4 points (not 5.0) | Verification_Log Panel G |  |
| v3                                                         | Cell-level corrections made during the v3 builds (citation attribution, dead links, count inconsistencies) | Logged one per row with the old value, the new value and the reason                                                                                 | V3_Change_Log            |  |

| 8. What this workbook still cannot do, and what would fix it |                                                                    |                                                                                               |                            |  |
|--------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------|--|
| Gap                                                          | Why it exists                                                      | The fix                                                                                       | Cost                       |  |
| No confidence interval on any HCUP trend                     | Weighted summary statistics ship without standard errors           | License NEDS or NIS microdata and apply AHRQ Methods 2015_09; HCUPnet is point estimates only | Purchase plus DUA          |  |
| No hospital-pair transfer flows                              | National HCUP files mask hospital identity                         | Purchase one state SID/SEDD with hospital IDs                                                 | Low four figures per state |  |
| No corporate ownership roll-up                               | An NPI is not a parent company                                     | Build the PECOS reassignment and ownership graph                                              | Free, engineering time     |  |
| No real commercial rate card                                 | FAIR Health publishes averages, not rates                          | Parse Transparency in Coverage machine-readable files for A042x by NPI                        | Free, heavy compute        |  |
| No Medicare Advantage transport volume                       | MA encounter data is not public                                    | Encounter data via ResDAC DUA, or a licensed claims panel                                     | Months, or paid            |  |
| No activation-level IFT flag nationally                      | NEMSIS public reports are aggregated                               | Request the NEMSIS research dataset                                                           | Free, paperwork            |  |
| No 2025 EMS data                                             | The NEMSIS 2025 annual report is not published                     | Expected around August 2026                                                                   | Wait                       |  |
| No Medicare ambulance margin                                 | MedPAC found GADCS reporting errors severe enough to disqualify it | MedPAC Recommendation 6 asks Congress to keep collecting, streamlined                         | Years                      |  |

9. How to verify any number in sixty seconds

Step 1 - read the colour. Blue: the value was typed from the cited document and nothing computed it. Black: a formula; every input is itself blue, black or green, so the arithmetic is fully inspectable. Green: a live link to another tab - follow it.

Step 2 - find the fact. Facts F01-F621 live on Fact\_Ledger: one row per fact with its value, unit, year, tier, source ID, the locator inside the source (table, page or cell) and the tab the fact lives on.

Step 3 - follow the source. Sources S01-S446 live on Source\_Index with publisher, document title, URL and access date. For API-backed cells, Pull\_Manifest carries the exact endpoint, dataset version UUID, filters and the SHA-256 of the cached payload that produced the number.

Step 4 - recompute. Open the workbook and let it recalculate: more than 50,000 formulas resolve with zero errors, and every carried v2.7 cell reproduces its original value. The Index tab links to all 335 tabs; the repository pipeline re-runs the pulls end to end.

*This tab was rebuilt in v3.3: same decision record, house-style layout. Carried sections 1-8 are v2.7 wording verbatim except three repairs - the data-api URL artifact ("revision 1" for "v1") was restored, the fact-ID range and the workbook formula count were updated to current truth - plus the v3 pipeline rows in section 5, the v3 family rows in section 3 and new section 9. Logged on V3\_Change\_Log.*

Verification log

This log records each verification pass, panel by panel, including the reconciliations that matched to the cent and the working hypotheses that were tested and rejected. The rejected ones stay visible on purpose: an audit trail that only records successes is not an audit trail.

Panel A. Values carried in this workbook, checked against the document that published them.

| Evidence layer                            | Values checked | Matched exactly | Corrections | How it was checked                                                                                                                                                                             | Primary document opened                                                                                 | Outcome                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|----------------|-----------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMS PSPS ambulance claims                 | 60             | 60              | -           | Downloaded 110,568 raw claim-summary rows for 2022-2024 from the CMS data.cms.gov API and re-aggregated every origin-destination pair, every acuity HCPCS and every annual total from scratch. | data.cms.gov/data-api/v1/dataset/(usid)/data, Physician/Supplier Procedure Summary, 2022, 2023 and 2024 | PERFECT: 12 origin-destination pairs across three years, the residual bucket, the annual totals, and all six hospital-to-hospital acuity codes across three years. Every cell matched to the unit.                                                                                                           |
| HCUP NEDS, DISP_ED                        | 68             | 67              | 1           | Downloaded 20 AHRQ weighted summary statistics PDFs (2007-2023, plus the revised 2007 and 2011 editions) and parsed the DISP_ED frequency table from each.                                     | hcup-us.ahrq.gov/db/nation/heds/stats/NEDS_(year)_(MaskedStats_Core_Weighted).PDF                       | 67 of 68 exact. The workbook correctly uses the REVISED (revision 2) editions for 2007 and 2011, where AHRQ restated transfers by up to 1.24%. The single gap is data year 2023 destination-unknown, and it is a source-mixing issue, not an error. See Cross_Source_Recon.                                  |
| HCUP NIS, DISPUNIFORM and TRAN_IN         | 74             | 73              | 1           | Downloaded 17 AHRQ NIS weighted summary statistics PDFs and parsed DISPUNIFORM, TRAN_IN, TRAN_OUT and ED_BIRTH.                                                                                | hcup-us.ahrq.gov/db/nation/nis/tools/stats/                                                             | 73 of 74 exact. 2008 through 2023 is flawless. TRAN_OUT = 1 equals DISPUNIFORM = 2 in every single year, confirming it is a reliable and not independent evidence. The one gap (the 2007 NIS total omits 16,542 discharged alive, destination unknown) records. 2007 is excluded from every trend statistic. |
| AHA community hospital admissions         | 4              | 4               | -           | Downloaded the AHA Fast Facts PDFs and read the admissions lines directly.                                                                                                                     | aha.org, Fast Facts on US Hospitals, 2025 and 2026 editions                                             | EXACT. 2023: total 34,436,650, community 32,345,846. 2024: total 35,658,563, community 33,053,725. The 2023 community figure is the denominator in the 2023 IFT ratio.                                                                                                                                       |
| NEMSIS 2024 interfacility counts          | 7              | 7               | -           | Downloaded the NEMSIS 2024 National EMS Data Report and parsed the eResponse.05 table.                                                                                                         | nemsis.org, NEMSIS End-of-Year Report 2024                                                              | EXACT. Total 60,298,684. Hospital-to-Hospital 5,510,664. Other Routine Medical Transport 4,467,204. Hospital to Non-Hospital 1,418,924. Non-Hospital to Non-Hospital 335,432. Non-Hospital to Hospital 247,636. Transports to a hospital ED 27,768,728.                                                      |
| Ambulance Inflation Factor, CY2020-CY2025 | 6              | 5               | 1           | Read the full AIF history table in the CMS Claims Processing Manual.                                                                                                                           | CMS Pub. 100-04, Ch.15 Sec.20.4, Transmittal 13464 (14 Nov 2025)                                        | CORRECTION. CY2024 is 2.6%, not the 3.0% the source market study carried. All other years match. CY2026 (2.0%) added as new.                                                                                                                                                                                 |
| Ambulance RVUs, seven ground levels       | 7              | 7               | -           | Read the RVU table in both the CMS public use file description and the MedPAC fact sheet.                                                                                                      | CMS Ambulance Fee Schedule public use file; MedPAC Payment Basics (Ambulance), Oct 2024, Table 1        | EXACT on all seven. BLS 1.00, BLS-emergency 1.60, ALS1 1.20, ALS1-emergency 1.90, paramedic intercept 1.75, ALS2 2.75, SCT 3.25. Upgraded from GOV-B to GOV-A.                                                                                                                                               |
| GADCS cost per transport                  | 3              | 2               | 1           | Downloaded the 201-page RAND report and read Table 6.1.                                                                                                                                        | RAND Corporation for CMS, GADCS Report: Year 1 and Year 2 Cohort Analysis, 19 Dec 2024                  | CORRECTION. Mean \$2.673 and government \$3.127 confirmed. Private for-profit is \$1,788, NOT the \$1,778 that circulates in the secondary coverage.                                                                                                                                                         |
| GADCS crew labor share of cost            | 1              | 1               | -           | Read Figure 5.1 and Table 5.1.                                                                                                                                                                 | Same report                                                                                             | CONFIRMED at 69.4%: labor is \$18,869M of the \$27.2B aggregated ground ambulance cost base. An earlier draft of this workbook flagged this as an unverifiable conflation. That flag was wrong and has been withdrawn.                                                                                       |
| Medicare FFS ambulance spend              | 1              | -               | 1           | Read p.1 of the fact sheet, then cross-checked against the raw claims.                                                                                                                         | MedPAC, Payment Basics: Ambulance Services Payment System, Oct 2024                                     | CORRECTION. Source workbook said \$4.08 for 2023. MedPAC reports \$3.9B of program spending in 2022, excluding beneficiary cost sharing. Reconciled against claims: carrier-billed ground NCH payments were \$3.72B in 2022; air and institutional claims close the gap.                                     |
| NEDS 2023 dual-table discrepancy          | 2              | 2               | -           | Downloaded the 2023 NEDS Introduction PDF (published 14 Jan 2026) and located both tables.                                                                                                     | hcup-us.ahrq.gov/db/nation/heds/NEDSIntroduction2023.pdf                                                | BOTH FIGURES CONFIRMED AS PRINTED. Table A7 reports 21,049,219 ED admissions and 2,552,895 transfers. Table D3, in the SAME document, reports 20,973,122, matching the Weighted Summary Statistics. Table A7's deltas sum to exactly zero: a reallocation, not a revision.                                   |
| CMS PSPS, 2010-2021 extension             | n/a            | n/a             | n/a         | Pulled every claim row carrying the HH origin-destination modifier for twelve additional data years, using the same API. Pre-2021 files drop the PSPS_ prefix on the amount fields.            | data.cms.gov, Physician/Supplier Procedure Summary, 2010 through 2021                                   | NEW SERIES. No values to check: this evidence did not exist in any source workbook. Produces the fifteen-year Medicare interfacility series on Medicare_IFT_Series. The 2022-2024 overlap reproduces the verified figures exactly.                                                                           |
| MedPAC June 2026 mandated report          | n/a            | n/a             | n/a         | Downloaded the 325-page June 2026 Report to the Congress and read Chapter 6, including Tables 6-6, 6-10, 6-11 and Recommendation 6.                                                            | MedPAC, Report to the Congress, June 2026, Chapter 6, published 15 June 2026                            | NEW SOURCE, PUBLISHED AFTER THE SOURCE WORKBOOKS. Supersedes the RAND GADCS unweighted means for operator benchmarking, quantifies the scale effect on cost, and states that GADCS-based revenue-to-cost ratios are not reliable indicators of payment adequacy today.                                       |
| CMS Timely and Effective Care             | n/a            | n/a             | n/a         | Queried the CMS provider-data API for the national, state and hospital files and re-derived every statistic from the returned rows (2,340 hospital records for OP-18d, 1,736 state records).   | data.cms.gov/provider-data, datasets ism-hgvy, apyc-v239, yv7e-xt69                                     | NEW SOURCE. Provides the only direct national measurement of interfacility transfer delay: OP-18d, median 287 minutes. Not present in any source workbook.                                                                                                                                                   |
| AHA Fast Facts, 2020 edition              | n/a            | n/a             | n/a         | Downloaded and read the PDF for the survey-year 2018 baseline.                                                                                                                                 | aha.org, Fast Facts on U.S. Hospitals, 2020 edition                                                     | NEW SOURCE. Community hospitals 5,198, in a system 3,491; staffed beds 792,417. Enables the 2018 to 2024 consolidation series.                                                                                                                                                                               |
| Peer-reviewed literature (Publbed)        | n/a            | n/a             | n/a         | Searched PubMed and retrieved article metadata and abstracts for four studies on interfacility transfer, travel time and ED boarding outcomes.                                                 | PubMed. DOIs recorded on Source_Register S35 through S38.                                               | NEW SOURCE. Quantifies travel time as a transfer driver (aOR 1.32) and establishes that the naive dwell-time to mortality correlation runs backwards.                                                                                                                                                        |
| TOTAL                                     | 233            | 228             | 5           | Match rate                                                                                                                                                                                     |                                                                                                         | 97.9% Five corrections, all applied in this file. Four are real errors in the source workbooks; the fifth is a source-mixing issue on one AHRQ cell.                                                                                                                                                         |

Panel B. The five corrections, stated plainly.

| # | What the source workbook said                 | What the primary document says                                      | Primary document                                 | Materiality                                                                                                                  |
|---|-----------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1 | CY2024 Ambulance Inflation Factor = 3.0%      | CY2024 Ambulance Inflation Factor = 2.6%                            | CMS Pub 100-04 Ch.15 Sec.20.4                    | Moves the trailing three-year average from 4.7% to 2.3%. Material to any rate-growth assumption.                             |
| 2 | Medicare FFS ambulance spend = \$4.08 in 2023 | \$3.98 in 2022, program spending excluding beneficiary cost sharing | MedPAC Payment Basics (Ambulance), Oct 2024, p.1 | Wrong value and wrong year. A separate MedPAC figure of \$5.38 for 2023 is TOTAL payments. Confusing the two is a 36% error. |

|                                                                                                                                                                                                                                                                                                                                                                               |                                                                     |                                                                     |                                                         |                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 3                                                                                                                                                                                                                                                                                                                                                                             | GADCS private for-profit mean cost per transport = \$1,778          | \$1,788                                                             | RAND GADCS Year 1 and Year 2 Cohort Analysis, Table 6.1 | Small. But it propagated from secondary coverage into the source workbook, which is the point.                                    |
| 4                                                                                                                                                                                                                                                                                                                                                                             | GADCS mean cost per transport of \$2,673, quoted without its median | Mean \$2,673, MEDIAN \$1,340                                        | Same report, Table 6.1                                  | Large. The mean is dragged up by thousands of tiny low-volume organizations. Quoting the mean as the operator benchmark is wrong. |
| 5                                                                                                                                                                                                                                                                                                                                                                             | NIS 2007 total discharges = 39,525,405                              | 39,541,947. The workbook omits 16,542 'destination unknown' records | AHRQ NIS 2007 weighted summary statistics               | Immaterial. 2007 precedes the 2012 NIS redesign and is excluded from every trend statistic in this workbook.                      |
| 6 This workbook claimed MedPAC's \$3.9B and \$5.3B ambulance spend figures 'both reconcile against the claims once air and institutional claims are added'. They do not. Carrier-billed ground plus air program payments were \$4,20B in 2022. ABOVE MedPAC's \$3.9B. Neither MedPAC document states the scope of its measure. CMS PSPS, air codes A0430, A0431, A0435, A0436 |                                                                     |                                                                     |                                                         |                                                                                                                                   |
| Material to anyone sizing the Medicare ambulance market. The claim is withdrawn and replaced with a bounded statement. See Medicare_PSPS Panel G.                                                                                                                                                                                                                             |                                                                     |                                                                     |                                                         |                                                                                                                                   |

Panel C. Evidence obtained during verification that exists in NO source workbook.

| Item                                                  | Value                                        | Where it came from                         | Why it matters                                                                                                 |
|-------------------------------------------------------|----------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| CY2024 ground ambulance conversion factor             | \$272.44                                     | MedPAC Payment Basics (Ambulance), Table 1 | The source workbook ASSUMED \$278.00 and labeled it ILLUSTRATIVE, which is why no dollar figure could survive. |
| CY2024 ground mileage rate                            | \$8.76 per statute mile                      | Same                                       | Confirmed independently by the claims: \$8.79 allowed per loaded mile.                                         |
| Allowed charges, hospital-to-hospital base rate, 2024 | \$319,615,284                                | CMS PSPS, PSPS_ALLOWED_CHARGE_AMT          | Real Medicare dollars, not an RVU-times-conversion-factor estimate.                                            |
| Allowed charges, hospital-to-hospital mileage, 2024   | \$267,413,039                                | CMS PSPS, HCPCS A0425, modifier HH         | Mileage is 44.6% of the revenue on an interfacility transport.                                                 |
| Loaded miles per hospital-to-hospital transport, 2024 | 33.6                                         | CMS PSPS                                   | Roughly three times a typical 911 scene run.                                                                   |
| Medicare payment as a share of allowed                | 77.7%                                        | CMS PSPS, PSPS_NCH_PAYMENT_AMT             | Statute sets Part B at 80% after the deductible. The realized 77.7% validates the entire extraction.           |
| GADCS median cost per transport                       | \$1,340                                      | RAND GADCS Table 6.1                       | The mean of \$2,673 is 1.99x the median. This changes the unit-economics conclusion.                           |
| GADCS cost per RVU                                    | \$1,624 mean, \$811 median                   | RAND GADCS Table 6.1                       | Directly comparable to the \$253 per RVU that Medicare allows on the HH book.                                  |
| GADCS cost composition                                | Labor 69.4%, vehicles 10.0%, facilities 3.5% | RAND GADCS Figure 5.1 and Table 5.1        | Replaces the excluded trade-benchmark cost split.                                                              |
| NEDS 2023 sampling structure                          | 4,489 target EDs; 3,185 frame; 963 sampled   | NEDS Introduction 2023, Table A6           | Quantifies the 2023 frame-coverage break.                                                                      |

Panel D. What remains unverifiable, by anyone, at any effort.

AHRQ's discharge weights are calibrated to the AHA Annual Survey Database, which is not public at the hospital level. The weighted counts in NEDS and NIS cannot be reproduced from public data by any external party. They are taken on AHRQ's authority, and everything downstream inherits that. AHRQ publishes no standard errors in the weighted summary statistics. NEDS clusters on emergency departments, and transfer propensity is a hospital-level attribute, so DISP\_ED = 2 carries a large design effect. The +0.86% per year total-episode CAGR is NOT demonstrably different from zero on this evidence. HCUPrnet returns design-based standard errors, free and without a data use agreement, and would settle it.

NEMESIS has no sampling frame and no weights. It is a voluntary submission census. No rational inference and no confidence interval is licensed from it, only mix and direction. CMS suppresses any PSPS cell with fewer than 11 services. In 2024, 24,781 origin-destination-coded rows are masked, hiding at most about 263,000 services, or 2.5% of the 10,637,727 counted. Every PSPS figure in this workbook is a floor by that margin. PSPS is carrier claims only. Hospital-owned ambulance departments bill institutionally and never appear, and Medicare Advantage is entirely absent. The commercial book is not measured by any source in this workbook.

Panel E. The one authoritative source published after this evidence base was assembled.

MedPAC delivered its Bipartisan Budget Act of 2018 mandated report on 15 June 2026: 'Chapter 6: Mandated report: Assessment of the Medicare Ground Ambulance Data Collection System', in the June 2026 Report to the Congress. It assesses the GADCS data and recommends continued but streamlined collection. It is the authoritative source on Medicare payment adequacy for ground ambulance. It IS reflected in this workbook on MedPAC\_2026\_Mandated [state sentence corrected in v3: Panel A row 14 and source S31 record the report as pulled, read, and incorporated].

Reproducibility. PSPS: GET [https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter={HCPCS\\_CD}={code}](https://data.cms.gov/data-api/v1/dataset/{uuid}/data?filter={HCPCS_CD}={code}). Dataset UUIDs: 2022 e4786716-10e9-4092-b936-7444177474d5, 2023 b72aeffb-22cf-4241-9c64-050ad9061e03, 2024 1646c736-4179-4100-9f79-592b69e41975. Ground base codes A0426, A0427, A0428, A0429, A0433, A0434; ground mileage A0425. Suppressed cells are marked with an asterisk and were excluded from sums, not imputed.

Panel F. Revision 2 verification (10 July 2026): new evidence opened, and the revision 1 extraction independently re-tested.

| Evidence layer                       | What was done                                                                                                                                                                  | Outcome                                                                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMS PSPS full O-D rebuild (S39, S40) | Pulled 392,409 raw rows across fifteen PSPS datasets (2019-2024 all twelve ambulance HCPCS; 2010-2018 air codes) and re-aggregated every origin-destination pair with dollars. | PERFECT on dollars: 2022-2024 ground totals reproduce Medicare. PSPS Panel F to the cent (allowed 4,753,662,348 / 4,992,692,676 / 5,073,090,003; paid 3,722,057,066 / 3,892,079,508 / 3,957,457,371). HH 2024 base allowed and mileage allowed match to the cent.    |
| Fractional service counts            | Compared initial hospital-pair transport counts against float-summed rebuild; isolated the gap.                                                                                | Correction 7. PSPS SUBMITTED_SERVICE_CNT contains fractional values (8 rows in the 2024 HH base set). The initial pipeline truncated: 871.753 against a float sum of 871.756 (+0.0003%). All revision 2 tabs sum floats. Original cells left as printed; immaterial. |
| PSPS field-name boundary             | Tested the 2020 dataset schema directly.                                                                                                                                       | Correction 8, to an earlier reproducibility note. The PSPS_ field prefix begins with DATA_YEAR 2020, not 2021 as an earlier note stated. 2010-2019 files carry unprefixd names.                                                                                      |
| Census NP2023-T2 (S50)               | Downloaded and parsed the xlsx from www2.census.gov.                                                                                                                           | EXACT values carried: 65+ 57,795k (2022), 71,183k (2030), 82,130k (2050), main series.                                                                                                                                                                               |
| NHAMCS 2022 Table 4 (S55)            | Downloaded the ED Summary Tables PDF and read Table 4.                                                                                                                         | EXACT: 17.0% (SE 0.9) ambulance arrival on 155,398k visits; age gradient carried. Final survey vintage.                                                                                                                                                              |
| Medicare Coverage Database (S56)     | Queried active LCDs and articles titled ambulance via the MCD connector.                                                                                                       | Zero active ambulance LCDs; one billing article, A52917 (Noridian Part A, rural air ambulance protocols, eff. 10/01/2015, updated 11/22/2023).                                                                                                                       |

|                                           |                                                                                                                                                                                        |                                                                                                                                                                    |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NPPES (S58)                               | Queried the registry live for ambulance taxonomy organizations.                                                                                                                        | Taxonomy 34160000X and 3416L0300X confirmed on active records (example NPIs on Code_Crosswalks).                                                                   |
| PubMed retrievals (S42, S53, S54)         | Retrieved metadata and abstracts for the three studies carried in v2.                                                                                                                  | PMIDs 359777218, 31989591, 41324958 with digital object identifiers recorded on the Source_Register.                                                               |
| Documents opened (S41, S43-S49, S51, S52) | CMS RSNAT page and press release; FAIR Health white papers and brief; GAPB report; Commonwealth Fund analysis; MACPAC Ch.5; HHS NEMT report; KFF 2026 enrollment brief; Sheps tracker. | Figures carried as printed, with each locator on the Source_Register. The Sheps and Charis closure counts use different conventions and are both recorded; DQ #31. |

Panel G. Revision 3 verification (10 July 2026).

| Evidence layer                    | What was done                                                                                                                                   | Outcome                                                                                                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Medicare Monthly Enrollment (S60) | Pulled national annual-average rows 2013-2025 from the API and carried four enrollment measures.                                                | EXACT values carried; A and B FFS denominator convention stated on the tab. Per-1,000 rates recompute inside the workbook by formula.                                                                                       |
| MUP-PHY supplier pull (S59)       | Pulled every NPI-by-code row for eight ambulance base HCPCS, 2019 and 2024 (44,614 rows), and aggregated counts, concentration and top billers. | Cross-check: MUP 2024 approved ground services 9,542,103 against PSPS submitted 10,637,766 + 1.1x definitional gap (final-action against submitted; suppression), logged as DQ #34.                                         |
| Acuity by channel (S39)           | Re-aggregated the existing PSPS pull by base HCPCS within each D-D channel, 2019 and 2024, six-code convention.                                 | Totals tie to Medicare_OD_Matrix (A0432 intercept excluded, 3,312 services in 2024). Two draft read-notes were corrected against computed values before release (HH per-capita minus 8.0%; HH emergency share +2.4 points). |
| Add-on statutory status (S61)     | Read the CMS AFS page and corroborating notices for the current expiration.                                                                     | CAA 2026 section 6203: extended through 31 December 2027; cliff 1 January 2028; Oct 2025 lapse history recorded on Payment_Rules.                                                                                           |

Panel H. Revision 4 verification (10 July 2026).

| Evidence layer                                | What was done                                                                                                                                                                                  | Outcome                                                                                                                                                                                                                           |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MUP-PHY decade series (S62)                   | Pulled every NPI-by-code row, six ground base codes, data years 2014-2018 and 2020-2023, plus 2024 city, state and ZIP fields. Dataset identifiers by data year are recorded under source S62. | Eleven-year supplier series on Supplier_Series_Raw. Endpoints reconcile exactly with the revision 3 two-point pull.                                                                                                               |
| Institutional ambulance route (Fault_Audit A) | Probed the Medicare Outpatient Hospitals by Provider and Service PUF with HCPCS filters for all twelve ambulance codes.                                                                        | Tested and closed: the file is APC-level and the HCPCS filter is silently ignored (identical 116,799-row universe returned for every code). Hospital-billed ambulance remains publicly uncountable; recorded on IFT_Alt_Measures. |
| OEWS May 2019 (S67)                           | Read the 29-2040 occupation page: national, industry, state, metro tables.                                                                                                                     | EXACT values carried; May 2024 taken from the OOH (S68). BLS bulk zips refused scripted access; full state files pending (P4).                                                                                                    |
| Scissors self-correction (Finding 39)         | The initial working hypothesis was that wages outran the AIP 2019-2024. Completed: EMT median +18.6% against AIF +18.6%.                                                                       | Hypothesis rejected on the five-year window and the finding rewritten to the sub-period and composition mechanisms. Rejected hypotheses are retained as part of the audit trail.                                                  |
| California APOT (S71)                         | Parsed the EMSA December 2024 commission report: statewide trend, LEMSA heat table, cumulative delay hours.                                                                                    | EXACT: 19.1% of offloads over 30 minutes; 13,364:40:19 cumulative hours, Apr-Sep 2024; San Diego alone 738:54:57.                                                                                                                 |
| Ambulance deserts (S65)                       | Verified the national figures on the Rural Health Research Gateway; the full chartbook PDF is pending retrieval.                                                                               | Headline figures pinned including the nine-missing-states undercount caveat; state and county tables pending (P2).                                                                                                                |
| MSA mapping coverage                          | Computed the share of national volume the principal-city method assigns to a CBSA.                                                                                                             | 21.8%. Printed on MSA_Landscape as the governing caveat; the full ZCTA crosswalk is pending item P3.                                                                                                                              |
| Access failures, logged not hidden            | Four bulk files could not yet be retrieved programmatically: the BLS OEWS archives, the Census ZCTA relationship file, the NASEMSO assessment, and the deserts chartbook PDF.                  | Each is named on Fault_Audit item L with its fallback source. Nothing was silently substituted.                                                                                                                                   |
| State cut cross-validation (S73)              | Pulled the official by-Geography state file and reconciled against the NPI-file aggregation.                                                                                                   | National gap +1.0% (9,637,009 vs 9,542,103), attributable to NPI-file small-cell suppression: the suppression tail is now BOUNDED, closing part of Fault_Audit item C and DQ #32. Official column carried on State_Saturation.    |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final consolidation, 10 July 2026 | Confirmed the distributed copy is byte-identical to this build; recounted the Source_Index tiers programmatically; corrected the stale 61-source title; repaired one gap-table cost cell; refreshed the HCUPnet claims across all tabs after the independent external review.                                                                                                                                              | The master is internally consistent at 47 tabs, 77 sources [corrected in v3: the rev-5 note undercounted the four client-alignment/connector tabs and sources S74-S77], facts through F165, findings through 41, zero formula errors. Open items live only on the README pending register (P1-P19). |
| Chart integrity audit             | Every chart series was programmatically resolved to its cell range and counted for numeric content; scale-mismatched two-series charts were rebuilt as dual-axis combinations so percentage series render visibly; all ten original charts received explicit titles.                                                                                                                                                       | 29 charts, all series resolving, palette-consistent. Percentage series that previously flatlined against count axes now plot on a right-hand axis.                                                                                                                                                  |
| Client-alignment layer            | Two tabs added from the engagement kickoff guidance: the two-build sizing structure with its segmentation, gates, and archetype specification, and the source-to-question map. No client-specific figures were ingested; every awaited input is a labeled empty cell.                                                                                                                                                      | Sizing_Playbook and Engagement_Data_Map live after Imbalance_Ledger; the README map carries the new section.                                                                                                                                                                                        |
| Connector layer                   | The workbook now draws on the live healthcare connectors attached to this engagement workspace. The PubMed connector retrieved and pinned three national NEDS transfer studies (S74 to S76) with full identifiers; the NPI Registry connector is loaded for on-demand supplier verification against NPPES; the PrognHIVE surveillance connector is available for a demand-seasonality layer pending approval to invoke it. | Condition_Transfer_Anchors added with three Tier A anchors; pending item P1 partially advanced.                                                                                                                                                                                                     |
| Analytical chart set              | Four charts added that read relationships rather than levels: the state structure scatter (supplier density against utilization), the concentration two-line showing movement confined to post-2021, the scheduled-transport teller exit, and the dual-axis pairing of the shrinking measurable book with the climbing dark share.                                                                                         | 33 charts, all series verified resolving.                                                                                                                                                                                                                                                           |
| National TAM model                | Built the three-scenario national model on a fully measured Medicare spine (claims, MedPAC, enrollment, code mix, all green-linked), a residual-based payer-lives panel (S77), a nine-row assumption register with <i>numbers and replacement notes, four notes</i>                                                                                                                                                        | Outputs live on TAM_Model_National Panel A; Fault_Audit item T governs quotation.                                                                                                                                                                                                                   |

|                                                                       |                                                                                                                                       |  |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--|
| Panel I (v3). Faithful-copy proof: v2.7 carried forward without loss. |                                                                                                                                       |  |
| Method                                                                | Every v2.7 sheet copied cell-by-cell (values, formulas, styles, comments, merges, widths, freeze panes) by the committed v3 pipeline. |  |
| Cells compared                                                        | 11,386                                                                                                                                |  |
| Differences                                                           | 0                                                                                                                                     |  |
| Excel error cells after recalc                                        | 0                                                                                                                                     |  |
| Charts                                                                | All 37 v2.7 chart objects re-created natively from their parsed XML (openpyxl loses chart objects on round-trip); series references   |  |

|                                                                                                                                                       |           |             |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------------------|
| Panel J (v3). Live-pull verification: every dataset pulled for the new tabs, with row counts and cross-checks. Full reproduce specs on Pull_Manifest. |           |             |                   |
| Dataset                                                                                                                                               | Artifacts | Rows cached | Access date (UTC) |
| (re-registered by reconcile_manifest after a concurrent-write race; original pull meta lost, payload hash is of the surviving canonical file)         |           | 26          | 24,813 2026-07-10 |
| BLS Occupational Employment and Wage Statistics (OEWS), May 2024, state file - EMS occupations                                                        |           | 1           | 144 2026-07-10    |
| BLS QCEW NAICS 621910 (Ambulance Services), annual averages                                                                                           |           | 12          | 1,208 2026-07-10  |
| BLS QCEW NAICS 621910 (Ambulance Services), county annual averages, private ownership                                                                 |           | 12          | 21,760 2026-07-10 |
| BLS QCEW NAICS 621910, quarterly                                                                                                                      |           | 48          | 4,832 2026-07-10  |

|                                                                                                                                                                           |     |           |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|------------|
| CDC PLACES: County Data (2023 release) - chronic kidney disease                                                                                                           | 1   | 3,077     | 2026-07-10 |
| CDC PLACES: Local Data for Better Health, County Data                                                                                                                     | 5   | 14,790    | 2026-07-10 |
| CMS Acute Hospital Care at Home Program - Approved List of Hospitals as of 4/5/2021 (only public machine-readable roster); current roster PENDING (JS portal + DUA-gated) | 1   | 27        | 2026-07-10 |
| CMS Hospital Enrollments (PECOS public enrollment file), Rural Emergency Hospital conversions                                                                             | 1   | 48        | 2026-07-10 |
| CMS Provider Data Catalog: Dialysis Facilities                                                                                                                            | 1   | 7,557     | 2026-07-10 |
| CMS Provider Data Catalog: Dialysis Facilities (full roster)                                                                                                              | 1   | 7,557     | 2026-07-10 |
| CMS Provider Data Catalog: Dialysis Facilities (rich fields)                                                                                                              | 1   | 7,557     | 2026-07-10 |
| CMS Provider Data Catalog: Dialysis Facility - Listing by Facility                                                                                                        | 1   | 7,557     | 2026-07-10 |
| CMS Provider Data Catalog: Home Health Agencies (rich fields)                                                                                                             | 1   | 12,392    | 2026-07-10 |
| CMS Provider Data Catalog: Home Health Care Agencies                                                                                                                      | 1   | 12,392    | 2026-07-10 |
| CMS Provider Data Catalog: Hospice Providers (rich fields)                                                                                                                | 1   | 6,852     | 2026-07-10 |
| CMS Provider Data Catalog: Hospice - General Information                                                                                                                  | 1   | 6,852     | 2026-07-10 |
| CMS Provider Data Catalog: Hospital General Information                                                                                                                   | 1   | 5,432     | 2026-07-10 |
| CMS Provider Data Catalog: Hospital General Information (rich fields)                                                                                                     | 1   | 5,432     | 2026-07-10 |
| CMS Provider Data Catalog: Inpatient Rehabilitation Facilities (rich fields)                                                                                              | 1   | 1,222     | 2026-07-10 |
| CMS Provider Data Catalog: Inpatient Rehabilitation Facility - General Information                                                                                        | 1   | 1,222     | 2026-07-10 |
| CMS Provider Data Catalog: Long-Term Care Hospital - General Information                                                                                                  | 1   | 311       | 2026-07-10 |
| CMS Provider Data Catalog: Long-Term Care Hospitals (rich fields)                                                                                                         | 1   | 311       | 2026-07-10 |
| CMS Provider Data Catalog: Nursing homes including rehab services (rich fields)                                                                                           | 1   | 14,695    | 2026-07-10 |
| CMS Provider Data Catalog: Nursing homes including rehab services - Provider Information                                                                                  | 1   | 14,695    | 2026-07-10 |
| CMS Provider Data Catalog: Rural Emergency Hospital Timely and Effective Care - Hospital (deduped roster)                                                                 | 1   | 41        | 2026-07-10 |
| CMS Provider Data Catalog: Skilled Nursing Facility Quality Reporting Program - Provider Data                                                                             | 1   | 14,695    | 2026-07-10 |
| CMS Provider Data Catalog: Timely and Effective Care - Hospital                                                                                                           | 3   | 79,220    | 2026-07-10 |
| CMS Zip Code to Carrier Locality File (ambulance fee schedule rural/super-rural ZIP designations)                                                                         | 1   | 42,956    | 2026-07-10 |
| Census Bureau 2024 Gazetteer Files - ZIP Code Tabulation Areas (national)                                                                                                 | 1   | 33,791    | 2026-07-10 |
| Census Bureau Vintage 2024 County Population Estimates by Age and Sex (CC-EST2024-AGESEX)                                                                                 | 1   | 15,720    | 2026-07-10 |
| Census Bureau Vintage 2024 State Population Estimates by Age and Sex, civilian (SC-EST2024-AGESEX-CIV)                                                                    | 1   | 4,524     | 2026-07-10 |
| EIA Weekly Retail On-Highway Diesel Prices (psw18vw*, all areas)                                                                                                          | 1   | 308       | 2026-07-10 |
| HHS OIG List of Excluded Individuals/Entities (LEIE)                                                                                                                      | 1   | 566       | 2026-07-10 |
| Hospital Service Area                                                                                                                                                     | 1   | 7,452     | 2026-07-10 |
| Hospital Service Area (top-15 ZIP corridors per hospital)                                                                                                                 | 1   | 106,799   | 2026-07-10 |
| Market Saturation & Utilization State-County                                                                                                                              | 1   | 2,340     | 2026-07-10 |
| Market Saturation & Utilization State-County (county grain)                                                                                                               | 6   | 54,756    | 2026-07-10 |
| Market Saturation & Utilization State-County (county grain, rolling window)                                                                                               | 9   | 82,540    | 2026-07-10 |
| Medicare Fee-For-Service Public Provider Enrollment                                                                                                                       | 2   | 20,930    | 2026-07-10 |
| Medicare Monthly Enrollment                                                                                                                                               | 2   | 767       | 2026-07-10 |
| Medicare Physician & Other Practitioners - by Geography and Service                                                                                                       | 156 | 6,315     | 2026-07-10 |
| Medicare Physician & Other Practitioners - by Provider and Service                                                                                                        | 144 | 389,793   | 2026-07-10 |
| Physician/Supplier Procedure Summary                                                                                                                                      | 210 | 1,794,540 | 2026-07-10 |
| UNC Sheps Center rural hospital closures list (2005-present, complete + converted closures)                                                                               | 1   | 197       | 2026-07-10 |
| USDA ERS 2010 Rural-Urban Commuting Area Codes - ZIP code file (approximation)                                                                                            | 1   | 41,164    | 2026-07-10 |
| USDA ERS 2010 Rural-Urban Commuting Area Codes - census tract file (revised 7/3/2019)                                                                                     | 1   | 74,002    | 2026-07-10 |

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Cross-check | PECOS enrolled ambulance suppliers: live found, rows 10,465 = vendored PPEF.                                                |
| Cross-check | NOTE: 24,010 rows in the hospital ID ARE EMPTY. MUP 2023 national AHA25 matches the independent probe made during connector |
|             |                                                                                                                             |

Panel K (v3). Recompute audit: every new DERIVED formula recomputed independently in python from the cached source data, and the whole workbook recalculated to zero error cells before release.

|                                                 |         |
|-------------------------------------------------|---------|
| Formulas recalculated (whole workbook)          | pending |
| Excel error cells                               | pending |
| New derived cells recomputed in python          | pending |
| Mismatches beyond 1e-9 relative tolerance       | pending |
| Printed-page estimate (landscape, fit-to-width) | pending |



Fact ledger

Every fact in this workbook carries an ID here (F01 to F165) with its metric, year, value, evidence tier, source, and the exact locator inside the source document. This is step two of the three-step audit path described on the README.

| Fact ID | Metric                                               | Year | Value                 | Unit | Basis   | Tier | Source ID         | Locator                            | Lives on          | Cross-check                                                                                                     |
|---------|------------------------------------------------------|------|-----------------------|------|---------|------|-------------------|------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| F01     | ED-origin acute-to-acute transfer episodes           | 2022 | 2,311,549 episodes    |      | GOV-A   | A    | S01               | NEDS Summary Stats, DISP_ED = 2    | Acute_IFT_Series  | VERIFIED against the AHRQ PDF. Agrees to the unit with NEDS Introduction 2022 Table A6                          |
| F02     | ED-origin acute-to-acute transfer episodes           | 2023 | 2,554,379 episodes    |      | GOV-A   | A    | S01               | NEDS Summary Stats, DISP_ED = 2    | Acute_IFT_Series  | VERIFIED. NEDS Introduction 2023 Table A7 prints 2,552,895. Gap of 1,484 (0.058%). See Cross_Source_Recon       |
| F03     | Inpatient-origin acute-to-acute transfer episodes    | 2022 | 617,724 episodes      |      | GOV-A   | A    | S04               | NIS Summary Stats, DISPUNIFORM = 2 | Acute_IFT_Series  | VERIFIED. TRAN_OUT = 1 equals this value in every year 2007-2023: a relabel, not independent evidence           |
| F04     | Inpatient-origin acute-to-acute transfer episodes    | 2023 | 630,765 episodes      |      | GOV-A   | A    | S04               | NIS Summary Stats, DISPUNIFORM = 2 | Acute_IFT_Series  | VERIFIED against the AHRQ PDF                                                                                   |
| F05     | Total acute-to-acute transfer episodes               | 2022 | 2,929,273 episodes    |      | DERIVED | A    | S01, S04          | Sum of F01 and F03                 | Acute_IFT_Series  | No overlap: an ED transfer-out is never also an inpatient discharge                                             |
| F06     | Total acute-to-acute transfer episodes               | 2023 | 3,185,144 episodes    |      | DERIVED | A    | S01, S04          | Sum of F02 and F04                 | Acute_IFT_Series  | Memo year. NEDS break in series                                                                                 |
| F07     | Mean acute-to-acute transfer episodes, 2019 to 2023  | 2023 | 3,061,156 episodes/yr |      | DERIVED | A    | S01, S04          | Five-year mean                     | Acute_IFT_Series  | Sample standard deviation 139,226. Range 2,929,273 (2022) to 3,234,266 (2019)                                   |
| F08     | IFT as % of AHA community hospital admissions        | 2022 | 9.28%                 | %    | DERIVED | A    | S01, S04, S09     | F05 over F18                       | Acute_IFT_Series  | Denominator choice moves this by 1.05 points. See Receiving_Side                                                |
| F09     | IFT as % of AHA community hospital admissions        | 2023 | 9.85%                 | %    | DERIVED | A    | S01, S04, S10     | F06 over 32,345,846                | Acute_IFT_Series  | Memo year. Denominator verified in AHA Fast Facts 2025 edition                                                  |
| F10     | Mean IFT % of admissions, 2019 to 2023               | 2023 | 9.49%                 | %    | DERIVED | A    | S01, S04, S06-S10 | Five-year mean                     | Acute_IFT_Series  | Band 9.24% (2021) to 9.85% (2023). Sample standard deviation 0.24 points                                        |
| F11     | ED-origin share of acute-to-acute episodes           | 2012 | 74.4%                 | %    | DERIVED | A    | S01, S04          |                                    | Acute_IFT_Series  | First comparable year after the 2012 NIS redesign                                                               |
| F12     | ED-origin share of acute-to-acute episodes           | 2022 | 78.9%                 | %    | DERIVED | A    | S01, S04          |                                    | Acute_IFT_Series  | Rises 4.5 points over ten years                                                                                 |
| F13     | ED-origin share of acute-to-acute episodes           | 2023 | 80.2%                 | %    | DERIVED | A    | S01, S04          |                                    | Acute_IFT_Series  | The clearest structural trend in the data                                                                       |
| F14     | ED-origin episode CAGR, 2012 to 2023                 | 2023 | 1.55%                 | %/yr | DERIVED | A    | S01               | 11 compounding years               | Acute_IFT_Series  | 2012 is the first comparable year (NIS redesign)                                                                |
| F15     | Inpatient-origin episode CAGR, 2012 to 2023          | 2023 | (1.48%)               | %/yr | DERIVED | A    | S04               |                                    | Acute_IFT_Series  | Negative. The inpatient channel is in structural decline                                                        |
| F16     | Total acute-to-acute episode CAGR, 2012 to 2023      | 2023 | 0.86%                 | %/yr | DERIVED | A    | S01, S04          |                                    | Acute_IFT_Series  | The measured market volume growth rate. NOT demonstrably different from zero: AHRQ publishes no standard errors |
| F17     | AHA community hospital admission CAGR, 2012 to 2023  | 2023 | (0.67%)               | %/yr | DERIVED | A    | S05-S10           |                                    | Acute_IFT_Series  | Negative. IFT grows as the denominator shrinks. This one is a hospital census, so it is real                    |
| F18     | AHA community hospital inpatient admissions          | 2022 | 31,555,807 admissions |      | GOV-A   | A    | S09               | Fast Facts, 2024 edition           | Acute_IFT_Series  | Identical in both uploaded AHA workbooks                                                                        |
| F19     | AHA community hospital inpatient admissions          | 2024 | 33,553,725 admissions |      | GOV-A   | A    | S11               | Fast Facts, 2026 edition           | Cost_and_Capacity | VERIFIED against the AHA PDF: 33,553,725. No NEDS 2024 file yet, so no 2024 ratio                               |
| F20     | Total admissions, all US hospitals                   | 2024 | 35,658,583 admissions |      | GOV-A   | A    | S11               | Fast Facts, 2026 edition           | Cost_and_Capacity | VERIFIED: 35,658,583. Still 1.9% below the 2018 peak of 36,353,946                                              |
| F21     | Receiving-side transfer admissions (NIS TRAN_IN = 1) | 2022 | 2,205,490 admissions  |      | GOV-A   | A    | S04               | NIS Summary Stats, TRAN_IN = 1     | Receiving_Side    | VERIFIED. Derived from UB-04 Point of Origin, independent of the sending-side field                             |
| F22     | Receiving-side transfer admissions (NIS TRAN_IN = 1) | 2023 | 2,463,771 admissions  |      | GOV-A   | A    | S04               | NIS Summary Stats, TRAN_IN = 1     | Receiving_Side    | VERIFIED against the AHRQ PDF                                                                                   |
| F23     | Sending-to-receiving conversion rate                 | 2022 | 75.3%                 | %    | DERIVED | A    | S01, S04          | F21 over F05                       | Receiving_Side    | Residual goes to psych, rehab, LTACH, ED treat-and-release, or observation                                      |
| F24     | Sending-to-receiving conversion rate                 | 2023 | 77.4%                 | %    | DERIVED | A    | S01, S04          |                                    | Receiving_Side    |                                                                                                                 |

|     |                                                                    |      |                        |         |   |          |                                                      |                        |                                                                                                             |
|-----|--------------------------------------------------------------------|------|------------------------|---------|---|----------|------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------|
| F25 | NEMSIS total EMS activations                                       | 2024 | 60,298,684 activations | GOV-A   | A | S12      | 2024 report, eResponse.05 table                      | EMS_Transports         | VERIFIED against the NEMSIS PDF: 60,298,684. Cover box ties exactly only in 2024                            |
| F26 | NEMSIS Hospital-to-Hospital Transfer activations                   | 2024 | 5,510,664 transports   | GOV-A   | A | S12      | 2024 report, eResponse.05, v3.5                      | EMS_Transports         | VERIFIED: 5,510,664. Includes psych, rehab and LTACH destinations plus return legs. Not the HCUP definition |
| F27 | NEMSIS all facility-to-facility transfer activations               | 2024 | 7,512,656 transports   | DERIVED | A | S12      | Four v3.5 transfer categories                        | EMS_Transports         | All four components verified. Excludes Other Routine Medical Transport                                      |
| F28 | NEMSIS Other Routine Medical Transport activations                 | 2024 | 4,467,204 transports   | GOV-A   | A | S12      | 2024 report, eResponse.05                            | EMS_Transports         | VERIFIED: 4,467,204. Dialysis, appointments, return home. NOT interfacility transfer                        |
| F29 | NEMSIS Interfacility Transport activations (v3.4)                  | 2019 | 3,309,316 activations  | GOV-A   | B | S12      | 2019 report, eResponse.05, v3.4                      | EMS_Transports         | Coverage was 47 states. Not comparable to 2024                                                              |
| F30 | NEMSIS Interfacility Transport activations (v3.4)                  | 2023 | 4,179,580 activations  | GOV-A   | B | S12      | 2023 report, eResponse.05, v3.4                      | EMS_Transports         | 2023 NEMSIS denominator is understated. See Data_Quality_Register                                           |
| F31 | NEMSIS transports to a hospital ED                                 | 2022 | 24,016,095 transports  | GOV-A   | B | S12      | 2022 report, eDisposition.21                         | EMS_Transports         | 17.5% of NEDS weighted ED visits, against the NHAMCS anchor of 16.2%                                        |
| F32 | NEDS weighted national ED visits                                   | 2022 | 136,974,618 visits     | GOV-A   | A | S02      | NEDS Introduction 2022, Table 1                      | EMS_Transports         | AHA Annual Survey reports the identical figure; it is the NEDS control total                                |
| F33 | NEDS weighted national ED visits                                   | 2023 | 142,193,558 visits     | GOV-A   | A | S03      | NEDS Introduction 2023, Table A6                     | EMS_Transports         | VERIFIED: 142,193,558. The nine DISP_ED categories sum to this exactly                                      |
| F34 | NHAMCS ambulance-arrival share of ED visits                        | 2016 | 16.15% %               | DERIVED | B | S25      | NHAMCS 2016 ED Summary Tables, Table 5               | EMS_Transports         | The only external anchor. NEDS carries no arrival-mode element                                              |
| F35 | Medicare FFS hospital-to-hospital ambulance transports             | 2022 | 825,178 transports     | GOV-A   | A | S13      | PSPS 2022, modifier HH                               | Medicare_PSPS          | REBUILT from raw CMS claims and matched to the unit. Carrier claims only                                    |
| F36 | Medicare FFS hospital-to-hospital ambulance transports             | 2024 | 871,753 transports     | GOV-A   | A | S13      | PSPS 2024, modifier HH                               | Medicare_PSPS          | REBUILT from raw CMS claims and matched to the unit                                                         |
| F37 | Medicare FFS hospital-to-hospital transport CAGR, 2022 to 2024     | 2024 | 2.78% %/yr             | DERIVED | A | S13      |                                                      | Medicare_PSPS          | Positive while total FFS ambulance volume is negative                                                       |
| F38 | All Medicare FFS ambulance transport CAGR, 2022 to 2024            | 2024 | (2.80%) %/yr           | DERIVED | A | S13      |                                                      | Medicare_PSPS          | Enrollment shift to Medicare Advantage                                                                      |
| F39 | Non-emergency BLS and ALS1 share of Medicare H2H transports        | 2024 | 62.3% %                | DERIVED | A | S13      | A0428 + A0426 over total                             | Medicare_PSPS          | Contradicts the excluded 56.4% high-acuity assumption                                                       |
| F40 | Specialty Care Transport share of Medicare H2H transports          | 2024 | 6.6% %                 | DERIVED | A | S13      | A0434 over total                                     | Medicare_PSPS          | 6.6% of transports. See F60 for the share of allowed dollars                                                |
| F41 | ALS2 share of Medicare H2H transports                              | 2024 | 0.7% %                 | DERIVED | A | S13      | A0433 over total                                     | Medicare_PSPS          | The only H2H acuity tier declining in absolute volume                                                       |
| F42 | Emergent share of Medicare H2H transports                          | 2024 | 30.3% %                | DERIVED | A | S13      | A0427 + A0429 over total                             | Medicare_PSPS          | Fastest-growing acuity tiers, still under a third of volume                                                 |
| F43 | Volume-weighted RVU of the Medicare H2H book                       | 2024 | 1.45 RVU               | DERIVED | A | S13, S14 | SUMPRODUCT of RVU and volume                         | Medicare_PSPS          | The book prices at 1.45 BLS runs, not 3.25                                                                  |
| F44 | Ambulance Inflation Factor, latest (CY2026)                        | 2026 | 2.0% %/yr              | GOV-A   | A | S29      | CMS Pub 100-04, Ch.15 Sec.20.4, Transmittal 13464    | Payment_Rules          | VERIFIED 09 Jul 2026 against the CMS manual                                                                 |
| F45 | Ambulance Inflation Factor, trailing 3-year average (2024 to 2026) | 2026 | 2.3% %/yr              | DERIVED | A | S29      |                                                      | Payment_Rules          | Source workbook had CY2024 at 3.0%. Actual is 2.6%. Corrected                                               |
| F46 | Ambulance Inflation Factor, 2020 to 2026 average                   | 2026 | 3.1% %/yr              | DERIVED | A | S29      |                                                      | Payment_Rules          | Sample standard deviation 2.9 points. The AIF is volatile                                                   |
| F47 | Specialty Care Transport RVU (A0434)                               | 2026 | 3.25 RVU               | GOV-A   | A | S14, S28 | CMS AFS public use file; MedPAC Table 1              | Payment_Rules          | VERIFIED. 3.25 times the BLS base of 1.00. All seven ground RVUs matched                                    |
| F48 | Mean ground ambulance cost per transport, all-in                   | 2024 | \$2,673 \$/transport   | GOV-A   | A | S30      | GADCS Table 6.1, all NPIs, N = 5,000                 | Cost_and_Capacity      | VERIFIED. The MEDIAN is \$1,340 (F73) and is the number to quote for a scaled operator                      |
| F49 | Mean cost per transport, private for-profit                        | 2024 | \$1,788 \$/transport   | GOV-A   | A | S30      | GADCS Table 6.1                                      | Cost_and_Capacity      | CORRECTED to \$1,788. The \$1,778 in circulation is a secondary-source transcription error                  |
| F50 | Crew labor share of aggregated ground ambulance cost               | 2024 | 69.4% %                | GOV-A   | A | S30      | GADCS Figure 5.1 and Table 5.1                       | Cost_and_Capacity      | VERIFIED at 69.4%; \$18,869M of \$27.2B. Distinct from the 70% labor-related portion of the base payment    |
| F51 | Crew labor share of cost, for-profit organizations                 | 2024 | 60.5% %                | GOV-A   | A | S30      | GADCS Table 5.1                                      | Cost_and_Capacity      | Government-owned run 73.6%. Ownership drives the cost structure                                             |
| F52 | National hospital inpatient occupancy                              | 2022 | 65.49% %               | SOURCED | C | S23      | HCRIS 2552-10, 5,888 filers                          | Cost_and_Capacity      | Analyst computation on a vendored CMS panel, not a published figure. Not externally verifiable              |
| F53 | Inpatient occupancy change, FY2020 to FY2022                       | 2022 | 3.42 pp                | DERIVED | C | S23      |                                                      | Cost_and_Capacity      | A lagging throughput proxy, not a transfer count                                                            |
| F54 | Total certified post-acute destinations                            | 2024 | 35,481 providers       | SOURCED | C | S24      | CMS provider rolls                                   | Cost_and_Capacity      | SNF 14,699; HHA 12,392; hospice 6,852; IRF 1,221; LTACH 317                                                 |
| F55 | ICD-10-CM codes in the clinical spine validated as real            | 2026 | 92 codes               | SOURCED | B | S27      | NLM Clinical Tables                                  | Clinical_Volumes_TierC | A coding-integrity check. It validates no volume figure                                                     |
| F56 | CY2024 ground ambulance conversion factor                          | 2024 | \$272.44 \$/RVU        | GOV-A   | A | S28      | MedPAC Payment Basics (Ambulance), Oct 2024, Table 1 | Payment_Rules          | VERIFIED. The source workbook assumed \$278.00 and labeled it ILLUSTRATIVE                                  |
| F57 | CY2024 ground mileage rate                                         | 2024 | \$8.76 \$/statute mile | GOV-A   | A | S28      | MedPAC Payment Basics (Ambulance), Oct 2024          | Payment_Rules          | Corroborated by the claims: \$8.79 allowed per loaded mile (F69)                                            |

|     |                                                                             |      |                       |         |   |              |                                                     |                         |                                                                                                                                                  |
|-----|-----------------------------------------------------------------------------|------|-----------------------|---------|---|--------------|-----------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| F58 | Base-rate allowed charges, Medicare FFS hospital-to-hospital                | 2024 | \$319,815,284 \$      | GOV-A   | A | S13          | PSPS 2024, PSPS_ALLOWED_CHARGE_AMT, modifier HH     | Medicare_PSPS           | Pulled from the CMS claims API. Not present in any source workbook                                                                               |
| F59 | TOTAL allowed charges (base + mileage), Medicare FFS hospital-to-hospital   | 2024 | \$577,228,322 \$      | DERIVED | A | S13          | Base rate plus A0425 mileage                        | Medicare_PSPS           | \$577.2M in 2024. The only assumption-free dollar figure in this workbook                                                                        |
| F60 | Specialty Care Transport share of H2H base-rate ALLOWED DOLLARS             | 2024 | 15.6% %               | DERIVED | A | S13          | A0434 allowed over total allowed                    | Medicare_PSPS           | 6.6% of transports, 15.6% of base-rate allowed dollars. Price weighting narrows the gap, it does not close it                                    |
| F61 | Ambulance Inflation Factor, CY2026                                          | 2026 | 2.0% %/yr             | GOV-A   | A | S29          | Transmittal 13464, 14 Nov 2025                      | Payment_Rules           | CPI-U 2.7% less a total factor productivity adjustment of 0.7%                                                                                   |
| F62 | Non-emergency BLS ESRD dialysis payment reduction                           | 2024 | 23.0% %               | GOV-A   | A | S28          | MedPAC Payment Basics (Ambulance), Oct 2024         | Payment_Rules           | Applies to the RJ, JR, NJ and JN modifier volumes on Medicare_PSPS Panel A, not to HH                                                            |
| F63 | Transfers to another facility type, ED plus inpatient origin                | 2022 | 6,509,053 legs        | DERIVED | A | S01, S04     | DISP_ED = 5 plus DISPUNIFORM = 5                    | Other_Transfer_Channels | 2.2 times the acute-to-acute count. Not all move by ambulance                                                                                    |
| F64 | Total transfers out, all destinations                                       | 2022 | 9,438,326 legs        | DERIVED | A | S01, S04     |                                                     | Other_Transfer_Channels | The widest sending-side count HCUP supports                                                                                                      |
| F65 | ED-originated share of all community hospital admissions                    | 2022 | 63.5% %               | DERIVED | A | S01, S05-S09 | DISP_ED = 9 over AHA admissions                     | Other_Transfer_Channels | 52.9% in 2007 rising to 63.5% in 2022. Confirms NEDS and AHA describe the same hospital universe                                                 |
| F66 | Allowed charges per Medicare FFS hospital-to-hospital transport             | 2024 | \$662.15 \$/transport | DERIVED | A | S13          | Total allowed over transports                       | Medicare_PSPS           | \$662.15, against a GADCS median cost per transport of \$1,340                                                                                   |
| F67 | Ground mileage allowed charges, hospital-to-hospital                        | 2024 | \$257,413,039 \$      | GOV-A   | A | S13          | PSPS 2024, HCPCS A0425, modifier HH                 | Medicare_PSPS           | 44.6% of total allowed dollars on the book                                                                                                       |
| F68 | Loaded miles per hospital-to-hospital transport                             | 2024 | 33.6 miles            | DERIVED | A | S13          | Mileage units over transports                       | Medicare_PSPS           | 33.6 miles. Roughly three times a typical 911 scene run                                                                                          |
| F69 | Allowed per loaded mile                                                     | 2024 | \$8.79 \$/mile        | DERIVED | A | S13          | Mileage allowed over mileage units                  | Medicare_PSPS           | \$8.79, against the published CY2024 rate of \$8.76. Independent confirmation of both                                                            |
| F70 | Total Medicare FFS carrier-billed ground ambulance allowed charges          | 2024 | \$5,073,090,003 \$    | GOV-A   | A | S13          | PSPS 2024, six base codes plus A0425, all modifiers | Medicare_PSPS           | Reconciles with MedPAC's \$5.3B (2023) once air and institutional claims are added                                                               |
| F71 | Total Medicare FFS carrier-billed ground ambulance payments (NCH)           | 2024 | \$3,957,457,371 \$    | GOV-A   | A | S13          | PSPS 2024, PSPS_NCH_PAYMENT_AMT                     | Medicare_PSPS           | The 2022 value of \$3.72B reconciles with MedPAC's \$3.9B program spending                                                                       |
| F72 | Hospital-to-hospital share of all Medicare carrier ground ambulance dollars | 2024 | 11.4% %               | DERIVED | A | S13          |                                                     | Medicare_PSPS           | Dollar share (11.4%) exceeds volume share (8.2%); interfacility runs are longer and higher acuity                                                |
| F73 | GADCS median total cost per ground ambulance transport                      | 2024 | \$1,340 \$/transport  | GOV-A   | A | S30          | GADCS Table 6.1                                     | Cost_and_Capacity       | The mean of \$2,673 is 1.99x the median. Quote the median for a scaled platform                                                                  |
| F74 | GADCS median total cost per RVU                                             | 2024 | \$811 \$/RVU          | GOV-A   | A | S30          | GADCS Table 6.1                                     | Cost_and_Capacity       | Mean \$1,624. Medicare allowed is about \$253 per RVU on the H2H book                                                                            |
| F75 | GADCS mean cost per transport, very high Medicare volume                    | 2024 | \$946 \$/transport    | GOV-A   | A | S30          | GADCS Table 6.1                                     | Cost_and_Capacity       | \$946 against \$3,652 for low-volume organizations. Scale is the entire story                                                                    |
| F76 | Medicare FFS ambulance program spending, excl. beneficiary cost sharing     | 2022 | \$3,900,000,000 \$    | GOV-A   | A | S28          | MedPAC Payment Basics (Ambulance), Oct 2024, p.1    | Payment_Rules           | CORRECTED. Source workbook said \$4.0B for 2023. About 1% of FFS spend; about 13% of FFS beneficiaries used an ambulance. ALL ambulance, not IFT |
| F77 | Medicare FFS ambulance total payments                                       | 2023 | \$5,300,000,000 \$    | GOV-B   | B | S28          | MedPAC ambulance session, Mar 2025                  | Payment_Rules           | Total payments INCLUDING beneficiary cost sharing, across 11.4M transports and about 10,500 organizations. Not the same measure as F76           |
| F78 | NEDS target universe, hospital-owned EDs                                    | 2023 | 4,489 EDs             | GOV-A   | A | S03          | NEDS Introduction 2023, Table A6                    | Data_Quality_Register   | Sampling frame 3,185 EDs and 101,915,572 visits; the 2023 sample is 963 EDs and 31,003,348 visits                                                |
| F79 | Medicare FFS hospital-to-hospital transports, peak year                     | 2017 | 1,139,524 transports  | GOV-A   | A | S32          | PSPS 2017, modifier HH                              | Medicare_IFT_Series     | 1,139,524. Every year since has been lower                                                                                                       |
| F80 | Medicare FFS hospital-to-hospital transports                                | 2010 | 937,317 transports    | GOV-A   | A | S32          | PSPS 2010, modifier HH                              | Medicare_IFT_Series     | Start of the fifteen-year series                                                                                                                 |
| F81 | Transport CAGR, 2010 to 2017                                                | 2017 | 2.83% %/yr            | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | Real pre-COVID growth                                                                                                                            |
| F82 | Transport CAGR, 2017 to 2024                                                | 2024 | (3.75%) %/yr          | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | The reversal. Medicare Advantage enrolment is the dominant driver                                                                                |
| F83 | Transports 2024 against the 2017 peak                                       | 2024 | (23.5%) %             | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | The Medicare FFS interfacility book is 23.5% smaller than at peak                                                                                |
| F84 | Total allowed CAGR, 2010 to 2024                                            | 2024 | 1.38% %/yr            | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | Dollars grew while transports shrank                                                                                                             |
| F85 | Allowed per transport CAGR, 2010 to 2024                                    | 2024 | 1.90% %/yr            | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | Below the 3.0% average AIF over the same window. Realised rate growth lags the statutory update                                                  |
| F86 | Volume-weighted RVU of the H2H book                                         | 2010 | 1.472 RVU             | DERIVED | A | S32, S14     |                                                     | Medicare_IFT_Series     | 1.472. Compare 1.450 in 2024: a 1.5% move in fifteen years                                                                                       |
| F87 | Mileage share of allowed dollars                                            | 2010 | 43.0% %               | DERIVED | A | S32          |                                                     | Medicare_IFT_Series     | 43.0%, against 44.6% in 2024. Stable for fifteen years                                                                                           |
| F88 | Medicare FFS ambulance transports, all types                                | 2024 | 11,300,000 transports | GOV-A   | A | S31          | MedPAC June 2026, Ch.6                              | MedPAC_2026_Mandated    | Carrier-billed claims account for about 10.76M, or 95%. The residual is institutionally billed ambulance                                         |

|      |                                                                        |      |                      |          |   |     |                                                                           |                      |                                                                                                                     |
|------|------------------------------------------------------------------------|------|----------------------|----------|---|-----|---------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| F89  | Total Medicare payments, AFS-covered transports                        | 2024 | \$5,300,000,000 \$   | GOV-A    | A | S31 | MedPAC June 2026, Ch.6                                                    | MedPAC_2026_Mandated | Sits between the claims-derived \$4.47B program payment and \$5.71B allowed charge. MedPAC does not state the scope |
| F90  | Mean cost per transport, lowest transport-volume quartile              | 2023 | \$2,878 \$/transport | GOV-A    | A | S31 | MedPAC June 2026, Table 6-6                                               | MedPAC_2026_Mandated | Mean 148 transports a year. Against \$918 in the highest quartile                                                   |
| F01  | Mean cost per transport, highest transport-volume quartile             | 2023 | \$918 \$/transport   | GOV-A    | A | S31 | MedPAC June 2026, Table 6-6                                               | MedPAC_2026_Mandated | Mean 15,291 transports a year. Average RVU only 5% lower than the lowest quartile: the gap is scale, not case mix   |
| F02  | Mean cost per transport, for-profit organizations                      | 2023 | \$563 \$/transport   | GOV-A    | A | S31 | MedPAC June 2026, Table 6-6                                               | MedPAC_2026_Mandated | Government-owned cost \$1,626, or 2.9x. Supersedes the RAND unweighted means for operator benchmarking              |
| F03  | Mean cost per transport, dynamic staffing model                        | 2023 | \$636 \$/transport   | GOV-A    | A | S31 | MedPAC June 2026, Table 6-6                                               | MedPAC_2026_Mandated | Static staffing costs \$1,524. The single largest controllable cost lever MedPAC identifies                         |
| F04  | Median revenue-to-cost ratio, lowest volume quartile                   | 2023 | 0.77 ratio           | GOV-A    | A | S31 | MedPAC June 2026, Table 6-11                                              | MedPAC_2026_Mandated | Highest quartile 1.10. NOT a Medicare margin: revenue includes non-Medicare sources                                 |
| F05  | Government organizations' non-billing revenue per transport            | 2023 | \$322 \$/transport   | GOV-A    | A | S31 | MedPAC June 2026, Ch.6                                                    | MedPAC_2026_Mandated | 30% of government organizations receive local funding. Nonprofit \$32, for-profit \$12                              |
| F06  | Carrier-billed AIR ambulance Medicare payments                         | 2024 | \$510,451,782 \$     | GOV-A    | A | S13 | PSPS 2024, A0430, A0431, A0435, A0436, A0432                              | Medicare_PSPS        | Needed to test MedPAC's spend figures. See Panel G                                                                  |
| F07  | Median ED time before transfer to another facility (OP-18d)            | 2025 | 287 minutes          | GOV-A    | A | S33 | CMS Timely and Effective Care, national                                   | Demand_Drivers       | 287 minutes. Measurement period 01 Jul 2024 to 30 Jun 2025                                                          |
| F08  | Median ED time before discharge (OP-18b)                               | 2025 | 161 minutes          | GOV-A    | A | S33 | CMS Timely and Effective Care, national                                   | Demand_Drivers       | 161 minutes. The comparison population                                                                              |
| F09  | Transfer penalty, OP-18d minus OP-18b                                  | 2025 | 126 minutes          | DERIVED  | A | S33 |                                                                           | Demand_Drivers       | 126 minutes, or 78% longer than a discharged patient                                                                |
| F100 | Top-decile hospital ED time before transfer                            | 2025 | 187 minutes          | GOV-A    | A | S33 | CMS Timely and Effective Care, national                                   | Demand_Drivers       | 187 minutes. One hundred minutes of the median delay is addressable                                                 |
| F101 | Median ED time before transfer, psychiatric patients (OP-18c)          | 2025 | 259 minutes          | GOV-A    | A | S33 | CMS Timely and Effective Care, national                                   | Demand_Drivers       | 259 minutes. Still faster than a medical transfer                                                                   |
| F102 | Hospitals reporting OP-18d                                             | 2025 | 2,340 hospitals      | GOV-A    | A | S33 | CMS Timely and Effective Care, hospital file                              | Demand_Drivers       | Median 294, 90th percentile 443, maximum 1,597 minutes                                                              |
| F103 | Hospitals with ED transfer delay at or above 8 hours                   | 2025 | 153 hospitals        | GOV-A    | A | S33 | CMS Timely and Effective Care, hospital file                              | Demand_Drivers       | 6.5% of reporting hospitals                                                                                         |
| F104 | Community hospitals in a system                                        | 2024 | 3,567 hospitals      | GOV-B    | B | S11 | AHA Fast Facts, 2026 edition                                              | Demand_Drivers       | 3,567 of 5,121 community hospitals                                                                                  |
| F105 | Independent community hospitals                                        | 2018 | 1,707 hospitals      | DERIVED  | B | S34 | AHA Fast Facts, 2020 edition                                              | Demand_Drivers       | 1,707. Fell to 1,554 by 2024, a 9.0% decline                                                                        |
| F106 | Independent community hospitals                                        | 2024 | 1,554 hospitals      | DERIVED  | B | S11 | AHA Fast Facts, 2026 edition                                              | Demand_Drivers       | 1,554                                                                                                               |
| F107 | System share of community hospitals                                    | 2024 | 69.7% %              | DERIVED  | B | S11 |                                                                           | Demand_Drivers       | 69.7%, up from 67.2% in 2018                                                                                        |
| F108 | Adjusted odds ratio of interfacility transfer, travel time 60+ minutes | 2021 | 1.32 aOR             | ACADEMIC | A | S35 | Clark et al., JAMA Netw Open 2025, doi:10.1001/jamanetworkopen.2024.55258 | Demand_Drivers       | 95% CI 1.15 to 1.51. n=190,311 adults, Florida and California, 2021                                                 |
| F109 | Adjusted additional charges, travel time 60+ minutes                   | 2021 | \$8,284 \$           | ACADEMIC | A | S35 | Same                                                                      | Demand_Drivers       | 95% CI \$5,532 to \$11,035                                                                                          |

Basis. GOV-A = publisher figure with document and table pinned, opened and read. GOV-B = publisher figure, locator not pinned. INDUSTRY-A = trade association census, opened and read. ACADEMIC = peer-reviewed, digital object identifier given. SOURCED = analyst computation on a vendored CMS file. DERIVED = an Excel formula whose every input is one of the above.

Tier. A = verified against the primary document on 09 July 2026. B = document named but not opened, or a locator that could not be pinned. C = analyst computation or unpinned publisher class, not externally reproducible.

Nothing labeled ILLUSTRATIVE, FRAMEWORK or INDUSTRY enters this ledger. Those 30 items are recorded on Excluded\_Not\_Sourced with what would make each citable.

Colour key. Blue = a hardcoded value taken from a source document. Green = a live link to the sheet that carries the number. Black = a formula computed on this sheet.

Facts F110 to F135, added in revision 2 (10 July 2026). Same schema, same rules; green values live on the named tab.

| Fact ID | Metric                                       | Year | Value         | Unit       | Basis    | Tier | Source ID | Locator                                               | Lives on              | Cross-check                                                                                                  |
|---------|----------------------------------------------|------|---------------|------------|----------|------|-----------|-------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------|
| F110    | All O-D coded Medicare FFS ground transports | 2019 | 14,289,866    | transports | GOV-A    | A    | S39       | PSPS 2019, valid O-D letter pairs, six base HCPCS     | Medicare_OD_Matrix    | Fractional service counts summed as floats                                                                   |
| F111    | All O-D coded Medicare FFS ground transports | 2024 | 10,637,766    | transports | GOV-A    | A    | S39       | PSPS 2024                                             | Medicare_OD_Matrix    | the initial extraction carried 10,637,727: a 39-service (0.0004%) integer-truncation gap. See DQ #27         |
| F112    | O-D coded transport decline, 2019 to 2024    | 2024 | (25.6%)       | %          | DERIVED  | A    | S39       |                                                       | Medicare_OD_Matrix    | The FFS shrinkage benchmark for every channel comparison                                                     |
| F113    | SNF-interface transports (NH + HN)           | 2024 | 2,398,253     | transports | GOV-A    | A    | S39       | PSPS 2024, NH and HN pairs                            | Medicare_OD_Matrix    | 2.75x the HH book. The largest facility-to-facility channel                                                  |
| F114    | Dialysis-pair Medicare FFS ground transports | 2019 | 2,332,855     | transports | GOV-A    | A    | S39       | All pairs containing J or G                           | Dialysis_ESRD_Channel | 2.5x the whole-book decline (F112). Largest single-year drop -34.2% in 2022, the RSNAT national rollout year |
| F115    | Dialysis-pair Medicare FFS ground transports | 2024 | 857,750       | transports | GOV-A    | A    | S39       |                                                       | Dialysis_ESRD_Channel |                                                                                                              |
| F116    | Dialysis-pair decline, 2019 to 2024          | 2024 | (63.2%)       | %          | DERIVED  | A    | S39       |                                                       | Dialysis_ESRD_Channel |                                                                                                              |
| F117    | Dialysis-pair total allowed charges          | 2024 | \$215,105,039 | \$         | GOV-A    | A    | S39       | Base plus mileage                                     | Dialysis_ESRD_Channel | \$506.9M in 2019                                                                                             |
| F118    | RSNAT effect on RSNAT expenditures           | 2019 | (77.0%)       | %          | ACADEMIC | A    | S42       | Contreary et al., JAMA Health Forum 2022;3(7):e222093 | Dialysis_ESRD_Channel | doi:10.1001/jamahealthforum.2022.2093, PMID 35977218. Also -61% use probability, -2.4% total cost            |

|      |                                                             |      |                    |            |   |     |                                                                                        |                        |                                                                                                                                     |
|------|-------------------------------------------------------------|------|--------------------|------------|---|-----|----------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| F119 | RSNAT four-year Medicare savings, model states              | 2020 | \$650,000,000 \$   | GOV-A      | A | S41 | CMS press release, 22 Sep 2020                                                         | Dialysis_ESRD_Channel  | With -63% use and -72% expenditures among ESRD and pressure-ulcer beneficiaries                                                     |
| F120 | Rotary-wing Medicare FFS transports, total                  | 2024 | 66,608 transports  | GOV-A      | A | S40 | PSPS 2024, A0431, all O-D pairs                                                        | Air_Ambulance_IFT      |                                                                                                                                     |
| F121 | Rotary-wing hospital-to-hospital transports                 | 2024 | 43,683 transports  | GOV-A      | A | S40 | PSPS 2024, A0431, HH                                                                   | Air_Ambulance_IFT      | 34,856 in 2010: +25% over the window in which ground HH fell 7%                                                                     |
| F122 | Rotary-wing HH share of rotary transports                   | 2024 | 65.6% %            | DERIVED    | A | S40 |                                                                                        | Air_Ambulance_IFT      | Air is the high-acuity interfacility book; ground SCT is 6.6%                                                                       |
| F123 | Fixed-wing transfer-point-to-transfer-point (II) share      | 2024 | 94.6% %            | DERIVED    | A | S40 | A0430, modifier II                                                                     | Air_Ambulance_IFT      | Fixed-wing O-D modifiers identify runways, not facilities. DQ #29                                                                   |
| F124 | Rotary-wing commercial allowed over Medicare, base          | 2020 | 5.0x x             | DERIVED    | A | S44 | FAIR Health air white paper: \$18,668 over \$3,739, same source, same year             | Air_Ambulance_IFT      | Pre-NSA benchmark                                                                                                                   |
| F125 | Commercial ALS-emergency average allowed, ground            | 2020 | \$758 \$           | INDUSTRY-A | A | S43 | FAIR Health ground white paper, Mar 2022                                               | Payer_Rates_Commercial | Up 56% from \$486 in 2017 while Medicare grew 4.8%                                                                                  |
| F126 | GAPB recommended patient cost-sharing cap                   | 2024 | \$100 \$           | GOV-A      | A | S46 | GAPB report, 29 Mar 2024: lesser of \$100 or 10% of allowed                            | Payer_Rates_Commercial | Interfacility transports named in scope; no federal enactment as of Jul 2026                                                        |
| F127 | States with own ground balance-billing protections          | 2024 | 14 states          | GOV-B      | B | S47 | Commonwealth Fund, Feb 2024                                                            | Payer_Rates_Commercial | Most set a payment standard; several tie to Medicare multiples                                                                      |
| F128 | Medicaid NEMT fee-for-service spending                      | 2018 | \$2,600,000,000 \$ | GOV-A      | A | S48 | MACPAC June 2021 report, Ch.5 (T-MSIS)                                                 | Payer_Rates_Commercial | 61.5M ride-days, 3.2M users. AMBULANCE IS EXCLUDED from this NEMT definition                                                        |
| F129 | Population 65 and over, projection                          | 2030 | 71,183,000 people  | GOV-A      | A | S50 | Census NP2023-T2, main series                                                          | Macro_Demand_Drivers   | 57.8M (2022), 82.1M (2050), 2025-2030 CAGR 2.4%/yr                                                                                  |
| F130 | Medicare Advantage share of eligible beneficiaries          | 2026 | 55.0% %            | INDUSTRY-A | A | S51 | KFF from CMS enrollment files: 35.2M of 64.2M                                          | Macro_Demand_Drivers   | 19% in 2007; CBO projects 63% by 2034. The deflator on every FFS series here                                                        |
| F131 | Rural hospital closures and conversions since 2010          | 2026 | 154 hospitals      | ACADEMIC   | A | S52 | Sheps Center tracker, accessed 10 Jul 2026                                             | Macro_Demand_Drivers   | 86 complete + 68 converted; REH conversions excluded. Chartis counts 182 under a different convention: never mix                    |
| F132 | Rural closure effect on EMS total activation time           | 2020 | 7 minutes          | ACADEMIC   | A | S53 | Miller et al., HSR 2020, doi:10.1111/1475-6773.13254                                   | Macro_Demand_Drivers   | PMID 31989591. Transport time +2.6 min; response time unchanged                                                                     |
| F133 | SMOCC immediate effect on interhospital transfer rate       | 2025 | 1.35 rate ratio    | ACADEMIC   | A | S54 | Richert et al., JAMA Netw Open 2025;8(12):e2546002                                     | Macro_Demand_Drivers   | doi:10.1001/jamanetworkopen.2025.46002, PMID 41324958. 95% CI 1.05-1.74; long-term RR 0.94; 441,709 NEMSIS transfers, 8 states      |
| F134 | NHAMCS ambulance-arrival share of ED visits (final vintage) | 2022 | 17.0% %            | GOV-A      | A | S55 | NHAMCS 2022 ED Summary Tables, Table 4                                                 | Macro_Demand_Drivers   | SE 0.9 points on 155,398,000 visits; 65+ 33.4%. NEMSIS 2022 coverage (17.5%) sits inside the 95% CI. Survey discontinued after 2022 |
| F135 | Active ambulance LCDs in the Medicare Coverage Database     | 2026 | - LCDs             | GOV-A      | A | S56 | MCD query, 09-10 Jul 2026; one billing article, A52917 (Noridian, rural air protocols) | Code_Crosswalks        | Ambulance coverage is national regulation (42 CFR 410.40), not MAC discretion                                                       |

Facts F136 to F147, revision 3 (10 July 2026).

| Fact ID | Metric                                                       | Year | Value              | Unit          | Basis   | Tier | Source ID | Locator                                                      | Lives on               | Cross-check                                                                                 |
|---------|--------------------------------------------------------------|------|--------------------|---------------|---------|------|-----------|--------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------|
| F136    | Original Medicare A and B enrollment (annual average)        | 2019 | 32,758,129         | beneficiaries | GOV-A   | A    | S60       | CMS Medicare Monthly Enrollment, national, Year rows         | Utilization_Normalized | The carrier-claims denominator                                                              |
| F137    | Original Medicare A and B enrollment (annual average)        | 2024 | 27,673,511         | beneficiaries | GOV-A   | A    | S60       |                                                              | Utilization_Normalized | Minus 15.5% from 2019. First increase in the series lands in 2025 (27,914,766)              |
| F138    | All O-D coded transports per 1,000 A and B FFS beneficiaries | 2024 | 384.4              | per 1,000     | DERIVED | A    | S39, S60  |                                                              | Utilization_Normalized | 436.2 in 2019: minus 11.9% per capita against minus 25.6% raw                               |
| F139    | HH transports per 1,000 A and B FFS beneficiaries            | 2024 | 31.5               | per 1,000     | DERIVED | A    | S39, S60  |                                                              | Utilization_Normalized | 34.2 in 2019: minus 8.0% per capita against minus 22.3% raw                                 |
| F140    | Dialysis-pair transports per 1,000 A and B FFS beneficiaries | 2024 | 31.0               | per 1,000     | DERIVED | A    | S39, S60  |                                                              | Utilization_Normalized | 71.2 in 2019: minus 56.5% per capita. The collapse survives normalization                   |
| F141    | HH emergency-coded share (ALS-E + BLS-E + ALS2)              | 2024 | 31.0% %            |               | DERIVED | A    | S39       | Six-base-code convention                                     | Acuity_by_Channel      | 28.6% in 2019: +2.4 points                                                                  |
| F142    | NH (SNF to hospital) emergency-coded share                   | 2024 | 94.5% %            |               | DERIVED | A    | S39       |                                                              | Acuity_by_Channel      | HN (hospital to SNF) is 98.4% BLS non-emergency: the two directions are clinical opposites  |
| F143    | Dialysis-pair BLS non-emergency share                        | 2024 | 95.6% %            |               | DERIVED | A    | S39       |                                                              | Acuity_by_Channel      | The RSNAT target profile                                                                    |
| F144    | Ground ambulance billing NPIs (base codes, MUP floor)        | 2024 | 8,721              | NPIs          | GOV-A   | A    | S59       | MUP-PHY 2024; rows with 10 or fewer beneficiaries suppressed | Supplier_Landscape     | 9,028 in 2019                                                                               |
| F145    | Top-10 billing NPIs, share of approved ground services       | 2024 | 6.2% %             |               | DERIVED | A    | S59       |                                                              | Supplier_Landscape     | 5.3% in 2019. Floor on corporate concentration: NPI is not parent company                   |
| F146    | NPIs billing BLS non-emergency (A0428)                       | 2024 | 2,683              | NPIs          | GOV-A   | A    | S59       |                                                              | Supplier_Landscape     | 3,075 in 2019: minus 12.7%, the supplier-side RSNAT exit                                    |
| F147    | Ground add-on payments statutory expiration                  | 2028 | 1 date: 1 Jan 2028 |               | GOV-A   | A    | S61       | CAA 2026, section 6203; CMS AFS page                         | Payment_Rules          | Extended through 31 Dec 2027 after an Oct 2025 lapse; 2% urban, 3% rural, 22.6% super-rural |

Facts F148 to F163, revision 4 (10 July 2026).

| Fact ID | Metric                                                       | Year | Value     | Unit               | Basis    | Tier | Source ID | Locator                                        | Lives on             | Cross-check                                                                            |
|---------|--------------------------------------------------------------|------|-----------|--------------------|----------|------|-----------|------------------------------------------------|----------------------|----------------------------------------------------------------------------------------|
| F148    | Paid EMT and paramedic employment (combined)                 | 2019 | 260,600   | jobs               | GOV-A    | A    | S67       | OEWS May 2019, 29-2040                         | Workforce_Supply     | Excludes volunteers and self-employed                                                  |
| F149    | Paid EMT and paramedic employment (combined)                 | 2024 | 282,900   | jobs               | DERIVED  | A    | S68       | 181,000 EMTs + 101,900 paramedics              | Workforce_Supply     | +8.6% vs 2019. The paid workforce grew                                                 |
| F150    | Paramedic median annual wage                                 | 2024 | \$58,410  | \$                 | GOV-A    | A    | S68       | OEWS May 2024 via OOH                          | Workforce_Supply     | No 2019 twin: occupation split in the 2018 SOC                                         |
| F151    | EMT median annual wage                                       | 2024 | \$41,340  | \$                 | GOV-A    | A    | S68       |                                                | Workforce_Supply     | \$35,400 combined median in 2019 (F152)                                                |
| F152    | EMT and paramedic combined median annual wage                | 2019 | \$35,400  | \$                 | GOV-A    | A    | S67       |                                                | Workforce_Supply     |                                                                                        |
| F153    | Ambulance Inflation Factor, compounded CY2020-CY2024         | 2024 | 18.6%     | %                  | DERIVED  | A    | S29       | 0.9, 0.2, 5.1, 8.7, 2.6 compounded             | Workforce_Supply     | Slightly ahead of EMT median wage growth (+16.8%) on this window; see the Panel B read |
| F154    | Ground ambulance billing NPIs                                | 2014 | 9,305     | NPIs               | GOV-A    | A    | S62       | MUP-PHY 2014, six ground base codes            | Supplier_Series_Raw  | 8,721 in 2024: minus 6.3% over the decade                                              |
| F155    | BLS non-emergency billers, 2014 to 2024                      | 2024 | (19.4%)   | %                  | DERIVED  | A    | S62       | 3,329 to 2,683 NPIs                            | Supplier_Trend       | Fell every single year; the scheduled-transport exit                                   |
| F156    | Population in ambulance deserts (25+ minutes from a station) | 2022 | 4,500,000 | people             | ACADEMIC | A    | S65       | 41 states, 2021-22; 2.3M rural                 | Imbalance_Ledger     | UNDERCOUNT: nine states lack data                                                      |
| F157    | California cumulative offload delay beyond 30 minutes        | 2024 | 13,365    | hours per 6 months | GOV-A    | A    | S71       | CEMSIS, Apr-Sep 2024                           | Imbalance_Ledger     | 19.1% of offloads exceed the standard (F158). CEMSIS participants only                 |
| F158    | California offloads exceeding the 30-minute standard         | 2024 | 19.1%     | %                  | GOV-A    | A    | S71       |                                                | Imbalance_Ledger     | Statewide mean offload 42.8 minutes, 2021-2023 (S72)                                   |
| F159    | Staffed hospital beds, pre-pandemic mean to post-PHE         | 2024 | 674,000   | beds               | ACADEMIC | A    | S63       | 802,000 (2009-2019 mean) to 674,000            | Supply_Stack         | Minus 16% on mandated reporting; AHA surveys say minus 2% (Fault_Audit F)              |
| F160    | National hospital occupancy, post-PHE steady state           | 2024 | 75.3%     | %                  | ACADEMIC | A    | S63       | 63.9% pre-pandemic mean                        | Demand_Stack         | Census flat at about 510,000: the rise is a bed-supply story                           |
| F161    | Hospital-at-home waiver statutory expiration                 | 2030 | 1         | date: 30 Sep 2030  | GOV-A    | A    | S64       | CAA 2026 sec. 6210                             | Demand_Stack         | About 400 approved hospitals; every H@H day creates home transport legs                |
| F162    | OP-18 boarding-measure removal                               | 2028 | 1         | reporting year     | GOV-A    | A    | S64       | CY2026 OPSPS final rule                        | Fault_Audit          | Successor ECAT eCQM measures boarding directly; mandatory CY2028                       |
| F163    | Medicare beneficiaries per ground billing NPI                | 2024 | 7,797     | beneficiaries      | DERIVED  | A    | S60, S62  | 68.0M over 8,721                               | Imbalance_Ledger     | 6,814 in 2019: +14.4%. Need per supplier is rising                                     |
| F164    | Official state-file national ground services (approved)      | 2024 | 9,637,009 | services           | GOV-A    | A    | S73       | MUP by Geography and Service, six ground codes | State_Saturation_Raw | 9,542,103 in the NPI file: the Geography file suppresses less                          |
| F165    | NPI-file suppression tail (Geography file over NPI file)     | 2024 | 1.0%      | %                  | DERIVED  | A    | S62, S73  |                                                | State_Saturation     | Bounds DQ #32: supplier-file undercount is about 1% of services nationally             |

| v3 additions: F166 onward - built 10 July 2026. Same contract: every fact resolves to a source and locator; DERIVED facts name their inputs; formulas are live. |                                                               |      |                     |                |         |   |     |                                                                   |                             |                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------|---------------------|----------------|---------|---|-----|-------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F166                                                                                                                                                            | Original Medicare (any part) beneficiaries, national          | 2024 | 33913847            | beneficiaries  | GOV     | A | S81 | Medicare Monthly Enrollment, National, MONTH=Year, 2024           | Enrollment_ESRD_State       | Consistent with the A&B denominator discussion on Utilization_Normalized (different definition: any-part vs A-and-B; both stated there)                         |
| F167                                                                                                                                                            | MA & other share of Medicare beneficiaries, national          | 2025 | 0.509016477810194   | %              | DERIVED | A | S81 | MA_AND_OTH_BENES / TOT_BENES, 2025                                | Enrollment_ESRD_State       | Tracks the KFF 54% (2025) MA share on Macro_Demand_Drivers (different definition: KFF excludes some "other" plans)                                              |
| F168                                                                                                                                                            | Medicare ESRD beneficiaries (aged + disabled), national       | 2025 | 571962              | beneficiaries  | DERIVED | A | S81 | AGED_ESRD_BENES + DSBLD_ESRD_AND_ESRD_ONLY_BENES, 2025            | Enrollment_ESRD_State       | Denominator context for the Dialysis_ESRD_Channel RSNAI collapse: the patient base did not shrink; the paid volume did                                          |
| F169                                                                                                                                                            | Medicare FFS BLS non-emergency services (A0428, final-action) | 2024 | 2,956,154.1         | services       | GOV     | A | S78 | MUP by Geography & Service 2024, National, A0428, Tot_Srvcs       | MUP_Ambulance_National      | MUP final-action basis; PSPS submitted basis differs by design (~1.10x in 2024, Data_Quality_Register #34)                                                      |
| F170                                                                                                                                                            | Rendering providers billing A0428                             | 2024 | 4,223               | NPIs           | GOV     | A | S78 | MUP by Geography & Service 2024, National, A0428                  | MUP_Ambulance_National      | The scheduled-transport biller base; see Supplier_Trend BLS-NE biller exit series                                                                               |
| F171                                                                                                                                                            | SCT (A0434) services CAGR, 2013-2024                          | 2024 | -0.0339576929557589 | %/yr           | DERIVED | A | S78 | (2024 services / 2013 services)^(1/11)-1, Panel B                 | MUP_Ambulance_National      | Directionally consistent with the SCT share-of-transports chart on the Charts tab (PSPS basis)                                                                  |
| F172                                                                                                                                                            | States/territories with ambulance MUP rows                    | 2024 | 55                  | geographies    | GOV     | A | S78 | MUP by Geography & Service 2024, State grain, A0425-A0436         | MUP_Ambulance_State         | Suppression makes absent state-code pairs floors, not zeros                                                                                                     |
| F173                                                                                                                                                            | BLS non-emergency (A0428) denial rate                         | 2024 | 0.129157061063293   | %              | DERIVED | A | S79 | Panel B, A0428 column, 2024 row (denied/submitted)                | PSPS_Denial_Series          | The medical-necessity screen on scheduled transport; compare the RSNAI natural experiment on Dialysis_ESRD_Channel and Payment_Integrity                        |
| F174                                                                                                                                                            | US counties with <3 measured FFS ambulance providers, share   | 2025 | 0.231707317073171   | % of counties  | DERIVED | A | S80 | Panel C total row: (suppressed/0 + 1-2) / counties, latest window | Market_Saturation_Ambulance | Whitespace measure; complements Imbalance_Ledger and the State_Saturation thin-supply screen                                                                    |
| F175                                                                                                                                                            | Private ambulance services employment (QCEW annual average)   | 2024 | 171,100             | jobs           | GOV     | A | S82 | QCEW NAICS 621910, US000, ownership 5, 2024 annual average        | QCEW_EMS_Employment         | Companion to the OEWS-based Workforce_Supply tab (occupation vs industry basis; do not mix)                                                                     |
| F176                                                                                                                                                            | Private ambulance establishments                              | 2024 | 5,820               | establishments | GOV     | A | S82 | QCEW NAICS 621910, US000, ownership 5, 2024                       | QCEW_EMS_Employment         | An establishment is a worksite, not a company: compare 10,465 PECOS-enrolled suppliers and 8,721 billing NPIs (Supplier_Landscape) - three different units      |
| F177                                                                                                                                                            | Private ambulance avg annual pay                              | 2024 | 59,511              | \$/yr          | GOV     | A | S82 | QCEW NAICS 621910, US000, ownership 5, 2024                       | QCEW_EMS_Employment         | Pay CAGR vs AIF: the scissors carried on this tab                                                                                                               |
| F178                                                                                                                                                            | Certified hospitals (Care Compare universe)                   | 2026 | 5432                | facilities     | DERIVED | A | S83 | Hospital General Information snapshot, sum over states            | Facility_Universe_State     | AHA counts ~6,100 hospitals total / ~5,200 community (Demand_Drivers): AHA includes non-Medicare-certified and federal units - different universes, both stated |
| F179                                                                                                                                                            | SNF certified beds (Care Compare universe)                    | 2026 | 1568348             | beds           | DERIVED | A | S83 | Nursing Home Provider Info, number_of_certified_beds summed       | Facility_Universe_State     | The SNF interface is 2.75x the hospital-to-hospital Medicare book (Medicare_OD_Matrix): this is the bed base behind it                                          |

|      |                                                                              |            |                         |         |   |            |                                                                                                         |                           |                                                                                                                                                                       |
|------|------------------------------------------------------------------------------|------------|-------------------------|---------|---|------------|---------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F180 | Hospitals: certified facilities (full registry)                              | 2026       | 5,432 facilities        | GOV     | A | S89        | Care Compare snapshot, Hosp_Registry tab row count                                                      | Hosp_Registry             | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F181 | Nursing homes (SNF): certified facilities (full registry)                    | 2026       | 14,695 facilities       | GOV     | A | S89        | Care Compare snapshot, SNF_Registry tab row count                                                       | SNF_Registry              | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F182 | Dialysis facilities: certified facilities (full registry)                    | 2026       | 7,557 facilities        | GOV     | A | S89        | Care Compare snapshot, Dialysis_Registry tab row count                                                  | Dialysis_Registry         | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F183 | Inpatient rehabilitation facilities: certified facilities (full registry)    | 2026       | 1,222 facilities        | GOV     | A | S89        | Care Compare snapshot, IRF_Registry tab row count                                                       | IRF_Registry              | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F184 | Long-term care hospitals: certified facilities (full registry)               | 2026       | 311 facilities          | GOV     | A | S89        | Care Compare snapshot, LTCH_Registry tab row count                                                      | LTCH_Registry             | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F185 | Hospices: certified facilities (full registry)                               | 2026       | 6,852 facilities        | GOV     | A | S89        | Care Compare snapshot, Hospice_Registry tab row count                                                   | Hospice_Registry          | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F186 | Home health agencies: certified facilities (full registry)                   | 2026       | 12,392 facilities       | GOV     | A | S89        | Care Compare snapshot, HHA_Registry tab row count                                                       | HHA_Registry              | Matches Facility_Universe_State state sums (same pull)                                                                                                                |
| F187 | Medicare-enrolled ambulance suppliers (named registry)                       | 2026       | 10,465 organizations    | GOV     | A | S90        | PECOS public enrollment, ambulance supplier type, 2026Q1                                                | PECOS_Registry            | EXACT match to vendored PPEF 2026.04.01 national count and to the live stats probe (10,465)                                                                           |
| F188 | Hospitals in the Hospital Service Area file                                  | 2025       | 7,452 hospitals         | GOV     | A | S91        | HSA 2025, distinct provider IDs                                                                         | HSA_Hospital_Catchment    | CCN-joinable to Hosp_Registry (Care Compare)                                                                                                                          |
| F189 | Hospital-measure rows in the ED throughput registry                          | 2026       | 13,980 rows             | GOV     | A | S94        | Timely & Effective Care hospital file, ED measures                                                      | ED_Timeliness_Registry    | National OP-18 statistics on Demand_Drivers were re-derived from this same file family in v2.7 (S33)                                                                  |
| F190 | Distinct NPIs billing Medicare ambulance codes, 2013 (provider-grain, floor) | 2013       | 9,746 NPIs              | GOV     | A | S95        | MUP by Provider & Service 2013, distinct Rndrng_NPI over A0425-A0436 rows                               | MUP_Providers_2013        | A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe                    |
| F191 | Distinct NPIs billing Medicare ambulance codes, 2019 (provider-grain, floor) | 2019       | 9,554 NPIs              | GOV     | A | S95        | MUP by Provider & Service 2019, distinct Rndrng_NPI over A0425-A0436 rows                               | MUP_Providers_2019        | A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe                    |
| F192 | Distinct NPIs billing Medicare ambulance codes, 2024 (provider-grain, floor) | 2024       | 9,318 NPIs              | GOV     | A | S95        | MUP by Provider & Service 2024, distinct Rndrng_NPI over A0425-A0436 rows                               | MUP_Providers_2024        | A floor (suppression); compare Supplier_Landscape 8,721 billing NPIs (MUP-geo basis) and PECOS 10,465 enrolled - three floors of the same universe                    |
| F193 | Hospital-ZIP corridor rows carried (top-15 per hospital)                     | 2025       | 106,799 rows            | GOV     | A | S96        | HSA latest year, per-hospital top-15 ZIPs by cases                                                      | HSA_Corridors             | Per-hospital totals reconcile to HSA_Hospital_Catchment (same pull family)                                                                                            |
| F194 | Counties with measured 65+ base carried                                      | 2024       | 3,144 counties          | GOV     | A | S97        | CC-EST2024-AGESEX, YEAR codes 2 and 6                                                                   | County_Age_65plus         | County 65+ sums reconcile to State_Age_65plus state rows (same vintage, civilian-vs-resident definitions stated)                                                      |
| F195 | County chronic-prevalence rows (4 measures)                                  | 2024       | 17,867 rows             | GOV     | A | S98        | PLACES county, KIDNEY/CHD/STROKE/DIABETES, CrdPrv                                                       | PLACES_County_Chronic     | Model-based estimates (stated); the ESRD claims denominator on Enrollment_ESRD_State is the measured companion                                                        |
| F196 | Quarterly QCEW rows carried                                                  | 2014-2025  | 4,832 rows              | GOV     | A | S99        | QCEW 621910 quarterly, US000 + statewide                                                                | QCEW_Quarterly            | Annual averages on QCEW_EMS_Employment are the same program aggregated by BLS                                                                                         |
| F197 | HCRIS hospital-year rows (FY2020-2022)                                       | 2020-2022  | 17,974 hospital-years   | GOV     | A | S100       | HCRIS extract, one row per CCN x fiscal year                                                            | HCRIS_Hospital_Panel      | The occupancy_trend statistics quoted in the repo modules derive from this same panel                                                                                 |
| F198 | Ambulance-related LEIE exclusions on file                                    | 2026       | 566 exclusions          | GOV     | A | S101       | LEIE UPDATED.csv, ambulance filter (manifest)                                                           | LEIE_Ambulance_Exclusions | Companion to the CERT/RSNAT payment-integrity evidence on Payment_Integrity                                                                                           |
| F199 | Medicare-enrolled ambulance suppliers, national (PECOS)                      | 2026       | 10465 suppliers         | GOV     | A | S102, S103 | PPEF_Enrollment_Extract_2026.04.01, PROVIDER_TYPE_DESC = "PART B SUPPLIER - AMBULANCE SERVICE SUPPLIER" | PECOS_Suppliers_State     | Live /data/stats probe 2026-07-10 returned the same 10,465 (MATCH cell on-sheet); MUP CY2024 A0425 renderers 10,061 (96%)                                             |
| F200 | Top state by enrolled ambulance suppliers (Ohio)                             | 2026       | 897 suppliers           | GOV     | A | S102       | State table, STATE_CD = OH                                                                              | PECOS_Suppliers_State     | OH 897 > TX 672 > NY 670: supplier counts track fragmentation, not population                                                                                         |
| F201 | Top-10 states share of national enrolled suppliers                           | 2026       | 0.474725274725275 share | DERIVED | A | S102       | SUM of top-10 sorted state rows / national                                                              | PECOS_Suppliers_State     | about 47% - a long-tail, unconsolidated supplier base                                                                                                                 |
| F202 | NE + IA Medicare-enrolled ambulance suppliers                                | 2026       | 583 suppliers           | DERIVED | A | S102       | SUMIF over state rows, NE (287) + IA (296)                                                              | PECOS_Suppliers_State     | NPPES_Registry_NE_IA holds 751 org NPIs for the same states - registry universe wider than enrollment                                                                 |
| F203 | Ambulance suppliers that billed ground mileage, CY2024 (MUP renderers)       | 2024       | 10061 suppliers         | GOV     | A | S104       | MUP by Geography & Service CY2024, National, A0425, Tot_Rndrng_Prvids                                   | PECOS_Suppliers_State     | 96% of the PECOS enrolled universe (formula on-sheet)                                                                                                                 |
| F204 | PECOS vendored-vs-live consistency check                                     | 2026 MATCH | flag                    | DERIVED | A | S102, S103 | IF(vendored national = live found_rows)                                                                 | PECOS_Suppliers_State     | Also: sum of 55 state rows = national (row below Panel B)                                                                                                             |
| F205 | NE+IA ambulance organization NPIs (NPPES registry)                           | 2026       | 751 organization NPIs   | DERIVED | A | S105       | COUNTIF over the 751-row roster (NE 400 + IA 351)                                                       | NPPES_Registry_NE_IA      | vs 583 PECOS-enrolled suppliers for NE+IA: the gap is the volunteer / non-Medicare-billing layer                                                                      |
| F206 | NE municipal/fire/volunteer squads                                           | 2026       | 328 organization NPIs   | DERIVED | A | S105       | COUNTIFS search_state=NE, category=municipal-fire-volunteer                                             | NPPES_Registry_NE_IA      | 328 of 400 NE org NPIs (82%) - the fragmented municipal base the consolidation thesis targets                                                                         |
| F207 | IA hospital-owned ambulance services                                         | 2026       | 40 organization NPIs    | DERIVED | A | S105       | COUNTIFS search_state=IA, category=hospital-owned                                                       | NPPES_Registry_NE_IA      | IA 40 vs NE 5: Iowa runs a hospital-owned model - the structural contrast between the two states                                                                      |
| F208 | Private/commercial ambulance operators, NE+IA                                | 2026       | 128 organization NPIs   | DERIVED | A | S105       | COUNTIFS category=private (NE 58 + IA 70)                                                               | NPPES_Registry_NE_IA      | The thin private layer an IFT specialist competes in                                                                                                                  |
| F209 | Distinct organization names behind the 751 NPIs                              | 2026       | 686 organizations       | DERIVED | A | S105       | SUMPRODUCT/COUNTIF distinct-count over org_name                                                         | NPPES_Registry_NE_IA      | Multi-NPI squads mean NPI counts overstate agency counts by ~9%                                                                                                       |
| F210 | Roster rows with out-of-state practice/mailling address                      | 2026       | 53 rows                 | DERIVED | A | S105       | SUMPRODUCT over addr-state column                                                                       | NPPES_Registry_NE_IA      | Search-state matches practice OR mailing address - 53 rows are out-of-state operators serving NE/IA                                                                   |
| F211 | Roster rows with non-ambulance primary taxonomy                              | 2026       | 21 rows                 | DERIVED | A | S105       | SUMPRODUCT ISERROR(SEARCH("Ambulance", primary_taxonomy))                                               | NPPES_Registry_NE_IA      | Hospitals/CAHs/NEMT vans with ambulance secondary taxonomy - 21 rows, kept and flagged                                                                                |
| F212 | Hospital CHOW events, 2016-2025 total                                        | 2025       | 755 events              | GOV     | A | S106       | Hospital_CHOW_2026.04.01, sum of state x year matrix                                                    | CHOW_Consolidation        | Matches report JSON total_chows 755 (check row); state sums = national series exactly                                                                                 |
| F213 | SNF CHOW events, 2016-2025 total                                             | 2025       | 5141 events             | GOV     | A | S107       | SNF_CHOW_2026.04.01, sum of state x year matrix                                                         | CHOW_Consolidation        | Matches report JSON total_chows 5,141 (check row); 6.8x the hospital CHOW count                                                                                       |
| F214 | SNF CHOWs, peak year 2023                                                    | 2023       | 882 events              | GOV     | A | S107       | National series, year 2023                                                                              | CHOW_Consolidation        | 3.6x the 2016 level (244) - post-acute ownership churn accelerated through 2023                                                                                       |
| F215 | Hospital CHOWs, 2024                                                         | 2024       | 101 events              | GOV     | A | S106       | National series, year 2024                                                                              | CHOW_Consolidation        | Volatile series: 38 (2022) to 127 (2019); companion to the Kaufman Hall M&A series on Growth_Evidence_Registry (announced deals vs recorded provider-agreement CHOWs) |
| F216 | SNF CHOW CAGR 2016>2024                                                      | 2024       | 0.142432182943028 %/yr  | DERIVED | A | S107       | (2024/2016)^(1/8)-1 live formula; 2025 excluded (partial accrual flag)                                  | CHOW_Consolidation        | 244 > 708 events/yr; approx +14%/yr                                                                                                                                   |

|      |                                                                                      |                 |                           |         |   |            |                                                                                                                                                                                 |                             |                                                                                   |
|------|--------------------------------------------------------------------------------------|-----------------|---------------------------|---------|---|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------|
| F217 | Hospital CHOW CAGR 2016>2024                                                         | 2024            | 0.0672625377760108 %/yr   | DERIVED | A | S106       | (2024/2016)^(1/8)-1 live formula; 2025 excluded                                                                                                                                 | CHOW_Consolidation          | 60 > 101 events/yr; approx +6.7%/yr on a volatile base - read with the YoY column |
| F218 | Top state by SNF CHOWs 2016-2025 (Texas)                                             | 2025            | 697 events                | DERIVED | A | S107       | State matrix row 1 (sorted by total), TX                                                                                                                                        | CHOW_Consolidation          | TX 697 > CA 393 > OH 310                                                          |
| F219 | CY2025 PFS payment localities carried (Addendum E)                                   | CY2025          | 109 localities            | GOV     | A | S108       | Addendum E, RVU25A package; vendored registry row cms_gpci_2025 (109 locality rows incl. DC/PR/VI)                                                                              | GPCI_Localities             | registry row_count=109; raw CSV 116 lines incl. headers/footnotes                 |
| F220 | National unweighted mean PE GPCI                                                     | CY2025          | 1.0434495412844 index     | DERIVED | A | S108       | AVERAGE over Addendum E PE GPCI column (109 localities)                                                                                                                         | GPCI_Localities             | python recompute 1.0434                                                           |
| F221 | National unweighted mean ambulance geographic factor (0.7 x PE + 0.3)                | CY2025          | 1.03041467889908 factor   | DERIVED | A | S108, S109 | 42 CFR 414.610(c)(4) 70/30 rule over Addendum E                                                                                                                                 | GPCI_Localities             | python recompute 1.0304                                                           |
| F222 | Lowest ambulance geographic factor (Mississippi)                                     | CY2025          | 0.8964 factor             | DERIVED | A | S108, S109 | MIN over derived col H; MS PE GPCI 0.852 -> 0.8964                                                                                                                              | GPCI_Localities             | 0.7*0.852+0.3 = 0.8964                                                            |
| F223 | Highest ambulance geographic factor (San Jose-Sunnyvale-Santa Clara, CA)             | CY2025          | 1.3045 factor             | DERIVED | A | S108, S109 | MAX over derived col H; Santa Clara/San Benito PE GPCI 1.435 -> 1.3045                                                                                                          | GPCI_Localities             | 0.7*1.435+0.3 = 1.3045                                                            |
| F224 | Max/min spread of the ambulance geographic factor                                    | CY2025          | 1.45526550647033 x        | DERIVED | A | S108, S109 | MAX/MIN over derived col H                                                                                                                                                      | GPCI_Localities             | 1.3045/0.8964 = 1.455x - geography moves the base rate by at most ~46% end to end |
| F225 | CY2026 AFS ground conversion factor                                                  | CY2026          | 284.56 \$/RVU             | GOV     | B | S110, S111 | CY2026 AFS public use file; MedPAC Jun 2026 Ch.6                                                                                                                                | Derived_Rate_Card           | v2.7 Payment_Rules CY2024 CF \$272.44 (vintage)                                   |
| F226 | CY2025 AFS ground conversion factor                                                  | CY2025          | 278.98 \$/RVU             | GOV     | B | S111, S112 | MedPAC Jun 2026 Ch.6 (prior-year CF)                                                                                                                                            | Derived_Rate_Card           |                                                                                   |
| F227 | CF update CY2025->CY2026                                                             | CY2026          | 0.0200014337945371 %/yr   | DERIVED | B | S114       | live formula (284.56/278.98)-1; equals published CY2026 AIF +2.0%                                                                                                               | Derived_Rate_Card           | CMS Transmittal 13464: AIF = CPI-U 2.7% - TFP 0.7% = 2.0%                         |
| F228 | SCT (A0434) CY2026 national unadjusted base                                          | CY2026          | 924.82 \$                 | DERIVED | B | S110, S111 | live formula 3.25 RVU x \$284.56 = \$924.82                                                                                                                                     | Derived_Rate_Card           | repo cy2026_base column, delta < \$0.005                                          |
| F229 | BLS non-emergency (A0428) super-rural base                                           | CY2026          | 348.87056 \$              | DERIVED | B | S115, S110 | live formula \$284.56 x 1.226 = \$348.87                                                                                                                                        | Derived_Rate_Card           | add-on % per 42 U.S.C. 1395m(l)(13)                                               |
| F230 | Ground mileage A0425, CY2026 national                                                | CY2026          | 9.15 \$/mile              | GOV     | B | S111, S110 | MedPAC Jun 2026 Ch.6: \$9.15/mile national                                                                                                                                      | Derived_Rate_Card           | v2.7 Payment_Rules CY2024 \$8.76 (vintage)                                        |
| F231 | Ground mileage A0425, rural miles 1-17                                               | CY2026          | 14.13 \$/mile             | DERIVED | B | S111, S115 | live formula \$9.42 x 1.5 = \$14.13                                                                                                                                             | Derived_Rate_Card           |                                                                                   |
| F232 | Paramedic Intercept (A0432) RVU                                                      | CY2026          | 1.75 RVU                  | SOURCED | B | S110       | AFS RVU ladder; carried from repo ift_analytics (cites the RVU table via a PYA Medicare-payment primer) - RE-VERIFY against the CY2026 AFS file                                 | Derived_Rate_Card           | flagged re-verify; PI base column derived from this RVU                           |
| F233 | CY2024 Medicare six-code ground services (supplier claims)                           | CY2024          | 9637068 services          | DERIVED | B | S113       | live SUM over the six A04xx code rows                                                                                                                                           | Service_Level_Economics     | 9,637,068 (python recompute)                                                      |
| F234 | ALS1-emergency (A0427) share of six-code services                                    | CY2024          | 0.383666588219571 %       | DERIVED | B | S113       | live formula services/total; 38.4%                                                                                                                                              | Service_Level_Economics     | module medicare_mix() share_pct 38.4                                              |
| F235 | Medicare FFS ground transports                                                       | 2024            | 11300000 transports       | GOV     | B | S111       | MedPAC Jun 2026 Ch.6: 11.3M transports, ~10,600 orgs                                                                                                                            | Service_Level_Economics     | supplier-file six-code total 9.64M (different universe - see reconciliation note) |
| F236 | Medicare AFS payments                                                                | 2024            | 5300000000 \$             | GOV     | B | S111       | MedPAC Jun 2026 Ch.6: \$5.3B                                                                                                                                                    | Service_Level_Economics     |                                                                                   |
| F237 | Average AFS payment per transport                                                    | 2024            | 469.026548672566 \$       | DERIVED | B | S111       | live formula \$5.3B / 11.3M = ~\$469                                                                                                                                            | Service_Level_Economics     | ift_unit_economics medicare_avg_payment \$469 (needs_reverify=False)              |
| F238 | GADCS all-payer BLS share of ground transports                                       | 2022-23 cohorts | 0.56 %                    | GOV     | B | S117       | GADCS Dec 2024 report, 3,694 reporting orgs; verbatim quote carried on-row                                                                                                      | Service_Level_Economics     | ALS1 42%; ALS2+SCT 3% on adjacent rows                                            |
| F239 | NEMSIS facility-to-facility share of EMS activations                                 | 2024            | 0.124590712460657 %       | DERIVED | B | S120       | live formula 7,512,656 / 60,298,684 = ~12.5%                                                                                                                                    | Service_Level_Economics     | hospital-to-hospital subset 5,510,664 (9.1%)                                      |
| F240 | GADCS mean cost per transport                                                        | 2022-23 cohorts | 2673 \$                   | GOV     | B | S117       | GADCS Dec 2024: mean \$2,673, median \$1,340, labor 69.4%                                                                                                                       | Service_Level_Economics     | GAO-13-6 2010 median \$429 (range \$224-\$2,204) on adjacent row                  |
| F241 | Ambulance Inflation Factor, CY2026                                                   | CY2026          | 0.02 %/yr                 | DERIVED | B | S114       | live formula CPI-U 2.7% - TFP 0.7% = 2.0%; transmittal quote on-row                                                                                                             | Service_Level_Economics     | matches Derived_Rate_Card CF update formula                                       |
| F242 | CO statewide commercial paid as multiple of Medicare - Inpatient/Outpatient combined | 2024            | 2.3856 x Medicare         | SOURCED | A | S124       | FY26 RBP public file, STATEWIDE row, IP/OP combined, 2024 (2.39x); facility/hospital claims, Colorado only - PATIENT for commercial multiples, NOT on multiplex rate            | Commercial_Context_APCD     | claims-weighted provider mean 2.42x (python)                                      |
| F243 | CO statewide commercial multiple - Outpatient                                        | 2024            | 2.5837 x Medicare         | SOURCED | A | S124       | FY26 RBP public file, STATEWIDE row (2.58x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT on multiplex rate                                  | Commercial_Context_APCD     |                                                                                   |
| F244 | CO statewide commercial multiple - Inpatient                                         | 2024            | 1.9116 x Medicare         | SOURCED | A | S124       | FY26 RBP public file, STATEWIDE row (1.91x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT on multiplex rate                                  | Commercial_Context_APCD     |                                                                                   |
| F245 | Median of the 12 CO statewide commercial-multiple readings                           | 2021-2024       | 2.35515 x Medicare        | DERIVED | A | S124       | live MEDIAN formula over staging block A; facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT on multiplex rate                                     | Commercial_Context_APCD     |                                                                                   |
| F246 | CO provider-level median commercial multiple - IP/OP combined                        | 2024            | 2.3845 x Medicare         | SOURCED | A | S124       | inclusive-quartile median over 86 provider rows of the vendored file (2.38x); facility/hospital claims, Colorado only - PATIENT for commercial multiples, NOT on multiplex rate | Commercial_Context_APCD     | statewide published 2.39x - consistent                                            |
| F247 | CO provider-level IQR of commercial multiple - IP/OP combined                        | 2024            | 0.9755 x Medicare (width) | DERIVED | A | S124       | live formula p75-p25 (1.79x to 2.77x); facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT an multiplex rate                                        | Commercial_Context_APCD     |                                                                                   |
| F248 | CO statewide commercial-multiple drift - IP/OP combined, 3-yr CAGR                   | 2021-2024       | 0.0137410578167461 %/yr   | DERIVED | A | S124       | live CAGR formula 2021->2024 window over staging block A; facility/hospital claims, Colorado only - CONTEXT for commercial multiples, NOT on multiplex rate                     | Commercial_Context_APCD     | 2.29x -> 2.39x                                                                    |
| F249 | Acute-MI inpatient stays (principal dx), US                                          | 2018            | 658600 stays/yr           | SOURCED | B | S125       | HCUP SB#277, 2018 NIS, principal-diagnosis stay counts                                                                                                                          | Condition_Transfer_Registry | CDC/AHA ~805,000 MI events/yr (registry note)                                     |
| F250 | Heart-failure inpatient stays (parent pool), US                                      | 2018            | 1135900 stays/yr          | SOURCED | B | S125       | HCUP SB#277, 2018 NIS - HF is the #2 US inpatient condition                                                                                                                     | Condition_Transfer_Registry | same pool reused by Cardiac recovery row - never double-count                     |
| F251 | Cerebral-infarction inpatient stays, US                                              | 2018            | 533400 stays/yr           | SOURCED | B | S125       | HCUP SB#277, 2018 NIS                                                                                                                                                           | Condition_Transfer_Registry | CDC >795,000 strokes/yr, ~87% ischemic                                            |
| F252 | Adults developing sepsis per year, US                                                | 2023            | 1700000 adults/yr         | SOURCED | B | S127       | CDC sepsis surveillance ("at least 1.7M adults")                                                                                                                                | Condition_Transfer_Registry | HCUP: septicemia #1 inpatient condition, 2,218,800 stays 2018                     |
| F253 | Mental-health-related ED visits, US                                                  | 2021            | 6000000 ED visits/yr      | SOURCED | B | S131       | CDC/SAMHSA ED utilization, 2021                                                                                                                                                 | Condition_Transfer_Registry | ~1 in 8 ED visits MH/SUD                                                          |



|      |                                                                      |           |                                 |          |   |                              |                                                                                                                                                    |                             |                                                                           |
|------|----------------------------------------------------------------------|-----------|---------------------------------|----------|---|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------|
| F254 | US births (all)                                                      | 2022      | 3667758 births/yr               | SOURCED  | B | S129                         | CDC/NCHS Births: Final Data for 2022                                                                                                               | Condition_Transfer_Registry | preterm row = 10.38% of this pool                                         |
| F255 | Medicare-covered SNF stays                                           | 2022      | 1800000 stays/yr                | SOURCED  | B | S136                         | MedPAC Payment Basics - SNF (2022 data)                                                                                                            | Condition_Transfer_Registry | ~14,700 SNFs cross-checks Post_Acute_Supply_State SNF count 14,699        |
| F256 | ED encounters transferred to another short-term hospital (2009)      | 2009      | 1900000 transfers/yr            | SOURCED  | B | S126                         | HCUP SB#155: 1.5% x 128,885,040 ED encounters, 2009 NEDS                                                                                           | Condition_Transfer_Registry | modern read: NEDS 2018-2022 ~1.97Myr (Nikolla 2025)                       |
| F257 | Registry rows with a GOV-cited national volume                       | 2026      | 23 rows (of 32)                 | DERIVED  | B | S140                         | live COUNTIF over the Basis column (GOV 23 / ACADEMIC 6 / FRAMEWORK 3)                                                                             | Condition_Transfer_Registry |                                                                           |
| F258 | Skilled nursing facility (SNF) - certified providers, US             | 2026      | 14699 providers                 | SOURCED  | A | S141                         | CMS Nursing Home Care Compare - Provider Information (NH_ProviderInfo), file vintage 2026-04-01, one row per certified provider (CPH)              | Post_Acute_Supply_State     | MedPAC ~14,700 SNFs; registry SNF row                                     |
| F259 | Inpatient rehabilitation facility (IRF) - certified providers, US    | 2026      | 1221 providers                  | SOURCED  | A | S142                         | CMS Inpatient Rehabilitation Facility Compare - General Information + Provider Data, file vintage 2026-02-13, one row per certified provider (CPH) | Post_Acute_Supply_State     | MedPAC ~1,180 IRFs (registry note)                                        |
| F260 | Long-term acute care hospital (LTACH/LTCH) - certified providers, US | 2026      | 317 providers                   | SOURCED  | A | S143                         | CMS Long-Term Care Hospital Compare - General Information + Provider Data, file vintage 2026-02-13, one row per certified provider (CPH)           | Post_Acute_Supply_State     | MedPAC ~320 LTCHs (registry note)                                         |
| F261 | Home health agency (HHA) - certified providers, US                   | 2026      | 12392 providers                 | SOURCED  | A | S144                         | CMS Provider Data Catalog - Home Health Care Agencies (dataset gpm-sake), file vintage 2026-05-23, one row per certified provider (CPH)            | Post_Acute_Supply_State     |                                                                           |
| F262 | Hospice - certified providers, US                                    | 2026      | 6852 providers                  | SOURCED  | A | S145                         | CMS Provider Data Catalog - Hospice General Information (dataset yc9t-dgpk), file vintage 2026-05-23, one row per certified provider (CPH)         | Post_Acute_Supply_State     |                                                                           |
| F263 | Total post-acute destinations, US (5 settings)                       | 2026      | 35481 providers                 | DERIVED  | A | S141, S142, S143, S144, S145 | live SUM of the five CMS provider-file counts (35,481)                                                                                             | Post_Acute_Supply_State     | matches ift_study.ecosystem().postacute_destinations                      |
| F264 | CA total post-acute destinations                                     | 2026      | 6474 providers                  | DERIVED  | A | S141, S142, S143, S144, S145 | row SUM across the five CMS provider files, state=CA                                                                                               | Post_Acute_Supply_State     |                                                                           |
| F265 | TX total post-acute destinations                                     | 2026      | 4306 providers                  | DERIVED  | A | S141, S142, S143, S144, S145 | row SUM across the five CMS provider files, state=TX                                                                                               | Post_Acute_Supply_State     |                                                                           |
| F266 | STEMI transfer median door-in-door-out time                          | 2011      | 68 minutes                      | ACADEMIC | B | S146                         | Wang et al., JAMA 2011, DOI 10.1001/jama.2011.862 (n=14,821)                                                                                       | Clinical_Benchmarks         | DIDO >30 min: mortality 5.9% vs 2.7%, aOR 1.56                            |
| F267 | STEMI in-hospital mortality when DIDO >30 min                        | 2011      | 0.059 share                     | ACADEMIC | B | S146                         | Wang et al., JAMA 2011, DOI 10.1001/jama.2011.862                                                                                                  | Clinical_Benchmarks         | vs 2.7% when <=30 min                                                     |
| F268 | Mean stroke-transfer DIDO                                            | 2024      | 171.4 minutes                   | ACADEMIC | B | S148                         | JAMA Netw Open 2024, DOI 10.1001/jamanetworkopen.2024.31183 (n=28,887)                                                                             | Clinical_Benchmarks         | GWGT median 174 min (JAMA 2023) is the companion read                     |
| F269 | EMS prenotification-associated stroke DIDO reduction                 | 2023      | 20.1 minutes                    | ACADEMIC | B | S149                         | GWtG-Stroke, JAMA 2023, DOI 10.1001/jama.2023.12739 (n=108,913)                                                                                    | Clinical_Benchmarks         |                                                                           |
| F270 | National median ambulance patient offload time                       | 2024      | 10.9 minutes                    | ACADEMIC | B | S150                         | Shaw et al., Prehosp Emerg Care 2025, DOI 10.1080/10903127.2025.2535576 (7,237,606 records)                                                        | Clinical_Benchmarks         |                                                                           |
| F271 | Hospitals detaining EMS crews >1 hour (CA 2017)                      | 2017      | 0.75 share                      | ACADEMIC | B | S151                         | Backer et al., Prehosp Emerg Care 2018, DOI 10.1080/10903127.2018.1525456 (830,637 transports)                                                     | Clinical_Benchmarks         | 40% >2h, 33% >3h                                                          |
| F272 | Admitted ED patients 65+ boarding >=2 hours                          | 2015-2022 | 0.852 share                     | ACADEMIC | B | S152                         | Lee et al., Ann Emerg Med 2026, DOI 10.1016/j.annemergmed.2026.03.011 (NHAMCS 2015-2022)                                                           | Clinical_Benchmarks         |                                                                           |
| F273 | Mean ED boarding, admitted 65+ (2022)                                | 2022      | 343 minutes                     | ACADEMIC | B | S152                         | Lee et al., Ann Emerg Med 2026, NHAMCS 2022 read                                                                                                   | Clinical_Benchmarks         | up from 138 min in 2018; 501 min with dementia                            |
| F274 | Discharges delayed - weighted mean across 64 studies                 | 2019      | 0.228 share                     | ACADEMIC | B | S154                         | Landeiro et al., The Gerontologist 2019, DOI 10.1093/geronl/gnz028                                                                                 | Clinical_Benchmarks         | range 1.6-91.3%; cost \$142-\$31,935 PPP/case                             |
| F275 | PACU delays caused by transport unavailability                       | 2022      | 0.111 share                     | ACADEMIC | B | S156                         | Ego et al., Ann Med Surg 2022, DOI 10.1016/j.amсу.2022.104680 (n=34 of 307; non-US)                                                                | Clinical_Benchmarks         | second to bed unavailability 22.5%                                        |
| F276 | Dialysis - cert-base CAGR of surviving stock, window 1981-2025       | 1981-2025 | 0.0674787558009695 %/yr         | DERIVED  | A | S158                         | live formula (cum 2025 / cum 1981)^(1/44)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F277 | SNF - cert-base CAGR of surviving stock, window 1968-2025            | 1968-2025 | 0.0511659229116082 %/yr         | DERIVED  | A | S159                         | live formula (cum 2025 / cum 1968)^(1/57)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F278 | Home health - cert-base CAGR of surviving stock, window 1978-2025    | 1978-2025 | 0.0643466141161573 %/yr         | DERIVED  | A | S160                         | live formula (cum 2025 / cum 1978)^(1/47)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F279 | Hospice - cert-base CAGR of surviving stock, window 1988-2025        | 1988-2025 | 0.0793692359442053 %/yr         | DERIVED  | A | S161                         | live formula (cum 2025 / cum 1988)^(1/37)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F280 | IRF - cert-base CAGR of surviving stock, window 1984-2025            | 1984-2025 | 0.0592294981076973 %/yr         | DERIVED  | A | S162                         | live formula (cum 2025 / cum 1984)^(1/41)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F281 | LTCH - cert-base CAGR of surviving stock, window 1976-2025           | 1976-2025 | 0.0642414341728086 %/yr         | DERIVED  | A | S163                         | live formula (cum 2025 / cum 1976)^(1/49)-1 over the certification-vintage grid; survivor-stock caveat travels                                     | Certification_Series        | matches analytics.supply_trend() to 1e-9                                  |
| F282 | Dialysis facilities in the certified roll                            | 2026      | 7557 facilities                 | SOURCED  | A | S158                         | live link to the grid's last cumulative cell (7,557 rows in DFC_FACILITY, source_date 2026-03-25)                                                  | Certification_Series        | equals State_Facility_Structure dialysis total                            |
| F283 | Home-health net adds in peak build year 2023                         | 2023      | 609 agencies certified          | SOURCED  | A | S160                         | certification-vintage grid, year 2023 (609 agencies of the surviving stock certified that year)                                                    | Certification_Series        |                                                                           |
| F284 | Hospice net adds in peak build year 2022                             | 2022      | 771 hospices certified          | SOURCED  | A | S161                         | certification-vintage grid, year 2022 (771 hospices of the surviving stock certified that year)                                                    | Certification_Series        |                                                                           |
| F285 | Dialysis chain HHI (all 30 named chains)                             | 2026      | 2767 HHI (0-10,000)             | SOURCED  | A | S158                         | chain_org column of DFC_FACILITY vendored roll (source_date 2026-03-25); share= sum over named operators, all 2,557 facility denominators          | State_Facility_Structure    | live top-6 formula computes 2,766; >2,500 = highly concentrated (DOJ/FTC) |
| F286 | Dialysis chain HHI - six largest chains (live)                       | 2026      | 2765.90655055805 HHI (0-10,000) | DERIVED  | A | S158                         | live Σ(share*100) <sup>2</sup> over the six named-chain rows                                                                                       | State_Facility_Structure    | module full-file HHI 2,767                                                |
| F287 | DaVita share of certified dialysis facilities                        | 2026      | 0.370517401085087 share         | DERIVED  | A | S158                         | live 2,800/7,557 from chain_org counts                                                                                                             | State_Facility_Structure    |                                                                           |
| F288 | Fresenius share of certified dialysis facilities                     | 2026      | 0.366812227074236 share         | DERIVED  | A | S158                         | live 2,772/7,557 from chain_org counts                                                                                                             | State_Facility_Structure    |                                                                           |
| F289 | DaVita + Fresenius combined facility share                           | 2026      | 0.737329628159323 share         | DERIVED  | A | S158                         | live sum of the two chain-share formulas (73.8%)                                                                                                   | State_Facility_Structure    | Koukounas et al. (JAMA/PMC12177639): 77.1% of facilities in 2019          |
| F290 | Dialysis for-profit share (all rows ownership-known)                 | 2026      | 0.895858144766442 share         | DERIVED  | A | S158                         | live ΣD/ΣB over the dialysis state panel (89.6%)                                                                                                   | State_Facility_Structure    |                                                                           |

|      |                                                                  |           |                                                                                    |          |   |            |                                                                                                                                         |                          |                                                                                                                                       |
|------|------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------|----------|---|------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| F291 | SNF certified facilities, US (live SUM of states)                | 2026      | 14699 facilities                                                                   | DERIVED  | A | S159       | live SUM over 53 state rows of NH_ProviderInfo (source_date 2026-04-01) = 14,699                                                        | State_Facility_Structure | matches Post_Acute_Supply_State SNF count and MedPAC ~14,700                                                                          |
| F292 | Home-health top-5 state concentration                            | 2026      | 0.605067785668173 share of agencies                                                | DERIVED  | A | S160       | live SUM of the five largest state shares (60.5%; CA alone 25.4%)                                                                       | State_Facility_Structure |                                                                                                                                       |
| F293 | Hospice facilities in CA (largest state)                         | 2026      | 2062 facilities                                                                    | SOURCED  | A | S161       | hospice state panel row 1 (CA 2,062 of 6,852 = 30.1%)                                                                                   | State_Facility_Structure |                                                                                                                                       |
| F294 | Hospital M&A announced transactions, 2025                        | 2025      | 46 transactions                                                                    | SOURCED  | B | S173       | Kaufman Hall 2025 Hospital and Health System M&A in Review (annual series 2015-2025)                                                    | Growth_Evidence_Registry | trough of the 11-year series; distress share 43.5% an all-time high                                                                   |
| F295 | Hospital M&A announced-transaction change 2015 -- 2025           | 2015-2025 | -0.589285714285714 % change                                                        | DERIVED  | B | S173       | live formula 46/112-1 over the series endpoints (-58.9%)                                                                                | Growth_Evidence_Registry |                                                                                                                                       |
| F296 | Distressed share of 2025 hospital M&A                            | 2025      | 0.435 share of transactions                                                        | SOURCED  | B | S173       | Kaufman Hall 2025 review: "43.5% of all transactions involving a distressed party"                                                      | Growth_Evidence_Registry |                                                                                                                                       |
| F297 | US community hospitals system-affiliated, FY2024                 | FY2024    | 0.696543643819567 share                                                            | DERIVED  | B | S170       | live 3,567/5,121 from AHA Fast Facts 2026 (2024 Annual Survey)                                                                          | Growth_Evidence_Registry | AHRQ CHSP: ~70% of non-federal general acute hospitals in a system (2022)                                                             |
| F298 | System-affiliation pp change 2005 -- FY2024                      | 2005-2024 | 0.166543643819566 percentage points                                                | DERIVED  | B | S170, S171 | live difference of the series endpoints (53% -- ~70%)                                                                                   | Growth_Evidence_Registry | do not CAGR - irregular multi-source series                                                                                           |
| F299 | Adult ED-to-ED interfacility transfers 2018-2022 (NEDS)          | 2018-2022 | 9,867,701 over 2018-2022 (~1.97M/yr); critical-care transfers (5 yrs)              | ACADEMIC | B | S165       | Nikolla et al., J Emerg Med 2025, DOI 10.1016/j.jemermed.2025.12.020 (HCUP NEDS 2018-2022); 0.967-70% critical care transfers QD 1,000+ | Growth_Evidence_Registry | Demand_Evidence_Quotes neds, ed, transfers record                                                                                     |
| F300 | ED visits arriving via interfacility EMS transfer - trend        | 2014-2020 | ~1.3M/yr; +15% (2017-19) and +35% (2020-22) vs 2014-16                             | ACADEMIC | B | S164       | Peters GA et al., Am J Emerg Med 2026;106:24-29, DOI 10.1016/j.ajem.2026.04.025 (verbatim)                                              | Growth_Evidence_Registry |                                                                                                                                       |
| F301 | Rural vs urban ED transfer rate multiplier                       | -         | 6.2% rural vs 2.0% urban - a >3x multiplier                                        | ACADEMIC | B | S166       | Greenwood-Ericksen et al., JAMA Netw Open 2021, DOI 10.1001/jamanetworkopen.2021.34980 (verbatim)                                       | Growth_Evidence_Registry |                                                                                                                                       |
| F302 | Rural hospital closures since 2005 (Sheps tracker)               | 2005-2022 | 194 since 2005 (151 after 2010); NE: 2 (latest March 2025 Online 2023)             | SOURCED  | B | S181       | UNC Sheps Center rural hospital closures tracker (needs re-verify)                                                                      | Growth_Evidence_Registry |                                                                                                                                       |
| F303 | Rural Emergency Hospital conversions since Jan 2023              | 2023-2025 | 40-50 conversions since Jan 2023; nearly half in KS/TX/NE/OK                       | ACADEMIC | B | S182       | J Rural Health 2026, DOI 10.1111/jrh.70112 (verbatim); REHS keep no inpatient beds - every admission becomes a transfer                 | Growth_Evidence_Registry |                                                                                                                                       |
| F304 | US ESRD prevalence (dialysis transport driver)                   | 2020      | >808,000 living with ESRD; ~68% on dialysis; prevalence dialysis                   | GOV      | B | S186       | NIDDK Kidney Disease Statistics (USRDS-based), Oct 2025 (needs re-verify)                                                               | Growth_Evidence_Registry |                                                                                                                                       |
| F305 | Medicare FFS ground ambulance transports / spend / orgs          | 2024      | 11.3 million transports; \$5.3 billion; ~10,600 organizations                      | GOV      | B | S197       | MedPAC Payment Basics "Ambulance Services Payment System", Oct 2024, p.1 (verbatim quote on-row)                                        | Demand_Evidence_Quotes   | MedPAC Mar 2025 report: ~11.4M transports (2023 data)                                                                                 |
| F306 | Adult ED-to-ED transfers, 2018-2022                              | 2018-2022 | 9,867,701 adult transfers over 2018-2022 (~1.97M/yr)                               | ACADEMIC | B | S165       | Am J Emerg Med (2025), HCUP NEDS 2018-2022 (verbatim)                                                                                   | Demand_Evidence_Quotes   | Growth_Evidence_Registry neds, transfers record                                                                                       |
| F307 | Acute-to-acute inter-hospital transfers per year                 | -         | ~1.5 million/yr = 3.5% of inpatient admissions                                     | ACADEMIC | B | S198       | Hernandez-Boussard T et al., J Patient Saf 2017 (2009 NIS; epub 2014), PubMed 25397857 (HCUP NIS; verbatim)                             | Demand_Evidence_Quotes   | 2009 data, epub 2014 - often miscited as Mueller 2014; corrected against PubMed                                                       |
| F308 | BLS share of ground ambulance transports                         | -         | 56% BLS (so ~44% ALS)                                                              | SOURCED  | B | S199       | CMS GADCS Year 1-2 cohort report (RAND 2024), cms.gov PDF (verbatim)                                                                    | Demand_Evidence_Quotes   |                                                                                                                                       |
| F309 | National EMS activations (2023)                                  | 2023      | 54,190,579 activations; 14,369 agencies; 54 agencies                               | SOURCED  | B | S200       | NEMSIS 2023 Public-Release Research Dataset (NHTSA Office of EMS) (verbatim)                                                            | Demand_Evidence_Quotes   |                                                                                                                                       |
| F310 | US health systems; hospital affiliation                          | 2022      | 640 health systems (2022); ~70% of non-federal general acute hospitals in a system | GOV      | B | S202       | AHRQ Compendium of US Health Systems, 2022 (verbatim)                                                                                   | Demand_Evidence_Quotes   | AHA Fast Facts FY2024: 3,567/5,121 = 69.6%                                                                                            |
| F311 | US inpatient discharges per year                                 | -         | ~35 million weighted discharges/yr (20% stratified census)                         | SOURCED  | B | S201       | AHRQ HCUP NIS overview page (quote carries editorial weighting note)                                                                    | Demand_Evidence_Quotes   |                                                                                                                                       |
| F312 | BLS non-emergency claim-line drift 2018 -- 2022 (GADCS)          | 2018-2022 | -0.0660000000000001 percentage points                                              | DERIVED  | B | S199       | live difference of the two published GADCS points (43.7% -- 37.1% = -6.6pp)                                                             | Demand_Evidence_Quotes   | 2-point pair - direction only                                                                                                         |
| F313 | US 65+ population CAGR 2025 -- 2030 (projection)                 | 2025-2030 | 0.0269022284200189 %/yr                                                            | DERIVED  | B | S188       | live (71.6/62.7)^(1/5)-1 over the Census projection pair (~2.69%/yr)                                                                    | Demand_Evidence_Quotes   | the Census 65+ projection pair is GOV; the per-condition blended ~2.7%/yr read is modeled and lives on Excluded_Not_Sourced, not here |
| F314 | US total MA enrollment (state-file sum)                          | 2022      | 29674598 enrollees                                                                 | DERIVED  | A | S205       | live SUM of 53 state rows; equals ma_geo_report.json total 29,674,598 (diff cell computes 0)                                            | MA_Geo_Variation         |                                                                                                                                       |
| F315 | MA IP stays per 1,000 enrollees, national (enrollment-weighted)  | 2022      | 199.291194877845 stays /1,000                                                      | DERIVED  | A | S205       | live SUMPRODUCT over the state table (ip_stays_per_1000 field)                                                                          | MA_Geo_Variation         | published bound for the TAM_Model_National MA-utilization TEAM-INPUT (0.75-1.0x)                                                      |
| F316 | MA SNF days per 1,000 enrollees, national (enrollment-weighted)  | 2022      | 783.260956019121 days /1,000                                                       | DERIVED  | A | S205       | live SUMPRODUCT over the state table (snf_days_per_1000 field)                                                                          | MA_Geo_Variation         |                                                                                                                                       |
| F317 | MA ER visits per 1,000 enrollees, national (enrollment-weighted) | 2022      | 354.956621244173 visits /1,000                                                     | DERIVED  | A | S205       | live SUMPRODUCT over the state table (er_visits_per_1000 field)                                                                         | MA_Geo_Variation         |                                                                                                                                       |
| F318 | Highest state SNF days per 1,000 MA enrollees                    | 2022      | 3237.767 days /1,000                                                               | DERIVED  | A | S205       | live MAX screen over the state table (companion INDEX/MATCH cell names the state)                                                       | MA_Geo_Variation         | Massachusetts, 3,237.8 - a published outlier >2x the next state (OH 1,502.6); outlier note on-sheet                                   |
| F319 | California MA enrollment (largest state)                         | 2022      | 3188104 enrollees                                                                  | SOURCED  | A | S205       | state table row 1 (CA 3,188,104)                                                                                                        | MA_Geo_Variation         |                                                                                                                                       |
| F320 | States/territories in the MA geo file                            | 2022      | 53 states                                                                          | DERIVED  | A | S205       | live COUNTA; report JSON states=53                                                                                                      | MA_Geo_Variation         |                                                                                                                                       |
| F321 | US 65+ civilian population                                       | 2024      | 61179918 persons                                                                   | GOV      | A | S206       | SC-EST2024-AGESEX-CIV, ages 65-85+, SEX=0, national                                                                                     | State_Age_65plus         | Measured estimate; the NP2023 projection series on Macro_Demand_Drivers is the forward companion                                      |
| F322 | US 65+ CAGR 2020-2024 (measured)                                 | 2020-2024 | 0.0295173468603869 %/yr                                                            | DERIVED  | A | S206       | (65+ 2024 / 65+ 2020)^(1/4)-1                                                                                                           | State_Age_65plus         | Brackets the projected ~2.7%/yr 65+ growth on Macro_Demand_Drivers; measured vs projected bases stated                                |
| F323 | EMT jobs (OEWS, sum of published state cells)                    | 2024      | 178900 jobs                                                                        | DERIVED  | A | S207       | OEWS May 2024 state file, OCC 29-2042, TOT_EMP summed                                                                                   | OEWS_EMS_Wages           | Occupation basis (all industries); QCEW industry employment (171,100 private, 2024) is a different universe - both stated             |
| F324 | Footprint hospitals across the 20 target metros                  | 2026      | 166 hospitals                                                                      | DERIVED  | B | S208       | SUM over 20 metro rows; geocoded Hospital General Information, city+state filtered                                                      | Metro_Structure_20       | footprint_rollup() returns 166; UNMC + Via Christi absences make this a floor                                                         |
| F325 | Footprint SNF certified beds across the 20 metros                | 2026      | 51108 beds                                                                         | DERIVED  | B | S208       | SUM over 20 metro rows; NH_ProviderInfo 2026-04-01 certified beds                                                                       | Metro_Structure_20       | footprint_rollup() returns 51,108                                                                                                     |
| F326 | Footprint HCRIS staffed beds across the 20 metros                | 2026      | 40762 beds                                                                         | DERIVED  | B | S100, S208 | SUM over 20 metro rows; HCRIS FY2020-22 latest per CCN, joined by CCN (152 of 166 hospitals matched - a floor)                          | Metro_Structure_20       | footprint_rollup() returns 40,762                                                                                                     |
| F327 | National geocoded hospital roll (denominator)                    | 2026      | 4630 hospitals                                                                     | SOURCED  | B | S208       | hospital_coords.csv full-US row count (4,630)                                                                                           | Metro_Structure_20       | Facility_Universe_State counts the same Care Compare universe from live pulls; AHA counts ~6,100 (broader universe - both stated)     |

|      |                                                                        |                 |                          |            |   |            |                                                                                                                                                    |                          |                                                                                                                |
|------|------------------------------------------------------------------------|-----------------|--------------------------|------------|---|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------|
| F328 | Footprint-state share of national hospitals (11 states)                | 2026            | 0.24902807775378 share   | DERIVED    | B | S208       | 1,153 footprint-state hospitals / 4,630 national (both from the geocoded roll) = 24.9%                                                             | Metro_Structure_20       | Numerator is state-level, not metro-level - stated on-sheet                                                    |
| F329 | Footprint-state share of national SNF certified beds                   | 2026            | 0.221535328511059 share  | DERIVED    | B | S208       | 347,674 footprint-state SNF beds / 1,569,384 national (NH_ProviderInfo 2026-04-01) = 22.2%                                                         | Metro_Structure_20       | National SNF bed base also carried on Facility_Universe_State (later snapshot)                                 |
| F330 | MMT footprint population (22 counties, 2020 Census)                    | 2020            | 1560299 residents        | DERIVED    | B | S209       | SUM of 22 county 2020 Census populations (= 1,560,299)                                                                                             | County_Demography        | lft_mmt.footprint_summary() pop_2020 = 1,560,299; CBSA panel SUM matches                                       |
| F331 | Douglas County NE population (2020 Census)                             | 2020            | 584526 residents         | SOURCED    | B | S209       | 2020 Decennial Census, Douglas County NE (FIPS 31055)                                                                                              | County_Demography        | MMT HQ county; largest footprint county                                                                        |
| F332 | Fastest-growing footprint county: Saunders NE CAGR 2020-2024           | 2024            | 0.0205375830415884 %/yr  | DERIVED    | B | S210, S209 | (23,406 / 21,578)^(1/4)-1 = +2.05%/yr (live formula)                                                                                               | County_Demography        | lft_mmt.county_growth() computes 2.054%/yr; re-verify flag travels                                             |
| F333 | Sarpy County NE population (Vintage-2024 estimate)                     | 2024            | 204828 residents         | SOURCED    | B | S210       | Vintage-2024 estimate 204,828 (captured 2026-07-10; re-verify queue)                                                                               | County_Demography        | +7.46% over 2020 - the fastest-growing large NE county                                                         |
| F334 | Footprint = complete OMB composition of 7 CBSAs (22 counties)          | 2023            | 22 counties              | DERIVED    | A | S211       | Live COUNTIF per CBSA against the vendored OMB Bulletin 23-01 crosswalk = 22 = footprint count                                                     | County_Demography        | 8+2+3+2+3+2+2 counties across CBSAs 36540/30700/24260/28260/25580/35820/18100                                  |
| F335 | Counties inside a CBSA (2023 OMB delineations)                         | 2023            | 1915 county rows         | DERIVED    | A | S211       | Live COUNTIF total over the vendored list1_2023 extract (= 1,915; meta cross-check delta 0)                                                        | CBSA_Crosswalk_Reference | cbsa_crosswalk_meta.json counties=1915                                                                         |
| F336 | Distinct CBSAs (2023 OMB delineations)                                 | 2023            | 935 CBSAs                | DERIVED    | A | S211       | SUM of first-occurrence helper column (= 935 = 393 MSA + 542 uSA)                                                                                  | CBSA_Crosswalk_Reference | cbsa_crosswalk_meta.json cbsas=935                                                                             |
| F337 | Metropolitan county rows (2023 delineations)                           | 2023            | 1252 county rows         | DERIVED    | A | S211       | COUNTIF area_type="Metropolitan" (= 1,252; micropolitan = 663)                                                                                     | CBSA_Crosswalk_Reference | Metro counties sit inside MSAs - the AFS urban proxy at county grain                                           |
| F338 | Footprint-state county rows inside CBSAs (11 states)                   | 2023            | 549 county rows          | DERIVED    | A | S211       | COUNTIF footprint-flag column (= 549 of 1,915)                                                                                                     | CBSA_Crosswalk_Reference | Flag formula is self-contained (SEARCH against the 11-state string)                                            |
| F339 | Designated primary-care HPSAs, national total                          | 2026            | 7635 designations        | DERIVED    | A | S213       | SUM over the 60 state/territory rows (= 7,635), HRSA snapshot 2026-05-25                                                                           | HPSA_Rural_Designations  | Aggregated from 19,696 raw designated rows (hrsa_hpsa_report.json)                                             |
| F340 | Designated primary-care HPSAs, footprint 11 states                     | 2026            | 1931 designations        | DERIVED    | A | S213       | SUMIF over footprint flag (= 1,931; 25.3% of national)                                                                                             | HPSA_Rural_Designations  | Share formula on row 78                                                                                        |
| F341 | Designated primary-care HPSAs, Nebraska                                | 2026            | 125 designations         | GOV        | A | S213       | NE row of hrsa_hpsa_pc_by_state.csv (= 125; median score 13, max 20)                                                                               | HPSA_Rural_Designations  | NE also has 80%+ volunteer EMS agencies (Workforce_Supply) - same thin-coverage geography                      |
| F342 | Super-rural ground ambulance base-rate bonus                           | 2026            | 0.226 share of base rate | SOURCED    | B | S214       | SSA sec. 1834(l)(12)(A); 42 CFR 414.610(c)(5)(ii); +22.6% for lowest-quartile rural density pickups, through 2027-12-31                            | HPSA_Rural_Designations  | Same figures carried on Payment_Rules / Service_Level_Economics; MedPAC "no empirical basis" caveat on-sheet   |
| F343 | Rural ground ambulance add-on (base + mileage)                         | 2026            | 0.03 share               | SOURCED    | B | S214, S215 | +3% rural (urban +2%); SSA sec. 1834(l)(13)(A), extended by CAA 2026 sec. 6203 through 2027-12-31                                                  | HPSA_Rural_Designations  | Sunset risk flagged: add-ons lapse 2028-01-01 absent re-extension                                              |
| F344 | Omaha hospitals (transfer origins)                                     | 2026            | 11 facilities            | SOURCED    | B | S217       | Hospital General Information geocoded roll (2026-05-23), city+state filtered to Omaha metro                                                        | Metro_Omaha              | Metro_Structure_20 row Omaha; Nebraska Medicine/UNMC absent from the geocoded roll (undercount caveat printed) |
| F345 | Omaha O/D nodes (hospitals + SNF + IRF + LTCH)                         | 2026            | 48 nodes                 | DERIVED    | B | S217       | Live formula over Panel A SOURCED counts                                                                                                           | Metro_Omaha              | module n_nodes = 48                                                                                            |
| F346 | Omaha SNF certified beds                                               | 2026            | 3584 beds                | SOURCED    | B | S217       | NH_ProviderInfo (2026-04-01), sum of certified_beds over Omaha-metro CCNs                                                                          | Metro_Omaha              | module snf_beds = 3,584                                                                                        |
| F347 | Omaha metro occupancy (HCRIS inpatient days / bed-days available)      | 2026            | 0.638464899080609 share  | DERIVED    | B | S217, S218 | HCRIS FY2020-22 panel, latest per CCN, 10 of 11 hospitals joined                                                                                   | Metro_Omaha              | python recompute 328,819/515,015 = 63.8%                                                                       |
| F348 | Kansas City (bi-state) O/D nodes - densest footprint metro             | 2026            | 105 nodes                | DERIVED    | B | S217       | Live formula over Panel A SOURCED counts                                                                                                           | Metro_KansasCity         | module n_nodes = 105, density tier very-dense                                                                  |
| F349 | Cincinnati O/D nodes                                                   | 2026            | 94 nodes                 | DERIVED    | B | S217       | Live formula over Panel A SOURCED counts                                                                                                           | Metro_Cincinnati         | module n_nodes = 94, density tier very-dense                                                                   |
| F350 | North Platte O/D nodes - thinnest footprint metro                      | 2026            | 4 nodes                  | DERIVED    | B | S217       | Live formula over Panel A SOURCED counts                                                                                                           | Metro_NorthPlatte        | module n_nodes = 4, tier thin/long-leg                                                                         |
| F351 | Footprint hospitals (20-metro live SUM)                                | 2026            | 166 facilities           | DERIVED    | B | S217       | SUM over the 20 Metro_tab hospital counts                                                                                                          | Metro_Index              | footprint_rollup() n_hospitals = 166                                                                           |
| F352 | Footprint SNF certified beds (20-metro live SUM)                       | 2026            | 51108 beds               | DERIVED    | B | S217       | SUM over the 20 Metro_tab SNF-bed rows                                                                                                             | Metro_Index              | footprint_rollup() snf_beds = 51,108                                                                           |
| F353 | Footprint O/D nodes (20-metro live SUM)                                | 2026            | 748 nodes                | DERIVED    | B | S217       | SUM over the 20 Metro_tab node totals                                                                                                              | Metro_Index              | python recompute = 748                                                                                         |
| F354 | Footprint HCRIS staffed beds (20-metro live SUM)                       | 2026            | 40762 beds               | DERIVED    | B | S217, S218 | SUM over the 20 Metro_tab HCRIS staffed-bed rows                                                                                                   | Metro_Index              | footprint_rollup() hcris_beds = 40,762; conservative floor (14 of 166 hospitals lack an HCRIS row)             |
| F355 | 20-metro share of national hospital universe                           | 2026            | 0.03585313174946 share   | DERIVED    | B | S217       | Footprint SUM / national roll total (4,630 hospitals)                                                                                              | Metro_Index              | Distinct from the 24.9% 11-STATE share on Metro_Structure_20                                                   |
| F356 | MMT active organizational NPIs (verified estate)                       | 2026            | 23 org NPIs              | SOURCED    | B | S220       | NPPES org-NPI sweep 2026-07-10; live COUNTA over the 23 Panel A rows                                                                               | MMT_NPI_Estate           | lft_company.SCALE_CLAIMS carries 23; module docstring says 24 - discrepancy flagged on-sheet                   |
| F357 | Distinct MMT NPI states (vs 13-state company claim)                    | 2026            | 11 states                | DERIVED    | B | S220       | Live SUMPRODUCT distinct-count over Panel A state column = 11 (NE IA SD MO OH IN WI CO RI NC VA; VA unconfirmed)                                   | MMT_NPI_Estate           | Company claims 13 states (Company_Dossier scale claims) - a 2-state gap the registry cannot verify             |
| F358 | MMT legacy-core NPIs (NE/IA/SD)                                        | 2026            | 15 org NPIs              | DERIVED    | B | S220       | COUNTIF estate class = "legacy core" (15: NE 4, IA 9, SD 2)                                                                                        | MMT_NPI_Estate           | lft_company.legacy_core_locations() returns 15                                                                 |
| F359 | MMT expansion-state NPIs (2023-24 wave)                                | 2026            | 8 org NPIs               | DERIVED    | B | S220       | COUNTIF estate class = "expansion" (8: MO OH IN WI CO RI NC VA, 1 each)                                                                            | MMT_NPI_Estate           | lft_company.expansion_locations() returns 8; VA record flagged unconfirmed                                     |
| F360 | Earliest MMT-lineage enumeration (Platte County Ambulance predecessor) | 2005 2005-11-03 | date                     | SOURCED    | B | S220       | NPI 1689665143, Platte County Ambulance Company DBA Midwest Medical Transport Co, enumerated 2005-11-03                                            | MMT_NPI_Estate           | Consistent with the 1987 founding story (Company_Dossier) - NPPES itself only began 2005                       |
| F361 | MMT web-listed stations outside the NPI estate                         | 2026            | 11 stations              | PUBLIC-WEB | B | S220       | 11 company-site station listings (8 NE + Iowa City, Tea SD, Cincinnati OH) with no own NPI - subparts typically bill under a parent NPI            | MMT_NPI_Estate           | lft_company.MMT_WEB_STATIONS (11); kept separate from SOURCED NPI counts                                       |
| F362 | MMT founded (Platte County Ambulance, Columbus NE)                     | 1,987           | 1,987 year               | PUBLIC-WEB | B | S226       | OWH/Columbus Telegram founding retrospective: "with one ambulance, doing a few transfers a week out of the Columbus Greenbelt (Rte. 6, 100 Miles)" | Company_Dossier          | Company site: "over 35 years" (2026); predecessor NPI enumerated 2005 (NPPES began 2005)                       |
| F363 | MMT claimed states of operation (company site)                         | 2026 13         | states (claim)           | PUBLIC-WEB | B | S224       | mmtamb.com About Us, captured 2026-07-10: 13 states                                                                                                | Company_Dossier          | NPPES verifies 11 NPI states (10 confirmed + VA flagged) - gap stated on both tabs                             |
| F364 | MMT claimed team members (company site)                                | 2026 2,800+     | people (claim, floor)    | PUBLIC-WEB | B | S224       | mmtamb.com About Us, captured 2026-07-10: "2,800+"                                                                                                 | Company_Dossier          | All three third-party estimators disagree (700-784 to 1,000-5,000) - conflict exhibit Panel E                  |

|      |                                                                                        |                                                                                                                |                                           |            |   |                  |                                                                                                                                                     |                     |                                                                                                                |
|------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------|---|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------|
| F365 | MMT claimed missions per year (deal release)                                           | 2022 200,000+                                                                                                  | missions/yr (claim)                       | PUBLIC-WEB | B | S222             | Harbour Point release, Jan 25 2022: "200,000+ missions/yr" (then-10-states framing)                                                                 | Company_Dossier     | Sell-side advisor framed the same deal as 7 states / ~1,000 employees - conflict carried, not blended          |
| F366 | MMT federal wage-and-hour suits on public dockets (2020-2024)                          | 2024                                                                                                           | 3 FLSA suits                              | PUBLIC-WEB | B | S227             | N.D. Ohio 1:20-cv-01548 (2020); E.D. Wis. 2:23-cv-00877 (2023); D. Neb. 4:2024-cv-03107 (2024); plus NLRB 14-CA-12-0001 (2014)                      | Company_Dossier     | Outcomes sealed without PACER - stated; no malpractice cases surfaced (recorded as not-found)                  |
| F367 | MMT claimed-headcount indicative CAGR 2015-2026                                        | 2026                                                                                                           | 0.208089444404447 %/yr (indicative)       | DERIVED    | B | S226, S223, S224 | (2,800/350)*(1/11)-1 as a live formula over the three claimed floors - self-reported endpoints, definition drift; indicative only, not a formula    | Company_Dossier     | Endpoints: 350+ (2015, 1 state) / ~1,000 (2022, 7 states) / 2,800+ (2026, 13-state claim)                      |
| F368 | NE+IA ambulance organizational NPIs (supply denominator)                               | 2026                                                                                                           | 751 org NPIs                              | SOURCED    | B | S221             | NPPES sweep (taxonomy Ambulance, NPI-2, NE+IA) vendored 2026-07-10; live SUM = 751 (NE 400 / IA 351)                                                | Competitor_Registry | lft_npi_landscape.summary() total_orgs = 751; 686 distinct org names (multi-NPI squads caveat)                 |
| F369 | NE municipal/fire/volunteer org NPIs                                                   | 2026                                                                                                           | 328 org NPIs                              | SOURCED    | B | S221             | NPPES sweep category matrix: 328 of 400 NE org NPIs (keyword-classified)                                                                            | Competitor_Registry | Consistent with the NE DHHS 80%+ all-volunteer finding (same tab)                                              |
| F370 | Hospital-owned ambulance org NPIs: IA vs NE                                            | 2026                                                                                                           | 40 org NPIs (IA)                          | SOURCED    | B | S221             | NPPES sweep: IA 40 vs NE 5 hospital-owned - the sharpest measured ownership-model contrast                                                          | Competitor_Registry | Live ratio formula = 8.0x on the contrast row; in-depth Q5 carries the same 40+ vs ~5 read                     |
| F371 | Ground ambulance organizations billing Medicare (national)                             | 2024                                                                                                           | 10600 organizations (approx.)             | GOV        | B | S229             | MedPAC Payment Basics: Ambulance, Oct 2024 - ~"10,600" organizations (11.3M transports, \$5.3B)                                                     | Competitor_Registry | PECOS ambulance-supplier extract (Group S tabs) counts 10,465 Part B ambulance suppliers                       |
| F372 | Competitor NPIs carried across the 7 curated profiles                                  | 2026                                                                                                           | 28 org NPIs                               | SOURCED    | B | S221             | Live SUM of per-profile COUNTIFs over the 28-row Panel B NPI list                                                                                   | Competitor_Registry | AMR 1 + AmeriPro 2 + Omaha locals 8 + Siouxland 1 + fire-based 6 + hospital-owned 3 + air 7 = 28               |
| F373 | NE EMS agencies all-volunteer (floor)                                                  | 2024 80%+                                                                                                      | share of agencies (floor)                 | GOV        | B | S191             | NE DHHS Statewide EMS Assessment 2023-24 (re-verify; excerpt-grade); companions: 31% adequately staffed, 28% cannot reliably operate within 5 years | Competitor_Registry | NPPES matrix: 328 of 400 NE org NPIs are municipal/fire/volunteer (82%)                                        |
| F374 | AmeriPro/Priority stated Nebraska counties (Feb 2025 entry)                            | 2025                                                                                                           | 7 counties                                | PUBLIC-WEB | B | S230             | PRNewsWire Feb 2025 release: Lincoln, Red Willow, Buffalo, Dawson, Adams, Dodge, Platte                                                             | Competitor_Registry | Includes Platte County - MMT's founding county (Company_Dossier)                                               |
| F375 | Ambulance improper-payment rate (CERT)                                                 | 2024                                                                                                           | 0.132 share of payments                   | GOV        | B | S233             | CERT 2024 supplemental data, ambulance service type: 13.2% (re-verify flag carried)                                                                 | Payment_Integrity   | Projected \$595.1M on the next row; 63.5/27.5 error split                                                      |
| F376 | Projected ambulance improper payments (CERT)                                           | 2024                                                                                                           | \$95.1 \$M                                | GOV        | B | S233             | CERT 2024 supplemental data: \$595.1M projected (re-verify)                                                                                         | Payment_Integrity   | 13.2% rate x the ambulance payment base; implied \$377.9M documentation / \$163.7M necessity as live formulas  |
| F377 | CERT error share from insufficient documentation                                       | 2024                                                                                                           | 0.635 share of improper \$                | GOV        | B | S233             | CERT 2024 error-category split: 63.5% (re-verify)                                                                                                   | Payment_Integrity   | Medical necessity 27.5%; residual 9.0% (live formula)                                                          |
| F378 | RSNAT effect on probability of RSNAT use                                               | 2022                                                                                                           | -0.61 change in probability               | ACADEMIC   | B | S234             | doi:10.1001/jamahealthforum.2022.2093 - verbatim "61% reduction in the probability of RSNAT use" (ESRD cohort, model in regression context)         | Payment_Integrity   | Paired with ~77% spend and +19% <sup>2</sup> /yr emergency dialysis on adjacent rows                           |
| F379 | RSNAT effect on RSNAT expenditures                                                     | 2022                                                                                                           | -0.77 change (\$1.136/beneficiary-yr)     | ACADEMIC   | B | S234             | Verbatim "77% reduction in RSNAT expenditures for a total of \$1136 per beneficiary-year"                                                           | Payment_Integrity   | Same paper as effects 1 and 3 (single DOI)                                                                     |
| F380 | RSNAT side-effect: emergency dialysis use                                              | 2022                                                                                                           | 0.19 annual change in probability         | ACADEMIC   | B | S234             | Verbatim "19% annual increase in the probability of emergency dialysis use" (ESRD cohort)                                                           | Payment_Integrity   | The cost-shift direction - carried with the suppression effects, never alone                                   |
| F381 | Medicare ambulance transport growth 2002-2011 (pre-RSNAT boom)                         | 2011                                                                                                           | 0.69 endpoint growth                      | GOV        | B | S236             | OEI-09-12-00350: transports +69% 2002-2011; dialysis-related +269%                                                                                  | Payment_Integrity   | Implied CAGRs 6.0%/15.7% as live formulas; series break flagged - RSNAT broke this trend                       |
| F382 | Suppliers with questionable-billing patterns (OIG)                                     | 2015                                                                                                           | 0.21 share of suppliers                   | GOV        | B | S237             | OEI-09-12-00351: 21% (about 1 in 5) met >=1 of 7 measures; 4 metros = 18% of transports / 52% of questionable (~\$207M total cost)                  | Payment_Integrity   | \$24M / \$30.2M / \$7.1M / \$4.3M half-year findings on adjacent rows                                          |
| F383 | Texas Medicaid NEMT on-time-pickup benchmark                                           | 2026                                                                                                           | 0.85 on-time share                        | GOV        | B | S239             | TX HHSC Medical Transportation Program contract terms: 85% benchmark, \$15,000 penalty cap (re-verify)                                              | Contract_Benchmarks | Broker-contract common form >=95% on adjacent row                                                              |
| F384 | San Diego / Falck per-response penalty                                                 | 2026                                                                                                           | 5,000 \$ per response beyond 24 min       | PUBLIC-WEB | B | S238             | San Diego municipal contract records via repo dossier (re-verify; no URL on file)                                                                   | Contract_Benchmarks | Same register carries Pasadena 8:59/90% cap 14:59 and Multnomah below-90% penalties                            |
| F385 | California AB 40 ambulance patient offload standard                                    | 2023                                                                                                           | 0.9 compliance threshold (30-min offload) | GOV        | B | S241             | AB 40 (2023) / EMSA APOT: <=30 minutes 90% of the time, monthly per-hospital monitoring (re-verify)                                                 | Contract_Benchmarks | emsa.ca.gov/apot program page cited; the one statutory offload clock in the register                           |
| F386 | Mississippi NEMT broker observed miss rate                                             | 2024 ~5.8%                                                                                                     | monthly trips late/missed                 | PUBLIC-WEB | B | S240             | Mississippi Today (Sep 2024): ~3,000 of >52,000 monthly trips, ~3x the contractual limit (re-verify)                                                | Contract_Benchmarks | Enforcement companions: NJ ~\$1.7M (2017-22), GA >\$1M (2018-20)                                               |
| F387 | NJ Medicaid NEMT enforcement fines vs Modivcare                                        | 2022 ~\$1.7M                                                                                                   | fines 2017-2022                           | PUBLIC-WEB | B | S240             | State enforcement records via press (re-verify; no URL on file)                                                                                     | Contract_Benchmarks | Modivcare later filed Ch.11 (Aug 2025) - context row on the same tab                                           |
| F388 | NEMT broker contract on-time standard (common form)                                    | 2026                                                                                                           | 0.95 on-time share                        | PUBLIC-WEB | B | S239             | State broker contract standards: >=95% on-time pickup with penalties (re-verify)                                                                    | Contract_Benchmarks | TX 85% benchmark is the enforce-with-cap variant                                                               |
| F389 | Medicare FFS ground ambulance universe (transports / payments / billing organizations) | 2024 11.3M transports · \$5.3B · ~10,600 organizations                                                         | mixed (see locator)                       | GOV        | B | S248             | MedPAC Ambulance Payment Basics, Oct 2024 - Q1 block q1-definitions: "11.3M transports, \$5.3B, ~10,600 organizations (2024)"                       | InDepth_Q01         | Same denominators power the \$469 DERIVED average (\$5.3B/11.3M) and the fragmentation read on InDepth_Q07/Q09 |
| F390 | Measured acute IFT base (adult ED-to-ED + inpatient interhospital transfers)           | 2022 ~1.97M/yr ED-to-ED (NEDS 2018-2022) + ~1.5M/yr inpatient (NEDS)                                           | transfers/yr                              | ACADEMIC   | B | S259             | Nikolla et al., J Emerg Med 2025 (doi:10.1016/j.jemermed.2025.12.020; HCUP NEDS 2018-2022) (re-verify)                                              | InDepth_Q01         | Feeds the lft_mmmt footprint demand band (3.47M legs/yr) on InDepth_Q10                                        |
| F391 | Ground ambulance out-of-network share                                                  | 2022 ~60% of rides                                                                                             | % of rides                                | SOURCED    | B | S256             | FAIR Health 2023 study of 2022 claims - Q1 q1-customer                                                                                              | InDepth_Q01         | Pairs with GADCS 19.7% collect-nothing and the NSA exclusion / GAPB rows on the same tab                       |
| F392 | CA hospital crew-detention distribution (2017, 830,637 transports)                     | 2017 75% of hospitals >1h · 40% >2h · 33% >3h                                                                  | % of CA hospitals                         | ACADEMIC   | B | S151             | Backer et al., Prehosp Emerg Care 2018 - Q2 crew-detention evidence                                                                                 | InDepth_Q02         | Consistent direction with Shaw 2025 national APOT tail (3.3% of agencies ≥25% >30 min)                         |
| F393 | National median ambulance patient offload time (APOT)                                  | 2024 10.9 min median (IQR 6.6-17.5)                                                                            | minutes                                   | ACADEMIC   | B | S150             | Shaw et al., Prehosp Emerg Care 2025, 7,237,606 records (2024) - Q2 offload evidence                                                                | InDepth_Q02         | CA AB 40 statutory standard (30 min at 90%) cited on the same tab                                              |
| F394 | ED boarding of admitted 65+ (mean minutes, 2018 ~2022)                                 | 2022 138 ~ 343 min (85.2% boarded ≥2h)                                                                         | minutes (endpoint growth)                 | ACADEMIC   | B | S152             | Lee et al., Ann Emerg Med 2026 (NHAMCS 2015-2022) - Q2 boarding evidence; 501 min with dementia                                                     | InDepth_Q02         | ACEP/Morning Consult poll (44%/16%) on InDepth_Q07 gives the public-experience read                            |
| F395 | Post-acute destination supply (national node count)                                    | 2026 35,481 destinations (14,699 SNF + 1,221 IRF + 317 LTCH + 12,293 LHA + 1,761 SNF + 12,293 LHA + 1,761 SNF) | facilities                                | SOURCED    | B | S276             | CMS Care Compare / Provider-of-Services files via lft_clinical_demand.destination_supply() - Q3 destination                                         | InDepth_Q03         | MedPAC payment-basics volumes (~1.8M SNF / 383K IRF / 90k LTCH stays) on the same tab                          |
| F396 | Rural ED visit volume (2005 ~ 2016)                                                    | 2016 16.7M ~ 28.4M visits                                                                                      | visits/yr (2-pt series)                   | ACADEMIC   | B | S260             | Greenwood-Ericksen & Kocher, JAMA Network Open 2019/2021 - Q3 rural demand evidence (~4.9%/yr endpoint growth)                                      | InDepth_Q03         | Rural transfer propensity 6.2% vs urban 2.0% (same authors) on InDepth_Q01/Q07                                 |
| F397 | STEMI transfer DIDO and mortality                                                      | 2011 median DIDO 68 min; 11% ≤30 min; mortality 5.9% vs 3.9% (adj. OR 1.6)                                     | minutes / % (n=14,821)                    | ACADEMIC   | B | S146             | Wang et al., JAMA 2011 - Q3 delay-consequence evidence                                                                                              | InDepth_Q03         | Stroke DIDO family (Ng 2017; JAMA Netw Open 2024; GWGT 2023) carried on InDepth_Q02                            |
| F398 | RSNAT prior-authorization natural experiment (ESRD cohort)                             | 2022 ~61% probability of RSNAT use ~77% RSNAT spend (83.1% beneficiaries vs 85.1% non-beneficiaries)           | effect sizes                              | ACADEMIC   | B | S268             | Contreary et al., JAMA Health Forum 2022 (doi:10.1001/jamahealthforum.2022.2093) - Q4 payment flow evidence                                         | InDepth_Q04         | RSNAT definitions (CMS rules) cited on InDepth_Q02; OIG utilization findings adjacent                          |
| F399 | OIG: payments with no Medicare service at origin or destination (H1-2012)              | 2015 \$30.2M in one half-year                                                                                  | USD (half-year)                           | GOV        | B | S237             | HHS OIG OEI-09-12-00351 (2015) - Q4 incentive evidence; companion findings (21% of suppliers, 4 metros 18%/52%) on InDepth_Q03                      | InDepth_Q04         | CERT 2024 split (13.2% / \$595.1M / 63.5% / 27.5%) carried on InDepth_Q02/Q07                                  |
| F400 | Bed-day value vs Medicare transport price mismatch                                     | 2024 ~\$3,132 adjusted expense per inpatient day vs \$469 Medicare average rate                                | USD                                       | DERIVED    | B | S266, S248       | KFF State Health Facts 2023 expense proxy (re-verify) + MedPAC Oct 2024 universe - equation stated in the Q4 claim                                  | InDepth_Q04         | Both inputs carried as separate cited lines on the same tab                                                    |
| F401 | Hospital-owned ambulance org NPIs - NE vs IA contrast                                  | 2026 NE 5 (of 400) vs IA 40+ (of 351)                                                                          | org NPIs                                  | SOURCED    | B | S285             | CMS NPPES NE+IA ambulance-org sweep, vendored 2026-07-10 (751 org NPIs) - Q5 prevalence evidence                                                    | InDepth_Q05         | Same registry powers Competitor_Registry and the InDepth_Q08 MMT estate                                        |

|      |                                                                                                       |                                                                                                 |                          |          |   |                                          |                                                                                                                                                                   |                                                                                                    |
|------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------|----------|---|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| F402 | CommonSpirit operating loss trajectory                                                                | 2025 <del>-\$875M (FY2024)</del> → <del>-\$225M (FY2025)</del>                                  | USD operating loss       | SOURCED  | B | S272                                     | CommonSpirit FY2025 results release (re-verify) - Q5 capital-<br>appetite evidence                                                                                | Context for why systems avoid insourcing capital: 2-pt trend, never extrapolated                   |
| F403 | Insourced-heavy comparators (scale of the alternative)                                                | 2024 Mayo Clinic Ambulance ~70 units; Allina Health EMS <del>24,249 ambulances nationwide</del> | units / requests         | SOURCED  | B | S283                                     | Operator public pages via footprint registry, captured 2026-07-10 (re-verify) - Q5 insourcing evidence                                                            | Contrast rows: NE systems own 5 ambulance NPIs total (same tab)                                    |
| F404 | Hospital system-affiliation share (buyer consolidation)                                               | 2024 70% (3,567 of 5,121 non-federal general acute hospitals)                                   | % of hospitals           | SOURCED  | B | S247                                     | AHA Fast Facts 2026 (FY2024 Annual Survey; re-verify) - Q6 procurement-ownership evidence                                                                         | Underpins system-level contracting logic on InDepth_Q09                                            |
| F405 | Ambulance Inflation Factor (contract escalator precedent)                                             | 2026 +2.4% (CY2025) - +2.0% (CY2026)                                                            | %/yr                     | GOV      | B | S246                                     | CMS Ambulance Fee Schedule PUF (re-verify against PUF) - Q6 contract-structure evidence                                                                           | 2-point escalator series; never trended forward                                                    |
| F406 | Modivcare Chapter 11 (vendor-stability precedent)                                                     | 2025 Filed Aug 20, 2025; emerged Dec 29, 2025 cutting >\$1.1B of debt                           | event                    | SOURCED  | B | S243                                     | S.D. Tex. bankruptcy docket + coverage (re-verify) - Q6 evaluation-criteria evidence                                                                              | GMR refinancing/IPO row on InDepth_Q10 gives the 911-side analogue                                 |
| F407 | Nebraska statewide transfer-center confirmation rate (measured shortfall)                             | 2021 146 of 234 confirmed (62%; Sep 2021) - 113 of 168 (67%; Oct 2021)                          | % of requested transfers | SOURCED  | B | S277                                     | Omaha World-Herald / Journal Star, Nov 2021 (re-verify; COVID-era context travels) - Q7 acceptance evidence                                                       | The only public IFT acceptance gauge in the corpus - 2-point episode, never trended                |
| F408 | PACU discharge delays attributable to transport unavailability                                        | 2022 11.1% of delays (2nd after-bed unavailability 22.5%); 64,000 discharges, 22,500 attributed | % of delays              | ACADEMIC | B | S156                                     | Ego et al., Ann Med Surg 2022, 307-patient non-US cohort (magnitude only - caveat carried) - Q7 flow evidence                                                     | Discharge-task 47% single-site study (PMC11023539) on InDepth_Q02 corroborates direction           |
| F409 | Average length of stay deterioration (2019 → 2022)                                                    | 2022 +19% overall; +24% for post-acute discharges                                               | % change (endpoint)      | SOURCED  | B | S267                                     | AHA Issue Brief, Dec 2022 (re-verify) - Q7 hospital-flow evidence                                                                                                 | Boarding trend 138 → 343 min (Lee 2026, InDepth_Q02) moves the same direction                      |
| F410 | MMT verified NPI estate vs company claim                                                              | 2026 23 active org NPIs across 11 NPI states vs a 13-state company claim                        | org NPIs / states        | SOURCED  | B | S285                                     | CMS NPPEs org-NPI sweep, pulled 2026-07-10 - Q8 geography evidence (VA link unconfirmed, flagged)                                                                 | Full row-level estate on MMT_NPI_Estate; conflict stated on both tabs                              |
| F411 | MMT claimed operating scale                                                                           | 2026 200,000+ missions/yr (Jan 2022 release) - 500+ missions, 9,000+ people                     | claims (floors)          | SOURCED  | B | S280, S224                               | Businesswire Harbour Point release Jan 25, 2022 + mmtamb.com About Us captured 2026-07-10 - Q8 scope evidence (full page, not redacted)                           | Advisor framed the 2022 deal as 7 states / ~1,000 employees - conflict carried, not blended        |
| F412 | MMT claimed headcount trajectory (self-reported floors)                                               | 2026 350+ (2015, 1 state) → ~1,000 (2022, 7 states) → 9,000, 13 states                          | people (claimed floors)  | SOURCED  | B | S280, S224                               | LJS Feb 2015 sale coverage; Lincoln International notice 2022; mmtamb.com 2026 - Q8 workforce evidence; definitions diff (company vs. team members) never trended | Same 3-point series charted on Company_Dossier with the identical caveats                          |
| F413 | GADCS mean cost per transport by ownership                                                            | 2025 \$2,673 all - \$3,127 governmental - \$1,778 private                                       | USD/transport            | SOURCED  | B | S245                                     | CMS/RAND GADCS Year 1-2 report via trade coverage (re-verify) - Q9 alternative-economics evidence                                                                 | MedPAC Dec 2025 volume-cost curve (same tab) explains the governmental/private gap via utilization |
| F414 | Unit-hour utilization benchmark (911 vs high-performance)                                             | 2024 911 band 0.30-0.50; AIMHI survey mean 0.508                                                | UHU ratio                | SOURCED  | B | S258                                     | AIMHI benchmarking via EMS1 (as captured 2026-07-10) - Q9 shared-fleet evidence                                                                                   | The capacity-productivity argument against shared 911 fleets for scheduled IFT                     |
| F415 | Omaha Fire Department 911 transport monopoly (city ALS fleet)                                         | 2026 18 ALS ambulances as primary city transport                                                | ambulances               | SOURCED  | B | S283                                     | City EMS pages via the NPPEs landscape sweep, 2026-07-10 (re-verify; public-web claim) - Q9 traditional-EMS evidence                                              | Municipal 911 posture rows corroborated by the muni-contract precedent family                      |
| F416 | Footprint demand band (measured national acute legs allocated to the MMT footprint)                   | 2026 3.47M measured national acute legs/yr (1.97M NEDS + 1.5M NEDS, additional 1.5M)            | legs/yr (national base)  | DERIVED  | B | S281, S259                               | ift_mmt footprint demand band - ACADEMIC transfer counts → GOV Census shares, equation stated in the Q10 claim text                                               | Inputs cited separately on InDepth_Q01 (NEDS/NIS) and Q8 (Census legacy core)                      |
| F417 | GMR capital-structure constraint (competitor context)                                                 | 2026 IPO valuation target cut to \$3.3B (May 2026) after the 6E 4B 2026 refinancing             | USD                      | SOURCED  | B | S278                                     | Deal/IPO coverage via the NPPEs landscape sweep (re-verify) - Q10 durability evidence                                                                             | Modivcare Ch.11 (InDepth_Q06) gives the NEMT-side analogue of vendor fragility                     |
| F418 | Public-API connectors in the repo data estate                                                         | 2026-07-10                                                                                      | 16 connectors            | DERIVED  | B | S286                                     | connectors.registry.catalog() n_connectors; pinned by connectors/tests/test_estate_invariants.py                                                                  | 16 Panel A rows rendered                                                                           |
| F419 | Registered datasets across the connector estate                                                       | 2026-07-10                                                                                      | 204 datasets             | DERIVED  | B | S286                                     | live SUM over Panel A dataset counts; catalog() n_datasets = len(all_registry_rows()) = 204 asserted at build                                                     | 204                                                                                                |
| F420 | Estate APIs verified live-reachable (smoke tests)                                                     | 2026-07-10 15 of 16                                                                             | APIs                     | DERIVED  | A | S286                                     | Panel A live-status column (COUNTIF formula on-sheet); only census_acs blocked (keyless)                                                                          | inv_connectors \$E smoke-test table, 2026-07-10                                                    |
| F421 | IFT probes in the connector probe registry                                                            | 2026-07-10                                                                                      | 18 probes                | DERIVED  | B | S287                                     | connector_estate_map() executed at build time; live COUNT formula over Panel B tier column                                                                        | estate_summary total=18 (design note said 17)                                                      |
| F422 | Live-pull artifacts cached for v3                                                                     | 2026-07-10                                                                                      | 669 artifacts            | DERIVED  | A | S288                                     | live SUM over Panel C cache-file counts; manifest.json has 669 entries, each with sha256 + UTC timestamp                                                          | 669                                                                                                |
| F423 | Raw rows cached across all v3 live pulls                                                              | 2026-07-10                                                                                      | 2956151 rows             | DERIVED  | A | S288                                     | live SUM over Panel C rows-cached column                                                                                                                          | python recompute 961,680 (897,270 of it PSPS carrier lines)                                        |
| F424 | Medicare-enrolled ambulance suppliers (PECOS live cross-check with corrected filter literal)          | 2026Q1                                                                                          | 10465 suppliers          | GOV      | A | S293, S288                               | PECOS /data/stats, filter PROVIDER_TYPE_DESC="PART B SUPPLIER - AMBULANCE SERVICE SUPPLIER" → <del>found 105,000, 197</del>                                       | matches vendored PPEF 2026.04.01 national row 10,465 exactly (PECOS_Suppliers_State)               |
| F425 | PSPS raw carrier line rows cached (2010-2024, 7 ground codes)                                         | 2010-2024                                                                                       | 897270 rows              | GOV      | A | S290, S288                               | 105 cached artifacts (15 vintages x 7 HCPCS); aggregated client-side on PSPS_Denial_Series                                                                        | 897,270 rows (manifest sum)                                                                        |
| F426 | Ambulance/IFT code-vocabulary rows carried (POS, revenue, condition, value, RARC, modifier, taxonomy) | current code sets                                                                               | 17 codes                 | DERIVED  | B | S297, S301                               | count of code rows in panels A-G (17)                                                                                                                             | 2+1+3+1+1+3+6 = 17                                                                                 |
| F427 | NUCC ambulance taxonomies (3416*)                                                                     | v24.1                                                                                           | 4 codes                  | GOV      | B | S301, S287                               | NUCC Transportation section: 341600000X + land/air/water children; ift_connectors_AMBULANCE_TAXONOMIES                                                            | NPPEs live search accepts the text description "Ambulance" (codes rejected as filters)             |
| F428 | NUCC NEMT-boundary taxonomies (3439*/3438*)                                                           | v24.1                                                                                           | 2 codes                  | GOV      | B | S301, S287                               | 343900000X NEMT van + 343800000X secured transport; ift_connectors_NEMT_TAXONOMIES                                                                                | NEMT excluded from IFT TAM throughout workbook                                                     |
| F429 | UB-04 revenue-center codes reserved for ambulance                                                     | current NUBC set 0540-0549 (10 codes)                                                           | code range               | GOV      | B | S299, S297                               | REVENUE_CATEGORIES range (540, 549, "Ambulance")                                                                                                                  | revenue_category("0540")=="Ambulance" verified at build                                            |
| F430 | MAC A/B jurisdictions in the repo seed table                                                          | seed (re-verify)                                                                                | 12 jurisdictions         | DERIVED  | B | S302                                     | live SUMPRODUCT over seed rows excluding the HHH national row (13 rows total)                                                                                     | 12 A/B jurisdictions + 1 HHH row                                                                   |
| F431 | Worked KPI definitions (metric/definition/formula/source/owner/why)                                   | n/a                                                                                             | 12 metrics               | DERIVED  | B | S303                                     | AUTHORED_SECTIONS["kpis"] "Sample KPI dictionary (worked metrics)" table, 12 rows x 6 cols                                                                        | COUNTA formula on-sheet                                                                            |
| F432 | Tier metric definitions across 5 KPI tiers                                                            | n/a                                                                                             | 32 definitions           | DERIVED  | B | S303                                     | AUTHORED_SECTIONS["kpis"] bullet tiers: access 6, speed 7, workflow 6, clinical 6, financial 7                                                                    | 6+7+6+6+7 = 32                                                                                     |
| F433 | Stakeholder scorecards (function-level KPI recommendations)                                           | n/a                                                                                             | 5 scorecards             | DERIVED  | B | S303                                     | AUTHORED_SECTIONS["kpis"] "Recommended metrics by stakeholder" table, 5 rows x 4 cols                                                                             |                                                                                                    |
| F434 | Benchmark values published on this tab                                                                | n/a                                                                                             | 0 benchmarks             | DERIVED  | B | S303                                     | source module states "template dictionary, not benchmarked values" - carried as-is, nothing invented                                                              | Excluded_Not_Sourced records what would make benchmarks citable                                    |
| F435 | Regulatory requirement rows binding IFT (statute-cited)                                               | current law                                                                                     | 11 rows                  | DERIVED  | B | S303, S306, S307, S308, S309, S310, S115 | AUTHORED_SECTIONS["regulatory"] table, 11 rows x 5 cols, per-row GOV basis                                                                                        | COUNTA formula on-sheet                                                                            |
| F436 | Service levels with verbatim 42 CFR 414.605 definitions                                               | current (2026)                                                                                  | 4 levels                 | DERIVED  | B | S304, S305                               | service_levels() BLS/ALS1/ALS2/SCT definition facts, each with quote + 3 sources                                                                                  | ALS2 blood-transfusion trigger effective CY2025 (89 FR 98559) noted                                |
| F437 | Level-determination edge-case scenarios                                                               | current rules                                                                                   | 19 scenarios             | DERIVED  | B | S304, S311, S305                         | edge_cases(): 19 rows, 14 GOV / 5 FRAMEWORK, each citing regs/scope model                                                                                         |                                                                                                    |
| F438 | Misconceptions corrected with sourced rebuttals                                                       | current rules                                                                                   | 12 rows                  | DERIVED  | B | S304                                     | misconceptions(): 12 myth/reality rows, each with source labels                                                                                                   | includes MedPOP CY2024-anchored corrections (30.7% BLS-NE share; SCT 0.7%)                         |



|      |                                                                                                               |                   |                                        |         |   |            |                                                                                                   |                        |                                                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------|---------|---|------------|---------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F439 | GOV facts in the underlying service-levels module (honesty audit)                                             | verified Jul 2026 | 103 facts                              | DERIVED | B | S304       | ift_service_levels.n_by_basis(); has_no_illustrative()==True; S3 URL-linked sources               | Regulatory_Register    | module summary() reports 109 facts total                                                                                                                                                                                                |
| F440 | Medicare beneficiaries 2024, sum of the 51 SP_Index state formulas                                            | 2024              | 66638829 beneficiaries                 | DERIVED | A | S312       | SP_Index US-total row: SUM of 51 INDEX/MATCH pulls from Enroll_State_2024 col C                   | SP_Index               | Must reconcile to the 2024 national row on Enrollment_ESRD_State (same CMS Monthly Enrollment file, national grain)                                                                                                                     |
| F441 | Ground base-code ambulance services 2024, 51-state formula sum (floor)                                        | 2024              | 9624703.4 services                     | DERIVED | A | S312       | SP_Index US-total row: SUM of 51 six-code SUMIFS over MUP_State_2024 (A0426/27/28/29/33/34)       | SP_Index               | Sits at or below the 2024 ground-base total on MUP_Ambulance_National (state-grain suppression makes the state sum a floor)                                                                                                             |
| F442 | PECOS-enrolled ambulance suppliers, 51-state COUNTIF sum                                                      | 2026              | 10153 suppliers                        | DERIVED | A | S312       | SP_Index US-total row: SUM of 51 COUNTIFs over PECOS_Registry col E (PPEF 2026-04-01)             | SP_Index               | PECOS_Registry national count (10,465) minus territory/other-jurisdiction rows                                                                                                                                                          |
| F443 | Ambulance organizations reporting facility-contract revenue (GADCS)                                           | 2025              | 0.186 share of organizations           | GOV     | A | S313       | GADCS Y1-4 appendix, Table 2.32, p.70                                                             | Facility_Pay_Layer     | Self-reported census; incidence, not an IFT share                                                                                                                                                                                       |
| F444 | Facility-contract revenue, conditional mean per organization (GADCS)                                          | 2025              | 922,555 USD                            | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Median \$48,393 on the same row: long right tail                                                                                                                                                                                        |
| F445 | Facility-contract revenue, median per organization (GADCS)                                                    | 2025              | 48,393 USD                             | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Mean/median gap is the skew caveat in-row                                                                                                                                                                                               |
| F446 | Organizations reporting any non-payer revenue (GADCS Question 5 categories)                                   | 2025              | 0.745 share of organizations           | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Includes subsidies, taxes, standby, memberships                                                                                                                                                                                         |
| F447 | Non-payer revenue, conditional mean (GADCS)                                                                   | 2025              | 2,177,569 USD                          | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Median \$162,265                                                                                                                                                                                                                        |
| F448 | Organizations with local-government contract revenue (GADCS)                                                  | 2025              | 0.154 share of organizations           | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Conditional mean \$556,091, median \$112,426                                                                                                                                                                                            |
| F449 | Organizations with EMS-earmarked local tax revenue (GADCS)                                                    | 2025              | 0.234 share of organizations           | GOV     | A | S313       | Table 2.32, p.70                                                                                  | Facility_Pay_Layer     | Conditional mean \$3.14M: the subsidy backbone of government services                                                                                                                                                                   |
| F450 | Mean payer revenue per ambulance NPI, all payers (GADCS)                                                      | 2025              | 3,914,787 USD                          | GOV     | A | S313       | Table 2.30, pp.64-65                                                                              | Facility_Pay_Layer     | FFS \$1.19M, MA \$1.24M, Medicaid \$0.73M, commercial \$1.32M on the same table                                                                                                                                                         |
| F451 | Medicare FFS share of ambulance transport revenue, mean per NPI (GADCS)                                       | 2025              | 0.29 share                             | GOV     | A | S313       | Table S.1, p.ix                                                                                   | Facility_Pay_Layer     | The one-sixth-of-market arithmetic elsewhere in this workbook uses dollars, not per-NPI means; bases differ and are never mixed                                                                                                         |
| F452 | Organizations that did not always bill in at least one payer category (GADCS, published as about one in five) | 2025              | 0.2 share, approximate as published    | GOV     | A | S313       | Table S.1, p.ix ("about one in five")                                                             | Facility_Pay_Layer     | The eligible-but-unbilled anchor for the sizing gross-up; approximate by publication                                                                                                                                                    |
| F453 | DocGo Transportation Services revenue FY2025                                                                  | 2025              | 200,765,608 USD                        | SEC-A   | A | S314       | 10-K FY2025, Note 2 (F-22) and Note 11 (F-37)                                                     | Facility_Pay_Layer     | FY2024 \$193.4M, FY2023 \$181.5M same tables                                                                                                                                                                                            |
| F454 | DocGo transportation revenue outside per-trip billing, FY2025                                                 | 2025              | 0.409 share                            | DERIVED | A | S314       | 1 - trips x ATP / segment revenue; inputs Note 2/MD&A; live formula on Facility_Pay_Layer Panel B | Facility_Pay_Layer     | FY2024 41.1% by the same arithmetic; whole-dollar ATP rounding gives about 0.1pp uncertainty                                                                                                                                            |
| F455 | VA ambulance procurement obligations (PSC V225), FY2025                                                       | 2025              | 246,082,814 USD                        | GOV     | A | S315       | USAspending spending_over_time, psc V225, VA awarding toptier (Pull_Manifest hash)                | Facility_Pay_Layer     | All-agency FY2025 \$292.7M; VA share 84%; Beneficiary Travel claims are a separate channel, never summed                                                                                                                                |
| F456 | Subject-company ground base transports billed to Medicare FFS (floor)                                         | 2024              | 69,684 services                        | GOV     | A | S316       | MUP by Provider & Service 2024, estate NPIs, base codes                                           | MMT_Medicare_Book      | Panel E per-NPI rows sum to this; flagship NPI A0428 row verified to the cent against the live API (Verification_Log)                                                                                                                   |
| F457 | Subject-company Medicare FFS allowed dollars, ground base codes (floor)                                       | 2024              | 22,004,764 USD                         | GOV     | A | S316       | Sum of services x average allowed over estate NPI rows                                            | MMT_Medicare_Book      | Paid share of allowed prints beside it (Panel A)                                                                                                                                                                                        |
| F458 | Subject-company SCT (A0434) share of base services                                                            | 2024              | 0.0084 share                           | DERIVED | A | S316       | Panel B live shares; SCT services / base services                                                 | MMT_Medicare_Book      | RUNS BELOW the 6.6% national SCT share (finding 6 lineage); the Medicare-visible book is the scheduled BLS interfacility product; high-acuity dedicated-unit work under facility contracts never appears in claims (Facility_Pay_Layer) |
| F459 | Subject-company mileage units per base transport (implied avg loaded miles)                                   | 2024              | 14.6 miles (floor/floor)               | DERIVED | A | S316       | Panel C, A0425 units / base services                                                              | MMT_Medicare_Book      | Interfacility books run long miles; compare national A0425-per-base on MUP_Ambulance_National                                                                                                                                           |
| F460 | Subject-company base-service CAGR 2019 to 2024 (Medicare FFS floor)                                           | 2024              | 0.1836 CAGR                            | DERIVED | A | S316       | Panel D live formula over Panel A                                                                 | MMT_Medicare_Book      | Read against FFS enrollment decline (Utilization_Normalized); per-beneficiary growth is stronger than nominal                                                                                                                           |
| F461 | Estate share of NE Medicare FFS ground base services (deepest footprint state)                                | 2024              | 0.6701 share of state base services    | DERIVED | A | S317       | Panel B spotlight, 2024 row; estate NE services / NE state total, base codes                      | Market_Share_Panels    | Panel A NE block: the top NE row is an estate NPI and its services match MMT_Medicare_Book Panel E                                                                                                                                      |
| F462 | Estate rank among NE Medicare ambulance billers                                                               | 2024              | 1 rank                                 | DERIVED | A | S317       | Panel B spotlight, 2024 row; estate combined vs per-NPI billers                                   | Market_Share_Panels    | Rank 1 in every manifested vintage back to 2013 (same column)                                                                                                                                                                           |
| F463 | Ambulance biller HHI, NE (NPI grain)                                                                          | 2024              | 4,572 HHI points (0-10000)             | DERIVED | A | S317       | Panel C NE row; squared percent shares over all NE billing NPIs, base codes                       | Market_Share_Panels    | Above the 2500 DOJ/FTC highly-concentrated line; NPI grain understates - parent roll-up would raise it                                                                                                                                  |
| F464 | Estate measured base services in the nine non-NE footprint states (registration-state basis)                  | 2024              | 0 services                             | GOV     | A | S317       | Panel B estate-services matrix, 2024 row, IA-KY columns                                           | Market_Share_Panels    | Zero in every vintage: MUP assigns volume to the NPI registration state, so border-market work books to NE; not evidence of no operations                                                                                               |
| F465 | Ambulance biller HHI, IA (second footprint state, NPI grain)                                                  | 2024              | 292 HHI points (0-10000)               | DERIVED | A | S317       | Panel C IA row; squared percent shares over all IA billing NPIs                                   | Market_Share_Panels    | Unconcentrated (under 1500): the estate-visible deep state (NE) is the outlier, not the rule                                                                                                                                            |
| F466 | Distinct US Medicare ambulance billing NPIs, ground base codes                                                | 2024              | 8,721 NPIs                             | GOV     | A | S318       | Panel A, 2024 row; distinct Rndrng_NPI over base codes                                            | Fragmentation_National | Panel B decile counts sum to it (live SUM row); overstates firms - NPI grain                                                                                                                                                            |
| F467 | National Medicare FFS ground base services (floor)                                                            | 2024              | 9,542,103 services                     | GOV     | A | S318       | Panel A, 2024 row                                                                                 | Fragmentation_National | Down about 36% from 2013 on the same basis; compare MUP_Ambulance_National                                                                                                                                                              |
| F468 | Top-100 NPI share of national base services (NPI grain)                                                       | 2024              | 0.2457 share                           | DERIVED | A | S318       | Panel A, 2024 row, top-100 share (live over blue totals)                                          | Fragmentation_National | Top-10 at the same grain holds 6.2%; parent roll-up raises only the top-10 view materially                                                                                                                                              |
| F469 | Top-10 PARENT share of national base services (name-pattern roll-up, printed layer)                           | 2024              | 0.1181 share                           | DERIVED | A | S318       | Panel C concentration table, 2024 row; pattern table printed above it                             | Fragmentation_National | 8.6% in 2013 by the same patterns; a floor - brands under unrelated names stay unrolled                                                                                                                                                 |
| F470 | Mean annual exit rate, smallest biller decile (one-year transitions only)                                     | 2024              | 0.226 share of decile exiting per year | DERIVED | A | S318       | Panel D, D1 column, mean over one-year transitions                                                | Fragmentation_National | Largest decile mean 1.4%: the exit gradient is monotone in size; tail exits include suppression dips (upper bound)                                                                                                                      |
| F471 | Hospital-billed share of Medicare FFS ground base services, FLOOR bound (H1 name AND H2 Care Compare linkage) | 2024              | 0.0144 share of base services          | DERIVED | B | S319, S320 | Insourcing_Bounds Panel B 2024 row, live formula F26 over blue service totals                     | Insourcing_Bounds      | Floor set is 85 of 8,721 billers; Panel A prints every rule count; suppression and the current-vintage hospital roster both push this DOWN                                                                                              |
| F472 | Hospital-billed share of Medicare FFS ground base services, CEILING bound (H1 OR H2)                          | 2024              | 0.2697 share of base services          | DERIVED | B | S319, S320 | Insourcing_Bounds Panel B 2024 row, live formula G26                                              | Insourcing_Bounds      | Place-name collisions (county/city agencies vs same-named hospitals) inflate this UP; billing-in-lieu also inflates it                                                                                                                  |
| F473 | Change in hospital-billed FLOOR share of ground base services, 2013 to 2024                                   | 2024              | 0.0021 share points                    | DERIVED | B | S319, S320 | Insourcing_Bounds Panel B change row F27                                                          | Insourcing_Bounds      | Direction: share ROSE while absolute floor volume FELL (about 184K to 138K services); the all-biller denominator shrank 36%, so the rise is denominator shrinkage                                                                       |
| F474 | SCT+ALS2 share of base services, hospital-affiliated FLOOR-set billers                                        | 2024              | 0.0228 share of base services          | DERIVED | B | S319, S320 | Insourcing_Bounds Panel E row 53, live formula col D                                              | Insourcing_Bounds      | Versus explicitly independent billers on the row two below; carrier-claims window only, so Part A bundled SCT is invisible to the test                                                                                                  |
| F475 | SCT+ALS2 share of base services, explicitly independent (H3) billers                                          | 2024              | 0.0152 share of base services          | DERIVED | B | S319, S320 | Insourcing_Bounds Panel E row 55, live formula col D                                              | Insourcing_Bounds      | The comparator for the acuity test; H3 names (AMBULANCE/EMS/MEDIC-/RESCUE/FIRE, no H1 term) cover 4,206 of 8,721 billers                                                                                                                |

|      |                                                                                                                                                                                 |      |                                                                                       |            |   |                  |                                                                                                                  |                             |                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------|------------|---|------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F476 | Medicare FFS ground base services, all billers (the classification denominator)                                                                                                 | 2024 | 9,542,103 services                                                                    | GOV        | A | S319             | Sum of Tot_Srvcs over every MUP 2024 provider-grain row, codes A0426-A0429/A0433/A0434; blue cell E26 on Panel B | Insourcing_Bounds           | Suppression makes it a floor; compare the national series tabs built from the same registry                                                                                                                   |
| F477 | Hospitals reporting an ambulance cost center (Worksheet A line 95), latest complete FY                                                                                          | 2023 | 865 hospitals                                                                         | GOV        | A | S321             | Panel A, FY end 2023 row; NMRC A000000/09500 nonzero cost, RPT-joined                                            | HCRIS_Ambulance_CostCenters | Panel B state counts sum to it (live All-states row); share of filers prints beside it                                                                                                                        |
| F478 | Share of all 2552-10 filers reporting ambulance cost                                                                                                                            | 2023 | 0.1433 share                                                                          | DERIVED    | A | S321             | Panel A live formula, FY2023 row (reporters / filers)                                                            | HCRIS_Ambulance_CostCenters | Stable 14.3-14.6% across all four complete FYs 2020-2023 in the same panel                                                                                                                                    |
| F479 | As-filed ambulance cost, all reporting hospitals                                                                                                                                | 2023 | 3,183,473,759 USD                                                                     | GOV        | A | S321             | Panel A, FY2023 row, column 3 totals as filed                                                                    | HCRIS_Ambulance_CostCenters | Panel B state dollars sum to it (live); gross Worksheet A cost before reclasses/adjustments                                                                                                                   |
| F480 | Net flow of ambulance cost-center reporters, latest complete pair (starts minus stops, 2022 to 2023)                                                                            | 2023 | 7 hospitals (net entry)                                                               | DERIVED    | A | S321             | Panel D live net formula, 2022 -> 2023 row (47 starts vs 40 stops)                                               | HCRIS_Ambulance_CostCenters | Direction REVERSED from 2021 -> 2022 (net -17); candidate events, filing-lag caveat in the same row                                                                                                           |
| F481 | Footprint-state hospitals reporting ambulance cost (NE IA KS MO OH WI VA MN IN KY)                                                                                              | 2023 | 259 hospitals                                                                         | DERIVED    | A | S321, S322       | Panel B footprint subtotal (live SUMIF over the FOOTPRINT flag column)                                           | HCRIS_Ambulance_CostCenters | Iowa (57) and Minnesota (38) sit 2nd and 3rd nationally by count; about 30% of all reporters sit in the ten footprint states                                                                                  |
| F482 | Research-cohort corridor cases, nine systems, top-15 origin ZIPs (Medicare FFS inpatient floor)                                                                                 | 2025 | 320,143 cases                                                                         | GOV        | A | S323, S324, S325 | HSA 2025 top-15 extract, CCNs per Panel A printed rules; Panel B total row                                       | Cohort_Corridors            | Sum of nine live system rows; per-CCN rows re-derivable from HSA_Corridors                                                                                                                                    |
| F483 | Largest cohort system by corridor volume (Ascension)                                                                                                                            | 2025 | 149,493 cases                                                                         | GOV        | A | S323, S324, S325 | Panel B first data row (systems sorted by cases)                                                                 | Cohort_Corridors            | 68 hospitals in the corridor file under the printed rule                                                                                                                                                      |
| F484 | Cohort corridor volume COVERED: originating in ZIPs with a registered subject-company estate location                                                                           | 2025 | 0.0102 share of cohort corridor cases                                                 | DERIVED    | A | S323, S326       | Panel C total row, covered share (live)                                                                          | Cohort_Corridors            | Registration presence only; understates operating coverage by construction                                                                                                                                    |
| F485 | Cohort corridor volume CONTESTED: participant ZIP present, no estate ZIP                                                                                                        | 2025 | 0.0256 share of cohort corridor cases                                                 | DERIVED    | A | S323, S326       | Panel C total row, contested share (live)                                                                        | Cohort_Corridors            | 400 participant ZIP5s from 38 resolved crosswalk queries                                                                                                                                                      |
| F486 | Cohort corridor volume OPEN: no estate or participant registration in the origin ZIP                                                                                            | 2025 | 0.9642 share of cohort corridor cases                                                 | DERIVED    | A | S323, S326       | Panel C total row, open share (live)                                                                             | Cohort_Corridors            | Covered + contested + open sum to 1 in the live row                                                                                                                                                           |
| F487 | Hub hospitals nationally (top-decile top-15 corridor volume AND breadth >= 8 of 15)                                                                                             | 2025 | 592 hospitals                                                                         | DERIVED    | A | S327             | Panel B HUB row; thresholds printed in Panel A (p90 = 4,694.9 cases, median = 206)                               | Hub_Spoke_Map               | 746 hospitals clear the volume bar alone; the breadth bar removes the narrow-shed fifth of them                                                                                                               |
| F488 | Hub hospitals in the 11 footprint states (NE OH WI IA KS IN KY MO MN VA WY)                                                                                                     | 2025 | 99 hospitals                                                                          | DERIVED    | A | S327, S328, S329 | Panel C footprint block, live total row                                                                          | Hub_Spoke_Map               | State attribution from rosters with SSA CCN-prefix fallback for 1,668 unrostered CCNs                                                                                                                         |
| F489 | Share of national top-15 corridor volume carried by hub hospitals                                                                                                               | 2025 | 0.4325 share of corridor cases                                                        | DERIVED    | A | S327             | Panel B HUB row, live share cell                                                                                 | Hub_Spoke_Map               | Hubs are 7.9% of hospitals: roughly a five-to-one concentration ratio                                                                                                                                         |
| F490 | Spoke hospitals nationally (at or below median volume, breadth <= 3 of 15)                                                                                                      | 2025 | 3,151 hospitals                                                                       | DERIVED    | A | S327             | Panel B SPOKE row                                                                                                | Hub_Spoke_Map               | Spokes carry 1.3% of national corridor volume despite being 42% of hospitals                                                                                                                                  |
| F491 | Hub CCNs belonging to research-cohort systems                                                                                                                                   | 2025 | 20 hospitals                                                                          | DERIVED    | A | S327, S328, S329 | Panel D cross-tag row; rules printed on Cohort_Corridors Panel A                                                 | Hub_Spoke_Map               | Name-match confounds carried from Cohort_Corridors apply here unchanged                                                                                                                                       |
| F492 | Public contract documents abstracted on the 6.4 field schema (E:2 corpus)                                                                                                       | 2026 | 12 documents                                                                          | GOV        | A | S330             | Contract_Corpus Panel A, one row per document                                                                    | Contract_Corpus             | Convenience sample of PUBLIC contracts; 11 further items parked with blockers on Panel D; private facility-to-operator agreements are not public records                                                      |
| F493 | USAspending PSC V225 award records pulled (FY2023-FY2025 activity window, top by amount)                                                                                        | 2025 | 300 award records                                                                     | GOV        | A | S331             | spending_by_award, psc V225, types A-D, 3 pages x 100, sorted by award amount                                    | Contract_Corpus             | 261 of 300 are Department of Veterans Affairs; top 15 flagged for PIID document lookup, blocked at SAM.gov (Panel D row 1)                                                                                    |
| F494 | Fixed annual escalator observed in the corpus, low end (five-town NH Pridestar agreement, renewal-term built-in)                                                                | 2024 | 0.025 share per year                                                                  | GOV        | A | S334             | Agreement renewal clause: USD 650,000 to USD 666,250 to USD 682,906.25 as printed                                | Contract_Corpus             | High end 3.0% (Charles City IA); range is live MIN/MAX on Panel C over n=2 documents                                                                                                                          |
| F495 | Fixed annual escalator observed in the corpus, high end (Charles City IA / Floyd County AMR agreement)                                                                          | 2023 | 0.03 share per year                                                                   | GOV        | A | S333             | Agency Contribution schedule: USD 415,000 to USD 427,450 to USD 440,273                                          | Contract_Corpus             | Low end 2.5% (NH Pridestar renewal); n=2, an observed range, not a market distribution                                                                                                                        |
| F496 | Dedicated-unit annual subsidy, five-town NH system (two dedicated ALS ambulances around the clock, Pridestar EMS)                                                               | 2024 | 650,000 USD per year                                                                  | GOV        | A | S334             | Subsidy paragraph: USD 32,500 per town per quarter, five towns                                                   | Contract_Corpus             | USD 130,000 per town per year; operator bills patients for EMS in addition                                                                                                                                    |
| F497 | Dedicated-unit annual subsidy, Charles City IA / Floyd County (one 24-hour ALS unit plus one 24-hour on-call BLS unit, AMR), Year 1                                             | 2023 | 415,000 USD per year                                                                  | GOV        | A | S333             | Agency Contribution: Year 1 USD 415,000, split 50/50 city and county                                             | Contract_Corpus             | Rises to USD 440,273 by Year 3; Floyd County Medical Center pledged USD 100,000 toward year one - a facility co-paying for ambulance capacity in a public document                                            |
| F498 | Per-trip rate observed, Nebraska DHHS client transport, Douglas and Sarpy counties (Camelot Transportation contract 104886-04)                                                  | 2025 | 39.02 USD per one-way trip (or USD 2.00 per mile, calculated as 19.01 miles per trip) | GOV        | A | S332             | Rate attachment Revised July 1, 2025; wholly within Douglas and Sarpy counties                                   | Contract_Corpus             | SCOPE: a state NON-AMBULANCE client-transport (NEMT-adjacent) rate under a child-welfare / client transportation contract - NOT an ambulance rate; recorded exactly as the document states, scope note in-row |
| F499 | Cohort systems resolved to public Form 990 filers                                                                                                                               | 2026 | 9 health systems (research cohort, public filings)                                    | GOV        | B | S336             | Cohort_990_Contractors Panel C EIN table                                                                         | Cohort_990_Contractors      | 24 EINs resolved; 23 with parsed XML; Cleveland Clinic East Region EIN parked with no e-file XML posted; mapping is analyst-selected from public registry facts (tier B)                                      |
| F500 | Form 990 filings parsed for Part VII Section B contractors                                                                                                                      | 2024 | 46 filings (latest two XML-available per EIN, tax years 2022-2024)                    | GOV        | A | S335             | Per-filing object_id in the register; per-system counts on Panel A                                               | Cohort_990_Contractors      | 11 newest-season filings XML-blocked (parked in-table); 6 parsed filings disclose no Section B contractors at all                                                                                             |
| F501 | Transport-matching contractors among the disclosed five-highest-paid contractors, all parsed cohort filings                                                                     | 2024 | 0 contractors (zero; stated as the information floor, not a threshold)                | GOV        | A | S335             | Panel A column G (live SUM in the total row); keyword regex in the Panel B note                                  | Cohort_990_Contractors      | Part VII Section B discloses only the top five contractors over \$100K; with up to 840 contractors over \$100K per filer, absence from the top five is NOT evidence of absence of spend                       |
| F502 | Research-cohort aggregate staffed beds (HCRIS latest filed FY, printed-rule matched CCNs)                                                                                       | 2022 | 22,582 beds                                                                           | GOV        | A | S337, S338       | System_Research_Cohort Panel A total row (live sum of nine system rows); HCRIS beds field, latest FY per CCN     | System_Research_Cohort      | FY2020-FY2022 filings; name-match confounds printed per panel                                                                                                                                                 |
| F503 | Research-cohort aggregate inpatient days (HCRIS latest filed FY, matched CCNs)                                                                                                  | 2022 | 5,322,510 patient days                                                                | GOV        | A | S337, S338       | System_Research_Cohort Panel A total row; HCRIS total_patient_days, latest FY per CCN                            | System_Research_Cohort      | Admissions are PENDING (not in the vendored extract; HCRIS S-3 Part 1 fills it)                                                                                                                               |
| F504 | Largest cohort system by measured corridor demand (Ascension)                                                                                                                   | 2025 | 149,493 top-15 corridor cases                                                         | GOV        | A | S339, S337, S338 | System_Research_Cohort Panel A first data row, col E (green-linked to Cohort_Corridors Panel B where built)      | System_Research_Cohort      | Matches the Cohort_Corridors roll-up by construction (same printed rules, same cache)                                                                                                                         |
| F505 | Cohort Form 990 filings parsed with ZERO ambulance / EMS / medical-transport contractors among the disclosed five highest-paid contractors                                      | 2024 | 46 filings                                                                            | PUBLIC-WEB | A | S340             | System_Research_Cohort Panel A 990 row (B19); 24 EINs, tax years 2022-2024, keyword screen printed in            | System_Research_Cohort      | Part VII Section B discloses only the top five over \$100K; an information floor, not evidence of no vendor                                                                                                   |
| F506 | Subject-company registered operating states (printed rule: at least one NPPES estate practice location)                                                                         | 2026 | 11 states                                                                             | GOV        | A | S342             | Footprint_Determination Panel E row 41 (states: IA NE OH SD CO IN MO NC RI VA WI)                                | Footprint_Determination     | Two states (IA, NE) independently corroborated by named V225 federal awards                                                                                                                                   |
| F507 | Provisional footprint states carried by earlier tabs WITHOUT a registered estate location (KS MN KY)                                                                            | 2026 | 3 states                                                                              | DERIVED    | A | S342             | Footprint_Determination Panel E row 44; reconciliation prose names the tabs                                      | Footprint_Determination     | Mirror gap: 4 registered states (CO NC RI SD) were absent from the provisional list; the live site publishes no state-grain claim, so claimed-not-registered states observable = 0                            |
| F508 | Top screened name family by measured transfer demand (Mercy)                                                                                                                    | 2025 | 80,732 top-15 corridor cases                                                          | DERIVED    | A | S339, S337       | Prospect_Landscape Panel B first data row (D15)                                                                  | Prospect_Landscape          | Family-caveat flag applies: a shared-name family can span unrelated organizations                                                                                                                             |
| F509 | Name families with 3+ hospitals screened in the registered footprint states (cohort excluded)                                                                                   | 2025 | 59 name families                                                                      | DERIVED    | A | S337, S339       | Prospect_Landscape Panel B count row (B35); grouping rule printed in Panel A                                     | Prospect_Landscape          | Short-term acute + CAH CCN ranges only; HCRIS FY2020-FY2022 name vintage                                                                                                                                      |
| F510 | Thinnest-supply large county in the footprint: Christian County, MO - 65+ residents per Medicare-billing ambulance provider (zero providers based in-county; ratio floor = 65+) | 2024 | 16,797 65+ residents per billing provider (floor)                                     | DERIVED    | A | S347, S348       | Panel L rank 1; census 65+ 7/1/2024 x MS_County_2025                                                             | County_Whitespace_Screens   | MS_County_2025 shows 105 users in the county, proving out-of-county service, not zero demand                                                                                                                  |
| F511 | Footprint counties with zero Medicare-billing ambulance providers and 5,000+ residents 65+                                                                                      | 2024 | 5 counties                                                                            | DERIVED    | A | S347, S348       | Panel M row 2; names printed in the same row                                                                     | County_Whitespace_Screens   | Each of the named counties shows positive Users in MS_County_2025 - served from out of county                                                                                                                 |
| F512 | Top-25 footprint whitespace counties with 0-1 dialysis facilities (recurring-transport overlay)                                                                                 | 2025 | 13 counties                                                                           | DERIVED    | A | S351, S348, S347 | Panel L overlay COUNTIF row; per-county counts in column O                                                       | County_Whitespace_Screens   | Dialysis_Registry carries the same registry at state/ZIP grain; county attribution is name-matched (0 can be a match miss)                                                                                    |

|      |                                                                                                                                        |      |                                                              |            |   |            |                                                                                                                               |                             |                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------|------------|---|------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F513 | Largest whitespace county by 65+ population (the metro consolidation read): Hennepin County, MN                                        | 2024 | 14,096 65+ residents per billing provider                    | DERIVED    | A | S347, S348 | Panel L; 211,444 65+ over 15 billing orgs                                                                                     | County_Whitespace_Screens   | Read as consolidation, not absence: users per provider prints in the same row                                                                                                      |
| F514 | PRICE component of the change in national ground base-code allowed dollars, 2019 to 2024 (Laspeyres, mix held constant)                | 2024 | 904,503,315 USD (signed)                                     | DERIVED    | A | S352       | Panel C, window 2019 to 2024, PRICE column                                                                                    | Growth_Decomposition        | Panel E code rows: price effect positive in all six codes; additivity check cell = 0                                                                                               |
| F515 | VOLUME component of the change in national ground base-code allowed dollars, 2019 to 2024 (services change at base-year average price) | 2024 | -1,138,151,885 USD (signed)                                  | DERIVED    | A | S352       | Panel C, window 2019 to 2024, VOLUME column                                                                                   | Growth_Decomposition        | Base services fell from about 13.0M to 9.6M (Panel B); FFS enrollment decline is the deflator caveat (Utilization_Normalized)                                                      |
| F516 | MIX component (residual) of the change in national ground base-code allowed dollars, 2019 to 2024                                      | 2024 | -107,082,292 USD (signed)                                    | DERIVED    | A | S352       | Panel C, window 2019 to 2024, MIX column                                                                                      | Growth_Decomposition        | A0428, the lowest-priced transport code, lost the most services; the residual nets code-shift against price-volume interaction, and the live check column proves additivity        |
| F517 | Medicare ambulance dollars denied, all seven PSPS codes (valuation: denied services x code-year average allowed per allowed service)   | 2024 | 553,269,323 USD                                              | DERIVED    | A | S354       | Panel B, 2024 row, all-codes column; live SUMPRODUCT over PSPS_Denial_Series rows 6-110                                       | Denial_Economics            | Panel C code rows sum to it; denominator (submitted minus denied) positive in every code-year, verified at build                                                                   |
| F518 | A0428 BLS non-emergency dollars denied (same valuation)                                                                                | 2024 | 115,666,690 USD                                              | DERIVED    | A | S354       | Panel B, 2024 row, A0428 column, live against PSPS_Denial_Series row 107                                                      | Denial_Economics            | A0428 denial rate 12.9% in 2024 vs 11.6% in 2010: rate UP while dollars fell - the volume-collapse divergence printed in Panel A                                                   |
| F519 | Fifteen-year trend in total ambulance denied dollars (2010 to 2024, nominal change)                                                    | 2024 | 0.0005 share (signed)                                        | DERIVED    | A | S354       | Panel B column C, 2010 vs 2024 rows                                                                                           | Denial_Economics            | Direction: flat nominal (declining real), peak 2012, trough 2020; A0428 denied dollars down roughly 40% over the same window while its rate rose                                   |
| F520 | Footprint median hospital OP-18b (median ED minutes before departure, discharged patients)                                             | 2025 | 134 minutes                                                  | DERIVED    | A | S356       | ED_Timeliness_Registry OP_18b rows, footprint states, numeric scores; period 07/2024 - 06/30/2025, median of hospital medians | Transfer_Delay_Burden       | Panel B live column re-counts hospitals at or above each decile against the registry                                                                                               |
| F521 | Footprint worst-decile OP-18b threshold (P90)                                                                                          | 2025 | 198 minutes                                                  | DERIVED    | A | S356       | Panel B P90 row; deciles over 1053 numeric footprint hospital medians                                                         | Transfer_Delay_Burden       | Panel C names the 20 hospitals above it, values green-linked to registry rows                                                                                                      |
| F522 | Research-cohort hospitals in the worst footprint OP-18b quartile                                                                       | 2025 | 23 hospitals (of 65 measured)                                | DERIVED    | A | S356       | Panel D, rules mirror Cohort_Corridors Panel A (footprint states only on this tab); worst quartile = OP-18b at or above 124   | Transfer_Delay_Burden       | Neutral placement read: cohort median 150 vs footprint 134 minutes; expected for referral-heavy systems                                                                            |
| F523 | Widest footprint paramedic-over-EMT median wage premium                                                                                | 2024 | 1.488 ratio (Minnesota)                                      | DERIVED    | A | S357       | Panel A live MAX over green-linked OEVS medians; Minnesota paramedic 68,000 vs EMT 45,690                                     | Workforce_Depth             | National extreme (WA, about 2.06x) noted in-row; footprint premiums span 1.26x-1.49x                                                                                               |
| F524 | Footprint private-ambulance average annual pay vs national                                                                             | 2025 | -0.0316 share gap (footprint below national)                 | DERIVED    | A | S358       | Panel B row 2025: live SUMPRODUCT footprint pay (about 59,733) vs QCEW_EMS_Employment Panel B national                        | Workforce_Depth             | Gap narrows from -4.9% (2019) to -3.2% (2025); footprint converging on national pay                                                                                                |
| F525 | Quarterly wage-vs-AIF scissors, 2019Q4 to 2025Q4                                                                                       | 2025 | 22.7 index points (2019Q4=100)                               | DERIVED    | A | S358, S361 | Panel D read 2, last row: wage index 144.1 minus AIF-compounded path 121.3                                                    | Workforce_Depth             | Annual version is finding 45 (QCEW_EMS_Employment Panel B): +5.0%/yr pay vs +3.1%/yr AIF over 2014-2025                                                                            |
| F526 | Tightest footprint EMS labor market (two-test rule: top-2 wage growth AND bottom-2 depth)                                              | 2025 | 0.701 wage growth 2019-2025 (Kansas; depth 1.45 per non-SEA) | DERIVED    | B | S358, S360 | Panel E ranks; rule printed in the header and banner - a matched pair of rankings, not a composite index                      | Workforce_Depth             | Iowa prints thinner depth but its employment floor is suppression-driven; Kansas tops wage growth outright                                                                         |
| F527 | Ambulance-tagged LEIE exclusions in footprint states (cumulative)                                                                      | 2025 | 126 exclusions (of 566 national)                             | GOV        | A | S363       | LEIE_Ambulance_Exclusions col G in (NE IA KS MO OH WI VA MN IN KY); live COUNTIF panel D                                      | LEIE_Read_Panel             | KY 36 + OH 43 carry most of the footprint total; dates span 1985-2025                                                                                                              |
| F528 | Business-entity share of ambulance-tagged LEIE exclusions (BUSNAME heuristic)                                                          | 2025 | 0.2102 share                                                 | DERIVED    | A | S363       | Panel C live COUNTIF on business-name column C (119 of 566)                                                                   | LEIE_Read_Panel             | Four in five excluded parties are individuals - the list is mostly people, not companies                                                                                           |
| F529 | Total daily cost per boarded acute stroke patient, med/surg boarding (TDABC)                                                           | 2024 | 1,856 USD per patient per day                                | ACADEMIC   | A | S364       | Ann Emerg Med 2024;84(4):376-385, abstract Results; PMID 38795079                                                             | Throughput_Economics_Public | Med/surg inpatient comparator \$993 and ICU pair \$2,267 vs \$2,165 in the same abstract; per-hour normalization exists only as live formulas in Panel D                           |
| F530 | Median door-in-door-out time at the referring hospital, stroke interhospital transfer                                                  | 2023 | 174 minutes                                                  | ACADEMIC   | A | S365       | JAMA 2023;330(7):636-649, abstract Results; PMID 37581671                                                                     | Throughput_Economics_Public | IQR 116 to 276; only 27.3% of 108,913 transfers from 1,925 hospitals met the recommended 120 minutes                                                                               |
| F531 | Forgone net revenue from ED capacity consumed by boarding, one 450-bed community hospital                                              | 2007 | 3,960,264 USD per year                                       | ACADEMIC   | A | S366       | Acad Emerg Med 2007;14(4):332-337, abstract Results; PMID 17331916                                                            | Throughput_Economics_Public | 10,397 boarding bed-hours and 3,175 forgone encounters in the same abstract; about \$381 per bed-hour derived live on Panel D                                                      |
| F532 | Increased risk of ICU death per hour of waiting for ICU admission                                                                      | 2011 | 0.015 risk increase per hour (HR 1.015)                      | ACADEMIC   | A | S367       | Crit Care 2011;15(1):R28, abstract Results; PMID 21244671                                                                     | Throughput_Economics_Public | 95% CI 1.006 to 1.023; attributable fraction of mortality risk 30% (CI 11.2 to 44.8)                                                                                               |
| F533 | Public records establishing EHR-integrated patient transport ordering (existence only)                                                 | 2026 | 6 PUBLIC-WEB records (4 distinct platform x EHR)             | PUBLIC-WEB | A | S368, S369 | Panel C rows on Throughput_Economics_Public; count is a live COUNTA on the tab; each row carries its URL and access data      | Throughput_Economics_Public | Covers both dominant US EHRs (Epic, Oracle Cerner) and extends to ambulance-level interfacility modes (Roundtrip 2025, VectorCare); adoption existence only, no performance claims |
| F534 | RSNAT prior authorization, estimated potential savings, 3 initial demonstration states (NJ, PA, SC)                                    | 2017 | 349,500,000 USD (nominal, implementation through March 2017) | GOV        | A | S370       | GAO-18-341, Table 2, p.16                                                                                                     | GAO_OIG_Shelf               | Plus \$38.0M in 6 expansion states on the same table; provisional affirmation rate rose from 28% to 66% (p.12); the model was later made national by statute                       |
| F535 | Growth in Medicare Part B dialysis-related ambulance transports, 2002 to 2011                                                          | 2011 | 2.69 growth over period (+269 percent)                       | GOV        | A | S371       | OEI-09-12-00350, Summary p.1 and dialysis finding section                                                                     | GAO_OIG_Shelf               | From 753,741 transports in 2002; fastest of any origin-destination category; all Part B transports +69% vs beneficiaries +7% over the same window                                  |
| F536 | Medicare Part B ambulance transport payments, 2012                                                                                     | 2012 | 5,800,000,000 USD (nominal 2012)                             | GOV        | A | S372       | OEI-09-12-00351, Executive Summary (Why We Did This Study)                                                                    | GAO_OIG_Shelf               | Almost double 2003; H1-2012 review found \$24M not meeting requirements plus \$30M with no Medicare services anywhere, and about 1 in 5 suppliers with questionable billing        |
| F537 | Dialysis-related ambulance claim allowances failing Medicare medical-necessity coverage review                                         | 1994 | 0.7 share of reviewed dialysis-related transports            | GOV        | A | S373       | OEI-03-90-02130, Findings p.4 of PDF                                                                                          | GAO_OIG_Shelf               | 1991 ESRD ambulance allowances were \$101M with 75% concentrated in fewer than 2% (2,573) of beneficiaries - the earliest marker of the dialysis integrity arc                     |
| F538 | Rural Emergency Hospitals enrolled (national)                                                                                          | 2026 | 48 facilities                                                | GOV        | A | S374       | CMS REH enrollment roster (Pull_Manifest)                                                                                     | REH_Closure_Flow            | Every REH admission-grade patient is a transfer by statute                                                                                                                         |
| F539 | Rural closures and conversions since 2010, footprint states                                                                            | 2026 | 19 events                                                    | GOV        | A | S181       | Sheps closure list, state field in footprint set                                                                              | REH_Closure_Flow            | The Sheps-vs-Chartis counting convention caveat from Methodology section 4 applies                                                                                                 |
| F540 | Nebraska Medicaid FFS ambulance fee schedule: A0428 base rate (1 Jul 2026)                                                             | 2026 | 154.89 USD per transport                                     | GOV        | A | S375       | 471-000-504.xlsx row 000A0428, MEDICAID ALLOWABLE                                                                             | Medicaid_Rate_Card          | Schedule cover states no increase for SFY2027                                                                                                                                      |
| F541 | Iowa Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2014, current 11 Jul 2026)                                        | 2026 | 84.67 USD per transport                                      | GOV        | A | S376       | C11.csv row A0428, Factor column                                                                                              | Medicaid_Rate_Card          | Lowest true-BLS A0428 in footprint; unchanged 12 years                                                                                                                             |
| F542 | Kansas Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2023, posted 7/1/2026)                                          | 2026 | 255.7 USD per transport                                      | GOV        | A | S377       | FeeSchedule_20260701_TXIX_DEF.txt row A0428 MAX FEE                                                                           | Medicaid_Rate_Card          | SFY2024 rebase                                                                                                                                                                     |
| F543 | Missouri Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2019, file 7 Jul 2026)                                        | 2026 | 105.62 USD per transport                                     | GOV        | A | S378       | Ambulance.xlsx rows A0428 HH / A0428 HD                                                                                       | Medicaid_Rate_Card          | Payable ONLY in HH/HD contexts; base code noncovered                                                                                                                               |
| F544 | Ohio Medicaid FFS ambulance fee schedule: A0428 base rate (1 Jan 2024)                                                                 | 2026 | 203.75 USD per transport                                     | GOV        | A | S379       | OAC 5160-15-28 appendix, row A0428                                                                                            | Medicaid_Rate_Card          | 2024 rebase; GEMT supplemental tops up public providers                                                                                                                            |
| F545 | Wisconsin Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jul 2008, report 11 Jul 2026)                                    | 2026 | 94.9 USD per transport                                       | GOV        | A | S380       | maxfee06_ambulance.pdf row A0428 MAX FEE                                                                                      | Medicaid_Rate_Card          | Unchanged 18 years; manager-coordinated NEMT paid off-schedule                                                                                                                     |
| F546 | Minnesota Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026)                                                        | 2026 | 284.56 USD per transport                                     | GOV        | A | S381       | MHCP Fee Schedule.xlsx row A0428, Max Fee column                                                                              | Medicaid_Rate_Card          | Highest A0428; exactly the CY2026 Medicare CF (parity)                                                                                                                             |
| F547 | Indiana Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026)                                                          | 2026 | 269.02 USD per transport                                     | GOV        | A | S382       | IHCP Professional Fee Schedule.xlsx row A0428, Rate Eff 01/01/2026                                                            | Medicaid_Rate_Card          | 100% of CY2025 Medicare (BT2025156); A0434 SCT noncovered                                                                                                                          |
| F548 | Kentucky Medicaid FFS ambulance fee schedule: A0428 base rate (eff 1 Jan 2026 (rev 1/9/2026))                                          | 2026 | 55 USD per transport                                         | GOV        | A | S383       | 2026 Transportation Rates PDF, PT56 Specialty 16 A0428                                                                        | Medicaid_Rate_Card          | Non-emergency STRETCHER-van line (formerly T2005), not a full BLS ambulance                                                                                                        |
| F549 | Lowest footprint Medicaid true-BLS A0428 rate as a multiple of the CY2026 Medicare national base (Iowa)                                | 2026 | 0.2975 x Medicare                                            | DERIVED    | A | S376       | Medicaid_Rate_Card Panel, MIN over the true-BLS A0428 ratios; 84.67 / 284.56 (KY stretcher excluded)                          | Medicaid_Rate_Card          | Vintage confound printed in-row: IA rate is a 2014 rate against a 2026 Medicare base                                                                                               |



|      |                                                                                                                                                      |      |                                                                                   |         |   |                                                      |                                                                                                                                    |                             |                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|---------|---|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F550 | Highest footprint Medicaid A0428 rate as a multiple of the CY2026 Medicare national base (Minnesota)                                                 | 2026 | 1 x Medicare                                                                      | DERIVED | A | S381                                                 | Medicaid_Rate_Card Panel, MAX over the true-BLS A0428 ratios; 284.56 / 284.56 = exact parity                                       | Medicaid_Rate_Card          | MN A0428 equals the CY2026 Medicare conversion factor to the cent                                                                                                                                   |
| F551 | Footprint states routing scheduled NEMT through a broker, transportation manager, or state-administered system                                       | 2026 | 8 of 9 states                                                                     | DERIVED | A | S375, S376, S377, S378, S379, S380, S381, S382, S383 | Medicaid_Rate_Card carve panel, COUNTIF over the carve column (OH is the only exception: county NET)                               | Medicaid_Rate_Card          | VA (extension) goes further: the broker pays FFS non-emergency ambulance itself                                                                                                                     |
| F552 | RSNAT baseline utilization, Year 1 cohort (share of ESRD and/or pressure-ulcer beneficiaries using RSNAT, baseline 2012-2014)                        | 2014 | 0.101 share of study population                                                   | GOV     | A | S389                                                 | Final report, Table II.2, p.9                                                                                                      | RSNAT_Series                | Year 2 cohort 4.7%, comparison states 4.9% on the same table: the model states were picked for high baseline use (FAQ Q10)                                                                          |
| F553 | GAO estimated potential RSNAT savings, 3 initial states, implementation through Mar 2017                                                             | 2017 | 349,500,000 USD                                                                   | GOV     | A | S370                                                 | GAO-18-341, Table 2, p.16                                                                                                          | RSNAT_Series                | Plus \$38.0M in the 6 expansion states, same table; "as high as" counterfactual estimates                                                                                                           |
| F554 | RSNAT provisional affirmation rate, initial PAR submissions                                                                                          | 2017 | 0.66 share of initial PARs affirmed, Oct 2016-Mar 2017                            | GOV     | A | S370                                                 | GAO-18-341, p.12                                                                                                                   | RSNAT_Series                | Up from 28% in the first six months (Dec 2014-May 2015): a documentation learning curve, not a loosening of the screen                                                                              |
| F555 | RSNAT use reduction, final evaluation (nine model states, ESRD and/or pressure-ulcer population)                                                     | 2021 | -0.72 change vs comparison                                                        | GOV     | A | S389                                                 | Final report, Table ES.1 and Ch. III                                                                                               | RSNAT_Series                | RSNAT expenditures -\$746M; total Medicare FFS -2.4%; no adverse quality or access findings                                                                                                         |
| F556 | RSNAT suppliers per 100,000 beneficiaries, change on implementation (model states)                                                                   | 2020 | -0.5 change                                                                       | GOV     | A | S388                                                 | Second interim report, Objective 3 (exec summary)                                                                                  | RSNAT_Series                | Concentrated almost entirely in Year 1 states; exiting suppliers were smaller and dialysis-dependent - the supply-side echo of finding 26 (dialysis transports -63%)                                |
| F557 | MA-to-FFS ambulance revenue ratio per NPI (the MA book calibrator)                                                                                   | 2025 | 1.038 x (ratio of means per NPI)                                                  | DERIVED | A | S313                                                 | GADCS Y1-4 appendix, Table 2.30, pp.64-65: \$1,238,482 / \$1,193,046; live formula on MA_Book_Calibrator Panel A                   | MA_Book_Calibrator          | Published cuts run 0.55x (super rural) to 1.19x (government); for-profit 1.08x; superseded the day MA encounter data (MA_Encounter_Recv) lands                                                      |
| F558 | MA dark share this calibrator prices (beneficiaries outside A and B FFS)                                                                             | 2024 | 0.593 share of Medicare beneficiaries                                             | DERIVED | A | S313                                                 | Imbalance_Ledger Panel C 2024 row (live link on MA_Book_Calibrator Panel C); enrollment inputs S60                                 | MA_Book_Calibrator          | Engagement_Data_Map quotes the same 59.3%; CBO projects 63% by 2034                                                                                                                                 |
| F559 | Emergency-capable acute hospitals, footprint states (receiving-capacity floor)                                                                       | 2026 | 575 hospitals                                                                     | GOV     | A | S391                                                 | Hospital General Information, acute type with emergency_services = Yes, footprint states                                           | Receiving_Center_Registry   | A floor: designated trauma/stroke/STEMI centers are a subset (PENDING Panel C)                                                                                                                      |
| F560 | Emergency-capable acute hospitals, national                                                                                                          | 2026 | 2,981 hospitals                                                                   | GOV     | A | S391                                                 | Hospital General Information, acute type with emergency_services = Yes                                                             | Receiving_Center_Registry   | Reconciles to the Hosp_Registry universe (same PDC source family)                                                                                                                                   |
| F561 | Footprint states with a statutory entry gate beyond licensure (CON, need test, endorsement gate, franchise power, department-set PSA)                | 2026 | 5 states of 10 (KY, MN, MO, VA, WI)                                               | GOV     | A | S392, S393                                           | Entry_Barrier_Register Panels A-B; live count Panel D                                                                              | Entry_Barrier_Register      | Each mechanism verified against fetched statute text; VA gate is at local OPTION and KS carries a flagged non-duplication rule short of a gate                                                      |
| F562 | Footprint licensure register completeness (agency + statute cite verified per state)                                                                 | 2026 | 1 share of 10 states                                                              | GOV     | A | S392                                                 | Entry_Barrier_Register Panel C (10 rows, each with fetched URL)                                                                    | Entry_Barrier_Register      | Two rows carry flagged ambiguities elsewhere (KS county power, IN EOA absence), none on licensure                                                                                                   |
| F563 | States with ground-ambulance balance-billing protections                                                                                             | 2026 | 22 states (some protection; includes partial regimes)                             | SOURCED | A | S394                                                 | Commonwealth Fund / CHIR, 13 Mar 2026 ("people in 22 states now have some protection"); register rows on Balance_Billing_Contracts | Balance_Billing_States      | This register prints all 22 with 16 verified against fetched statute text and 6 marked partial/ambiguous (DE, FL, MD, NV, WV, VT-flag); 13 of 22 also cover non-emergency transport per the tracker |
| F564 | Highest state Medicare peg for out-of-network ground ambulance (Indiana, footprint state)                                                            | 2024 | 4 x Medicare (fallback cap absent a local rate; lesser of this or billed charges) | GOV     | A | S395                                                 | IC 27-1-2.3 as amended by HEA 1385-2024, eff. 1 Jul 2024                                                                           | Balance_Billing_States      | Peg range across verified rows 1.5x (NM) to 4.0x (IN); modal peg 3.25x (CO, LA, TX, NH, OR)                                                                                                         |
| F565 | States with V225 ambulance obligations, FY2025                                                                                                       | 2025 | 49 states                                                                         | GOV     | A | S397                                                 | spending_by_geography, place_of_performance, FY2025                                                                                | Federal_Ambulance_Contracts | 10 footprint states present; subtotal is a live SUMIF on the tab                                                                                                                                    |
| F566 | Distinct top-50 V225 recipients appearing in any of FY2023-2025                                                                                      | 2025 | 77 recipients                                                                     | GOV     | A | S397                                                 | spending_by_category/recipient, three fiscal years                                                                                 | Federal_Ambulance_Contracts | Recipient names are as registered in SAM; parent roll-ups not applied (stated)                                                                                                                      |
| F567 | National distinct Medicare ambulance billing NPIs, ground base codes - net change 2013 to 2024                                                       | 2024 | -597 NPIs                                                                         | DERIVED | A | S399                                                 | Annual_Market_Structure Panel A col B, 2024 row minus 2013 row                                                                     | Annual_Market_Structure     | 9,318 (2013) to 8,721 (2024), -6.4%; gross churn of 330-490 entries and exits per year dwarfs the net drift                                                                                         |
| F568 | Share of the 2014 bottom size quartile (by base services) still billing in 2024                                                                      | 2024 | 0.6505 share                                                                      | DERIVED | A | S399                                                 | Annual_Market_Structure Panel C, Q1 row, 2024 column                                                                               | Annual_Market_Structure     | OVERSTATED exit by construction: the 11-beneficiary suppression floor binds hardest on the smallest quartile, so a floor-crossing reads as an exit                                                  |
| F569 | Share of the 2014 top size quartile (by base services) still billing in 2024                                                                         | 2024 | 0.6867 share                                                                      | DERIVED | A | S399                                                 | Annual_Market_Structure Panel C, Q4 row, 2024 column                                                                               | Annual_Market_Structure     | Below the middle quartiles (81.8% / 81.8%) - the U-shape reads as consolidation retiring large NPIs, not failure                                                                                    |
| F570 | Subject-company estate share of NE Medicare ground base services, 2022-2024 smoothed average                                                         | 2024 | 0.6285 share                                                                      | DERIVED | A | S399                                                 | Annual_Market_Structure Panel D smoothed row (live AVERAGE over the annual share column)                                           | Annual_Market_Structure     | vs 24.1% smoothed across 2013-2015; the single-year series rises in all eleven year-over-year steps                                                                                                 |
| F571 | Top-100 NPI share of national Medicare ground base services                                                                                          | 2024 | 0.2457 share                                                                      | DERIVED | A | S399                                                 | Annual_Market_Structure Panel A, top-100 column, 2024 row                                                                          | Annual_Market_Structure     | 22.3% in 2013; essentially flat through 2021, then rising three years running                                                                                                                       |
| F572 | Widest state SCT-to-BLS realized multiple, 2024 (A0434/A0428 avg allowed, 30+ services both codes): GA                                               | 2024 | 3.81 x                                                                            | DERIVED | A | S400                                                 | Realized_Price_Ladders Panel B, GA row, live multiple column                                                                       | Realized_Price_Ladders      | vs the frozen 3.25x AFS RVU ratio and SD at 3.16x on the narrow end; deviation is geographic mix, not payment policy                                                                                |
| F573 | National volume-weighted realized allowed per A0428 service, CAGR 2013-2024                                                                          | 2024 | 0.0166 CAGR                                                                       | DERIVED | A | S400                                                 | Realized_Price_Ladders Panel A, A0428 NATIONAL row, 2013 and 2024 columns                                                          | Realized_Price_Ladders      | \$218.22 to \$261.47; an administered-price series - compare AFS conversion-factor updates on Payment_Rules                                                                                         |
| F574 | Footprint A0428 realized-price dispersion, 2024: max state over min state                                                                            | 2024 | 1.173 x                                                                           | DERIVED | A | S400                                                 | Realized_Price_Ladders Panel C, A0428 row, footprint max/min live column                                                           | Realized_Price_Ladders      | MN \$288.42 over KY \$245.78; national max/min is wider only because AK and territories enter                                                                                                       |
| F575 | Registered organizational ambulance NPIs, national (NPPES 3416 taxonomy)                                                                             | 2026 | 20,377 NPIs                                                                       | GOV     | A | S401                                                 | NPPES roster, Entity Type 2, 3416+ taxonomy                                                                                        | Registered_vs_Billing       | More than double the Medicare-billing universe; the gap bounds the non-Medicare layer                                                                                                               |
| F576 | Registered-to-billing ratio, national ambulance supply                                                                                               | 2026 | 2.34 ratio                                                                        | DERIVED | A | S401                                                 | National registered NPIs / distinct MUP 2024 ground base billers                                                                   | Registered_vs_Billing       | Bound not headcount: dormant/air/parent-billed NPIs inflate; suppression deflates the billing count                                                                                                 |
| F577 | Total mission-demand line over the 20 target metros (HSA catchment cases x measured transfer rate, proxy-based floor)                                | 2025 | 69,767 transfer episodes per year (floor)                                         | DERIVED | A | S402, S403                                           | Metro_TAM_Panels Panel S, live SUM over 20 mission-demand lines                                                                    | Metro_TAM_Panels            | Medicare-only cases x all-payer rate: a floor by construction; 20-metro HSA cases 735,507.0 x mean rate 0.0949                                                                                      |
| F578 | Target metros with complete Medicare-anchored pricing (catchment proxy + transfer rate + CY2026 base rate all live)                                  | 2026 | 20 metros (of 20)                                                                 | DERIVED | A | S404, S402                                           | Metro_TAM_Panels Panel S, live COUNT over the 20 priced-total cells                                                                | Metro_TAM_Panels            | Medicaid/commercial/gross-up adders are PENDING in all 20 panels; only the Medicare anchor is complete                                                                                              |
| F579 | Footprint-10-state Medicare-anchored assembly floor (HSA cases x transfer rate x CY2026 A0428 base)                                                  | 2025 | 69,431,382 USD per year (floor)                                                   | DERIVED | A | S402, S403, S404                                     | TAM_Assembly_State Panel B FOOTPRINT row, col D (live SUM)                                                                         | TAM_Assembly_State          | 2,572,301.0 footprint HSA cases; Medicare-only volume x base-only rate keeps it a floor                                                                                                             |
| F580 | National HSA service-area Medicare FFS inpatient case roll-up (all CCNs)                                                                             | 2025 | 14,144,025 cases                                                                  | DERIVED | A | S402                                                 | TAM_Assembly_State Panel B NATIONAL row, col B (live SUM over HSA_Hospital_Catchment)                                              | TAM_Assembly_State          | x the 9.5% rate gives about 1.34M proxy episodes vs about 3.06M measured all-payer HCUP episodes - the Medicare-slice gap the bridge panel prints                                                   |
| F581 | Claims-visible x Medicare-anchored grid cell (the hardest floor): 2024 Medicare FFS hospital-to-hospital transports x measured allowed per transport | 2024 | 577,228,322.31 USD (allowed, 2024)                                                | DERIVED | A | S405                                                 | Scenario_Matrix Panel B, claims-visible row, Medicare-anchored column (live product)                                               | Scenario_Matrix             | Equals Medicare_IFT_Series 2024 total allowed (base + mileage) by construction: 871,753 transports                                                                                                  |
| F582 | Live cells in the 3x3 scenario grid (honesty metric)                                                                                                 | 2026 | 2 grid cells live (of 9)                                                          | DERIVED | A | S405, S406                                           | Scenario_Matrix Panel B honesty-metric row (live COUNT over the nine grid cells)                                                   | Scenario_Matrix             | 7 PENDING cells, each naming the dataset that closes it                                                                                                                                             |
| F583 | US 65+ population CAGR, 2020-2024                                                                                                                    | 2024 | 0.0295 share per year                                                             | DERIVED | A | S407                                                 | State_Age_65plus Panel A, United States row, G column (live formula over Census PEP levels)                                        | Growth_Outlook_Shell        | Median county 65+ CAGR prints beside it live; 85+ CAGR runs lower on the same row                                                                                                                   |
| F584 | MA & other share of Medicare beneficiaries, 2024                                                                                                     | 2024 | 0.5012 share                                                                      | DERIVED | A | S408                                                 | Enrollment_ESRD_State Panel A, 2024 row, E/C columns                                                                               | Growth_Outlook_Shell        | 2013 share 28.3% on the same panel; the series rises every single year                                                                                                                              |
| F585 | Transport vendors among five highest-paid 990 contractors, cohort systems (count across filings)                                                     | 2024 | 0 vendors                                                                         | SOURCED | A | S410                                                 | Form 990 Part VII Section B, latest two XML-available filings per resolved EIN; cohort_990.json headline_result                    | Vendor_Share_Stack          | Floor-censored: disclosure covers only the top five contractors over \$100K per filer, so zero bounds vendor scale, not vendor existence                                                            |
| F586 | Form 990 filings parsed for the contractor sweep                                                                                                     | 2024 | 46 filings                                                                        | SOURCED | A | S410                                                 | cohort_990.json: 9 systems, 24 EINs, latest two XML filings per EIN                                                                | Vendor_Share_Stack          | IRS XML availability lags: latest filings are tax years 2022-2023 for most EINs                                                                                                                     |

|      |                                                                                                                                                 |      |                                                      |            |   |            |                                                                                                                    |                               |                                                                                                                                                                                                        |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------|------------|---|------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F587 | Longest stated initial term in the public contract corpus                                                                                       | 2026 | 120 months                                           | SOURCED    | A | S411       | contract_corpus.json public_contracts[],term_and_renewal_parsed; Sickiness_Evidence Panel A prints every parse     | Sickiness_Evidence            | Shortest stated term 12 months; 8 of 12 documents state a parseable initial term; documents dated 2018-2025                                                                                            |
| F588 | Documents with explicit exclusivity or first-call language in the public contract corpus                                                        | 2026 | 3 documents of 12                                    | SOURCED    | A | S411       | contract_corpus.json public_contracts[],exclusivity_or_first_call; live COUNTIF on                                 | Sickiness_Evidence            | 1 documents are explicitly NONexclusive; corpus skews to 911/government forms where exclusivity is a franchise feature                                                                                 |
| F589 | Footprint nonprofit health systems resolved to public Form 990 filers (beyond the 9-system cohort)                                              | 2026 | 64 health systems (public filings)                   | GOV        | B | S414       | Footprint_990_Sweep Panel C roster; total row                                                                      | Footprint_990_Sweep           | 78 attempted, 14 parked (Panel D); mapping is analyst-selected from public registry facts (tier B)                                                                                                     |
| F590 | Form 990 filings parsed for Part VII Section B contractors (footprint sweep)                                                                    | 2024 | 128 filings (latest two XML-available per filer, tax | GOV        | A | S413       | Panel C filings column (live SUM in the total row); per-filing object_id in footprint_990_sweep.json               | Footprint_990_Sweep           | 64 filers; latest two filings each where XML available                                                                                                                                                 |
| F591 | Classified transport contractors (ground + air) observed in the disclosed five-highest-paid contractors (footprint sweep, ED-staffing excluded) | 2024 | 3 filing appearances (2 distinct operators across 2  | GOV        | A | S413       | Panel A (GROUND + AIR only), live COUNTA in the transport total row; Panel B frequency; Panel C classified column  | Footprint_990_Sweep           | Observed amounts \$2,058,817 to \$2,786,213; a ground-ambulance and an air-medical operator; the four ED-staffing keyword hits are quarantined in the exclusion panel and excluded from this count     |
| F592 | Maximum contractors over \$100,000 disclosed at a single filer (the five-slot information floor)                                                | 2024 | 2,090 contractors over \$100K (only five named)      | GOV        | A | S413       | Panel C max column (live MAX in the total row); CntrctRcvdGreaterThan100KCnt as filed                              | Footprint_990_Sweep           | Part VII Section B discloses only the top five over \$100K; with this many undisclosed, absence from the top five is NOT evidence of absence of spend                                                  |
| F593 | Footprint-state pooled median risk-standardized SNF potentially preventable 30-day post-discharge readmission rate (SNF QRP S_004_01)           | 2026 | 10.59 percent of stays                               | DERIVED    | A | S415       | SNF_Return_Leg_Quality Panel A, Footprint pooled median of PPR_PD_RSRR over reporting footprint SNFs               | SNF_Return_Leg_Quality        | National median 10.66% (Panel A National row); risk-standardized, case-mix adjusted                                                                                                                    |
| F594 | Footprint SNFs in the worst national decile for potentially preventable rehospitalization (bounce-back transport-demand nodes)                  | 2026 | 286 SNFs                                             | DERIVED    | A | S415       | SNF_Return_Leg_Quality Panel A, SUM of per-state worst-decile counts (PPR at/above the national 90th percentile =  | SNF_Return_Leg_Quality        | Out of 3,997 reporting footprint SNFs                                                                                                                                                                  |
| F595 | National median risk-standardized SNF potentially preventable readmission rate, for-profit SNFs (return-leg bounce-back)                        | 2026 | 10.72 percent of stays                               | DERIVED    | A | S415, S416 | SNF_ReturnLeg_Structure Panel A, For profit row, national median of PPR_PD_RSRR                                    | SNF_ReturnLeg_Structure       | Non-profit median 10.41% (Panel A); for-profit bounces back more                                                                                                                                       |
| F596 | National median risk-standardized SNF potentially preventable readmission rate, 1-star vs 5-star SNFs (quality gradient)                        | 2026 | 10.74 percent of stays (1-star)                      | DERIVED    | A | S415, S416 | SNF_ReturnLeg_Structure Panel C, rating 1 row, national median of PPR_PD_RSRR                                      | SNF_ReturnLeg_Structure       | 5-star median 10.42%; monotonic gradient - lower-rated SNFs bounce back more                                                                                                                           |
| F597 | Footprint AHCAH-approved hospital rows on the CMS public roster (4/5/2021 snapshot)                                                             | 2021 | 27 hospital rows                                     | GOV        | A | S417       | Hospital_at_Home_Participants Panel A/B; footprint rows of the 4/5/2021 CMS Approved List                          | Hospital_at_Home_Participants | Stale by construction; current roster PENDING (Panel C, naming QualityNet + ResDAC); participation has grown materially since but no public machine-readable roster states the current count           |
| F598 | Footprint states with at least one AHCAH-approved hospital on the 4/5/2021 CMS public roster                                                    | 2021 | 7 of 10 states                                       | GOV        | A | S417       | Hospital_at_Home_Participants Panel B; states with a non-zero 2021 count (OH, MN, WI, IN, IA, MO, VA)              | Hospital_at_Home_Participants | NE, KS, KY had no 2021 approval on the public list                                                                                                                                                     |
| F599 | Private ambulance average pay growth minus compounded AIF, 2014-2025 (the annual full-basket scissors)                                          | 2025 | 0.4316 percentage points, as share                   | DERIVED    | A | S422, S421 | Input_Cost_Index Panel C cumulative block (live); pay legs QCEW US000 own 5 avg_annual_pay 2014/2025; AIF legs     | Input_Cost_Index              | Pay 70.9% vs compounded AIF 27.8%; the quarterly wage scissors on QCEW_EMS_Employment is the same phenomenon on a shorter window                                                                       |
| F600 | US on-highway diesel retail, peak single-year swing (2022 annual mean vs 2021)                                                                  | 2022 | 0.5179 share change YoY                              | GOV        | A | S419       | EIA psw18bvwall.xls, EMD_EPD2D_PTE_NUS_DPG weekly, annual means 4.989 vs 3.287 \$/gal; Input_Cost_Index Panel A    | Input_Cost_Index              | The CY2023 AIF of 8.7% (largest since 2002) is the lagged CPI response to this shock                                                                                                                   |
| F601 | PPI diesel fuel cumulative change 2014-2025 minus compounded AIF (the fuel-leg gap)                                                             | 2025 | -0.7304 percentage points, as share                  | DERIVED    | A | S420, S421 | WP0U531 annual means 99.8 (2025) vs 182.4 (2014); Input_Cost_Index Panels B and C                                  | Input_Cost_Index              | Fuel deflated over the full window while the update compounded +27.8%; the intended PPI ambulance industry series (PCUC621910621910) does not exist and is parked with catalog proof                   |
| F602 | ModivCare service revenue, FY2024 (latest annual filing in companyfacts)                                                                        | 2024 | 2,787,586,000 USD                                    | SEC-A      | A | S423       | companyfacts CIK 1220754, RevenueFromContractWithCustomerExcludingAssessedTax,                                     | Public_Operator_Benchmarks    | FY2025 annual is PENDING (not in companyfacts 11 Jul 2026); 9M-2025 YTD 1,989,914,000 per Q3 10-Q - NEMT brokerage, outside the ambulance boundary                                                     |
| F603 | DocGo Transportation Services share of total revenue, FY2025                                                                                    | 2025 | 0.6231 share                                         | DERIVED    | A | S423       | Numerator: Facility_Pay_Layer Panel B (10-K Note 11, F-37); denominator: companyfacts Revenues FY2025 322,196,000, | Public_Operator_Benchmarks    | Share rose from about a third in FY2023-24 because total revenue FELL (Mobile Health wind-down), not because transport grew                                                                            |
| F604 | Missouri licensed ground ambulance services (DHSS directory)                                                                                    | 2026 | 455 services                                         | GOV        | A | S425       | data.mo.gov e7p8-a69d full pull, rows updated 2026-05-14; Panel A roll-up row (live COUNTIF)                       | State_EMS_Licensure           | Live COUNTIF over the printed panel equals the API row count; air (26) and stretcher van (5) carried separately                                                                                        |
| F605 | Missouri licensed-but-not-Medicare-ground-billing services (licensed minus CY2024 ground-billing NPIs)                                          | 2024 | 278 services (unit caveat: licenses vs NPIs)         | DERIVED    | A | S425, S431 | Panel E MO row, col H (live)                                                                                       | State_EMS_Licensure           | Confound printed in the same row; license and NPI are different objects, one company can hold several of either; MUP suppression makes the biller count a floor, so the gap is a ceiling on that basis |
| F606 | Iowa authorized EMS service programs (official roll-up)                                                                                         | 2025 | 724 programs                                         | GOV        | A | S427       | Iowa HHS EMS Snapshot (June 2025), page 1; Panel B roll-up                                                         | State_EMS_Licensure           | Locations (901) split 502 ambulance / 375 non-transport / 24 air on the same page; 2013 full roster panel shows 411 transporting of 779 parsed - same order of magnitude                               |
| F607 | Iowa CY2024 ground-billing NPIs (MUP, distinct, floor)                                                                                          | 2024 | 195 NPIs                                             | GOV        | A | S431       | mup_provider_2024_* caches, IA org NPIs with ground base rows; Panel E col F                                       | State_EMS_Licensure           | PECOS_Suppliers_State A0425-renderer count for IA prints beside it (live link) and runs higher because mileage renderers include air and individuals                                                   |
| F608 | Subject-company website "Currently Serving" enumerated entries (self-claim), 2023-03-24 capture                                                 | 2023 | 11 entries (10 states + DC)                          | PUBLIC-WEB | B | S435       | Common Crawl capture 20230324134522 of mmtamb.com homepage; Panel B row + Panel C                                  | Press_Footprint_Registry      | Self-claim, labeled; the CY2024 Medicare-billing footprint of the NPI estate (MMT_Medicare_Book Panel E) is the measured comparator, and 6 further states were claimed only as "Coming Soon"           |
| F609 | Subject-company website staffing self-claim, later endpoint ("2,800+ ... professionals")                                                        | 2025 | 2,800 staff (self-claimed floor)                     | PUBLIC-WEB | B | S435       | Common Crawl captures 20250522221503 and 20260114034214 of mmtamb.com/about-us/; Panel B                           | Press_Footprint_Registry      | Earlier endpoint on the same site: "4,000+ providers" (capture 20230324121017); both are issuer self-claims, not payroll data                                                                          |
| F610 | Provider-of-services flagged share of Medicare FFS ground transports (QN+QM), floor                                                             | 2024 | 0.0012 share of transport services                   | DERIVED    | A | S436       | Modifier_QM_QN_Series Panel B 2024 row, live G27                                                                   | Modifier_QM_QN_Series         | Carrier file only; Part A bundled and SNF consolidated-billing transports absent, suppression removes small rows - both push this DOWN                                                                 |
| F611 | QN (furnished directly by a provider of services) share of ground transports                                                                    | 2024 | 0.0012 share of transport services                   | DERIVED    | A | S436       | Modifier_QM_QN_Series Panel B 2024 row, live F27                                                                   | Modifier_QM_QN_Series         | QN 12,804 services dwarfs QM 52; direct furnishing is the dominant hospital-billed mode                                                                                                                |
| F612 | QM (under arrangement by a provider of services) services, ground transports                                                                    | 2024 | 52 services                                          | GOV        | A | S436       | Modifier_QM_QN_Series Panel A QM row C8                                                                            | Modifier_QM_QN_Series         | Billing-in-lieu volume: the arrangement the insourcing ceiling warns inflates a name-based share                                                                                                       |
| F613 | Provider-flagged share of Medicare FFS ground transports, 2010 (trend base)                                                                     | 2010 | 0.0001 share of transport services                   | DERIVED    | A | S436       | Modifier_QM_QN_Series Panel B 2010 row, live G13                                                                   | Modifier_QM_QN_Series         | Rises to 0.12% by 2024; the flag was thinly used early                                                                                                                                                 |
| F614 | Provider-flagged share of SCT (A0434) transports                                                                                                | 2024 | 0.0012 share of SCT services                         | DERIVED    | A | S436       | Modifier_QM_QN_Series Panel C SCT row, live col F                                                                  | Modifier_QM_QN_Series         | About the same as BLS (0.123%) and ALS: the flagged share is flat across service levels, so there is no acuity skew in the flagged data                                                                |
| F615 | ESO EMS records platform install base (vendor-claimed)                                                                                          | 2026 | 10,000 agencies (CLAIMED, floor "more than")         | CLAIMED    | C | S440       | IFT_Software_Landscape Panel A, ESO row                                                                            | IFT_Software_Landscape        | Vendor self-reported on eso.com; not independently verified; carried as a public claim, not a measurement                                                                                              |
| F616 | Central Logic / ABOUT transfer-center reach (vendor-claimed)                                                                                    | 2021 | 0.14 share of US hospital patients touched           | CLAIMED    | C | S441       | IFT_Software_Landscape Panel A, Central Logic row                                                                  | IFT_Software_Landscape        | Vendor self-reported (800 hospitals; "14% of U.S. hospital patients each year"); a marketing figure, not an independent measurement                                                                    |
| F617 | Share of hospital cost-report filers booking a line-95 ambulance cost centre (HCRIS floor)                                                      | 2023 | 0.1433 share of hospital filers                      | GOV        | A | S442       | Hospital_Ambulance_Prevalence Panel A 2023 row, live D9                                                            | Hospital_Ambulance_Prevalence | 865 of 6,035 filers; steady near 14-15% across FY2021-FY2023; a slight floor (filers only)                                                                                                             |
| F618 | Share of short-term acute + CAH hospitals name-matching an ambulance-taxonomy NPI (loose registry ceiling)                                      | 2026 | 0.5177 share of STACH+CAH (loose ceiling)            | DERIVED    | B | S443       | Hospital_Ambulance_Prevalence Panel B, live D13                                                                    | Hospital_Ambulance_Prevalence | Collision-inflated upper bound, not a point estimate; the HCRIS floor is the defensible number                                                                                                         |
| F619 | Institutional (off-carrier) share of Medicare FFS ground transports, interim top-down wedge                                                     | 2023 | 0.0515 share of all-claims transports (interim)      | DERIVED    | B | S445       | Institutional_Ambulance_Wedge Panel A residual row, live C9                                                        | Institutional_Ambulance_Wedge | MedPAC 11.3M minus carrier 10.7M; interim, scope-approximate, ResDAC route replaces it                                                                                                                 |
| F620 | Institutional (off-carrier) Medicare FFS ground transports, interim top-down wedge                                                              | 2023 | 581,532 transports (interim)                         | DERIVED    | B | S445       | Institutional_Ambulance_Wedge Panel A residual row, live B9                                                        | Institutional_Ambulance_Wedge | Bundles hospital OP, IP-bundled and SNF CB; not separated at this stage                                                                                                                                |
| F621 | Hospitals booking a Worksheet A line-95 ambulance cost centre                                                                                   | 2023 | 875 hospitals                                        | GOV        | A | S446       | Institutional_Ambulance_Wedge Panel B 2023 row                                                                     | HCRIS_Institutional_Roster    | \$3,267M operating cost; implied 1,222-2,438K operation transports (all payers)                                                                                                                        |

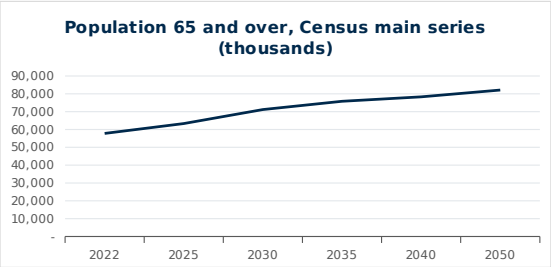
The measured macro drivers. Census, payer mix, supply exits, and the coordination lever.

Four measured forces, each pinned to a primary document: the aging population (Census projections), the payer-mix migration that shrinks every fee-for-service series, rural hospital closures and their measured effect on transport times, and the final external anchor for how often patients arrive by ambulance.

Panel A. The population that rides. Census Bureau 2023 National Population Projections, Table NP2023-T2, main series (S50). Thousands.

| Measure                   | 2022 | 2025    | 2030    | 2035    | 2040    | 2050    | CAGR 2025-2030 | CAGR 2030-2040 |      |
|---------------------------|------|---------|---------|---------|---------|---------|----------------|----------------|------|
| Population 65 and over    |      | 57,795  | 63,327  | 71,183  | 75,828  | 78,294  | 82,130         | 2.4%           | 1.0% |
| Total resident population |      | 333,288 | 338,016 | 345,074 | 350,861 | 355,309 | 360,639        |                |      |
| 65+ share of population   |      | 17.3%   | 18.7%   | 20.6%   | 21.6%   | 22.0%   | 22.8%          |                |      |

The 65+ population adds 7.9 million people between 2025 and 2030, a 2.4% per year compounding rate, then decelerates to 1.0% per year in the 2030s. NHAMCS puts the 65+ ambulance-arrival share at 33.4% against 17.0% for all ages (Panel E), so the riding population grows roughly twice as fast as total ED demand through 2030. This is the only measured tailwind in the demand equation; the Acute\_IFT\_Series episode CAGR of +0.86% already contains its past effect.



Panel B. The payer-mix migration that shrinks every fee-for-service series in this workbook. KFF from CMS enrollment files (S51).

| Year | MA share of eligible beneficiaries | Enrolled (millions) | Eligible (millions) | Note                             |
|------|------------------------------------|---------------------|---------------------|----------------------------------|
| 2007 | 19.0%                              |                     |                     | Series start as published by KFF |
| 2025 | 54.0%                              | 34.1                | 62.8                |                                  |
| 2026 | 55.0%                              | 35.2                | 64.2                | Current                          |
| 2034 | 63.0%                              |                     |                     | CBO projection as cited by KFF   |

Read together with Medicare\_IFT\_Series: carrier-billed FFS hospital-to-hospital transports peaked in 2017 and are 23.5% smaller, while MA crossed half of beneficiaries. The demand did not leave; the measurable window narrowed, and it will keep narrowing toward the CBO 63%. Every PSPS series in this workbook must be deflated mentally by this migration before being read as demand.

Panel C. Supply exits on the sending side. Sheps Center rural closure tracker, accessed 10 July 2026 (S52), and the measured transport-time mechanism (S53).

| Measure                                                    | Value | Detail                                                                                                  |
|------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------|
| Rural hospital closures and conversions since January 2005 | 197   | 109 complete closures plus 88 converted closures (inpatient services ended, some services remain)       |
| Closures and conversions since 2010                        | 154   | 86 complete plus 68 converted. Rural Emergency Hospital conversions are excluded and tracked separately |

Convention warning

The Chartis Center for Rural Health counts 182 since 2010 under a broader convention that treats REH conversion differently. The two counts are not interchangeable; cite one tracker and its convention.

Mechanism, measured. Rural hospital closures raised mean EMS transport time by 2.6 minutes (P=.09) and total activation time by 7.2 minutes (P=.02), with no change in response time: the ambulance reaches the patient as fast, then drives farther. Miller et al., Health Services Research 2020;55(2):288-300, doi:10.1111/1475-6773.13254, PMID 31989591, NEMSIS 2010-2016 (S53). Retrieved from PubMed 10 July 2026.

Panel D. Coordination is a measured lever, not a slide claim.

State medical operations coordination centers (SMOCCs), stood up in eight states during the pandemic, were associated with an IMMEDIATE 35% increase in interhospital transfer rates (rate ratio 1.35, 95% CI 1.05 to 1.74) followed by a long-term rate ratio of 0.94, across 441,709 NEMSIS transfers, June 2020 to December 2022. Transfer throughput responds to coordination infrastructure on a measurable scale. Richert et al., JAMA Network Open 2025;8(12):e2546002, doi:10.1001/jamanetworkopen.2025.46002, PMID 41324958 (S54). Retrieved from PubMed 10 July 2026. Read with Demand\_Drivers Panel B: the 287-minute median transfer delay is the friction this lever attacks.

Panel E. The final external anchor. NHAMCS 2022 was the last vintage of the survey.

| Measure                              | Value   | Standard error | Source                                       |
|--------------------------------------|---------|----------------|----------------------------------------------|
| ED visits, 2022 (thousands)          | 155,398 |                | NHAMCS 2022 ED Summary Tables, Table 4 (S55) |
| Ambulance arrival share, all ages    | 17.0%   | 0.9%           | NHAMCS 2022 ED Summary Tables, Table 4 (S55) |
| Ambulance arrival share, 65 and over | 33.4%   | 1.6%           | NHAMCS 2022 ED Summary Tables, Table 4 (S55) |
| Ambulance arrival share, 75 and over | 39.6%   | 2.0%           | NHAMCS 2022 ED Summary Tables, Table 4 (S55) |
| Ambulance arrival share, under 15    | 5.0%    | 0.8%           | NHAMCS 2022 ED Summary Tables, Table 4 (S55) |

The 2016 anchor of 16.2% carried on EMS\_Transports updates to 17.0% with a standard error of 0.9 points. NEMSIS 2022 coverage of EMS-to-ED transport (17.5% of NEDS visits) sits INSIDE the 95% interval of this final anchor: at full state participation, NEMSIS captures effectively all EMS-to-ED transport. NCHS discontinued NHAMCS after 2022 (successor: National Hospital Care Survey), so this anchor is frozen. Logged as Data\_Quality\_Register #30.

Table metrics

CAGR of 65+ population per five-year interval (from the projection row above)

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 1.85% | 2.37% | 1.27% | 0.64% | 0.96% |
|-------|-------|-------|-------|-------|

EMS activations and transports, NEMSIS 2019 to 2024

Hospital data counts patients moved; EMS data counts the vehicles moving them. This tab carries the national EMS activation series, its interfacility categories, and the external anchor that validates its coverage.

Coverage. Read every count against this.

| Metric                                 | 2019       | 2020       | 2021       | 2022       | 2023       | 2024       | Note                                                                                                                                                                                    |
|----------------------------------------|------------|------------|------------|------------|------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total activations (eResponse.05 table) | 36,119,261 | 44,798,443 | 50,192,875 | 53,572,097 | 46,040,873 | 60,298,684 | Use the table total, not the cover box. The 2021 cover reprints the 2020 figures verbatim. The 2023 cover prints 54,190,579 against a table total of 46,040,873, a gap of 8.1M records. |
| Participating states and territories   | 47         | 50         | 50         | 54         | 54         | 54         | 2021: DE, ID, American Samoa and Puerto Rico did not participate.                                                                                                                       |
| Number of agencies                     | 11,025     | 12,319     | 12,319     | 13,778     | 14,369     | 14,756     | The 2021 agency count repeats 2020.                                                                                                                                                     |
| Data standard                          | v3.4       | v3.4       | v3.4       | v3.4       | v3.5       | v3.5       | v3.3,4 retired August 2021.                                                                                                                                                             |

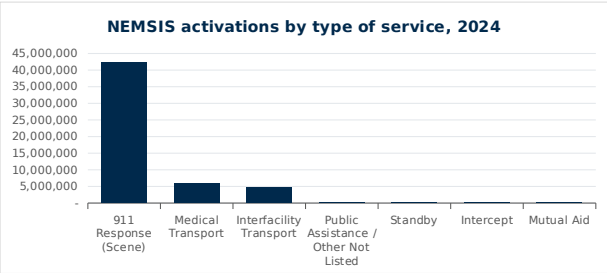
Activation growth over this window is part demand and part coverage expansion (47 states to 54, 11,025 agencies to 14,756). Never read the activation CAGR as EMS volume growth.

Type of service requested. eResponse.05. The value set expands from 7 categories (v3.4) to 18 (v3.5) in 2024, so the two blocks are NOT one series.

| Category (v3.4)                      | 2019       | 2020       | 2021       | 2022       | 2023       | Share 2019 | Share 2023 | CAGR 2019-2023 |
|--------------------------------------|------------|------------|------------|------------|------------|------------|------------|----------------|
| 911 Response (Scene)                 | 27,925,669 | 34,718,261 | 39,057,108 | 42,302,358 | 36,367,261 | 77.3%      | 79.0%      | 6.83%          |
| Medical Transport                    | 4,408,022  | 5,430,872  | 5,813,908  | 5,758,167  | 4,765,743  | 12.2%      | 10.4%      | 1.97%          |
| Interfacility Transport              | 3,309,316  | 4,070,413  | 4,613,080  | 4,697,465  | 4,179,580  | 9.2%       | 9.1%       | 6.01%          |
| Public Assistance / Other Not Listed | 182,470    | 238,734    | 266,525    | 278,152    | 243,225    | 0.5%       | 0.5%       | 7.45%          |
| Standby                              | 140,532    | 139,488    | 176,882    | 204,649    | 137,636    | 0.4%       | 0.3%       | (0.52%)        |
| Intercept                            | 90,829     | 112,426    | 130,416    | 158,524    | 151,884    | 0.3%       | 0.3%       | 13.72%         |
| Mutual Aid                           | 62,423     | 88,249     | 134,956    | 172,782    | 195,544    | 0.2%       | 0.4%       | 33.04%         |
| TOTAL                                | 36,119,261 | 44,798,443 | 50,192,875 | 53,572,097 | 46,040,873 |            |            |                |

Type of service requested. v3.5 value set, 2024 only.

| Category (v3.5) | 2024 | Implied share | Facility-to-facility? |
|-----------------|------|---------------|-----------------------|
|-----------------|------|---------------|-----------------------|



|                                                         |            |          |
|---------------------------------------------------------|------------|----------|
| Emergency Response (Primary Response Area)              | 46,733,668 | 77.5% No |
| Hospital-to-Hospital Transfer                           | 5,510,664  | 9.1% Yes |
| Other Routine Medical Transport                         | 4,467,204  | 7.4% No  |
| Hospital to Non-Hospital Facility Transfer              | 1,418,924  | 2.4% Yes |
| Public Assistance                                       | 630,530    | 1.0% No  |
| Non-Hospital Facility to Non-Hospital Facility Transfer | 335,432    | 0.6% Yes |
| Emergency Response (Mutual Aid)                         | 310,027    | 0.5% No  |
| Non-Hospital Facility to Hospital Transfer              | 247,636    | 0.4% Yes |
| Emergency Response (Intercept)                          | 243,105    | 0.4% No  |
| Standby                                                 | 240,680    | 0.4% No  |
| Support Services                                        | 58,179     | 0.1% No  |
| Crew Transport Only                                     | 53,257     | 0.1% No  |
| Mobile Integrated Health Care Encounter                 | 26,441     | 0.0% No  |
| Evaluation for Special Referral / Intake Programs       | 10,885     | 0.0% No  |
| Administrative Operations                               | 5,022      | 0.0% No  |
| Transport of Organs or Body Parts                       | 2,879      | 0.0% No  |
| Non-Patient Care Rescue / Extrication                   | 2,201      | 0.0% No  |
| Mortuary Services                                       | 1,950      | 0.0% No  |
| TOTAL                                                   | 60,298,684 |          |

The 2024 interfacility book, isolated

| Measure                                                               | 2024 activations | % of all activations | Basis |
|-----------------------------------------------------------------------|------------------|----------------------|-------|
| Hospital-to-Hospital Transfer                                         | 5,510,664        | 9.14% GOV-A          |       |
| All facility-to-facility transfers (four categories)                  | 7,512,656        | 12.46% DERIVED       |       |
| Other Routine Medical Transport (dialysis, appointments, return home) | 4,467,204        | 7.41% GOV-A          |       |

Other Routine Medical Transport is not interfacility transfer and is excluded from every IFT figure in this workbook.

NEMSIS Hospital-to-Hospital Transfer is acute-to-acute PLUS transfers to psychiatric, rehabilitation and long-term acute care hospitals, and it includes return legs. HCUP codes those destinations as DISP\_ED = 5 or DISPUNIFORM = 5, not = 2. This is the largest single driver of the wedge on the Cross\_Source\_Recon tab.

Type of destination. eDisposition.21. Stable value set across all six years, the cleanest series in the NEMSIS reports.

| Destination                             | 2019       | 2020       | 2021       | 2022       | 2023       | 2024       | Share 2019 | Share 2024 | CAGR 2019-2024 |
|-----------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|----------------|
| Hospital, Emergency Department          | 16,267,916 | 19,719,180 | 22,054,717 | 24,016,095 | 20,914,310 | 27,706,728 | 87.9%      | 92.0%      | 11.24%         |
| Hospital, Non-Emergency Department Bed  | 1,682,714  | 1,666,293  | 1,736,151  | 1,863,294  | 1,502,001  | 1,738,980  | 9.1%       | 5.8%       | 0.66%          |
| Other EMS Responder (ground)            | 141,030    | 150,330    | 175,514    | 198,770    | 194,664    | 188,683    | 0.8%       | 0.6%       | 5.99%          |
| Freestanding Emergency Department       | 43,070     | 70,736     | 104,199    | 130,988    | 117,984    | 156,346    | 0.2%       | 0.5%       | 29.41%         |
| Home                                    | 56,731     | 73,593     | 78,952     | 69,811     | 52,980     | 73,293     | 0.3%       | 0.2%       | 5.26%          |
| Other                                   | 116,748    | 134,198    | 161,668    | 188,520    | 189,280    | 69,367     | 0.6%       | 0.2%       | (9.89%)        |
| Nursing Home / Assisted Living Facility | 90,418     | 65,250     | 60,901     | 51,031     | 44,553     | 66,934     | 0.5%       | 0.2%       | (5.84%)        |
| Morgue / Mortuary                       | 12,500     | 19,160     | 19,618     | 19,648     | 15,775     | 48,756     | 0.1%       | 0.2%       | 31.29%         |
| Other EMS Responder (air)               | 25,532     | 37,097     | 44,356     | 45,011     | 39,053     | 40,502     | 0.1%       | 0.1%       | 9.67%          |
| Medical Office / Clinic                 | 54,281     | 37,921     | 38,829     | 34,223     | 25,436     | 24,752     | 0.3%       | 0.1%       | (14.53%)       |
| Urgent Care                             | 3,179      | 2,964      | 4,614      | 5,670      | 4,581      | 6,993      | 0.0%       | 0.0%       | 17.08%         |
| Police / Jail                           | 4,013      | 4,264      | 4,432      | 4,523      | 3,772      | 1,940      | 0.0%       | 0.0%       | (13.53%)       |

|                  |            |            |            |            |            |            |
|------------------|------------|------------|------------|------------|------------|------------|
| TOTAL transports | 18,498,132 | 21,980,986 | 24,483,951 | 26,627,584 | 23,104,389 | 30,123,274 |
|------------------|------------|------------|------------|------------|------------|------------|

NEMSIS coverage of EMS-to-ED transport, benchmarked against a probability sample

| Year          | NEMSIS transports to a hospital ED | NEDS weighted US ED visits | Ratio  | Participating states | Read                                                                                                                 |
|---------------|------------------------------------|----------------------------|--------|----------------------|----------------------------------------------------------------------------------------------------------------------|
| 2019          | 16,267,916                         | 143,432,284                | 11.34% | 47                   | Only 47 states participate. The gap is the missing seven.                                                            |
| 2020          | 19,719,180                         | 123,278,165                | 16.00% | 50                   | Three states added. Ratio jumps 4.7 points.                                                                          |
| 2021          | 22,054,717                         | 126,968,321                | 17.37% | 50                   | Denominator falls with COVID. Ratio keeps climbing.                                                                  |
| 2022          | 24,016,095                         | 136,974,618                | 17.53% | 54                   | Full participation. Ratio lands on the independent NHAMCS anchor.                                                    |
| 2023          | 20,914,310                         | 142,193,558                | 14.71% | 54                   | Ratio falls 2.8 points with no coverage change. This is the 2023 NEMSIS denominator defect, independently confirmed. |
| 2024          | 27,706,728                         |                            |        | 54                   | NEDS 2024 is not yet released.                                                                                       |
| NHAMCS anchor | 22,936,000                         | 142,006,000                | 16.15% |                      | NEDS carries no arrival-mode element, so the anchor has to come from NHAMCS.                                         |

Verdict. NEMSIS coverage of EMS-to-ED transport rises from roughly 11% to roughly 17.5% of US ED visits as participating states go from 47 to 54, then converges on the independent NHAMCS ambulance-arrival anchor of about 16%. The 2019 undercount is measurable, not hypothetical. The 2023 dip is a defect, not a market event.



Transport mode from scene. eResponse.23. All EMS transports, not IFT-specific.

| Category                            | 2019       | 2020       | 2021       | 2022       | 2023       | 2024       | Share 2019 | Share 2024 |
|-------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Non-Emergent                        | 12,514,692 | 15,099,744 | 16,331,143 | 17,585,742 | 15,203,669 | 18,620,902 | 71.7%      | 65.7%      |
| Emergent (Immediate Response)       | 4,562,512  | 5,399,074  | 6,541,606  | 7,190,541  | 6,217,885  | 9,350,263  | 26.1%      | 33.0%      |
| Emergent Downgraded to Non-Emergent | 290,274    | 297,186    | 353,517    | 345,407    | 259,039    | 277,726    | 1.7%       | 1.0%       |
| Non-Emergent Upgraded to Emergent   | 84,185     | 90,818     | 92,328     | 101,260    | 76,794     | 114,898    | 0.5%       | 0.4%       |
| TOTAL                               | 17,451,663 | 20,886,822 | 23,318,594 | 25,222,950 | 21,757,387 | 28,363,789 |            |            |

Roughly two thirds of all EMS transports run non-emergent. Non-emergent is the transport mode an interfacility book is built on, but this exhibit covers all EMS transports and cannot be cut to IFT alone in the public reports.

Table metrics: withheld.

No CAGR is computed on the activation series above: agency participation in the national EMS registry grew across the window, so a raw growth rate would measure reporting adoption, not demand (Fault\_Audit item M). Coverage-adjusted reads only.

Medicare interfacility ambulance, 2010 to 2024. Fifteen years, rebuilt from raw CMS claims.

No published table contains this series, so it was built by pulling fifteen annual claim files and filtering the hospital-to-hospital modifier. The overlap years reproduce the verified totals exactly.

| Year | Transports | Base-rate allowed \$ | Mileage allowed \$ | TOTAL allowed \$ | Medicare paid \$ | Allowed per transport | Loaded miles per transport | Allowed per loaded mile | Volume-weighted RVU | SCT share of transports | Non-emergency BLS share of transports |
|------|------------|----------------------|--------------------|------------------|------------------|-----------------------|----------------------------|-------------------------|---------------------|-------------------------|---------------------------------------|
| 2010 | 937,317    | \$271,512,215        | \$205,209,891      | \$476,722,106    | \$377,800,254    | \$508.60              | 31.6                       | \$6.92                  | 1.472               | 7.3%                    | 38.5%                                 |
| 2011 | 978,653    | \$285,751,131        | \$215,437,944      | \$501,189,075    | \$397,462,255    | \$512.12              | 31.8                       | \$6.93                  | 1.472               | 7.4%                    | 38.5%                                 |
| 2012 | 1,017,527  | \$293,374,396        | \$227,989,494      | \$521,363,889    | \$413,920,312    | \$512.38              | 31.9                       | \$7.03                  | 1.470               | 7.2%                    | 38.1%                                 |
| 2013 | 1,027,279  | \$302,811,761        | \$236,054,612      | \$538,866,373    | \$421,104,143    | \$524.56              | 32.4                       | \$7.09                  | 1.467               | 7.2%                    | 38.0%                                 |
| 2014 | 1,059,917  | \$315,707,983        | \$249,259,086      | \$564,967,069    | \$439,573,166    | \$533.03              | 32.7                       | \$7.19                  | 1.463               | 7.1%                    | 37.7%                                 |
| 2015 | 1,092,979  | \$327,742,623        | \$263,664,308      | \$591,406,931    | \$460,416,892    | \$541.10              | 33.3                       | \$7.24                  | 1.460               | 6.8%                    | 36.8%                                 |
| 2016 | 1,118,830  | \$333,278,408        | \$274,475,823      | \$607,754,231    | \$472,736,126    | \$543.21              | 34.0                       | \$7.22                  | 1.454               | 6.7%                    | 36.9%                                 |
| 2017 | 1,139,524  | \$341,340,142        | \$286,291,780      | \$627,631,921    | \$488,162,542    | \$550.78              | 34.4                       | \$7.30                  | 1.456               | 6.8%                    | 36.1%                                 |
| 2018 | 1,126,991  | \$342,671,425        | \$291,471,830      | \$634,143,255    | \$493,288,616    | \$562.69              | 35.0                       | \$7.38                  | 1.454               | 6.8%                    | 36.1%                                 |
| 2019 | 1,121,589  | \$347,484,350        | \$298,263,508      | \$645,747,857    | \$502,566,930    | \$575.74              | 35.2                       | \$7.56                  | 1.453               | 6.9%                    | 36.2%                                 |
| 2020 | 968,422    | \$304,478,810        | \$261,815,462      | \$566,294,272    | \$446,252,976    | \$584.76              | 35.5                       | \$7.62                  | 1.455               | 7.2%                    | 36.7%                                 |
| 2021 | 887,312    | \$273,572,051        | \$233,530,831      | \$507,102,882    | \$402,209,258    | \$571.50              | 35.2                       | \$7.48                  | 1.440               | 6.9%                    | 38.7%                                 |
| 2022 | 825,178    | \$267,031,071        | \$221,125,370      | \$488,156,441    | \$382,518,128    | \$591.58              | 34.4                       | \$7.80                  | 1.433               | 6.5%                    | 40.2%                                 |
| 2023 | 837,422    | \$301,395,424        | \$246,150,021      | \$547,545,444    | \$426,054,991    | \$653.85              | 34.2                       | \$8.60                  | 1.446               | 6.7%                    | 39.2%                                 |
| 2024 | 871,753    | \$319,815,284        | \$257,413,039      | \$577,228,322    | \$449,935,989    | \$662.15              | 33.6                       | \$8.79                  | 1.450               | 6.6%                    | 38.8%                                 |

What the fifteen-year view says that the three-year view does not.

| Measure                                             | Value           | Basis   | Read                                                                                                                                              |
|-----------------------------------------------------|-----------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Peak year for Medicare FFS interfacility transports | 2017            | DERIVED | 1,139,524 transports. Every year since has been lower.                                                                                            |
| Transports, 2024 against the 2017 peak              | (23.5%) DERIVED |         | The Medicare fee-for-service interfacility book is a quarter smaller than it was in 2017.                                                         |
| Transport CAGR, 2010 to 2017                        | 2.83% DERIVED   |         | Real growth in the pre-COVID, pre-Medicare-Advantage-surge era.                                                                                   |
| Transport CAGR, 2017 to 2024                        | (3.75%) DERIVED |         | The reversal. Medicare Advantage enrolment crossing 50% is the dominant driver, not demand.                                                       |
| Transport CAGR, 2022 to 2024                        | 2.78% DERIVED   |         | CAUTION. This is the recovery off the 2022 trough, not a growth rate. Quoting it as trend is the single easiest mistake to make with this series. |
| Total allowed CAGR, 2010 to 2024                    | 1.38% DERIVED   |         | Dollars grew while transports shrank.                                                                                                             |
| Total allowed CAGR, 2022 to 2024                    | 8.74% DERIVED   |         | Again a trough-to-peak artefact. Never annualise it forward.                                                                                      |
| Allowed per transport CAGR, 2010 to 2024            | 1.90% DERIVED   |         | Below the 3.0% average Ambulance Inflation Factor over the same window. Realised rate growth lags the statutory update.                           |
| Allowed per loaded mile CAGR, 2010 to 2024          | 1.72% DERIVED   |         | \$6.92 in 2010 to \$8.79 in 2024.                                                                                                                 |
| Mileage as a share of allowed dollars, 2010         | 43.0% DERIVED   |         | Mileage has been roughly 43-45% of Medicare interfacility revenue for fifteen straight years.                                                     |
| Mileage as a share of allowed dollars, 2024         | 44.6% DERIVED   |         | Stable. Any model that prices only the base rate has been wrong for fifteen years, not just recently.                                             |
| Volume-weighted RVU, 2010                           | 1.472 DERIVED   |         | 1.472.                                                                                                                                            |
| Volume-weighted RVU, 2024                           | 1.450 DERIVED   |         | 1.450. The Medicare interfacility acuity mix has moved by 1.5% in fifteen years.                                                                  |
| SCT share of transports, range 2010 to 2024         | 6.5% to 7.4%    | DERIVED | A 0.9 point band across fifteen years. There is no acuity escalation in the Medicare interfacility book, and there never was.                     |
| Medicare paid as a share of allowed, 2024           | 77.9% DERIVED   |         | Statute sets Part B at 80% after the deductible. 77.9% realised. Stable across the whole series.                                                  |

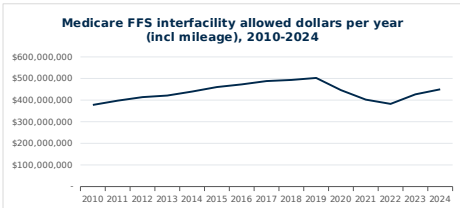
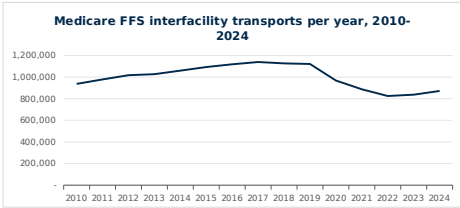
Suppression. CMS masks any cell with fewer than 11 services. Suppressed rows were excluded from the sums, not imputed, so every figure here is a floor. The masked share of hospital-to-hospital volume is on the order of 2%.

Scope. Carrier claims only. Hospital-owned ambulance departments bill institutionally and never enter PPS. Medicare Advantage is entirely absent. The commercial book is not measured.

Reproduce: GET [Table metrics](https://data.cms.gov/data-api/v1/dataset/(uuid)/data?filter(HCPCS_INITIAL_MODIFIER_CD)=HH,one dataset per year, then aggregate A0426,A0427,A0428,A0429,A0433,A0434(base) and A0425(mileage). Pre-2021 files drop the PPS_ prefix on the amount fields.</a></p></div><div data-bbox=)

|                                                  |          |
|--------------------------------------------------|----------|
| CAGR, transports, 2010 to 2024                   | (0.52%)  |
| CAGR, transports, 2019 to 2024                   | (4.92%)  |
| CAGR, allowed dollars, 2010 to 2024              | 1.90%    |
| Peak year, transports                            | 2017     |
| Decline from peak to 2024                        | (23.5%)  |
| HH share of ground, change 2010 to 2024 (points) | -0.7 pts |

All figures computed from the series above. Book-level CAGRs are payer-window figures: the per-beneficiary rates on Utilization\_Normalized carry the demand signal.



Input-cost index: measured operating inputs against the Medicare payment update

Sources: EIA weekly on-highway diesel by PADD (annual means), BLS PPI diesel fuel (WPU0531) and ECEC private total compensation, QCEW private ambulance pay (green-linked from QCEW\_EMS\_Employment), and the AIF (CY2014-2019 verified against the CMS Pub. 100-04 Ch.15 chart; CY2020+ green-linked from Payment\_Rules). Join key: calendar year. House rule applied: every series is printed separately; there is NO blended input basket on this tab.

DATA QUALITY: diesel is the retail pump price including taxes (fleet and bulk purchase prices differ in level; the trend is the signal). WPU0531 is a producer price index (1982=100), not a retail price. ECEC is ALL private industry, not EMS-specific. QCEW pay excludes volunteers and the self-employed. The AIF is a payment update factor (CPI-U less a productivity offset), a different basis from any measured input series - gaps are printed in percentage points, never blended. The PPI ambulance services industry index (PCU621910621910) DOES NOT EXIST; parked, with the catalog gap as proof. 2026 rows are partial years and excluded from every cumulative figure.

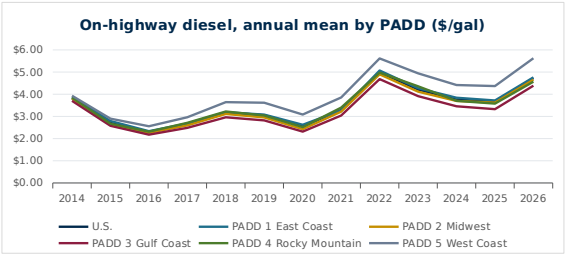
| Panel A. Diesel at the pump: EIA weekly on-highway retail price, annual means by PADD (dollars per gallon) |        |             |                |             |                  |                   |         |      |  |
|------------------------------------------------------------------------------------------------------------|--------|-------------|----------------|-------------|------------------|-------------------|---------|------|--|
| Year                                                                                                       | U.S.   | PADD 1 East | PADD 2 Midwest | PADD 3 Gulf | PADD 4 Rocky Mtn | PADD 5 West       | US YoY  | Note |  |
| 2014                                                                                                       | \$3.82 | \$3.88      | \$3.81         | \$3.70      | \$3.85           | \$3.93 <i>n/a</i> |         |      |  |
| 2015                                                                                                       | \$2.71 | \$2.79      | \$2.64         | \$2.58      | \$2.67           | \$2.90            | (29.2%) |      |  |
| 2016                                                                                                       | \$2.30 | \$2.33      | \$2.26         | \$2.18      | \$2.31           | \$2.56            | (14.9%) |      |  |
| 2017                                                                                                       | \$2.65 | \$2.68      | \$2.60         | \$2.48      | \$2.71           | \$2.96            | 15.0%   |      |  |
| 2018                                                                                                       | \$3.18 | \$3.19      | \$3.11         | \$2.96      | \$3.22           | \$3.64            | 19.9%   |      |  |
| 2019                                                                                                       | \$3.06 | \$3.08      | \$2.95         | \$2.82      | \$3.04           | \$3.62            | (3.8%)  |      |  |
| 2020                                                                                                       | \$2.55 | \$2.62      | \$2.43         | \$2.31      | \$2.52           | \$3.08            | (16.5%) |      |  |
| 2021                                                                                                       | \$3.29 | \$3.27      | \$3.21         | \$3.04      | \$3.40           | \$3.85            | 28.9%   |      |  |
| 2022                                                                                                       | \$4.99 | \$5.07      | \$4.90         | \$4.68      | \$4.97           | \$5.62            | 51.8%   |      |  |
| 2023                                                                                                       | \$4.21 | \$4.27      | \$4.12         | \$3.91      | \$4.36           | \$4.94            | (15.5%) |      |  |
| 2024                                                                                                       | \$3.76 | \$3.84      | \$3.70         | \$3.46      | \$3.70           | \$4.42            | (10.8%) |      |  |
| 2025                                                                                                       | \$3.66 | \$3.72      | \$3.62         | \$3.33      | \$3.58           | \$4.37            | (2.6%)  |      |  |
| 2026                                                                                                       | \$4.73 | \$4.76      | \$4.66         | \$4.39      | \$4.57           | \$5.62            | 29.2%   |      |  |

Every cell is the mean of that calendar year's weekly prints (52-53 weeks; 2026 partial). Sub-PADD splits (New England, Central Atlantic, Lower Atlantic, California) are in the cached artifact eia\_diesel\_padd.

| Panel B. BLS input series: PPI diesel fuel and ECEC private total compensation (annual means of monthly / quarterly observations) |                                    |         |                               |          |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------|-------------------------------|----------|------|
| Year                                                                                                                              | PPI diesel fuel WPU0531 (1982=100) | PPI YoY | ECEC private total comp \$/hr | ECEC YoY | Note |
| 2014                                                                                                                              | 182.4 <i>n/a</i>                   |         | 30.44 <i>n/a</i>              |          |      |
| 2015                                                                                                                              | 105.4                              | (42.2%) | 31.57                         | 3.7%     |      |
| 2016                                                                                                                              | 96.2                               | (8.7%)  | 32.35                         | 2.5%     |      |
| 2017                                                                                                                              | 118.7                              | 23.3%   | 33.41                         | 3.3%     |      |
| 2018                                                                                                                              | 112.1                              | (5.5%)  | 34.24                         | 2.5%     |      |
| 2019                                                                                                                              | 85.5                               | (23.7%) | 34.61                         | 1.1%     |      |
| 2020                                                                                                                              | 67.5                               | (21.1%) | 35.87                         | 3.7%     |      |
| 2021                                                                                                                              | 150.0                              | 122.2%  | 37.15                         | 3.6%     |      |
| 2022                                                                                                                              | 245.3                              | 63.6%   | 39.34                         | 5.9%     |      |
| 2023                                                                                                                              | 93.1                               | (62.0%) | 41.62                         | 5.8%     |      |
| 2024                                                                                                                              | 72.8                               | (21.8%) | 44.20                         | 6.2%     |      |
| 2025                                                                                                                              | 99.8                               | 37.0%   | 45.81                         | 3.6%     |      |
| 2026                                                                                                                              | 125.2                              | 25.4%   | 46.60                         | 1.7%     |      |

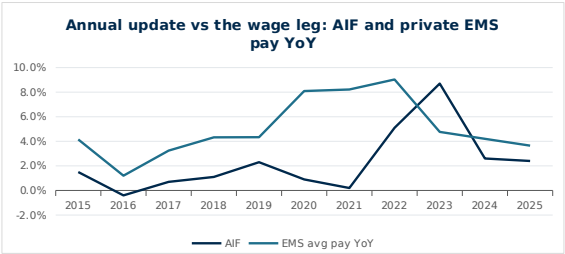
PARKED: PCU621910621910, the PPI industry index for NAICS 621910 ambulance services, does not exist - the BLS pc.series catalog jumps from 621610 (home health) to 621991 (blood and organ banks). The API confirms "Series does not exist". What would fill it: a BLS PPI industry index for NAICS 621910, if ever published. The fuel PPI (WPU0531) and the ECEC compensation series above are the closest published legs.

| Panel C. The AIF against each input series, year by year - each series its own column, gaps in percentage points, no blended basket |        |                            |         |                    |               |                       |                |            |                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------|---------|--------------------|---------------|-----------------------|----------------|------------|--------------------------------------------------------------------------------------------------------|
| Calendar year                                                                                                                       | AIF    | EMS avg pay \$ (QCEW link) | Pay YoY | Pay minus AIF (pp) | Diesel US YoY | Diesel minus AIF (pp) | PPI diesel YoY | ECEC YoY   | Basis note (same row, house rule)                                                                      |
| 2014                                                                                                                                | 1.0%   | \$36,087 <i>n/a</i>        |         | <i>n/a</i>         | <i>n/a</i>    | <i>n/a</i>            | <i>n/a</i>     | <i>n/a</i> | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2015                                                                                                                                | 1.5%   | \$37,585                   | 4.2%    | 2.7%               | (29.2%)       | (30.7%)               | (42.2%)        | 3.7%       | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2016                                                                                                                                | (0.4%) | \$38,037                   | 1.2%    | 1.6%               | (14.9%)       | (14.5%)               | (8.7%)         | 2.5%       | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2017                                                                                                                                | 0.7%   | \$39,271                   | 3.2%    | 2.5%               | 15.0%         | 14.3%                 | 23.3%          | 3.3%       | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2018                                                                                                                                | 1.1%   | \$40,968                   | 4.3%    | 3.2%               | 19.9%         | 18.8%                 | (5.5%)         | 2.5%       | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2019                                                                                                                                | 2.3%   | \$42,744                   | 4.3%    | 2.0%               | (3.8%)        | (6.1%)                | (23.7%)        | 1.1%       | AIF re-carried blue from CMS Pub. 100-04 Ch.15 AIF chart (p.23 of PDF); gaps are measured minus update |
| 2020                                                                                                                                | 0.9%   | \$46,201                   | 8.1%    | 7.2%               | (16.5%)       | (17.4%)               | (21.1%)        | 3.7%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |
| 2021                                                                                                                                | 0.2%   | \$49,998                   | 8.2%    | 8.0%               | 28.9%         | 28.7%                 | 122.2%         | 3.6%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |
| 2022                                                                                                                                | 5.1%   | \$54,514                   | 9.0%    | 3.9%               | 51.8%         | 46.7%                 | 63.6%          | 5.9%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |
| 2023                                                                                                                                | 8.7%   | \$57,112                   | 4.8%    | (3.9%)             | (15.5%)       | (24.2%)               | (62.0%)        | 5.8%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |
| 2024                                                                                                                                | 2.6%   | \$59,511                   | 4.2%    | 1.6%               | (10.8%)       | (13.4%)               | (21.8%)        | 6.2%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |
| 2025                                                                                                                                | 2.4%   | \$61,683                   | 3.6%    | 1.2%               | (2.6%)        | (5.0%)                | 37.0%          | 3.6%       | AIF green-linked from Payment_Rules (S29); gaps are measured-series change minus payment update        |



Peak year of the window: the 2022 diesel shock

YTD mean of 27 weekly prints through 2026-07-06; partial year, excluded from cumulatives



| Cumulative, 2014 to 2025 (2026 excluded: partial)                 | Value   | Note                                                                                                                                                                    |
|-------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compounded AIF, CY2015-CY2025                                     | 27.8%   | Eleven annual updates compounded; a factor path, not an index                                                                                                           |
| Private ambulance avg pay growth, 2014-2025 (QCEW)                | 70.9%   | Measured level change; excludes volunteers                                                                                                                              |
| THE SCISSORS: pay growth minus compounded AIF (pp)                | 43.2%   | The annual full-basket extension of the quarterly scissors on QCEW_EMS_Employment; bases differ (level vs update factor), which is why this is a gap in pp, not a ratio |
| ECEC private total comp growth, 2014-2025                         | 50.5%   | All private industry, not EMS-specific                                                                                                                                  |
| Diesel US retail change, 2014-2025                                | (4.3%)  | 2025 pump price sits BELOW 2014; the 2022 shock washed out                                                                                                              |
| PPI diesel fuel change, 2014-2025                                 | (45.3%) | Producer index change; same wash-out as retail                                                                                                                          |
| Input series that outgrew the compounded AIF, of the four printed | 2       | Both labor legs outgrew the update; both fuel legs did not - and labor is the dominant ambulance cost line                                                              |

**Read panel**

What is measured: the payment update and the input series diverge on the labor side, not the fuel side. Compounded CY2015-CY2025 AIF is about 27.8%; private ambulance pay grew about 70.9% over the same span (gap about 43 pp) and economy-wide private total compensation about 50.5%. Fuel cuts the other way across the full window: 2025 diesel sits below its 2014 level at retail and in the PPI - but the path is violent, with a 52% single-year US retail jump in 2022 that the CY2023 AIF (8.7%, the largest since 2002) chased with a lag. What is NOT measured: any blended cost index - the series have different bases and are never combined; and the PPI for the ambulance industry itself, which BLS does not publish.

Subject-company Medicare FFS ambulance book by NPI, 2013 / 2019 / 2024

Roll-up of the 23 registered NPIs on MMT\_NPI\_Estate across MUP\_Providers\_2013/2019/2024. Medicare FFS is roughly one-sixth of the market by this workbook's own payer arithmetic (GADCS Table S.1: FFS 29% of revenue per NPI on average, and enrollment splits); this tab is the measured PUBLIC FLOOR of the book, not the book.

DATA QUALITY: per-NPI code rows with 10 or fewer beneficiaries are suppressed at source, so every figure is a floor; final-action basis; NPIs enumerated after a vintage year cannot appear in it (Panel F states which and why); billing under a parent or billing-agent NPI is invisible here.

| Panel A. Consolidated book by vintage (ground base codes A0426-A0429, A0433, A0434) |  |              |  |                       |        |                               |        |              |              |                             |                       |                                                          |
|-------------------------------------------------------------------------------------|--|--------------|--|-----------------------|--------|-------------------------------|--------|--------------|--------------|-----------------------------|-----------------------|----------------------------------------------------------|
| Vintage                                                                             |  | NPIs present |  | Base services (floor) |        | Beneficiary-code rows (floor) |        | Allowed \$   | Paid \$      | Allowed per base service \$ | Paid share of allowed | Note                                                     |
| 2013                                                                                |  |              |  | 1                     | 10,162 |                               | 8,210  | \$3,484,246  | \$2,736,343  | \$342.87                    | 78.5%                 | beneficiaries summed over code rows: not unique patients |
| 2019                                                                                |  |              |  | 1                     | 29,999 |                               | 25,227 | \$8,327,999  | \$6,594,789  | \$277.61                    | 79.2%                 |                                                          |
| 2024                                                                                |  |              |  | 2                     | 69,684 |                               | 53,773 | \$22,004,764 | \$17,292,232 | \$315.78                    | 78.6%                 |                                                          |

| Panel B. Acuity mix by vintage (share of base services) and the volume-weighted RVU |               |               |               |            |            |            |         |                                                         |
|-------------------------------------------------------------------------------------|---------------|---------------|---------------|------------|------------|------------|---------|---------------------------------------------------------|
| Level (HCPCS)                                                                       | 2013 services | 2019 services | 2024 services | 2013 share | 2019 share | 2024 share | AFS RVU |                                                         |
| ALS non-emergency (A0426)                                                           | 265           | 4,643         | 2,927         | 2.6%       | 15.5%      | 4.2%       | 1.20    | RVUs frozen since 2002; same constants as Payment_Rules |
| ALS emergency (A0427)                                                               | 5,793         | 5,929         | 8,845         | 57.0%      | 19.8%      | 12.7%      | 1.90    |                                                         |
| BLS non-emergency (A0428)                                                           | 2,865         | 18,077        | 52,513        | 28.2%      | 60.3%      | 75.4%      | 1.00    |                                                         |
| BLS emergency (A0429)                                                               | 1,078         | 959           | 4,546         | 10.6%      | 3.2%       | 6.5%       | 1.60    |                                                         |
| ALS level 2 (A0433)                                                                 | 84            | 177           | 268           | 0.8%       | 0.6%       | 0.4%       | 2.75    |                                                         |
| SCT (A0434)                                                                         | 77            | 214           | 585           | 0.8%       | 0.7%       | 0.8%       | 3.25    | acuity in one number: rises only if the mix shifts up   |
| Volume-weighted RVU (live)                                                          |               |               |               | 1.61       | 1.25       | 1.19       |         |                                                         |

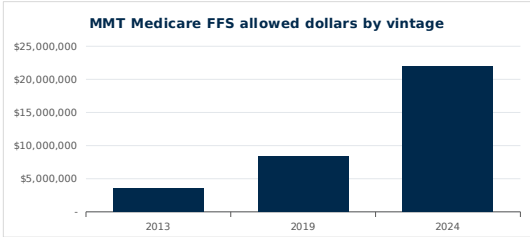
| Panel C. Mileage economics (A0425 ground mileage) |                       |                      |                                        |                    |                                |
|---------------------------------------------------|-----------------------|----------------------|----------------------------------------|--------------------|--------------------------------|
| Vintage                                           | Mileage units (floor) | Base services (link) | Implied avg loaded miles per transport | Mileage allowed \$ | Mileage share of total allowed |
| 2013                                              | 266,373               | 10,162               | 26.2                                   | \$2,056,338        | 37.1%                          |
| 2019                                              | 773,370               | 29,999               | 25.8                                   | \$6,024,712        | 42.0%                          |
| 2024                                              | 1,014,607             | 69,684               | 14.6                                   | \$9,037,972        | 29.1%                          |

Implied miles are mileage UNITS divided by base transports: a floor over a floor, so read the level cautiously and the trend directionally. Air codes are excluded from every ground panel; any air rows in the registry are shown in Panel F.

| Panel D. Trajectory (live CAGRs over Panel A) |                   |                   |                   |
|-----------------------------------------------|-------------------|-------------------|-------------------|
| Metric                                        | 2013 to 2019 CAGR | 2019 to 2024 CAGR | 2013 to 2024 CAGR |
| Base services                                 | 19.8%             | 18.4%             | 19.1%             |
| Allowed \$                                    | 15.6%             | 21.4%             | 18.2%             |
| Allowed per service \$                        | (3.5%)            | 2.6%              | (0.7%)            |

| Panel E. Per-NPI book, 2024 (the granular floor) |                                       |       |                  |               |              |              |                        |               |  |
|--------------------------------------------------|---------------------------------------|-------|------------------|---------------|--------------|--------------|------------------------|---------------|--|
| NPI                                              | Registered name (NPPEs)               | State | City             | Base services | Allowed \$   | Paid \$      | Allowed per service \$ | Mileage units |  |
| 1871991125                                       | Midwest Medical Transport Company Llc | NE    | Columbus         | 69,643        | \$21,993,137 | \$17,282,968 | \$315.80               | 1,013,484     |  |
| 1629831102                                       | Midwest Medical Transport Company Llc | CO    | Colorado Springs | 41            | \$11,627     | \$9,264      | \$283.59               | 1,123         |  |

| Panel F. NPI resolution by vintage (which registered NPIs appear, and why absences are expected) |                                                                  |         |         |         |                                                                                                                          |  |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------|---------|---------|--------------------------------------------------------------------------------------------------------------------------|--|
| NPI                                                                                              | Registered name                                                  | In 2013 | In 2019 | In 2024 | Reading                                                                                                                  |  |
| 1003266339                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1053162420                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1073296893                                                                                       | MMT Co LLC DBA Southeast Iowa Ambulance                          | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1134670219                                                                                       | MMT Co LLC - Huron, SD                                           | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1205387545                                                                                       | MMT Co LLC - Aberdeen, SD                                        | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1235681354                                                                                       | MMT Co LLC DBA Fraser Transportation Services                    | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1235998626                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1346792371                                                                                       | MMT Co LLC DBA Southeast Iowa Ambulance                          | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1356115562                                                                                       | Midwest Medical Transport Co LLC                                 | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1477490381                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1538612320                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1548034754                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1548047152                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1588115893                                                                                       | MMT Co LLC - Sioux City, IA (subpart)                            | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1629831102                                                                                       | MMT Co LLC                                                       | -       | -       | Y       | Absent early vintages: enumerated later, below the suppression floor, or volume consolidated on another NPI              |  |
| 1689665143                                                                                       | Platte County Ambulance Company DBA Midwest Medical Transport Co | Y       | -       | -       | Resolves across vintages                                                                                                 |  |
| 1740645332                                                                                       | Midwest Medical Transport Co LLC (subpart)                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1760832695                                                                                       | MMT Co LLC                                                       | -       | -       | -       | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |  |
| 1871991125                                                                                       | Midwest Medical Transport Co LLC (flagship)                      | -       | Y       | Y       | Absent early vintages: enumerated later, below the suppression floor, or volume consolidated on another NPI              |  |



|            |                               |   |   |   |                                                                                                                          |
|------------|-------------------------------|---|---|---|--------------------------------------------------------------------------------------------------------------------------|
| 1922872134 | Midwest Medical Transport LLC | - | - | - | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |
| 1932650272 | MMT Co LLC - Albia, IA        | - | - | - | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |
| 1982069134 | MMT Co LLC                    | - | - | - | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |
| 1992579338 | MMT Co LLC                    | - | - | - | Never resolves: suppression floor (every code under 11 beneficiaries), enumerated after 2024, or bills under another NPI |

Read panel

The measured floor and its shape: the registered NPI estate bills Medicare FFS about 69,684 ground base transports in 2024 (allowed about \$22.0M), roughly seven times the 2013 floor, and the mix moved decisively to BLS non-emergency (28.2% of base services in 2013 to 75.4% in 2024) - the scheduled interfacility contract product - while implied loaded miles per transport roughly halved as the book scaled and densified. SCT and ALS2 stay under 1.5% of the MEDICARE-VISIBLE book. What this is NOT: the company's book. Medicare FFS is roughly one-sixth of the market by payer arithmetic, MA and commercial volumes are invisible here, and dedicated-unit facility contracts (Facility\_Pay\_Layer) never generate these claims at all - which is exactly where high-acuity dedicated work would sit. Direction of bias: every number on this tab understates.

The facility-pay layer: ambulance revenue that never touches a payer claim

Three primaries, verified against the documents on 11 Jul 2026: the CMS/RAND GADCS Year 1-4 census (the only national revenue-source survey), the DocGo 10-K (the only US-listed pure ambulance P&L), and USAspending PSC V225 (the federal direct-contract channel). This tab bounds incidence and magnitude; the IFT-specific facility-pay share does not exist publicly and stays PENDING.

DATA QUALITY: GADCS values are self-reported survey responses from the Year 1-4 collection cohorts (MedPAC judged the COST side unusable for margins; revenue-source incidence is carried here as the census it is, with that caveat). DocGo figures are audited but one operator. USAspending rows are procurement obligations only; the VA Beneficiary Travel claims channel is separate and larger and must NEVER be summed with these.

| Panel A. GADCS: revenue sources of US ambulance organizations (share reporting any; conditional mean; median) |                 |                     |           |                  |
|---------------------------------------------------------------------------------------------------------------|-----------------|---------------------|-----------|------------------|
| Revenue source (GADCS Question 5)                                                                             | Share reporting | Conditional mean \$ | Median \$ | Locator          |
| Facility contracts (hospitals, SNFs, other facilities)                                                        | 18.6%           | \$922,555           | \$48,393  | Table 2.32, p.70 |
| Any non-payer revenue (all Question 5 categories)                                                             | 74.5%           | \$2,177,569         | \$162,265 | Table 2.32, p.70 |
| Local-government contracts                                                                                    | 15.4%           | \$556,091           | \$112,426 | Table 2.32, p.70 |
| EMS-earmarked local taxes                                                                                     | 23.4%           | \$3,135,660         | \$352,117 | Table 2.32, p.70 |
| Standby event revenue (share reporting)                                                                       | 24.5%           |                     |           | Table 2.32, p.70 |
| Membership programs (share reporting)                                                                         | 7.6%            |                     |           | Table 2.32, p.70 |

| Payer revenue per NPI (GADCS mean)                                    | Mean \$     | Locator                              |
|-----------------------------------------------------------------------|-------------|--------------------------------------|
| All payers                                                            | \$3,914,787 | Table 2.30, pp.64-65                 |
| Medicare FFS                                                          | \$1,193,046 | Table 2.30, pp.64-65                 |
| Medicare Advantage                                                    | \$1,238,482 | Table 2.30, pp.64-65                 |
| Medicaid                                                              | \$732,436   | Table 2.30, pp.64-65                 |
| Commercial                                                            | \$1,322,170 | Table 2.30, pp.64-65                 |
| Medicare FFS share of transport revenue (average per NPI)             | 29.0%       | Table S.1, p.ix                      |
| Medicare Advantage additional share                                   | 18.0%       | Table S.1, p.ix                      |
| Organizations that did not always bill in at least one payer category | 20.0%       | Table S.1, p.ix; "about one in five" |
| MA-to-FFS revenue ratio per NPI (the MA book calibrator)              | 1.04x       | DERIVED from Table 2.30 rows above   |

| Interfacility share of transport volume (GADCS)   | Share of organizations | Locator         |
|---------------------------------------------------|------------------------|-----------------|
| For-profit organizations at 81-100% interfacility | 20.2%                  | Table 2.6, p.33 |
| For-profit organizations at 21-80% interfacility  | 42.8%                  | Table 2.6, p.33 |
| All organizations at 81-100% interfacility        | 7.4%                   | Table 2.6, p.33 |

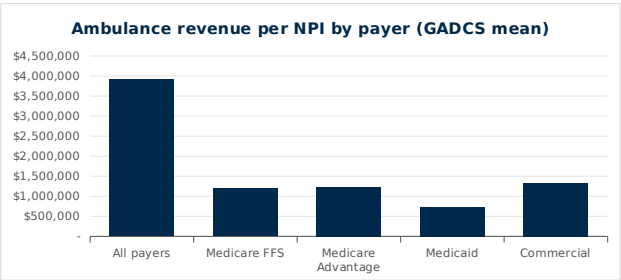
Read: the interfacility-heavy operator is overwhelmingly a FOR-PROFIT phenomenon (20.2% of for-profits vs 7.4% of all organizations run 81-100% interfacility books), which is why claims-only market reads systematically under-see the private IFT segment: it is exactly the segment with facility contracts.

| Panel B. DocGo 10-K: the measured share of transportation revenue billed outside per-trip claims |               |               |               |                                               |
|--------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-----------------------------------------------|
| Metric                                                                                           | FY2025        | FY2024        | FY2023        | Locator                                       |
| Transportation Services segment revenue \$                                                       | \$200,765,608 | \$193,429,092 | \$181,495,105 | Note 2 (F-22); Note 11 (F-37)                 |
| Trips (per-trip billed only, by the filing definition)                                           | 296,014       | 283,570       | 250,114       | MD&A pp.62/66                                 |
| Average trip price \$                                                                            | \$401         | \$402         | \$407         | MD&A pp.57/62/66; FY23 ATP from Note 2        |
| Implied per-trip billed revenue \$                                                               | \$118,701,614 | \$113,995,140 | \$101,796,398 | trips x ATP                                   |
| Revenue OUTSIDE per-trip billing (dedicated-unit programs billed hc                              | 40.9%         | 41.1%         | 43.9%         | MD&A p.57 exclusion language; arithmetic live |

Guardrail: one operator, and the plus-or-minus 0.1pp rounding from whole-dollar ATP is real. What it establishes is EXISTENCE and scale at a listed operator: roughly 41 cents of every transportation dollar arrives outside a per-trip claim, and the filing itself names healthcare facilities as a billing counterparty class (Note 2, F-22).

| Panel C. USAspending PSC V225: the federal direct-contract ambulance channel (obligations, pulled live) |                   |                           |          |                              |
|---------------------------------------------------------------------------------------------------------|-------------------|---------------------------|----------|------------------------------|
| Fiscal year                                                                                             | VA obligations \$ | All-agency obligations \$ | VA share | Note                         |
| 2023                                                                                                    | \$173,527,508     | \$210,994,517             | 82.2%    | spending_over_time, psc V225 |
| 2024                                                                                                    | \$190,909,874     | \$247,491,935             | 77.1%    |                              |
| 2025                                                                                                    | \$246,082,814     | \$292,654,678             | 84.1%    |                              |
| CAGR FY2023-FY2025 (VA)                                                                                 | 19.1%             | 17.8%                     |          | two-year CAGR                |

| Top FY2025 V225 recipients                     | Obligations \$ |
|------------------------------------------------|----------------|
| EXCELSIOR AMBULANCE SERVICE INC                | \$19,132,382   |
| RELIANCE MEDICAL TRANSPORT, LLC                | \$15,611,296   |
| MEDSHORE AMBULANCE SERVICE, INC.               | \$11,461,265   |
| JAN-CARE AMBULANCE OF NORTH CENTRAL W.VA., INC | \$11,443,449   |
| JOURNEY VIA GURNEY, LLC                        | \$10,256,714   |



|                                          |             |
|------------------------------------------|-------------|
| HAWAII LIFE FLIGHT LLC                   | \$8,614,952 |
| BRADSHER, JAMES E                        | \$8,485,576 |
| PROFESSIONAL HEALTHCARE ASSOCIATES, INC. | \$8,305,345 |
| MEDEX MEDICAL TRANSPORT SERVICE INC      | \$7,980,186 |
| NAVARRE CORPORATION                      | \$7,737,577 |

Guardrail: procurement obligations only. The VA Beneficiary Travel program pays ambulance CLAIMS through a separate and larger channel; the two must never be summed. IHS and DoD V225 obligations exist and are additive to VA only within this procurement frame.

| Panel D. What this layer does to the sizing math (live links)         |           |                                                                                                                                                 |
|-----------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| NEMSIS interfacility-scale activations (EMS_Transports)               | 5,510,664 | vehicle legs, all payers                                                                                                                        |
| Medicare FFS hospital-to-hospital services 2024 (Medicare_IFT_Seri    | 871,753   | claims-visible floor                                                                                                                            |
| The identity this tab prices: activations minus claims-visible volume | 4,638,911 | wedge, not a market size                                                                                                                        |
| IFT-specific facility-pay share of that wedge                         | PENDING   | No public source; would take a claims-vendor panel with facility-remit flags or primary buyer research (bordered slot; see Engagement_Data_Map) |

Read panel

What is measured: (1) 18.6 percent of US ambulance organizations report facility-contract revenue, with a conditional mean near \$0.92M and a long right tail (median \$48K); (2) 74.5 percent report some non-payer revenue; (3) a listed pure-play operator books about 41 percent of transportation revenue outside per-trip claims, naming facilities as a payer class in its revenue-recognition note; (4) the federal government alone obligates about a quarter-billion dollars a year buying ambulance service by contract. Together these are the public floor and existence proof for the facility-pay layer that claims-based sizing cannot see. What is NOT measured: the IFT-specific share of that layer; it stays PENDING by design.



### Press and footprint registry: what issuers announced, and what the subject company's own site claimed over time

Sources: issuer newsrooms/wire pages (URL + date per row); Internet Archive availability API; Common Crawl WARC captures. This tab is a REGISTRY feeding Company\_Dossier timelines; every row is a PUBLIC-WEB claim by its issuer, not a verified event.

DATA QUALITY: rows restate release text only - a release saying two organizations partner is recorded as exactly that claim by the issuer. Newsroom coverage is uneven (PARK's press host returned 503s, so several known items are omitted rather than carried updated); the subject company's site has NO press section, so its two rows come from wire postings). Panel A row counts measure what was HARVESTED under a 3-8-items-per-org relevance screen, not relevance volume - so NO per-year count table is provided for Panel A. Archive panels: Wayback content is egress-blocked here (rows recorded and FALCK); page-content rows come from Common Crawl, an independent public archive.

| Panel A. Press registry - dated public announcements, 2020-2026 (PUBLIC-WEB; URL + date per row) |            |                                                                                                                |                                                                                                                                                                                                           |                       |                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Issuer / org                                                                                     | Date       | Headline (as published)                                                                                        | What it establishes (issuer claim only)                                                                                                                                                                   | Label                 | URL                                                                                                                                                                                                                                                                                                     |
| AMR / Global Medical Response                                                                    | 2025-09-23 | Global Medical Response Solidifies Leadership in EMS with Strategic Refinancing                                | Global Medical Response states it completed a \$5.4 billion refinancing, including a \$3.6 billion term loan B and \$1.0 billion of senior notes                                                          | PUBLIC-WEB self-claim | <a href="https://www.globalmedicalresponse.com/news/global-medical-response-solidifies-leadership-in-ems-with-strategic-refinancing">https://www.globalmedicalresponse.com/news/global-medical-response-solidifies-leadership-in-ems-with-strategic-refinancing</a>                                     |
| AMR / Global Medical Response                                                                    | 2026-05-04 | GMR Solutions Inc. Announces Launch of its Initial Public Offering                                             | Global Medical Response states GMR Solutions Inc. launched an initial public offering of approximately 31.9 million shares of Class A common stock, with a planned NYSE listing under ticker GMRS         | PUBLIC-WEB self-claim | <a href="https://www.globalmedicalresponse.com/news/gmr-solutions-inc-announces-launch-of-its-initial-public-offering">https://www.globalmedicalresponse.com/news/gmr-solutions-inc-announces-launch-of-its-initial-public-offering</a>                                                                 |
| AMR / Global Medical Response                                                                    | 2026-05-12 | GMR Solutions Inc. Announces Pricing of its Initial Public Offering                                            | Global Medical Response states GMR Solutions Inc. priced an initial public offering of 31,914,893 shares of Class A common stock for NYSE trading under ticker GMRS                                       | PUBLIC-WEB self-claim | <a href="https://www.globalmedicalresponse.com/news/gmr-solutions-inc-announces-pricing-of-its-initial-public-offering">https://www.globalmedicalresponse.com/news/gmr-solutions-inc-announces-pricing-of-its-initial-public-offering</a>                                                               |
| AMR / Global Medical Response                                                                    | 2026-06-08 | Global Medical Response + Avel eCare Partner to Reimagine 911 Access for Rural America                         | Global Medical Response states it formed a partnership with Avel eCare to launch 911 REACT, described as an integrated 911-to-clinical-care access model for rural America                                | PUBLIC-WEB self-claim | <a href="https://www.globalmedicalresponse.com/news/global-medical-response-avel-ecare-partner-to-reimagine-911-access-for-rural-america">https://www.globalmedicalresponse.com/news/global-medical-response-avel-ecare-partner-to-reimagine-911-access-for-rural-america</a>                           |
| Acadian Ambulance / Acadian Companies                                                            | 2020-08-03 | Blaise Zuschlag promoted to Vice President and Chief Administrative Officer of Acadian Companies               | Acadian Companies states Blaise Zuschlag was promoted to vice president and chief administrative officer                                                                                                  | PUBLIC-WEB self-claim | <a href="https://acadian.com/blaise-zuschlag-promoted-to-vice-president-and-chief-administrative-officer-of-acadian-companies/">https://acadian.com/blaise-zuschlag-promoted-to-vice-president-and-chief-administrative-officer-of-acadian-companies/</a>                                               |
| Acadian Ambulance / Acadian Companies                                                            | 2023-09-30 | Acadian Ambulance to finalize strategic transaction with St. Landry EMS                                        | Acadian Ambulance states it agreed to acquire substantially all assets of St. Landry EMS in Louisiana, with closing expected on or before November 30, 2023                                               | PUBLIC-WEB self-claim | <a href="https://acadian.com/acadian-ambulance-to-finalize-strategic-transaction-with-st-landry-ems-14/">https://acadian.com/acadian-ambulance-to-finalize-strategic-transaction-with-st-landry-ems-14/</a>                                                                                             |
| Acadian Ambulance / Acadian Companies                                                            | 2024-01-15 | Acadian Ambulance acquires SouthernCross Ambulance                                                             | Acadian Ambulance states it acquired substantially all assets of New Braunfels, Texas-based SouthernCross Ambulance effective January 15, 2024, adding 12 ambulances, 10 wheelchair vans and 75 employees | PUBLIC-WEB self-claim | <a href="https://acadian.com/acadian-ambulance-acquires-southerncross-ambulance/">https://acadian.com/acadian-ambulance-acquires-southerncross-ambulance/</a>                                                                                                                                           |
| Acadian Ambulance / Acadian Companies                                                            | 2024-08-20 | Remembering Richard E. Zuschlag                                                                                | Acadian Companies states its founder, chairman and CEO Richard E. Zuschlag died on June 5, 2024                                                                                                           | PUBLIC-WEB self-claim | <a href="https://acadian.com/richard/">https://acadian.com/richard/</a>                                                                                                                                                                                                                                 |
| Acadian Ambulance / Acadian Companies                                                            | 2025-12-04 | Acadian Companies Announces Leadership Transition in Human Resources                                           | Acadian Companies states Joe Lightfoot retired as vice president of human resources after 25 years, with Krista Gravouia moving into the role and Allyson Duck becoming director of human resources       | PUBLIC-WEB self-claim | <a href="https://acadian.com/acadian-companies-announces-leadership-transition-in-human-resources/">https://acadian.com/acadian-companies-announces-leadership-transition-in-human-resources/</a>                                                                                                       |
| Acadian Ambulance / Acadian Companies                                                            | 2026-06-11 | Acadian Ambulance to provide additional EMS support to the City of Houston                                     | Acadian Ambulance states it will provide additional EMS support to the City of Houston                                                                                                                    | PUBLIC-WEB self-claim | <a href="https://acadian.com/acadian-ambulance-to-provide-additional-ems-support-to-the-city-of-houston/">https://acadian.com/acadian-ambulance-to-provide-additional-ems-support-to-the-city-of-houston/</a>                                                                                           |
| DocGo                                                                                            | 2021-11-05 | Motion Acquisition Corp. and DocGo Announce Closing of Business Combination, will Begin Trading on Nasdaq U... | Motion Acquisition Corp. and DocGo state their business combination closed, raising approximately \$158 million, with DocGo common stock to begin Nasdaq trading as DCGO on November 8, 2021              | PUBLIC-WEB self-claim | <a href="https://www.prnewswire.com/news-releases/motion-acquisition-corp-and-docgo-announce-closing-of-business-combination-will-begin-trading-on...">https://www.prnewswire.com/news-releases/motion-acquisition-corp-and-docgo-announce-closing-of-business-combination-will-begin-trading-on...</a> |
| DocGo                                                                                            | 2023-09-18 | DocGo Appoints Lee Bienstock as Chief Executive Officer                                                        | DocGo states its board appointed Lee Bienstock, previously president and chief operating officer, as chief executive officer effective September 15, 2023                                                 | PUBLIC-WEB self-claim | <a href="https://www.businesswire.com/news/home/20230918906310/en/DocGo-Appoints-Lee-Bienstock-as-Chief-Executive-Officer">https://www.businesswire.com/news/home/20230918906310/en/DocGo-Appoints-Lee-Bienstock-as-Chief-Executive-Officer</a>                                                         |
| DocGo                                                                                            | 2025-02-10 | DocGo Acquires PTI Health To Expand Proactive Healthcare Offering with Mobile Lab Collection and Mobile Phl... | DocGo states it acquired PTI Health, a mid-Atlantic mobile lab collection and phlebotomy services company, with PTI Health president Wayne Meadows joining DocGo                                          | PUBLIC-WEB self-claim | <a href="https://www.businesswire.com/news/home/20250210820237/en/DocGo-Acquires-PTI-Health-To-Expand-Proactive-Healthcare-Offering-with-Mobile-La...">https://www.businesswire.com/news/home/20250210820237/en/DocGo-Acquires-PTI-Health-To-Expand-Proactive-Healthcare-Offering-with-Mobile-La...</a> |
| DocGo                                                                                            | 2025-10-20 | DocGo Acquires Virtual Care Platform SteadyMD, Expands Telehealth Services Across all 50 States                | DocGo states it acquired virtual care platform SteadyMD, Inc., with SteadyMD co-founders Guy Friedman and Yarone Goren joining DocGo's leadership team                                                    | PUBLIC-WEB self-claim | <a href="https://docgo.com/press/docgo-acquires-virtual-care-platform-steadymd-expands-telehealth-services-across-all-50-states/">https://docgo.com/press/docgo-acquires-virtual-care-platform-steadymd-expands-telehealth-services-across-all-50-states/</a>                                           |
| Falck US / Falck                                                                                 | 2022-04-22 | Falck generated strong revenue growth but lower earnings in Q1 2022                                            | Falck states Q1 2022 Emergency Health and Safety revenue increased due in part to its new ambulance contract in San Diego and generally higher activity in its US ambulance business                      | PUBLIC-WEB self-claim | <a href="https://www.falck.com/news/latest-news/falck-generated-strong-revenue-growth-but-lower-earnings-in-q1-2022/">https://www.falck.com/news/latest-news/falck-generated-strong-revenue-growth-but-lower-earnings-in-q1-2022/</a>                                                                   |
| Falck US / Falck                                                                                 | 2026-04-22 | New directors for Scandinavia and Europe join Falck's Group Executive Management                               | Falck states new directors for Scandinavia and Europe joined its Group Executive Management                                                                                                               | PUBLIC-WEB self-claim | <a href="https://www.falck.com/news/latest-news/new-directors-for-scandinavia-and-europe-join-falcks-group-executive-management/">https://www.falck.com/news/latest-news/new-directors-for-scandinavia-and-europe-join-falcks-group-executive-management/</a>                                           |
| Falck US / Falck                                                                                 | 2026-06-02 | Falck appoints Catalina Giraldo Head of Region Latam                                                           | Falck states it appointed Catalina Giraldo as Head of Region Latam                                                                                                                                        | PUBLIC-WEB self-claim | <a href="https://www.falck.com/news/latest-news/falck-appoints-catalina-giraldo-head-of-region-latam/">https://www.falck.com/news/latest-news/falck-appoints-catalina-giraldo-head-of-region-latam/</a>                                                                                                 |
| MedSpeed                                                                                         | 2023-05-18 | Allina Health Names MedSpeed as Partner to Provide System-wide Healthcare Logistics                            | MedSpeed states Allina Health named MedSpeed to provide system-wide healthcare logistics                                                                                                                  | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/allina-health-names-medspeed-as-partner-to-provide-system-wide-healthcare-logistics/">https://www.medspeed.com/news/allina-health-names-medspeed-as-partner-to-provide-system-wide-healthcare-logistics/</a>                                                     |
| MedSpeed                                                                                         | 2024-05-15 | MedSpeed Promotes Wesley Crampton to President                                                                 | MedSpeed states it promoted Wesley Crampton to president                                                                                                                                                  | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/medspeed-promotes-wesley-crampton-to-president/">https://www.medspeed.com/news/medspeed-promotes-wesley-crampton-to-president/</a>                                                                                                                               |
| MedSpeed                                                                                         | 2024-07-11 | MedSpeed Promotes Matt Gutwein to COO                                                                          | MedSpeed states it promoted Matt Gutwein to chief operating officer                                                                                                                                       | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/medspeed-promotes-matt-gutwein-to-coo/">https://www.medspeed.com/news/medspeed-promotes-matt-gutwein-to-coo/</a>                                                                                                                                                 |
| MedSpeed                                                                                         | 2024-08-13 | MedSpeed Announces Partnership with Strategic Health Care Investor                                             | MedSpeed states it partnered with strategic health care investor Water Street Healthcare Partners, which committed capital and resources to expand MedSpeed's services                                    | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/medspeed-announces-partnership-with-strategic-health-care-investor/">https://www.medspeed.com/news/medspeed-announces-partnership-with-strategic-health-care-investor/</a>                                                                                       |
| MedSpeed                                                                                         | 2024-11-12 | WakeMed Selects MedSpeed to Build Integrated Logistics Network                                                 | MedSpeed states WakeMed selected MedSpeed to build an integrated logistics network                                                                                                                        | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/wakemed-selects-medspeed-to-build-integrated-logistics-network/">https://www.medspeed.com/news/wakemed-selects-medspeed-to-build-integrated-logistics-network/</a>                                                                                               |
| MedSpeed                                                                                         | 2025-01-14 | TriCore Renews Contract with MedSpeed                                                                          | MedSpeed states TriCore renewed its contract with MedSpeed                                                                                                                                                | PUBLIC-WEB self-claim | <a href="https://www.medspeed.com/news/tricore-renews-contract-with-medspeed/">https://www.medspeed.com/news/tricore-renews-contract-with-medspeed/</a>                                                                                                                                                 |

|                                          |            |                                                                                                                |                                                                                                                                                                                                                              |                       |                                                                                                                                              |
|------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| MedSpeed                                 | 2026-03-26 | MedSpeed Appoints Ross Lieb as Chief Financial Officer                                                         | MedSpeed states it appointed Ross Lieb as chief financial officer                                                                                                                                                            | PUBLIC-WEB self-claim | https://www.medspeed.com/news/medspeed-appoints-ross-lieb-as-chief-financial-officer/                                                        |
| Midwest Medical Transport Company (MM... | 2022-01-25 | Harbour Point Capital Completes Investment in Midwest Medical Transport                                        | Harbour Point Capital states it completed an investment in Midwest Medical Transport in partnership with executive Kevin Ketzel, with prior investors Panorama Point Partners, Dixon Midland Company and ORIX selling the... | PUBLIC-WEB self-claim | https://www.businesswire.com/news/home/20220125006174/en/Harbour-Point-Capital-Completes-Investment-in-Midwest-Medical-Transport             |
| Midwest Medical Transport Company (MM... | 2023-04-05 | MMT Ambulance Appoints Chris Ciatto as New CEO                                                                 | MMT Ambulance states Chris Ciatto was appointed chief executive officer and board member effective immediately, joining from Phoenix Physical Therapy                                                                        | PUBLIC-WEB self-claim | https://www.businesswire.com/news/home/20230405005703/en/MMT-Ambulance-Appoints-Chris-Ciatto-as-New-CEO                                      |
| Priority Ambulance                       | 2021-10-04 | Priority OnDemand welcomes First Call Ambulance to national family of companies                                | Priority OnDemand states First Call Ambulance joined its national family of companies                                                                                                                                        | PUBLIC-WEB self-claim | https://priorityambulance.com/2021/10/04/priority-ambulance-welcomes-first-call-ambulance-to-national-family-of-companies/                   |
| Priority Ambulance                       | 2024-12-31 | Priority Ambulance to Welcome Mississippi-Based LifeCare EMS Team to National Family of Companies as Magnol... | Priority Ambulance states it is acquiring certain assets of Mississippi-based LifeCare EMS and will operate in seven central Mississippi counties as Magnolia Ambulance, with the transaction set to close January 13, 2025  | PUBLIC-WEB self-claim | https://priorityambulance.com/priority-ambulance-to-welcome-mississippi-based-lifecare-ems-team-to-national-family-of-companies-as-magnol... |
| Priority Ambulance                       | 2025-01-08 | Priority OnDemand Announces Key Executive Appointments to Support Accelerated Growth Strategy                  | Priority OnDemand states it made key executive appointments in support of its growth strategy                                                                                                                                | PUBLIC-WEB self-claim | https://priorityambulance.com/priority-ondemand-announces-key-executive-appointments-to-support-accelerated-growth-strategy/                 |
| Priority Ambulance                       | 2026-03-17 | Priority Ambulance Launches Sunstate Ambulance in Florida                                                      | Priority Ambulance states it launched Sunstate Ambulance in Florida                                                                                                                                                          | PUBLIC-WEB self-claim | https://priorityambulance.com/priority-ambulance-launches-sunstate-ambulance-in-florida/                                                     |
| Priority Ambulance                       | 2026-06-16 | First Call Ambulance Receives Extension of Humphreys County 911 Contract                                       | Priority Ambulance states its First Call Ambulance operation received an extension of the Humphreys County, Tennessee 911 contract                                                                                           | PUBLIC-WEB self-claim | https://priorityambulance.com/first-call-ambulance-receives-extension-of-humphreys-county-911-contract/                                      |
| Priority Ambulance                       | 2026-06-17 | Priority Ambulance Names Ray Gayk Pacific Region Regional President                                            | Priority Ambulance states Ray Gayk was named Pacific Region regional president                                                                                                                                               | PUBLIC-WEB self-claim | https://priorityambulance.com/priority-ambulance-names-ray-gayk-pacific-region-regional-president-strengthening-commitment-to-fire-servic... |
| Royal Ambulance                          | 2025-04-21 | 10 Years with Donor Network West: A Legacy of Compassion                                                       | Royal Ambulance states it marked ten years of partnership with Donor Network West                                                                                                                                            | PUBLIC-WEB self-claim | https://www.royalambulance.com/post/ten-years-and-counting-honoring-a-decade-of-partnership-with-donor-network-west                          |
| Royal Ambulance                          | 2025-09-10 | Working Side by Side with Berkeley Fire                                                                        | Royal Ambulance states it operates a dedicated BLS unit embedded within Berkeley's 911 system focused on psychiatric transports                                                                                              | PUBLIC-WEB self-claim | https://www.royalambulance.com/post/working-side-by-side-with-berkeley-fire                                                                  |
| Royal Ambulance                          | 2026-07-02 | Royal Ambulance Named a Best Place to Work in EMS - 5th Year Running                                           | Royal Ambulance states it ranked No. 6 on the 2026 Inspiring Workplaces Top 100 in North America, its fifth consecutive year on the list                                                                                     | PUBLIC-WEB self-claim | https://www.royalambulance.com/post/royal-ambulance-named-a-best-place-to-work-in-ems-5th-year-running                                       |
| Superior Air-Ground Ambulance            | 2020-04-22 | Beaumont Health signs definitive agreement to sell ambulance division to Superior Air Ground Ambulance Serv... | Beaumont Health states it signed a definitive agreement to sell its ambulance division, including Community Emergency Medical Service, Parastar and Beaumont Mobile Medicine, to Superior Air-Ground Ambulance Service, w... | PUBLIC-WEB self-claim | https://newsroom.corewellhealth.org/2020-04-22-Beaumont-Health-signs-definitive-agreement-to-sell-ambulance-division-to-Superior-Air-Grou... |
| Superior Air-Ground Ambulance            | 2020-11-17 | Beaumont Health sells ambulance division to Superior Air-Ground Ambulance Service of Michigan, Inc.            | Beaumont Health states it sold its ambulance division to Superior Air-Ground Ambulance Service of Michigan following Michigan Attorney General approval, with closing expected in mid-December 2020                          | PUBLIC-WEB self-claim | https://newsroom.corewellhealth.org/2020-11-17-Beaumont-Health-sells-ambulance-division-to-Superior-Air-Ground-Ambulance-Service-of-Michi... |
| Superior Air-Ground Ambulance            | 2026-05-21 | Oakton and Superior Ambulance Launch "Earn While You Learn" Partnership During National EMS Week               | Superior Air-Ground Ambulance states it launched an 'Earn While You Learn' EMT training partnership with Oakton College                                                                                                      | PUBLIC-WEB self-claim | https://superiorambulance.com/oakton-and-superior-ambulance-launch-earn-while-you-learn-partnership-during-national-ems-week/                |
| Superior Air-Ground Ambulance            | 2026-06-04 | Superior Ambulance Awarded Wisconsin Fast Forward Grant                                                        | Superior Air-Ground Ambulance states it was awarded a \$287,690 Wisconsin Fast Forward grant                                                                                                                                 | PUBLIC-WEB self-claim | https://superiorambulance.com/superior-ambulance-awarded-wisconsin-fast-forward-grant/                                                       |

Rows harvested (live)39

| Panel B. Subject-company website self-claim series (archived captures; self-claims recorded verbatim) |                                    |                                  |                                                                                                                                                                                                                                                                    |                                               |                      |                                          |
|-------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------|------------------------------------------|
| Capture date                                                                                          | Archive                            | Page                             | Self-claimed footprint / claim text (verbatim excerpt)                                                                                                                                                                                                             | Enumerated entries                            | States if enumerable | Note                                     |
| 2014-05-17                                                                                            | Wayback (recorded; content PARKED) | midwestmedicaltransport.com      | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2014-12-17                                                                                            | Wayback (recorded; content PARKED) | midwestmedicaltransport.com      | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2023-03-07                                                                                            | Wayback (recorded; content PARKED) | mmtamb.com                       | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2023-06-07                                                                                            | Wayback (recorded; content PARKED) | mmtamb.com                       | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2024-06-25                                                                                            | Wayback (recorded; content PARKED) | mmtamb.com                       | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2026-06-13                                                                                            | Wayback (recorded; content PARKED) | mmtamb.com                       | (no content retrievable)                                                                                                                                                                                                                                           |                                               |                      | Snapshot recorded via archive.org        |
| 2023-03-24                                                                                            | Common Crawl WARC                  | https://mmtamb.com/              | 'The Communities We Serve - Click on a state below to learn more. Currently Serving / Coming Soon' ... state panels: Missouri ('Proudly Serving St. Louis and the Kansas City Metro Area'), Iowa ('Proudly Serving Iowa City, Sioux City, and the Des Moines Metro | 11 MO, IA, NE, WI, OH, VA, MA, RI, DC, KS, NC |                      | n_states_claim ed counts DC as one of 11 |
| 2023-03-24                                                                                            | Common Crawl WARC                  | https://mmtamb.com/about-us/     | With a growing fleet of 500+ vehicles and a workforce of 4,000+ providers, MMT strives to provide the absolute best patient care 24/7/365.                                                                                                                         |                                               |                      |                                          |
| 2023-03-24                                                                                            | Common Crawl WARC                  | https://mmtamb.com/partnerships/ | MMT has operations spread out across multiple states and offers premier, reliable ambulance transportation services to dozens of communities and thousands of medical centers and healthcare organizations.                                                        |                                               |                      |                                          |

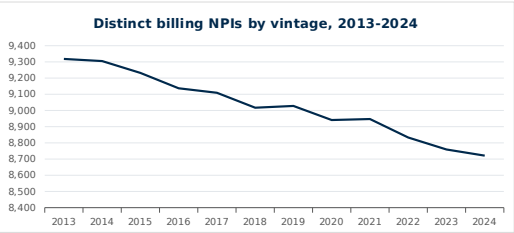
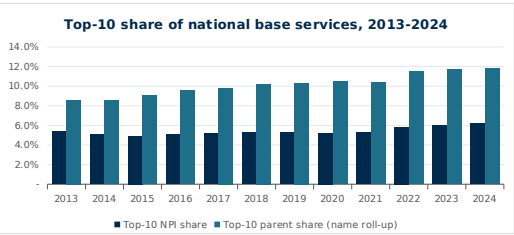


Fragmentation: the US Medicare ambulance biller universe, 12 vintages 2013-2024

Source: MUP by Provider and Service, ground base codes A0426-A0429/A0433/A0434, all states and territories, 12 manifested vintages 2013-2024; join keys Rndmg\_NPI x HCPCS\_Cd x vintage. NPI-grain panels first, then a printed, auditable name-pattern roll-up to parents.

DATA QUALITY: NPI grain - one organization can bill under several NPIs, so biller counts OVERSTATE firms and top-N shares UNDERSTATE concentration; the name-pattern parent layer below corrects only what names reveal and is itself a floor. Per-NPI code rows under 11 beneficiaries are suppressed at source: the smallest billers are partly invisible, and an NPI "exiting" can be a dip below the suppression floor rather than a true exit. Final-action claims, Medicare FFS only. Vintages are independent annual snapshots, not a continuous series (until the annual series lands); every consecutive transition below is one year apart.

| Panel A. Biller count and top-N NPI concentration by vintage (distinct NPIs, ground base codes) |                       |                                |                     |              |                     |              |                      |               |                                                                     |
|-------------------------------------------------------------------------------------------------|-----------------------|--------------------------------|---------------------|--------------|---------------------|--------------|----------------------|---------------|---------------------------------------------------------------------|
| Vintage                                                                                         | Distinct billing NPIs | National base services (floor) | Top-10 NPI services | Top-10 share | Top-50 NPI services | Top-50 share | Top-100 NPI services | Top-100 share | Note                                                                |
| 2013                                                                                            | 9,318                 | 14,938,429                     | 801,682             | 5.4%         | 2,239,654           | 15.0%        | 3,329,992            | 22.3%         | counts overstate firms (NPI grain); services are suppression floors |
| 2014                                                                                            | 9,305                 | 15,037,862                     | 766,032             | 5.1%         | 2,177,226           | 14.5%        | 3,293,063            | 21.9%         |                                                                     |
| 2015                                                                                            | 9,232                 | 14,235,792                     | 700,232             | 4.9%         | 2,085,546           | 14.7%        | 3,159,170            | 22.2%         |                                                                     |
| 2016                                                                                            | 9,137                 | 13,783,259                     | 700,237             | 5.1%         | 2,010,282           | 14.6%        | 3,043,316            | 22.1%         |                                                                     |
| 2017                                                                                            | 9,109                 | 13,713,824                     | 710,395             | 5.2%         | 1,990,181           | 14.5%        | 3,019,137            | 22.0%         |                                                                     |
| 2018                                                                                            | 9,017                 | 13,358,909                     | 701,154             | 5.2%         | 1,957,262           | 14.7%        | 2,970,069            | 22.2%         |                                                                     |
| 2019                                                                                            | 9,028                 | 12,890,235                     | 678,782             | 5.3%         | 1,870,212           | 14.5%        | 2,827,987            | 21.9%         |                                                                     |
| 2020                                                                                            | 8,941                 | 11,531,316                     | 601,885             | 5.2%         | 1,648,686           | 14.3%        | 2,505,563            | 21.7%         |                                                                     |
| 2021                                                                                            | 8,947                 | 10,902,433                     | 573,348             | 5.3%         | 1,575,526           | 14.5%        | 2,379,182            | 21.8%         |                                                                     |
| 2022                                                                                            | 8,833                 | 10,084,704                     | 586,576             | 5.8%         | 1,549,571           | 15.4%        | 2,338,034            | 23.2%         |                                                                     |
| 2023                                                                                            | 8,759                 | 9,660,141                      | 579,232             | 6.0%         | 1,555,694           | 16.1%        | 2,329,728            | 24.1%         |                                                                     |
| 2024                                                                                            | 8,721                 | 9,542,103                      | 588,683             | 6.2%         | 1,578,097           | 16.5%        | 2,344,426            | 24.6%         |                                                                     |



| Panel B. Size distribution by services decile (2024 vs 2013; deciles of each vintage's own biller distribution) |              |               |            |                      |              |               |            |                                                            |
|-----------------------------------------------------------------------------------------------------------------|--------------|---------------|------------|----------------------|--------------|---------------|------------|------------------------------------------------------------|
| Decile (by base services)                                                                                       | Billers 2024 | Services 2024 | Share 2024 | Median services 2024 | Billers 2013 | Services 2013 | Share 2013 | Note                                                       |
| D1 (smallest tenth)                                                                                             | 872          | 18,546        | 0.2%       | 20                   | 931          | 21,051        | 0.1%       | suppression truncates this decile: the true tail is larger |
| D2                                                                                                              | 872          | 41,255        | 0.4%       | 47                   | 932          | 49,497        | 0.3%       |                                                            |
| D3                                                                                                              | 872          | 75,013        | 0.8%       | 85                   | 932          | 89,816        | 0.6%       |                                                            |
| D4                                                                                                              | 872          | 123,907       | 1.3%       | 141                  | 932          | 149,667       | 1.0%       |                                                            |
| D5                                                                                                              | 872          | 196,112       | 2.1%       | 223                  | 932          | 246,899       | 1.7%       |                                                            |
| D6                                                                                                              | 872          | 306,319       | 3.2%       | 350                  | 931          | 397,345       | 2.7%       |                                                            |
| D7                                                                                                              | 872          | 464,442       | 4.9%       | 529                  | 932          | 636,388       | 4.3%       |                                                            |
| D8                                                                                                              | 872          | 732,547       | 7.7%       | 826                  | 932          | 1,087,182     | 7.3%       |                                                            |
| D9                                                                                                              | 872          | 1,343,066     | 14.1%      | 1,463                | 932          | 2,266,436     | 15.2%      |                                                            |
| D10 (largest tenth)                                                                                             | 873          | 6,240,896     | 65.4%      | 4,466                | 932          | 9,994,148     | 66.9%      | the whole market story lives here                          |
| All billers                                                                                                     | 8,721        | 9,542,103     |            |                      | 9,318        | 14,938,429    |            | ties to Panel A rows for the same vintages                 |

| Panel C. Name-normalization layer: NPIs rolled to parents by name pattern - the FULL pattern table prints below in blue so the layer is auditable |                   |             |                    |                   |                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|--------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parent family                                                                                                                                     | NPIs matched 2024 | States 2024 | Base services 2024 | Share of national | Match patterns (case-insensitive substring / word)                                                                                                                                               |
| AMR / American Medical Response                                                                                                                   | 65                | 17          | 423,252            | 4.44%             | AMERICAN MEDICAL RESPONSE; word "bAMRb" - mandated pattern; the ground flagship of the GMR family                                                                                                |
| Global Medical Response / GMR                                                                                                                     | -                 | -           | -                  | -                 | GLOBAL MEDICAL RESPONSE; word "bGMRb" - mandated pattern; no NPI bills under the holding-company name - GMR-family volume rides brand rows (AMR, Lifeguard), so any GMR-level roll-up is a floor |
| Falck                                                                                                                                             | 4                 | 3           | 72,805             | 0.76%             | FALCK - mandated pattern                                                                                                                                                                         |
| Acadian                                                                                                                                           | 4                 | 3           | 151,225            | 1.58%             | ACADIAN - mandated pattern                                                                                                                                                                       |
| PatientCare                                                                                                                                       | -                 | -           | -                  | -                 | PATIENTCARE; PATIENT CARE EMS - mandated pattern; zero matches - the brand bills under legacy local names, another reason the roll-up is a floor                                                 |
| Priority Ambulance                                                                                                                                | 4                 | 4           | 6,496              | 0.07%             | PRIORITY AMBULANCE - mandated pattern; subsidiaries billing under local brands stay unrolled                                                                                                     |
| DocGo / Ambulnz                                                                                                                                   | 6                 | 5           | 59,315             | 0.62%             | DOCGO; AMBULNZ - mandated pattern                                                                                                                                                                |
| Superior Air-Ground                                                                                                                               | 5                 | 5           | 112,351            | 1.18%             | SUPERIOR AIR-GROUND; SUPERIOR AIR GROUND - mandated pattern                                                                                                                                      |
| Royal Ambulance                                                                                                                                   | 1                 | 1           | 10,909             | 0.11%             | ROYAL AMBULANCE - mandated pattern                                                                                                                                                               |
| Lifeguard Ambulance Service                                                                                                                       | 13                | 8           | 36,269             | 0.38%             | LIFEGUARD AMBULANCE - detected same-name multi-state family (8+ states); publicly a GMR brand                                                                                                    |
| Cal-Ore Life Flight                                                                                                                               | 8                 | 3           | 3,321              | 0.03%             | CAL-ORE; CAL ORE LIFE - detected same-name multi-state family; ground legs of the base codes only                                                                                                |

Roll-up rules, printed for audit: an NPI joins a family only when its NPPES organization name matches a pattern above; every unmatched NPI stays its own parent. Municipal and generic same-names (CITY OF X, COUNTY OF X, COMMUNITY AMBULANCE SERVICE) are NOT rolled - they are distinct local entities sharing a name. Direction of bias: this layer can only RAISE measured concentration toward truth, never reach it - brands billing under unrelated legacy names stay unrolled, so top-10 parent share is still a floor.

| Vintage | Top-10 NPI share (link) | Top-10 parent services | Top-10 parent share | Parent minus NPI (pp) | Note                                                                                                   |
|---------|-------------------------|------------------------|---------------------|-----------------------|--------------------------------------------------------------------------------------------------------|
| 2013    | 5.4%                    | 1,282,256              | 8.6%                | 3.2%                  | top-10 entities after roll-up (families plus unrolled NPIs); denominator is the Panel A national total |
| 2014    | 5.1%                    | 1,281,223              | 8.5%                | 3.4%                  |                                                                                                        |
| 2015    | 4.9%                    | 1,291,353              | 9.1%                | 4.2%                  |                                                                                                        |
| 2016    | 5.1%                    | 1,318,994              | 9.6%                | 4.5%                  |                                                                                                        |
| 2017    | 5.2%                    | 1,343,974              | 9.8%                | 4.6%                  |                                                                                                        |
| 2018    | 5.2%                    | 1,365,039              | 10.2%               | 5.0%                  |                                                                                                        |
| 2019    | 5.3%                    | 1,332,350              | 10.3%               | 5.1%                  |                                                                                                        |
| 2020    | 5.2%                    | 1,206,349              | 10.5%               | 5.2%                  |                                                                                                        |
| 2021    | 5.3%                    | 1,131,272              | 10.4%               | 5.1%                  |                                                                                                        |
| 2022    | 5.8%                    | 1,157,512              | 11.5%               | 5.7%                  |                                                                                                        |
| 2023    | 6.0%                    | 1,131,747              | 11.7%               | 5.7%                  |                                                                                                        |
| 2024    | 6.2%                    | 1,126,848              | 11.8%               | 5.6%                  |                                                                                                        |

| Panel D. Exit by size decile: NPIs present in one vintage and absent from the next (decile of the START vintage) |                         |      |      |      |      |      |      |      |      |                         |             |                                                                                                     |
|------------------------------------------------------------------------------------------------------------------|-------------------------|------|------|------|------|------|------|------|------|-------------------------|-------------|-----------------------------------------------------------------------------------------------------|
| Transition                                                                                                       | Exit rate D1 (smallest) | D2   | D3   | D4   | D5   | D6   | D7   | D8   | D9   | Exit rate D10 (largest) | Exited NPIs | Note                                                                                                |
| 2013 to 2014                                                                                                     | 20.5%                   | 4.9% | 2.4% | 2.6% | 2.5% | 2.7% | 2.8% | 4.2% | 2.9% | 1.3%                    | 435         | absence includes dips below the suppression floor, so tail exit rates are upper bounds on true exit |
| 2014 to 2015                                                                                                     | 17.1%                   | 3.2% | 1.9% | 2.0% | 3.0% | 3.4% | 3.3% | 4.3% | 6.0% | 3.2%                    | 443         |                                                                                                     |
| 2015 to 2016                                                                                                     | 22.6%                   | 4.8% | 2.5% | 2.3% | 3.5% | 2.0% | 2.5% | 4.1% | 2.8% | 1.8%                    | 451         |                                                                                                     |
| 2016 to 2017                                                                                                     | 21.9%                   | 4.6% | 3.1% | 1.8% | 1.8% | 2.0% | 1.9% | 3.6% | 2.4% | 1.5%                    | 406         |                                                                                                     |
| 2017 to 2018                                                                                                     | 25.3%                   | 4.0% | 3.7% | 2.3% | 1.9% | 2.0% | 2.5% | 3.0% | 1.6% | 1.1%                    | 431         |                                                                                                     |
| 2018 to 2019                                                                                                     | 19.1%                   | 4.3% | 3.4% | 3.3% | 2.3% | 2.3% | 2.9% | 2.5% | 1.8% | 1.1%                    | 389         |                                                                                                     |
| 2019 to 2020                                                                                                     | 25.5%                   | 3.9% | 2.8% | 1.9% | 2.4% | 2.2% | 2.7% | 2.3% | 1.9% | 1.1%                    | 421         |                                                                                                     |
| 2020 to 2021                                                                                                     | 18.5%                   | 4.6% | 2.2% | 1.1% | 1.5% | 1.9% | 1.6% | 3.7% | 2.8% | 1.3%                    | 350         |                                                                                                     |
| 2021 to 2022                                                                                                     | 23.4%                   | 2.8% | 3.6% | 3.4% | 2.7% | 2.3% | 2.6% | 4.1% | 4.5% | 1.3%                    | 453         |                                                                                                     |
| 2022 to 2023                                                                                                     | 27.4%                   | 6.3% | 2.7% | 2.5% | 3.1% | 2.8% | 2.1% | 3.4% | 3.7% | 1.0%                    | 487         |                                                                                                     |
| 2023 to 2024                                                                                                     | 27.8%                   | 7.0% | 3.9% | 4.0% | 2.3% | 2.3% | 1.8% | 1.4% | 2.4% | 1.0%                    | 471         |                                                                                                     |

| Transition   | Entered NPIs (absent start, present end) | Entry rate of end-vintage billers | Note                                                                   |
|--------------|------------------------------------------|-----------------------------------|------------------------------------------------------------------------|
| 2013 to 2014 | 422                                      | 4.5%                              | entry runs below exit in most transitions: the universe shrinks on net |
| 2014 to 2015 | 370                                      | 4.0%                              |                                                                        |
| 2015 to 2016 | 356                                      | 3.9%                              |                                                                        |
| 2016 to 2017 | 378                                      | 4.1%                              |                                                                        |
| 2017 to 2018 | 339                                      | 3.8%                              |                                                                        |
| 2018 to 2019 | 400                                      | 4.4%                              |                                                                        |
| 2019 to 2020 | 334                                      | 3.7%                              |                                                                        |
| 2020 to 2021 | 356                                      | 4.0%                              |                                                                        |
| 2021 to 2022 | 339                                      | 3.8%                              |                                                                        |
| 2022 to 2023 | 413                                      | 4.7%                              |                                                                        |
| 2023 to 2024 | 433                                      | 5.0%                              |                                                                        |

**Read panel**

Measured, twice over: the universe is extraordinarily fragmented - 8,721 distinct billing NPIs in 2024, with the top 10 holding 6.2% of national base services, the top 100 holding 24.6%, and even after rolling brands to parents by name the top 10 parents hold only 11.8% (up from 8.6% in 2013: consolidation is real but glacial, and the roll-up is itself a floor). And the long tail is thinning: the smallest decile of billers exits at roughly 23% per year against about 1% for the largest decile, entry runs below exit, and the biller count fell from 9,318 to 8,721 while national Medicare FFS base volume fell about 36%. Read every level as a floor: NPI grain, suppression, and Medicare-FFS-only visibility all push the same direction - the true market is somewhat more concentrated and the true tail somewhat larger than shown.

Medicaid ground ambulance rate card: footprint states

Sources: nine published state fee schedules named on each panel, each fetched and parsed from the primary file or portal (NE/IA/KS/MO/OH/WI on 11 Jul 2026; MN/IN/KY on 12 Jul 2026); Virginia is PARKED (only 2009-vintage public files). Join key is the HCPCS code. Comparison prices each state against the CY2026 Medicare national unadjusted base on Derived\_Rate\_Card.

DATA QUALITY: Medicaid FFS schedule rates ONLY - managed-care plans negotiate their own rates and broker-care states route scheduled volume outside these schedules, so the real price of scheduled transport is unobservable here. Effective dates differ by 18 years (WI 7/1/2008 vs MN/IN 1/1/2026); the cross-state comparison mixes vintages by construction. Rates are NOT applies-to-applies product-for-product. MO A0428 is HH+HD-context only; KY has NO ALS1+non-emergency, ALS2, or SCT line and its A0428 is a non-emergency STRETCHER-van line (formerly T2005), not a full BLS ambulance; IN and MO do not cover SCT (A0434). KY is therefore excluded from the A0428 BLS comparison range.

| Panel A. Nebraska - Nebraska DHHS fee schedule 471-000-504, Ambulance Services, July 1 2026 (xlsx). Sc |                                                                                                                                                                             |                 |      |            |                                                                                                                  |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|------------------------------------------------------------------------------------------------------------------|
| HCPCS                                                                                                  | Service level                                                                                                                                                               | Rate \$         | Unit | Rate eff.  | In-row note                                                                                                      |
| A0428                                                                                                  | BLS non-emergency                                                                                                                                                           | \$154.89 base   |      | 07/01/2026 | schedule comment: includes bariatric transfers                                                                   |
| A0429                                                                                                  | BLS emergency                                                                                                                                                               | \$189.93 base   |      | 07/01/2026 |                                                                                                                  |
| A0426                                                                                                  | ALS1 non-emergency                                                                                                                                                          | \$387.24 base   |      | 07/01/2026 |                                                                                                                  |
| A0427                                                                                                  | ALS1 emergency                                                                                                                                                              | \$387.24 base   |      | 07/01/2026 |                                                                                                                  |
| A0433                                                                                                  | ALS2                                                                                                                                                                        | \$387.24 base   |      | 07/01/2026 |                                                                                                                  |
| A0434                                                                                                  | Specialty Care Transport                                                                                                                                                    | \$387.24 base   |      | 07/01/2026 | flat ALS rate: A0426+A0427+A0433=A0434                                                                           |
| A0425                                                                                                  | Ground mileage                                                                                                                                                              | \$6.35 per mile |      | 07/01/2026 |                                                                                                                  |
| Source file                                                                                            |                                                                                                                                                                             |                 |      |            | per statute mile                                                                                                 |
| PA rule                                                                                                | PA column blank on every ground code (no PA flag); policy 471 NAC 4.                                                                                                        |                 |      |            | https://dhhs.ne.gov/Medicaid%20Practitioner%20Fee%20Schedules/471-000-504%20Ambulance%20-%20July%20%26%2726.xlsx |
| Broker carve                                                                                           | YES - Heritage Health MCO brokers (MTM for NE Total Care and Molina; ModivCare for UHC, MTM Health from 1 Jan 2026); DHHS staff arrange residual FFS NET under 471-000-503. |                 |      |            |                                                                                                                  |
| Dialysis policy                                                                                        | No dialysis restriction printed on the ambulance schedule; scheduled dialysis rides route through the NET benefit.                                                          |                 |      |            |                                                                                                                  |

| Panel B. Iowa - Iowa Medicaid fee schedule portal extract, provider type 11 AMBULANCE (C11.csv), a |                                                                                                                                          |                 |      |            |                                                           |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|-----------------------------------------------------------|
| HCPCS                                                                                              | Service level                                                                                                                            | Rate \$         | Unit | Rate eff.  | In-row note                                               |
| A0428                                                                                              | BLS non-emergency                                                                                                                        | \$84.67 base    |      | 07/01/2014 | lowest true BLS A0428 in the footprint                    |
| A0429                                                                                              | BLS emergency                                                                                                                            | \$114.30 base   |      | 10/01/2015 |                                                           |
| A0426                                                                                              | ALS1 non-emergency                                                                                                                       | \$101.60 base   |      | 10/01/2015 |                                                           |
| A0427                                                                                              | ALS1 emergency                                                                                                                           | \$127.01 base   |      | 10/01/2015 |                                                           |
| A0433                                                                                              | ALS2                                                                                                                                     | \$232.84 base   |      | 05/01/2020 |                                                           |
| A0434                                                                                              | Specialty Care Transport                                                                                                                 | \$232.84 base   |      | 12/01/2020 | same rate as ALS2                                         |
| A0425                                                                                              | Ground mileage                                                                                                                           | \$2.61 per mile |      | 07/01/2014 | per statute mile                                          |
| Source file                                                                                        |                                                                                                                                          |                 |      |            | https://secureapp.dhs.state.ia.us/MedicaidFeeSchedC11.csv |
| PA rule                                                                                            | No PA; non-emergency requires bed-confinement or equivalent necessity criteria (Ambulance Services Provider Manual Ch. III, 1 Oct 2015). |                 |      |            |                                                           |
| Broker carve                                                                                       | YES - NEMT broker Access2Care renamed MTM Health 18 Sep 2025 (IA Health Link MCOs); MTM administers FFS NEMT.                            |                 |      |            |                                                           |
| Dialysis policy                                                                                    | Dialysis NEMT rides schedulable 90 days ahead; ambulance claim destination modifiers G/J are dialysis facilities.                        |                 |      |            |                                                           |

| Panel C. Kansas - KMAP TXIX fee schedule files, 7/1/2026 versions (FeeSchedule_20260701_TXIX_AMB.txt |                                                                                                                |                  |      |            |                                                                                               |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------|------|------------|-----------------------------------------------------------------------------------------------|
| HCPCS                                                                                                | Service level                                                                                                  | Rate \$          | Unit | Rate eff.  | In-row note                                                                                   |
| A0428                                                                                                | BLS non-emergency                                                                                              | \$255.70 base    |      | 07/01/2023 | SFY2024 rebase                                                                                |
| A0429                                                                                                | BLS emergency                                                                                                  | \$409.12 base    |      | 07/01/2023 |                                                                                               |
| A0426                                                                                                | ALS1 non-emergency                                                                                             | \$306.84 base    |      | 07/01/2023 |                                                                                               |
| A0427                                                                                                | ALS1 emergency                                                                                                 | \$485.83 base    |      | 07/01/2023 |                                                                                               |
| A0433                                                                                                | ALS2                                                                                                           | \$703.18 base    |      | 07/01/2023 |                                                                                               |
| A0434                                                                                                | Specialty Care Transport                                                                                       | \$831.03 base    |      | 07/01/2023 | Ambulance rate-type file: legacy \$3.00 row (eff 07/01/2022) persists on the general schedule |
| A0425                                                                                                | Ground mileage                                                                                                 | \$13.20 per mile |      | 07/01/2023 |                                                                                               |
| Source file                                                                                          |                                                                                                                |                  |      |            | https://portal.kmap-state-ks.us/PublicPage/ProviderPricing/FeeSchedules                       |
| PA rule                                                                                              | No PA, but ALL nonemergent transports need a Medical Necessity form attached to the claim (KMAP manual).       |                  |      |            |                                                                                               |
| Broker carve                                                                                         | YES - KanCare is near-total managed care; NEMT via MCO broker ModivCare (taxi to stretcher van and ambulance). |                  |      |            |                                                                                               |
| Dialysis policy                                                                                      | Dialysis named among critical services eligible for NEMT in the Kansas NEMT provider manual.                   |                  |      |            |                                                                                               |

| Panel D. Missouri - MHD Ambulance.xlsx, file dated 7 Jul 2026 (provider files updated 07/08/2026) |                                                                                        |                 |      |            |                                                                              |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------|------|------------|------------------------------------------------------------------------------|
| HCPCS                                                                                             | Service level                                                                          | Rate \$         | Unit | Rate eff.  | In-row note                                                                  |
| A0428                                                                                             | BLS non-emergency                                                                      | \$105.62 base   |      | 07/01/2019 | HH and HD (hospital-to-diagnostic) contexts ONLY; unmodified code noncovered |
| A0429                                                                                             | BLS emergency                                                                          | \$379.69 base   |      | 07/01/2024 |                                                                              |
| A0426                                                                                             | ALS1 non-emergency                                                                     | \$172.26 base   |      | 07/01/2019 |                                                                              |
| A0427                                                                                             | ALS1 emergency                                                                         | \$474.75 base   |      | 07/01/2024 |                                                                              |
| A0433                                                                                             | ALS2                                                                                   | \$215.20 base   |      | 07/01/2019 |                                                                              |
| A0434                                                                                             | Specialty Care Transport                                                               | NONCOVERED base |      | 07/01/2002 | HH (hospital-to-hospital) modifier ONLY; unmodified code noncovered          |
| A0425                                                                                             | Ground mileage                                                                         | \$7.04 per mile |      | 07/01/2024 |                                                                              |
| Source file                                                                                       |                                                                                        |                 |      |            | NONCOVERED under MO HealthNet FFS                                            |
| PA rule                                                                                           | PA = No on all covered rows; A0428 billable ONLY with HH or HD modifier (MHD hot tip). |                 |      |            | per statute mile                                                             |
| Broker carve                                                                                      | YES - statewide NEMT broker MTM (mydss.mo.gov/mhd/nemt).                               |                 |      |            | https://apps.dss.mo.gov/fms/feeschedules/dfiles/Ambulance.xlsx               |

Dialysis policy

Dialysis runs are NOT billable as FFS A0428 (H+H/D only); scheduled dialysis travel is an MTM broker trip.

| Panel E. Ohio - Appendix to OAC rule 5160-15-28 (file dated 22 Dec 2023), rule effective 1 Jan 202 |                                                                                                                                                                     |                 |      |            |                  |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|------------------|
| HCPCS                                                                                              | Service level                                                                                                                                                       | Rate \$         | Unit | Rate eff.  | In-row note      |
| A0428                                                                                              | BLS non-emergency                                                                                                                                                   | \$203.75 base   |      | 01/01/2024 |                  |
| A0429                                                                                              | BLS emergency                                                                                                                                                       | \$244.00 base   |      | 01/01/2024 |                  |
| A0426                                                                                              | ALS1 non-emergency                                                                                                                                                  | \$244.50 base   |      | 01/01/2024 |                  |
| A0427                                                                                              | ALS1 emergency                                                                                                                                                      | \$289.75 base   |      | 01/01/2024 |                  |
| A0433                                                                                              | ALS2                                                                                                                                                                | \$349.50 base   |      | 01/01/2024 |                  |
| A0434                                                                                              | Specialty Care Transport                                                                                                                                            | \$413.00 base   |      | 01/01/2024 |                  |
| A0425                                                                                              | Ground mileage                                                                                                                                                      | \$5.05 per mile |      | 01/01/2024 | per statute mile |
| Source file                                                                                        |                                                                                                                                                                     |                 |      |            |                  |
| PA rule                                                                                            | No PA in rule; necessity per OAC 5160-15-23(B); EMT care, oxygen, or supervised restraint during transport, or qualifying hospital-to-hospital transfer.            |                 |      |            |                  |
| Broker carve                                                                                       | NO statewide broker - county CDJFS NET programs (5160-15-10) plus MCO rides; ambulance stays a direct ODM/MCO claim. GEMT supplemental for public EMS (5160-15-30). |                 |      |            |                  |
| Dialysis policy                                                                                    | Rules retrieved are silent on dialysis transport.                                                                                                                   |                 |      |            |                  |

[https://codes.ohio.gov/assets/laws/administrative-code/pdfs/5160/015/5160-15-28\\_PH\\_FF\\_A\\_APP1\\_20231222\\_0942.pdf](https://codes.ohio.gov/assets/laws/administrative-code/pdfs/5160/015/5160-15-28_PH_FF_A_APP1_20231222_0942.pdf)

| Panel F. Wisconsin - ForwardHealth AMBULANCE MAXIMUM ALLOWABLE FEE SCHEDULE, live portal report accesse |                                                                                                                                                                         |                 |      |            |                                   |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|-----------------------------------|
| HCPCS                                                                                                   | Service level                                                                                                                                                           | Rate \$         | Unit | Rate eff.  | In-row note                       |
| A0428                                                                                                   | BLS non-emergency                                                                                                                                                       | \$94.90 base    |      | 07/01/2008 | unchanged for 18 years            |
| A0429                                                                                                   | BLS emergency                                                                                                                                                           | \$151.84 base   |      | 07/01/2008 |                                   |
| A0426                                                                                                   | ALS1 non-emergency                                                                                                                                                      | \$113.88 base   |      | 07/01/2008 |                                   |
| A0427                                                                                                   | ALS1 emergency                                                                                                                                                          | \$180.31 base   |      | 07/01/2008 |                                   |
| A0433                                                                                                   | ALS2                                                                                                                                                                    | \$260.97 base   |      | 07/01/2008 |                                   |
| A0434                                                                                                   | Specialty Care Transport                                                                                                                                                | \$308.42 base   |      | 07/01/2008 |                                   |
| A0425                                                                                                   | Ground mileage                                                                                                                                                          | \$5.56 per mile |      | 07/01/2008 | GM multi-patient half-rate \$2.78 |
| Source file                                                                                             |                                                                                                                                                                         |                 |      |            |                                   |
| PA rule                                                                                                 | No code-level PA flag; non-emergency needs ALS/BLS care or supine transport, and manager-coordinated trips need a trip ID within two business days (Update 2014-32).    |                 |      |            |                                   |
| Broker carve                                                                                            | YES - statewide NEMT manager (MTM Inc.; Veyo from 11/2021 until MTM acquisition). Manager-coordinated NEMT ambulance claims are paid by the manager, NOT ForwardHealth. |                 |      |            |                                   |
| Dialysis policy                                                                                         | Recurring dialysis rides are standing rides with the NEMT manager; stretcher vans sit on the separate SMV schedule.                                                     |                 |      |            |                                   |

[https://www.forwardhealth.wi.gov/WiPortal/Tab/42/csscontent/provider/max/fee/pdf/maxfee06\\_ambulance.pdf.spaga](https://www.forwardhealth.wi.gov/WiPortal/Tab/42/csscontent/provider/max/fee/pdf/maxfee06_ambulance.pdf.spaga)

| Panel G. Minnesota - Minnesota MHCP Fee Schedule.xlsx (mn.gov/dhs), downloaded 12 Jul 2026 |                                                                                                                                                                                                                                               |                 |      |            |                                                                                                                 |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|-----------------------------------------------------------------------------------------------------------------|
| HCPCS                                                                                      | Service level                                                                                                                                                                                                                                 | Rate \$         | Unit | Rate eff.  | In-row note                                                                                                     |
| A0428                                                                                      | BLS non-emergency                                                                                                                                                                                                                             | \$284.56 base   |      | 01/01/2026 | highest A0428 in the footprint - exactly the C-2026 Medicare conversion factor (\$284.56), i.e. Medicare parity |
| A0429                                                                                      | BLS emergency                                                                                                                                                                                                                                 | \$455.29 base   |      | 01/01/2026 |                                                                                                                 |
| A0426                                                                                      | ALS1 non-emergency                                                                                                                                                                                                                            | \$341.47 base   |      | 01/01/2026 |                                                                                                                 |
| A0427                                                                                      | ALS1 emergency                                                                                                                                                                                                                                | \$540.66 base   |      | 01/01/2026 |                                                                                                                 |
| A0433                                                                                      | ALS2                                                                                                                                                                                                                                          | \$782.54 base   |      | 01/01/2026 |                                                                                                                 |
| A0434                                                                                      | Specialty Care Transport                                                                                                                                                                                                                      | \$924.82 base   |      | 01/01/2026 | highest SCT in the footprint                                                                                    |
| A0425                                                                                      | Ground mileage                                                                                                                                                                                                                                | \$9.51 per mile |      | 04/01/2026 | per statute mile (ground codes eff 01/01/2026; mileage 04/01/2026)                                              |
| Source file                                                                                |                                                                                                                                                                                                                                               |                 |      |            |                                                                                                                 |
| PA rule                                                                                    | No PA flag on the ambulance rows; MHCP covers emergency and non-emergency ground/air ambulance billed directly to MHCP by enrolled providers (ambulance is not routed through the administered NEMT system).                                  |                 |      |            |                                                                                                                 |
| Broker carve                                                                               | MIXED / STATE-ADMINISTERED (no single MCO broker) - split between local county/tribal agency-administered NEMT and State-Administered NEMT (Acentra Health / Kepro); ambulance itself stays FFS. NEMT providers need MnDOT STS certification. |                 |      |            |                                                                                                                 |
| Dialysis policy                                                                            | Explicit RETURN-LEG rule: for dialysis members eligible for local NEMT, State-Administered NEMT covers the RETURN trip - Kepro enters a single-day certification for dialysis return trips only.                                              |                 |      |            |                                                                                                                 |

[https://mn.gov/dhs/assets/mhcp-fee-schedule\\_tcm1053-293968.xlsx](https://mn.gov/dhs/assets/mhcp-fee-schedule_tcm1053-293968.xlsx)

| Panel H. Indiana - Indiana IHCP Professional Fee Schedule.xlsx (in-session portal download), last upd |                                                                                                                                                                                      |                 |      |            |                                                                        |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------------|------------------------------------------------------------------------|
| HCPCS                                                                                                 | Service level                                                                                                                                                                        | Rate \$         | Unit | Rate eff.  | In-row note                                                            |
| A0428                                                                                                 | BLS non-emergency                                                                                                                                                                    | \$269.02 base   |      | 01/01/2026 | 100% of CY2025 Medicare (BT2025156)                                    |
| A0429                                                                                                 | BLS emergency                                                                                                                                                                        | \$430.44 base   |      | 01/01/2026 |                                                                        |
| A0426                                                                                                 | ALS1 non-emergency                                                                                                                                                                   | \$322.83 base   |      | 01/01/2026 |                                                                        |
| A0427                                                                                                 | ALS1 emergency                                                                                                                                                                       | \$511.14 base   |      | 01/01/2026 |                                                                        |
| A0433                                                                                                 | ALS2                                                                                                                                                                                 | \$739.81 base   |      | 01/01/2026 |                                                                        |
| A0434                                                                                                 | Specialty Care Transport                                                                                                                                                             | NONCOVERED base |      | 04/01/2004 | NONCOVERED (NOCOV) under IHCP                                          |
| A0425                                                                                                 | Ground mileage                                                                                                                                                                       | \$9.15 per mile |      | 01/01/2026 | per statute mile; reduced-modifier variants (U3/U5, MRT/SE) pay \$1.67 |
| Source file                                                                                           |                                                                                                                                                                                      |                 |      |            |                                                                        |
| PA rule                                                                                               | No PA flag on covered ambulance rows; A0434 SCT is Non-Covered (PA=Y, \$0.00). ALS/BLS ambulance is EXEMPT from the FFS NEMT brokerage and billed FFS directly.                      |                 |      |            |                                                                        |
| Broker carve                                                                                          | PARTIAL - non-ambulance FFS NEMT brokered statewide through Verida (formerly Southeastrans); managed care via the MCE. ALS/BLS ambulance is carved OUT of the broker and billed FFS. |                 |      |            |                                                                        |
| Dialysis policy                                                                                       | Recurring dialysis trips are standing orders through the Verida FFS broker (or the MCE); the ambulance schedule is silent at code level.                                             |                 |      |            |                                                                        |

<https://provider.indianamedicaid.com/ihcp/Publications/MaxFee/Professional%20Fee%20Schedule.xlsx>

Panel I. Kentucky - KY Medicaid Transportation Fee Schedule 2026, revised 1/9/2026 (no changes vs prio

|                                                                                                                                                                                                     | HCPCS | Service level            | Rate \$         | Unit | Rate eff.  | In-row note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------|-----------------|------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| A0428                                                                                                                                                                                               |       | BLS non-emergency        | \$55.00 base    |      | 01/09/2026 | NON-EMERGENCY STRETCHER line (PT56 Specialty 16), flat \$55 - a stretcher-van rate, NOT a full BLS ambulance; formerly T2005, renamed A0428 eff. 1/1/2025<br><br>Other (independent) \$60.00 / Hospital-based \$82.50; A0429 GM add-patient \$20<br><br>NOT LISTED - no ALS1 non-emergency line on the KY transportation schedule<br><br>Other (independent) \$60.00 / Hospital-based \$110.00; A0427 GM add-patient \$25<br><br>NOT LISTED - no ALS2 line<br><br>NOT LISTED - no SCT line<br><br>BLS mileage (A0425 UB) Other \$2.50 / Hospital \$3.00; ALS mileage \$2.50/\$4.00; stretcher and return-trip mileage \$2.00<br><br><a href="https://www.chfs.ky.gov/agencies/dms/DMSFeeRateSchedules/2026TransportationRates.pdf">https://www.chfs.ky.gov/agencies/dms/DMSFeeRateSchedules/2026TransportationRates.pdf</a> |  |
| A0429                                                                                                                                                                                               |       | BLS emergency            | \$60.00 base    |      | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| A0426                                                                                                                                                                                               |       | ALS1 non-emergency       | NOT LISTED      | base | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| A0427                                                                                                                                                                                               |       | ALS1 emergency           | \$60.00 base    |      | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| A0433                                                                                                                                                                                               |       | ALS2                     | NOT LISTED      | base | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| A0434                                                                                                                                                                                               |       | Specialty Care Transport | NOT LISTED      | base | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| A0425                                                                                                                                                                                               |       | Ground mileage           | \$2.50 per mile |      | 01/09/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| Source file                                                                                                                                                                                         |       |                          |                 |      |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| PA rule                                                                                                                                                                                             |       |                          |                 |      |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| No code-level PA on the schedule; emergency ambulance (PT55) billed FFS. Only non-emergency ambulance line is the PT56 stretcher; broader NEMT goes through the HSTD regional broker, off-schedule. |       |                          |                 |      |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |

**Broker carve** YES - Human Service Transportation Delivery (HSTD): DMS contracts with the KY Transportation Cabinet, which assigns a regional broker per county; providers contract with the broker, trips booked 72h ahead. Five MCOs administer NEMT for their members.

**Dialysis policy** Recurring dialysis trips route through the HSTD regional broker (or the MCO); the ambulance/stretcherschedule is silent on dialysis.

| HCPCS                   | Service level                                                                                                                                                                                                                     | Rate \$ | Why parked / what fills it                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A0427/A0429             | Emergency codes                                                                                                                                                                                                                   | PENDING | The only public DMS emergency ground file is the 2009 sliding scale (VERIFIED "DOS October 31 2009 and Before"): 1-5 mi \$75, 6-10 mi \$150, then +\$2.50/mi. No current-effective FFS rate file is published; the DMS procedure fee file search portal (interactive per-code lookup) would fill it. |
| A0428                   | BLS non-emergency                                                                                                                                                                                                                 | PENDING | Not payable on a public FFS schedule: DMS Transportation manual Ch.5 (rev 12/5/2025) requires A0428 to be preauthorized and billed to the FFS NEMT broker.                                                                                                                                           |
| A0425/A0426/A0433/A0434 | Mileage / ALS / SCT                                                                                                                                                                                                               | PENDING | Appear only on the 2009-2012 crossover rate table "for crossover calculation only"; no current standalone FFS rate is published.                                                                                                                                                                     |
| Source (verified)       | <a href="https://www.dmas.virginia.gov/media/1569/fee-schedules-for-emergency-ground-ambulance-a0427-and-a0429.pdf">https://www.dmas.virginia.gov/media/1569/fee-schedules-for-emergency-ground-ambulance-a0427-and-a0429.pdf</a> |         |                                                                                                                                                                                                                                                                                                      |

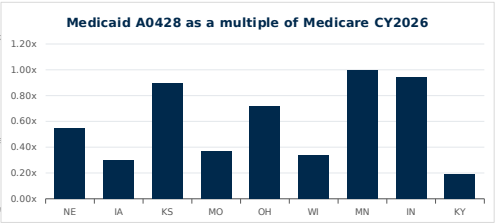
| Panel K. Comparison - each state as a multiple of the CY2026 Medicare national unadjusted base (live formulas)                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Medicare column is a GREEN link to Derived_Rate_Card (located by scanning that tab for the code labels; base codes column D, mileage cell B23). |

| HCPCS (service)                                       | Medicare CY26 \$ | NE    | IA    | KS    | MO    | OH    | WI    | MN    | IN    | KY    | Confound printed with the ratio                                                                                   |
|-------------------------------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------------------------------------------------------------------------------------------------------------------|
| A0428 (BLS non-emergency)                             | \$284.56         | 0.54x | 0.30x | 0.90x | 0.37x | 0.72x | 0.33x | 1.00x | 0.95x | 0.19x | IA lowest true-BLS floor, MN at Medicare parity (1.00x); KY excluded (its A0428 is a stretcher-van line, not BLS) |
| A0429 (BLS emergency)                                 | \$455.30         | 0.42x | 0.25x | 0.90x | 0.83x | 0.54x | 0.33x | 1.00x | 0.95x | 0.13x | Same-code comparison; state schedules do not adjust for geography the way Medicare does                           |
| A0426 (ALS1 non-emergency)                            | \$341.47         | 1.13x | 0.30x | 0.90x | 0.50x | 0.72x | 0.33x | 1.00x | 0.95x | n/a   | MO A0426 is HH-modifier context; KY has no ALS1-nonE line                                                         |
| A0427 (ALS1 emergency)                                | \$540.66         | 0.72x | 0.23x | 0.90x | 0.88x | 0.54x | 0.33x | 1.00x | 0.95x | 0.11x | Same-code comparison; state schedules do not adjust for geography the way Medicare does                           |
| A0433 (ALS2)                                          | \$782.54         | 0.49x | 0.30x | 0.90x | 0.28x | 0.45x | 0.33x | 1.00x | 0.95x | n/a   | KY has no ALS2 line (n/a)                                                                                         |
| A0434 (Specialty Care Transport)                      | \$924.82         | 0.42x | 0.25x | 0.90x | n/a   | 0.45x | 0.33x | 1.00x | n/a   | n/a   | MO and IN NONCOVERED - no ratio; KY has no SCT line                                                               |
| A0425 (Ground mileage)                                | \$9.15           | 0.69x | 0.29x | 1.44x | 0.77x | 0.55x | 0.61x | 1.04x | 1.00x | 0.27x | KS mileage from the ambulance rate-type file, not the legacy \$3.00 row; KY mileage is the BLS/Other rate         |
| Footprint A0428 range - LOWEST multiple (Iowa)        |                  |       | 0.30x |       |       |       |       |       |       |       | min over the eight true-BLS A0428 ratios (KY stretcher excluded)                                                  |
| Footprint A0428 range - HIGHEST multiple (Minnesota)  |                  |       | 1.00x |       |       |       |       |       |       |       | max over the eight true-BLS A0428 ratios; MN = Medicare parity                                                    |
| Dispersion: highest / lowest A0428 rate               |                  |       | 3.36x |       |       |       |       |       |       |       | MN pays about 3.4x what IA pays for the same code                                                                 |
| Memo - KY non-emergency STRETCHER (not BLS ambulance) |                  | 0.19x |       |       |       |       |       |       |       |       | A0428 stretcher-van line (formerly T2005), \$55 flat - a different product; shown for context, excluded from FFS  |

| State                                                         | Carve? | Broker / manager                               | What is carved                                                                                     |
|---------------------------------------------------------------|--------|------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Nebraska                                                      | YES    | MTM / ModivCare (MTM Health from 1/1/26)       | NEMT via Heritage Health MCO brokers; FFS NET by DHHS staff                                        |
| Iowa                                                          | YES    | MTM Health (was Access2Care to 9/18/25)        | NEMT for MCO and FFS members; ambulance stays FFS-billable                                         |
| Kansas                                                        | YES    | ModivCare (KanCare MCOs)                       | NEMT incl. stretcher van and NEMT ambulance coordination                                           |
| Missouri                                                      | YES    | MTM (statewide)                                | All scheduled NEMT; FFS ambulance only in HH/HHD contexts                                          |
| Ohio                                                          | NO     | County CDJFS NET + MCO vendors                 | No statewide broker; ambulance is a direct ODM/MCO claim                                           |
| Wisconsin                                                     | YES    | MTM Inc. (NEMT manager; Veyo 2021-22)          | Manager-coordinated NEMT ambulance is paid by the MANAGER, not ForwardHealth                       |
| Minnesota                                                     | YES    | Acentra/Kepro (State-Admin) + county/tribal    | State-administered + local-agency NEMT (no single MCO broker); ambulance stays FFS                 |
| Indiana                                                       | YES    | Veridia (formerly Southeastrans)               | Non-ambulance FFS NEMT brokered; ALS/BLS ambulance EXEMPT and billed FFS                           |
| Kentucky                                                      | YES    | HSTD regional brokers (via KY Transp. Cabinet) | NEMT via county regional brokers; emergency ambulance + PT56 stretcher billed FFS                  |
| Broker/administered-carve states of the nine footprint states | 8      |                                                | live count over the column above (OH is the only NO)                                               |
| Memo - Virginia (extension state)                             | YES    | DMAS FFS NEMT broker                           | Broker pays FFS NON-EMERGENCY AMBULANCE itself; no public FFS price for scheduled ambulance exists |

**Read panel**

What is measured: nine published state Medicaid FFS ambulance schedules, fetched and parsed from the primary file or portal (NE/IA/KS/MO/OH/WI on 11 Jul 2026; MN/IN/KY on 12 Jul 2026); Virginia is parked because its only public rate files are the 2009 / 2009-2012 vintage. The BLS non-emergency base (A0428), the workforce code of scheduled interfacility transport, ranges from \$84.67 (IA, unchanged since 2014) to \$284.56 (MN, exactly the Medicare conversion factor - full parity) - roughly 0.30x to 1.00x the CY2026 Medicare national base, a 3.4x spread, with WI frozen at \$94.90 since 2008 and IN pinned to 100% of CY2025 Medicare. Kentucky is the outlier: it has NO ALS1-nonemergency, ALS2, or SCT line at all, and its only non-emergency ground line is a \$55 stretcher-van rate (formerly T2005), so it is excluded from the BLS comparison. Eight of the nine states route scheduled NEMT through a broker, transportation manager, or a state-administered system (only OH runs county NET instead), and Missouri will not pay A0428 outside HH/HHD contexts nor SCT at all (IN also does not cover SCT). What is NOT measured: managed-care negotiated rates, broker-paid trip prices, and GEMT supplemental payments - the real price of most scheduled Medicaid transport is set inside those channels and is not public.





State TAM assembly: the same row math at state grain, reconciled to the carried top-down model

Run the metro row math at state grain for the ten footprint states plus a national roll-up - does the bottom-up assembly reconcile to TAM\_Model\_National, and where exactly do the two differ? Sources: CMS HSA 2025 (cases by CCN, state = SSA CCN prefix), Acute\_IFT\_Series rate, Derived\_Rate\_Card CY2026 base, Insourcing\_Bounds Panel C state shares. Join key: CCN prefix -> state. The bridge panel lists every difference with green links to both sides; the SAM filter row states its grouping rule.

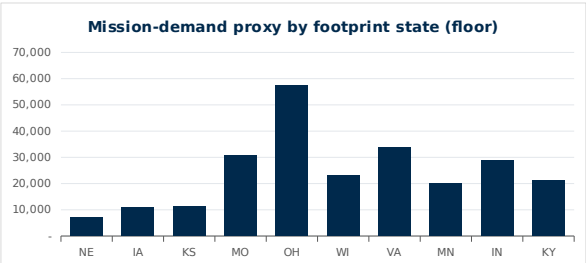
DATA QUALITY: state volume is the same Medicare FFS HSA proxy as the metro panels (floor of all-payer admissions; VA/psych/children's hospitals absent). The transfer rate is national, not state-specific - state case mix and rurality shift the true rate and that confound is printed on every state row. Insourcing shares are name-rule bounds, not a census. Nothing on this tab is a TAM; the national roll-up is labeled the proxy floor it is.

| Panel A. The shared spine (green anchors used by every state row) |       |                                                                                                                                              |
|-------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Measured transfer rate, mean IFT % of admissions 2019-2023        | 9.49% | Denominator: AHA all-payer community-hospital admissions (HCUP NEDS+NIS numerator) - disclosed here and applied to Medicare-only cases below |
| Medicare-anchored \$ per transport (A0428 base, CY2026)           | \$285 | Base only; mileage excluded (43-45% of Medicare IFT allowed \$)                                                                              |

| Panel B. State-grain assembly (footprint states + national roll-up) |                          |                      |                            |                        |                                                                        |      |
|---------------------------------------------------------------------|--------------------------|----------------------|----------------------------|------------------------|------------------------------------------------------------------------|------|
| State                                                               | HSA cases (Medicare FFS) | Mission-demand proxy | Medicare-anchored \$ floor | Insourcing floor share | Insourcing ceiling share                                               | Note |
| NE                                                                  | 74,208                   | 7,039                | \$2,003,018                | 0.0%                   | 18.5% CCN prefix "28"; national rate applied - state case mix confound |      |
| IA                                                                  | 113,036                  | 10,722               | \$3,051,060                | 1.5%                   | 30.2% CCN prefix "16"; national rate applied - state case mix confound |      |
| KS                                                                  | 120,270                  | 11,408               | \$3,246,320                | 0.6%                   | 20.3% CCN prefix "17"; national rate applied - state case mix confound |      |
| MO                                                                  | 321,719                  | 30,517               | \$8,683,818                | -                      | 14.4% CCN prefix "26"; national rate applied - state case mix confound |      |
| OH                                                                  | 606,331                  | 57,514               | \$16,366,047               | 0.1%                   | 20.6% CCN prefix "36"; national rate applied - state case mix confound |      |
| WI                                                                  | 242,525                  | 23,005               | \$6,546,219                | 3.3%                   | 27.3% CCN prefix "52"; national rate applied - state case mix confound |      |
| VA                                                                  | 356,673                  | 33,832               | \$9,627,295                | 2.9%                   | 30.1% CCN prefix "49"; national rate applied - state case mix confound |      |
| MN                                                                  | 212,018                  | 20,111               | \$5,722,776                | 26.6%                  | 67.7% CCN prefix "24"; national rate applied - state case mix confound |      |
| IN                                                                  | 301,874                  | 28,634               | \$8,148,163                | 1.2%                   | 37.7% CCN prefix "15"; national rate applied - state case mix confound |      |
| KY                                                                  | 223,647                  | 21,214               | \$6,036,665                | 2.5%                   | 35.0% CCN prefix "18"; national rate applied - state case mix confound |      |
| FOOTPRINT (10 states)                                               | 2,572,301                | 243,996              | \$69,431,382               |                        | provisional footprint; matches Insourcing_Bounds Panel C               |      |
| NATIONAL roll-up (all CCNs in the HSA file)                         | 14,144,025               | 1,341,631            | \$381,774,607              | 1.4%                   | 27.0% the bottom-up side of the bridge below                           |      |

| Panel C. Bridge: this bottom-up assembly vs the carried top-down scenario model (row per difference, green links both sides) |                           |                                                          |                 |                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Difference                                                                                                                   | Bottom-up side (this tab) | Top-down side (TAM_Model_National)                       | Direction       | Why (stated, public)                                                                                                                                                           |
| Segment scope                                                                                                                | \$381,774,607             | \$20,595,460,675                                         | top-down higher | The model prices ALL reimbursed ground ambulance (911 + IFT, five payer books); this assembly prices acute-to-acute IFT mission demand only                                    |
| The model's own scheduled/IFT segment                                                                                        | \$381,774,607             | \$12,973,695,487                                         | top-down higher | Model segment = Medicare CODE mix (non-emergency + SCT + ALS2) applied across payers; this tab = HCUP transfer EPISODE definition - different segment definitions, never mixed |
| Volume basis and payer scope                                                                                                 | 14,144,025                | 41.3                                                     | assembly lower  | Bottom-up counts Medicare FFS inpatient CASES (HSA); top-down builds all-payer TRANSPORTS from lives (top-down cell is in millions) - payer scope and unit both differ         |
| Price basis                                                                                                                  | \$285                     | \$475                                                    | assembly lower  | A0428 base only vs measured allowed per transport incl mileage (mileage share green: see Medicare_IFT_Series)                                                                  |
| Episode vs leg vs billed-transport counting                                                                                  | 3,061,156                 | 5,510,664                                                | differs by unit | HCUP counts sending-side episodes; NEMSIS counts vehicle legs; claims count billed transports - one transfer can appear once, twice or zero times depending on the counter     |
| Gross-up layers (unbilled; facility direct-pay)                                                                              | PENDING                   | not in the model (reimbursed both understate spend only) |                 | No public IFT-specific volume share exists for either layer; anchors live on Facility_Pay_Layer; PENDING GADCS microdata / claims-vendor panel (Claims_Vendor_Recv)            |

| Panel D. SAM filter: multi-hospital-system share of the footprint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |  |         |                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grouping rule (printed): a system family = hospitals sharing a health-system crosswalk ID (AHRQ Compendium of US Health Systems); interim name-family rule = shared leading corporate                                                                                                                                                                                                                                                                                                                                                            | PENDING |  | PENDING | PENDING dataset: AHRQ Compendium hospital-to-system crosswalk (built on the AHA Annual Survey license); Hosp_Registry carries name/city/state/ownership but no system field, so the share is not computed from a weaker proxy |
| Read panel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |  |         |                                                                                                                                                                                                                               |
| What the bridge shows: the bottom-up assembly and the top-down model are different instruments and the panel prints why, difference by difference - segment scope (IFT mission demand vs all ground), payer scope (Medicare FFS proxy vs five payer books), price basis (base-only vs measured allowed including mileage), counting unit (episodes vs legs vs billed transports), and the gross-up layers both sides leave unpriced. The assembly is the auditable floor; the model is the scenario envelope; neither is quoted as a single TAM. |         |  |         |                                                                                                                                                                                                                               |



Scenario matrix: three volume bases x three price bases, live where measured, PENDING where not

Across volume bases (claims-visible / NEMSIS-anchored / gross-up-full) and price bases (Medicare-anchored / blended-observed / MRF-forward), which sizing cells are live from public evidence today, and what would close each gate? Sources: PSPS-derived Medicare\_IFT\_Series (claims floor), NEMSIS 2024 (EMS\_Transports), Insourcing\_Bounds, Facility\_Pay\_Layer, TAM\_Model\_National. No cell of this grid is "the TAM"; the grid IS the answer, and its gates are printed below it.

DATA QUALITY: the claims-visible basis is Medicare FFS carrier claims only (hospital institutional billing excluded); the NEMSIS basis counts all-payer vehicle legs from a voluntary-participation registry (54 states/territories, coverage grew over time); gross-up volume shares have no public source and stay PENDING. Blended and MRF price bases await Medicaid\_Rate\_Card, Contract\_Corpus rate extraction and Transparency-in-Coverage files. Mixing any cell across rows or columns re-introduces the exact double-counting this grid exists to prevent.

| Panel A. The bases (each a green link to its measured cell, or PENDING naming its dataset) |           |                                                                                                                                                                                                                               |                                                                                      |
|--------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Basis                                                                                      | Value     | Source / dataset named                                                                                                                                                                                                        | Note                                                                                 |
| V1 claims-visible volume: Medicare FFS hospital-to-hospital transpor                       | 871,753   | Medicare_IFT_Series 2024 row (PSPS-derived carrier claims)                                                                                                                                                                    | the hardest, fully measured volume                                                   |
| V2 NEMSIS-anchored volume: hospital-to-hospital activations, 2024                          | 5,510,664 | EMS_Transports v3.5 type-of-service table (NEMSIS 2024 End-of-Year report)                                                                                                                                                    | all-payer vehicle LEGS, not billed transports                                        |
| V3 gross-up-full volume: claims-visible + unbilled + facility-paid                         | PENDING   | PENDING datasets: IFT-specific unbilled volume share (GADCS microdata beyond the published one-in-five incidence) + facility-pay volume share (claims-vendor panel, Claims_Vendor_Recv; NEMSIS_State_IFT for the state wedge) | anchors are live on Facility_Pay_Layer; the SHARES are not public                    |
| P1 Medicare-anchored price: measured allowed per transport, 2024 (incl mileage)            | \$662     | Medicare_IFT_Series 2024 row, allowed per transport                                                                                                                                                                           | reference only: A0428 base-only anchor used at metro grain sits on Derived_Rate_Card |
| P2 blended-observed price                                                                  | PENDING   | PENDING datasets: state Medicaid ambulance fee schedules (Medicaid_Rate_Card_B.3) + observed public contract rates                                                                                                            | (blend weights would also need payer-mix evidence)                                   |
| P3 MRF-forward price                                                                       | PENDING   | PENDING dataset: Transparency-in-Coverage negotiated-rate machine-readable files (landing shape Commercial_Rate_MRF, shipped as an empty schema)                                                                              |                                                                                      |

| Panel B. The 3x3 grid (LIVE formula where both bases exist; PENDING names the missing input) |                         |                  |             |                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------|-------------------------|------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume basis \ price basis                                                                   | Medicare-anchored       | Blended-observed | MRF-forward | Gate / note                                                                                                                                                                                                                                             |
| Claims-visible                                                                               | \$577,228,322 PENDING   | PENDING          | PENDING     | THE HARDEST FLOOR: fully measured (equals 2024 Medicare FFS IFT allowed dollars) applies the Medicare price to ALL-PAYER legs - a scale read under gate G2, NOT a TAM PENDING: unbilled + facility-pay volume shares (datasets named in Panel A row V3) |
| NEMSIS-anchored                                                                              | \$3,648,867,667 PENDING | PENDING          | PENDING     |                                                                                                                                                                                                                                                         |
| Gross-up-full                                                                                | PENDING                 | PENDING          | PENDING     |                                                                                                                                                                                                                                                         |
| Gate G1: NEMSIS x Medicare cell contains the claims floor                                    | PASS                    |                  |             | containment gate: legs must be at least billed transports                                                                                                                                                                                               |
| Gate G2: NEMSIS legs vs claims transports (volume containment)                               | PASS                    |                  |             | if this fails, no scenario arithmetic is run on this grid                                                                                                                                                                                               |
| Honesty metric: LIVE grid cells (of 9)                                                       |                         | 2                |             | PENDING cells = 9 minus this count; each names its dataset                                                                                                                                                                                              |

| Panel B2 (4.3). Two price columns per price basis: BASE-ONLY vs MILEAGE-LOADED, Medicare-anchored (both live formulas; loading derived on Derived_Rate_Card) |                           |                                |                    |                         |                       |                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------|--------------------|-------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume basis                                                                                                                                                 | Base-only price/transport | Mileage-loaded price/transport | Base-only floor \$ | Mileage-loaded floor \$ | Loading (loaded/base) | Note                                                                                                                                                                  |
| V1 claims-visible (Medicare FFS, 2024)                                                                                                                       | \$367                     | \$662                          | \$319,815,284      |                         | \$577,228,322         | 1.80 base-only omits A0425 mileage (43-45% of allowed \$); the loaded column is the measured total per transport. Metro_TAM_Panels and TAM_Assembly_State carry both. |
| V2 NEMSIS-anchored (all-payer legs, 2024)                                                                                                                    | \$367                     | \$662                          | \$2,021,667,342    |                         | \$3,648,867,667       | 1.80 same two prices applied to the NEMSIS leg count                                                                                                                  |

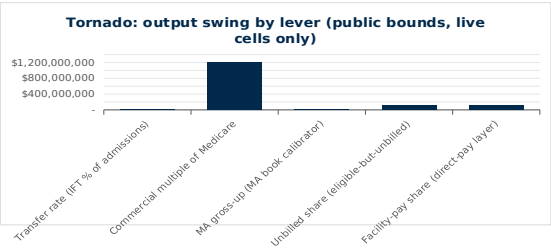
4.3: every dollar cell now ships as a base-only / mileage-loaded PAIR. The base-only column multiplies the measured A0428-anchored base allowed per transport (Derived\_Rate\_Card B34); the mileage-loaded column multiplies the measured total allowed per transport (base + A0425 mileage). The loading factor is live from Derived\_Rate\_Card, where its two-way derivation (MMT book and national registry) is printed.

| Panel C. Tornado: five levers, public bounds only, swings live against the grid and assembly cells |            |           |            |                                                                                                                                                                                                                       |                 |                                                                                                            |
|----------------------------------------------------------------------------------------------------|------------|-----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------|
| Lever                                                                                              | Base value | Low bound | High bound | Public source of the bounds                                                                                                                                                                                           | Output swing \$ | Swings which output / confound                                                                             |
| Transfer rate (IFT % of admissions)                                                                | 9.49%      | 9.24%     | 9.85%      | HCUP NEDS+NIS / AHA annual ratio band, 2019-2023 (Acute_IFT_Series min/max rows)                                                                                                                                      | \$4,426,562     | swings the bottom-up FOOTPRINT assembly (TAM_Assembly_State), not the claims floor - the floor is measured |
| Commercial multiple of Medicare                                                                    | 1.80x      | 1.60x     | 2.00x      | Low = pinned cross-year FAIR Health ground multiple (Payer_Rates_Commercial); high = the model's published-bounds high (TAM_Model_National assumption register)                                                       | \$1,193,320,088 | swings the top-down commercial book at base volume (TAM_Model_National Panel F)                            |
| MA gross-up (MA book calibrator)                                                                   | 1.04x      | 1.00x     | 1.04x      | High = GADCS Table 2.30 MA-to-FFS revenue per NPI, 1.038x (Facility_Pay_Layer calibrator); low = fee-schedule parity (TAM_Model_National MA-multiple low)                                                             | \$21,983,181    | MA-book adder bound on the claims floor; a per-NPI revenue ratio, not a price index - confound printed     |
| Unbilled share (eligible-but-unbilled)                                                             | 20.0%      | -         | 20.0%      | GADCS Table S.1 "about one in five" organizations did not always bill (Facility_Pay_Layer); low = definitional zero (all bill)                                                                                        | \$115,445,664   | incidence of ORGANIZATIONS used as an upper VOLUME bound - stated, not measured                            |
| Facility-pay share (direct-pay layer)                                                              | 18.6%      | -         | 18.6%      | GADCS Table 2.32 facility-contract incidence 18.6% (Facility_Pay_Layer Panel A); DocGo books about 41% of transport revenue outside per-trip claims - existence at one operator, not a bound; low = definitional zero | \$107,364,468   | incidence used as an upper bound on the claims floor; the IFT-specific share has no public source          |

Tornado read: swings are LIVE formulas against the grid floor cell (B16), the footprint assembly and the carried model - never against a typed number. A lever without public bounds would stay a PENDING row; the chart renders only when every lever row is live.

Read panel

The grid is the answer. The one fully measured cell - claims-visible volume x the measured Medicare allowed per transport - is the hardest floor, and it equals 2024 Medicare FFS interfacility allowed dollars by construction. The NEMSIS-anchored row shows the all-payer volume scale under gate G2. Every other cell is PENDING and names what closes it: Medicaid\_Rate\_Card and Contract\_Corpus rate extraction (blended-observed), Transparency-in-Coverage files (MRF-forward), and GADCS microdata or a claims-vendor panel (gross-up-full). No cell of this grid is "the TAM".



Investor QA: twelve tearsheets, every number a live cell or a named PENDING slot

The twelve questions an investor asks about this market, each answered in three sentences of plain words, wired to 2-4 live green cells on the home tabs, with the guardrail printed under every block. Sources: the home tabs named per block (this tab wires; it does not create). Join keys: none - green links only.

DATA QUALITY: answers inherit every caveat of their home tabs; where a home tab lands in a parallel module the number row is a bordered PENDING slot that names it, and the link goes live at assembly order. Nothing in this tab is an assertion without a cell behind it.

| Tearsheets wired (live count of Q blocks below) | Count |                          |
|-------------------------------------------------|-------|--------------------------|
|                                                 | 12    | the twelve banners below |

Q1. Is IFT the same business as 911 or NEMT?

THE ANSWER IN THREE SENTENCES: Interfacility transport is scheduled clinical logistics between licensed facilities, paid per transport under the same Medicare fee schedule as 911 but with a different code and acuity mix. The hospital-to-hospital channel runs about 31 percent emergency while residence and scene channels run above 98 percent, and nearly all specialty care transport lives in the facility channels. NEMT is a separate Medicaid benefit of brokered vans and sedans that sometimes shares vehicles but not the ambulance fee schedule.

|                                                             |                                                                     |                           |
|-------------------------------------------------------------|---------------------------------------------------------------------|---------------------------|
| SCT services in the hospital-to-hospital channel, 2024      | 57,906                                                              | Acuity_by_Channel Panel A |
| Hospital-to-hospital emergency share, 2024                  | 31.0%                                                               | Acuity_by_Channel Panel B |
| Hospital-to-SNF BLS non-emergency services, 2024            | 1,221,981                                                           | Acuity_by_Channel Panel A |
| The institutional claims fingerprint (revenue centers 054x) | Ambulance (category name for the full 054x range, verbatim from the | Code_Vocabulary section B |

THE GUARDRAIL: Channel shares are Medicare FFS origin-destination coded services only; MA, Medicaid and commercial mixes are not publicly published at this grain.

Q2. Why do claims undercount this market, and by how much?

THE ANSWER IN THREE SENTENCES: Claims-based sizing sees only transports that become payer claims, and a measured share of ambulance revenue never does. Three quarters of ambulance organizations report some non-payer revenue, about one in five did not always bill at least one payer category, and a listed pure-play operator books roughly 41 percent of transport revenue outside per-trip claims. The activation-versus-claims reconciliation prices the wedge; the IFT-specific undercount inside it has no public number and stays PENDING.

|                                                          |            |                                      |
|----------------------------------------------------------|------------|--------------------------------------|
| Organizations reporting any non-payer revenue (GADCS)    | 74.5%      | Facility_Pay_Layer Panel A           |
| Listed-operator revenue outside per-trip billing, FY2025 | 40.9%      | Facility_Pay_Layer Panel B           |
| Five-universe reconciliation (activations vs claims)     | 60,298,684 | Universe_Reconciliation (title pull) |

THE GUARDRAIL: The wedge is a mixed pool of unbilled, facility-paid and non-FFS volume; no public source splits it, so no cell here claims to.

Q3. What does insourced really mean, and how much exists?

THE ANSWER IN THREE SENTENCES: Insourced means the hospital bills the transport itself or carries the ambulance cost center on its own books. Medicare claims bound hospital-billed ground base transports between about 1.4 and 27 percent (a strict-linkage floor against a name-collision ceiling), and the HCRIS panel shows the share of cost-report filers carrying an ambulance cost center. Between the bounds sits the mixed reality of owned services, JVs and contracted units billed under hospital numbers.

|                                                                      |       |                                     |
|----------------------------------------------------------------------|-------|-------------------------------------|
| Hospital-billed share of ground base transports, 2024 FLOOR          | 1.4%  | Insourcing_Bounds Panel B           |
| Same share, 2024 CEILING                                             | 27.0% | Insourcing_Bounds Panel B           |
| Cost-report filers with an ambulance cost center, latest complete FY | 14.3% | HCRIS_Ambulance_CostCenters Panel A |

THE GUARDRAIL: The floor is a floor twice over (roster gaps understate linkage) and the ceiling is padded by place-name collisions; the AHA ambulance-ownership flag would turn bounds into a measured split (AHA\_Recv schema).

Q4. How big is the facility direct-bill shadow?

THE ANSWER IN THREE SENTENCES: Some ambulance revenue is billed to the facility, not a payer, so it never appears in any claims dataset. In the national census 18.6 percent of organizations report facility-contract revenue with a long right tail, and public contracts price the form by the dedicated unit, the hour and the annual subsidy. The corpus proves the channel and its pricing forms; its IFT-specific dollar share is a named PENDING slot.

|                                                              |           |                                                    |
|--------------------------------------------------------------|-----------|----------------------------------------------------|
| Organizations reporting facility-contract revenue (GADCS)    | 18.6%     | Facility_Pay_Layer Panel A                         |
| Facility-contract revenue, conditional mean per organization | \$922,555 | same row: median \$48K prints beside it            |
| Corpus documents with dedicated-unit or unit-hour pricing    | 2         | contract_corpus.json; abstracts on Contract_Corpus |

THE GUARDRAIL: GADCS incidence covers ALL ambulance work, not IFT alone; the IFT-specific facility-pay share stays PENDING on Facility\_Pay\_Layer Panel D by design.

Q5. How much of the market is invisible because of MA?

THE ANSWER IN THREE SENTENCES: Medicare Advantage ambulance claims are not published at provider grain, so the MA half of Medicare is dark to every public claims series. The MA and other share crossed 50 percent of beneficiaries in 2024 and rises every year, mechanically shrinking the FFS window all public series stand on. The GADCS census shows MA revenue per organization at parity or above FFS, so the dark half is not smaller business, just unmeasured.

|                                                   |             |                                 |
|---------------------------------------------------|-------------|---------------------------------|
| MA book calibrator                                | \$1,193,046 | MA_Book_Calibrator (title pull) |
| MA & other share of beneficiaries, 2024           | 50.1%       | Enrollment_ESRD_State Panel A   |
| MA-to-FFS revenue ratio per ambulance NPI (GADCS) | 1.04x       | Facility_Pay_Layer Panel A      |

THE GUARDRAIL: MA encounter data exists at ResDAC under DUA (MA\_Encounter\_Recv schema); until licensed, MA volume is inferred from enrollment and the GADCS revenue census, never from claims.

Q6. What do commercial payers actually pay?

THE ANSWER IN THREE SENTENCES: Commercial allowed amounts run well above Medicare for the same service, and dispersion across states is wide. FAIR Health puts ALS emergency in-network allowed at 758 dollars nationally in 2020, state means span roughly 1.7x, and ground ambulance stayed outside the No Surprises Act. Machine-readable negotiated rates exist in Transparency-in-Coverage MRF files; the receiving schema is built and the pull is the named gap.

|                                                               |         |                                                                                                  |
|---------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------|
| ALS emergency average allowed, in-network, 2020               | \$758   | Payer_Rates_Commercial Panel A (FAIR Health)                                                     |
| State dispersion: high-state over low-state ALS allowed, 2022 | 1.75x   | Payer_Rates_Commercial Panel B                                                                   |
| Negotiated-rate slices (5 payers x footprint states)          | PENDING | Commercial_Rate_MRF receiving schema lands with B.14; TIC MRF slice pull is the named public gap |

THE GUARDRAIL: FAIR Health figures are national/state means from a contributor panel, not a census; negotiated-rate truth arrives only with the MRF slices.

Q7. Where does growth come from without a hockey stick?

THE ANSWER IN THREE SENTENCES: Growth is presented as measured components, never a blended headline: the 65+ base compounds near 3 percent, the current-law price update is 2 percent, and regionalization keeps concentrating sending and receiving. The offsets are equally measured: MA migration shrinks the visible FFS base and the statutory add-ons expire at the end of 2027. Anyone needing one growth number must blend the components and own the blend.

|                                              |                  |                              |
|----------------------------------------------|------------------|------------------------------|
| Demographic component (US 65+ CAGR, live)    | 3.0%             | Growth_Outlook_Shell Panel A |
| Price component (AIF CY2026, live)           | 2.0%             | Growth_Outlook_Shell Panel A |
| MA migration component (MA share 2024, live) | 50.1%            | Growth_Outlook_Shell Panel A |
| Add-on cliff date (per statute)              | 31 DECEMBER 2027 | Growth_Outlook_Shell Panel B |

THE GUARDRAIL: No blended growth number exists in this workbook; the conversion-rate slot on Growth\_Outlook\_Shell is bordered PENDING, not assumed.

Q8. How fragmented is this market, and who do we actually compete with?

THE ANSWER IN THREE SENTENCES: The national biller base is thousands of mostly small organizations, and the top of the distribution holds a modest share, so the real competitive set is local, not national. Concentration only means something per state and per corridor, which is why the workbook prints share panels and HHI at state and NPI grain. Parent roll-ups raise measured concentration, so every printed top share is a floor.

|                                      |                                                                              |                                   |
|--------------------------------------|------------------------------------------------------------------------------|-----------------------------------|
| National concentration (top-10 view) | 931800.0%                                                                    | Fragmentation_National top-10 row |
| State share panels and per-state HHI | Market share: Medicare FFS ambulance billers in the ten footprint states, by | Market_Share_Panels (title pull)  |

THE GUARDRAIL: NPI-grain shares understate parent concentration; roll-up panels print the correction where public ownership is traceable.

Q9. What makes these relationships sticky?

THE ANSWER IN THREE SENTENCES: Public contracts show multi-year initial terms with renewal options, explicit exclusivity or first-call language in a minority of documents, and enforceable SLAs with liquidated damages in the 911 and NEMT forms. Registry presence in the cohort-corridor states churns slowly, with most 2019 billers still present in 2024. This is an evidence panel, not a stickiness score, and the corpus is a convenience sample.

|                                                                 |       |                                     |
|-----------------------------------------------------------------|-------|-------------------------------------|
| Longest observed initial term (months)                          | 120   | Stickiness_Evidence Panel A summary |
| Documents with explicit exclusivity language (live count)       | 3     | Stickiness_Evidence Panel A summary |
| Survivor share of the 2019 org-NPI cohort, cohort states (live) | 84.3% | Stickiness_Evidence Panel C         |

THE GUARDRAIL: Hospital IFT agreements are private and absent from the corpus; churn is registry presence, not contract retention.

Q10. How bad is the labor problem?

THE ANSWER IN THREE SENTENCES: The binding input is EMT and paramedic labor: wages, depth and churn. Over 2019-2024 the compounded payer update roughly matched national EMT median wage growth, so the scissors story is window-dependent, while county-level depth varies by an order of magnitude. Labor risk prices locally: thin-depth counties with hub demand are where service fails first.

|                                                    |          |                              |
|----------------------------------------------------|----------|------------------------------|
| Workforce depth panel                              | \$54,170 | Workforce_Depth (title pull) |
| EMT median wage growth 2019-2024 (live)            | 16.8%    | Workforce_Supply Panel B     |
| Compounded AIF over the same window (live)         | 18.5%    | Workforce_Supply Panel B     |
| Private ambulance employment (QCEW, latest annual) | 171,100  | QCEW_EMS_Employment Panel A  |

THE GUARDRAIL: National medians hide the local market; wage-vs-depth tightness is printed state by state on the workforce tabs, and the 2019-2024 scissors read reverses on other windows.

Q11. What is the reimbursement cliff risk?

THE ANSWER IN THREE SENTENCES: Current-law price growth is the AIF, 2.0 percent for CY2026, and the temporary add-ons of 2, 3 and 22.6 percent expire on 31 December 2027 under current statute. A Medicare-heavy ground book therefore carries a dated statutory cliff inside a normal hold period, largest for super-rural mileage. Input costs are tracked separately; the spread between the AIF and measured input inflation is the margin risk to watch.

|                                           |                  |                               |
|-------------------------------------------|------------------|-------------------------------|
| Latest AIF (CY2026, live)                 | 2.0%             | Payment_Rules                 |
| Add-on expiry (per statute, carried live) | 31 DECEMBER 2027 | Growth_Outlook_Shell Panel B  |
| Input cost index                          | \$3.82           | Input_Cost_Index (title pull) |

THE GUARDRAIL: The cliff is statutory, not speculative: the add-ons lapsed once already in October 2025 before CAA 2026 s.6203 restored them through 2027.

Q12. What regulatory barriers protect incumbents?

THE ANSWER IN THREE SENTENCES: Entry is regulated locally: state licensure, certificates of need in some states, franchise and exclusive-operating-area statutes, and municipal contracts with performance bonds. Ground ambulance sits outside the federal No Surprises Act, and a growing list of states has enacted its own balance-billing limits, reshaping out-of-network economics state by state. Barriers are real but jurisdictional; there is no single national moat number.

|                                                                                                                       |                                                                   |                                     |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------|
| Entry barrier register                                                                                                | 5                                                                 | Entry_Barrier_Register (title pull) |
| State balance-billing pegs                                                                                            | 22                                                                | Balance_Billing_States (title pull) |
| Ground exclusion from the No Surprises Act (statutory anchor, live pull)                                              | Ground ambulance is EXCLUDED from federal out-of-network balance- | Payment_Rules statutory anchors     |
| THE GUARDRAIL: Barrier strength varies by state and municipality; the register prints citations, not a barrier score. |                                                                   |                                     |

**Read panel**

Twelve tearsheets, and every number in them is either a green live cell pulled from its home tab or a bordered PENDING slot that names the public dataset or landing schema that would fill it. Nothing on this tab is an assertion: the tab wires evidence that exists elsewhere in this workbook, and where a home tab lands later in assembly order the slot says so and goes live on the next build.

Slide feed: every planned exhibit mapped to live cells, with an honest evidence status

Status rule: GREEN = the home tab exists with live cells and no named PENDING input; AMBER = the tab exists but a named input is partial or PENDING; RED = the tab or data is absent from the workbook at this build. Statuses recompute at every assembly because this module scans the workbook it is given. Sources: pointers only - this tab creates no evidence.

OPERATIONAL TAB: this feed powers the deck build and the remaining work plan; it does NOT feed the findings ledger, and honest RED/AMBER statuses are the deliverable. Every non-GREEN row names its gap: either a public dataset/receiving schema or the parallel module that lands the tab in assembly order.

| The exhibit register                    |                             |                                     |                                                                                                                                                      |        |                                                                                         |
|-----------------------------------------|-----------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------|
| Planned exhibit                         | Home tab (resolved live)    | Anchor                              | Live pull (green = tab title, proves the tab exists)                                                                                                 | Status | Named gap (every non-GREEN row)                                                         |
| Market definition: IFT vs 911 vs NEMT   | Acuity_by_Channel           | Acuity_by_Channel!C21               | Each channel has its own clinical character, and the two directions of one interface are opposites.                                                  | GREEN  | none - live                                                                             |
| Code vocabulary and claims fingerprint  | Code_Vocabulary             | Code_Vocabulary!C11                 | Code vocabulary: ambulance/IFT codes beyond HCPCS: place of service, revenue centers, condition/value codes, remittance, modifiers, taxonomies, MACs | GREEN  | none - live                                                                             |
| Five-universe reconciliation            | Universe_Reconciliation     | Universe_Reconciliation!A1          | Universe reconciliation: why the national ambulance numbers differ, in one ladder                                                                    | GREEN  | none - live                                                                             |
| TAM floor grid (Medicare-visible floor) | TAM_Model_National          | TAM_Model_National!A1               | National TAM model: United States ground ambulance, reimbursed spend                                                                                 | AMBER  | MA and Medicaid legs need licensed extracts (MA_Encounter_Recv, TAF_Ambulance_Recv)     |
| Metro TAM panel (20 target metros)      | Metro_TAM_Panels            | Metro_TAM_Panel!A1                  | Metro TAM panels: the mission-demand line and its priceable floor, one panel per target metro                                                        | GREEN  | none - live                                                                             |
| Gross-up waterfall (floor to all-payer) | Sizing_Playbook             | Sizing_Playbook!A1                  | Sizing playbook: how the team converges on a defensible market number                                                                                | AMBER  | gross-up legs beyond Medicare FFS carry PENDING inputs (MA encounter, TAF, MRF)         |
| Facility-pay triangulation              | Facility_Pay_Layer          | Facility_Pay_Layer!B8               | The facility-pay layer: ambulance revenue that never touches a payer claim                                                                           | AMBER  | IFT-specific facility-pay share PENDING (claims-vendor panel with facility-remit flags) |
| Insourcing bounds + flow                | Insourcing_Bounds           | Insourcing_Bounds Panel B           | Insourcing bounds: the hospital-billed share of Medicare FFS ground ambulance, floor and ceiling                                                     | AMBER  | AHA ambulance-ownership flag would turn bounds into a split (AHA_Recv)                  |
| HCRIS ambulance cost centers            | HCRIS_Ambulance_CostCenters | HCRIS_Ambulance_CostCenters Panel A | Hospital-run ambulance: the Worksheet A line 95 cost center, FY2019 - FY2024                                                                         | AMBER  | FY2024 year-file not yet published (filing lag)                                         |
| Fragmentation curve                     | Fragmentation_National      | Fragmentation_National!A1           | Fragmentation: the US Medicare ambulance biller universe, 12 vintages 2013-2024                                                                      | GREEN  | none - live                                                                             |
| Biller survival curves                  | Annual_Market_Structure     | Annual_Market_Structure!A1          | Annual market structure, 2013-2024: census, churn, concentration, survival - twelve vintages, one series                                             | GREEN  | none - live                                                                             |
| Market share panels (state grain)       | Market_Share_Panels         | Market_Share_Panels!A1              | Market share: Medicare FFS ambulance billers in the ten footprint states, by vintage                                                                 | GREEN  | none - live                                                                             |
| HHI / concentration bands               | Market_Share_Panels         | Market_Share_Panels Panel C         | Market share: Medicare FFS ambulance billers in the ten footprint states, by vintage                                                                 | GREEN  | none - live                                                                             |
| Cohort corridor map                     | Cohort_Corridors            | Cohort_Corridors!A1                 | Cohort corridors: where the research-cohort systems draw their inpatients, and who is registered there                                               | GREEN  | none - live                                                                             |
| Hub counts and hub share                | Hub_Spoke_Map               | Hub_Spoke_Map Panel B               | Hub and spoke: the measured regionalization of hospital corridor volume                                                                              | GREEN  | none - live                                                                             |
| Whitespace counties                     | County_Whitespace_Screens   | County_Whitespace_Screens!A1        | County whitespace screens: heavy 65+ demand over thin Medicare-billing ambulance supply, ten footprint states                                        | GREEN  | none - live                                                                             |
| Growth decomposition (price x volume)   | Growth_Decomposition        | Growth_Decomposition!A1             | Growth decomposition: price, volume and mix in the national Medicare ground ambulance book, 2013-2024                                                | GREEN  | none - live                                                                             |
| Growth outlook components               | Growth_Outlook_Shell        | Growth_Outlook_Shell Panel A        | Growth outlook shell: measured components only, no blended headline                                                                                  | AMBER  | forward conversion rate is a bordered PENDING slot (AHA flag or claims-vendor panel)    |
| Denial dollars                          | Denial_Economics            | Denial_Economics!A1                 | Denial economics: what the Medicare medical-necessity screen filters out, in dollars, 2010-2024                                                      | GREEN  | none - live                                                                             |



|                                       |                             |                                |                                                                                                                        |       |                                                                                     |
|---------------------------------------|-----------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|
| Wage scissors                         | Workforce_Depth             | Workforce_Supply!B17           | Workforce depth: the EMS labor base of the footprint, and the wage-vs-AIF scissors at quarterly grain                  | GREEN | none - live                                                                         |
| Boarding-cost shelf                   | Throughput_Economics_Public | Throughput_Economics_Public!A1 | Throughput economics, public spine: what a boarded bed and a delayed transfer cost, and where transport ordering lives | GREEN | none - live                                                                         |
| DIDO / transfer delay friction        | Transfer_Delay_Burden       | Transfer_Delay_Burden!A1       | Transfer delay burden: the measured ED-flow environment of the footprint, hospital by hospital                         | GREEN | none - live                                                                         |
| REH conversions and rural closures    | REH_Closure_Flow            | REH_Closure_Flow Panel A/B     | Rural Emergency Hospitals and the closure flow: transfer demand created by statute and by exit                         | GREEN | none - live                                                                         |
| Receiving centers                     | Receiving_Side              | Receiving_Centers!A1           | Independent validation from the receiving hospital's record, 2012 to 2023                                              | GREEN | none - live                                                                         |
| Medicaid rate floor                   | Medicaid_Rate_Card          | Medicaid fee-schedule tab!A1   | Medicaid ground ambulance rate card: footprint states                                                                  | GREEN | none - live                                                                         |
| Balance-billing pegs                  | Balance_Billing_States      | Balance_Billing_States!A1      | State ground-ambulance balance-billing protections: the payment standards, verified statute by statute                 | GREEN | none - live                                                                         |
| Entry barrier register                | Entry_Barrier_Register      | Entry_Barrier_Register!A1      | Entry barrier register: where ambulance market entry is legally gated in the ten footprint states                      | GREEN | none - live                                                                         |
| Input cost index                      | Input_Cost_Index            | Input_Cost_Index!A1            | Input-cost index: measured operating inputs against the Medicare payment update                                        | GREEN | none - live                                                                         |
| Cohort dossiers                       | System_Research_Cohort      | per-system dossier tabs        | System research cohort: what the public record establishes, system by system                                           | GREEN | none - live                                                                         |
| Footprint determination               | Press_Footprint_Registry    | Press_Footprint_Registry!A1    | Press and footprint registry: what issuers announced, and what the subject company's own site claimed over time        | GREEN | none - live                                                                         |
| Prospect screen                       | Prospect_Landscape          | Prospect screen tab!A1         | Landscape screen: additional multi-hospital systems in the footprint, screened by measured transfer demand             | GREEN | none - live                                                                         |
| Scenario grid (no tornado without it) | Scenario_Matrix             | C.3 scenario tab!A1            | Scenario matrix: three volume bases x three price bases, live where measured, PENDING where not                        | GREEN | none - live                                                                         |
| Vendor share stack                    | Vendor_Share_Stack          | Vendor_Share_Stack Panel A     | The vendor-share evidence stack: six labeled legs, none blended                                                        | AMBER | legs 5-6 PENDING: licensed buyer research; claims-vendor panel (Claims_Vendor_Recv) |
| Stickiness evidence panel             | Stickiness_Evidence         | Stickiness_Evidence Panel A    | Stickiness evidence: observed terms, exclusivity language, SLA precedents, registry churn - a panel, NOT a score       | AMBER | hospital IFT agreements are private; corpus is a convenience sample                 |
| Subject-company measured book         | MMT_Medicare_Book           | MMT_Medicare_Book!A1           | Subject-company Medicare FFS ambulance book by NPI, 2013 / 2019 / 2024                                                 | GREEN | none - live                                                                         |
| Investor QA tearsheets                | Investor_QA                 | Investor_QA (12 blocks)        | Investor QA: twelve tearsheets, every number a live cell or a named PENDING slot                                       | GREEN | none - live                                                                         |
| MA invisibility calibrator            | MA_Book_Calibrator          | Imbalance_Ledger!B39           | The MA book calibrator: Medicare Advantage revenue per NPI at 1.04x Medicare FFS (GADCS Table 2.30)                    | AMBER | MA encounter extract under DUA is the named gap (MA_Encounter_Recv)                 |
| Commercial rate benchmarks            | Payer_Rates_Commercial      | Payer_Rates_Commercial!C6      | What everyone other than Medicare fee-for-service pays. Published evidence only.                                       | AMBER | TIC MRF negotiated-rate slices PENDING (Commercial_Rate_MRF schema)                 |

|                             |       |       |     |       |
|-----------------------------|-------|-------|-----|-------|
| Status counts (live)        | GREEN | AMBER | RED | Total |
| Exhibits by evidence status | 28    | 10    | -   | 38    |

Read panel

This tab is the deck feed and the remaining work plan in one: every planned exhibit of the study points at its home tab and anchor, GREEN rows are ready to render, AMBER rows render with their named PENDING input printed on the slide, and RED rows are not exhibits yet - they are work items whose gap column names the module or public dataset that unblocks them. The statuses are recomputed live at every assembly, so this page is only ever as optimistic as the workbook it sits in. It feeds the deck, not the findings ledger.

SNF return-leg quality: discharge-to-community vs bounce-back rehospitalization

Source: CMS SNF Quality Reporting Program provider data (dataset fykj-qjee), risk-standardized measures S\_005\_02 (DTC) and S\_004\_01 (PPR-PD). Join key: CMS Certification Number (CCN) and state; both are national, risk-adjusted rates in percent.

DATA QUALITY: these are CLAIMS-BASED, RISK-STANDARDIZED, SNF-REPORTED facility measures - not counts of transports. Transport demand is INFERRED (a readmission implies a SNF-to-hospital transport leg), not measured. CMS suppresses facilities below the minimum eligible-stay count (they arrive as "Not Available" and are dropped here). Risk standardization already adjusts for case mix, so residual variation is facility signal; the worst-decile floor is the NATIONAL 90th-percentile PPR rate. Rates mix a ~2-year claims window (Start/End Date), not a single year.

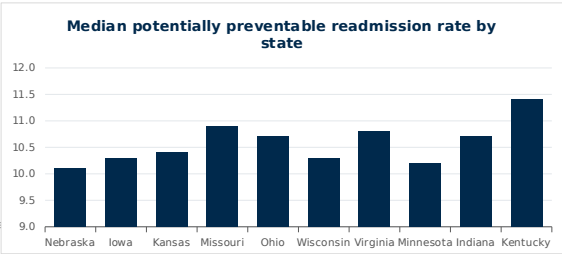
| Panel A. Footprint state distribution - discharge-to-community and rehospitalization (risk-standardized, percent) |          |                     |                     |                               |                    |                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|----------|---------------------|---------------------|-------------------------------|--------------------|-------------------------------------------------------------------------------------|
| State                                                                                                             | SNFs (n) | Median DTC rate (%) | Median PPR rate (%) | SNFs in worst natl PPR decile | Worst-decile share | Note                                                                                |
| Nebraska                                                                                                          | 179      | 43.9                | 10.1                | 2                             | 1.1%               | higher PPR / lower DTC = more bounce-back transport                                 |
| Iowa                                                                                                              | 389      | 49.6                | 10.3                | 15                            | 3.9%               |                                                                                     |
| Kansas                                                                                                            | 296      | 50.1                | 10.4                | 13                            | 4.4%               |                                                                                     |
| Missouri                                                                                                          | 487      | 48.9                | 10.9                | 44                            | 9.0%               |                                                                                     |
| Ohio                                                                                                              | 921      | 53.5                | 10.7                | 56                            | 6.1%               |                                                                                     |
| Wisconsin                                                                                                         | 323      | 51.8                | 10.3                | 18                            | 5.6%               |                                                                                     |
| Virginia                                                                                                          | 289      | 55.4                | 10.8                | 27                            | 9.3%               |                                                                                     |
| Minnesota                                                                                                         | 338      | 54.5                | 10.2                | 5                             | 1.5%               |                                                                                     |
| Indiana                                                                                                           | 507      | 54.2                | 10.7                | 58                            | 11.4%              |                                                                                     |
| Kentucky                                                                                                          | 268      | 47.2                | 11.4                | 48                            | 17.9%              |                                                                                     |
| Footprint (pooled)                                                                                                | 3,997    | 52.0                | 10.6                | 286                           | 7.2%               | pooled over all footprint SNFs                                                      |
| National (all SNFs)                                                                                               | 14,695   | 51.5                | 10.7                | 1,287                         | 8.8%               | worst decile = top 10% of PPR-reporting SNFs; as a share of ALL SNFs it is lower bc |

| Panel B. Footprint vs national gap (live formulas) - is the bounce-back worse here? |                  |                 |                 |                                                   |
|-------------------------------------------------------------------------------------|------------------|-----------------|-----------------|---------------------------------------------------|
| Measure                                                                             | Footprint median | National median | Gap (fp - natl) | Read                                              |
| Discharge to Community (higher is better)                                           | 52.0             | 51.5            | 0.5             | negative = footprint sends fewer home             |
| Potentially Preventable Readmission (lower is better)                               | 10.6             | 10.7            | (0.1)           | positive = footprint bounces back more            |
| Footprint SNFs in the worst NATIONAL PPR decile                                     | 286              |                 | 22.2%           | footprint share of the national worst-decile tail |

| Panel C. Worst-decile SNFs in footprint states - highest rehospitalization (top 22 of 286 at/above the national 90th-percentile PPR) |       |                |              |                    |                                          |             |        |
|--------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|--------------|--------------------|------------------------------------------|-------------|--------|
| SNF (provider name)                                                                                                                  | State | City           | PPR rate (%) | PPR eligible stays | DTC rate (%)                             | vs national | CCN    |
| Spring Mill Health Campus                                                                                                            | IN    | Merrillville   | 19.0         | 227                | 42.4 Worse than the National Rate        |             | 155764 |
| August Healthcare At Iliff                                                                                                           | VA    | Dunn Loring    | 18.3         | 197                | 21.4 Worse than the National Rate        |             | 495205 |
| August Healthcare At Leewood                                                                                                         | VA    | Ammandale      | 17.4         | 263                | 55.8 Worse than the National Rate        |             | 495337 |
| Eastgate Health Care Center                                                                                                          | OH    | Cincinnati     | 16.6         | 272                | 50.0 Worse than the National Rate        |             | 365772 |
| Austinwoods Rehab Health Care                                                                                                        | OH    | Austintown     | 16.4         | 181                | 56.9 Worse than the National Rate        |             | 365654 |
| Autumn Woods Health Campus                                                                                                           | IN    | New Albany     | 16.2         | 343                | 59.6 Worse than the National Rate        |             | 155681 |
| Lincoln Hills Of New Albany                                                                                                          | IN    | New Albany     | 16.1         | 163                | 45.2 Worse than the National Rate        |             | 155614 |
| Sikeston Convalescent Center                                                                                                         | MO    | Sikeston       | 16.1         | 117                | 31.5 Worse than the National Rate        |             | 265479 |
| Nhc Healthcare, Kennett                                                                                                              | MO    | Kennett        | 15.9         | 162                | 44.5 Worse than the National Rate        |             | 265168 |
| Westminster Village Kentuckiana                                                                                                      | IN    | Clarksville    | 15.8         | 128                | 54.4 Worse than the National Rate        |             | 155191 |
| Carriage Hill Health & Rehab Center                                                                                                  | VA    | Fredericksburg | 15.6         | 442                | 49.2 Worse than the National Rate        |             | 495396 |
| Gasconade Manor Nursing Home                                                                                                         | MO    | Owensville     | 15.5         | 65                 | 52.0 No Different than the National Rate |             | 265546 |
| Altercare Cambridge Inc.                                                                                                             | OH    | Cambridge      | 15.5         | 134                | 50.7 Worse than the National Rate        |             | 366128 |
| Princeton Nursing & Rehabilitation                                                                                                   | KY    | Princeton      | 15.5         | 114                | 27.5 Worse than the National Rate        |             | 185316 |
| Signature Healthcare Of Bowling Green                                                                                                | KY    | Bowling Green  | 15.5         | 203                | 43.9 Worse than the National Rate        |             | 185089 |
| Signature Healthcare Of Terre Haute                                                                                                  | IN    | Terre Haute    | 15.4         | 253                | 44.1 Worse than the National Rate        |             | 155426 |
| Charlestown Place At New Albany                                                                                                      | IN    | New Albany     | 15.4         | 197                | 44.3 Worse than the National Rate        |             | 155668 |
| Our Lady Of The Valley                                                                                                               | VA    | Roanoke        | 15.1         | 278                | 63.2 Worse than the National Rate        |             | 495357 |
| Cumberland Trace Health & Living Community                                                                                           | IN    | Plainfield     | 15.1         | 247                | 67.9 Worse than the National Rate        |             | 155836 |
| Signature Healthcare Of Elizabethtown                                                                                                | KY    | Elizabethtown  | 14.9         | 318                | 51.5 Worse than the National Rate        |             | 185118 |
| Greendale Park Nursing And Rehab                                                                                                     | WI    | Greendale      | 14.9         | 527                | 64.2 Worse than the National Rate        |             | 525549 |
| Park Terrace Health Campus                                                                                                           | KY    | Louisville     | 14.9         | 478                | 67.3 Worse than the National Rate        |             | 185462 |

Read panel

What is measured: two risk-standardized SNF QRP claims measures across every reporting footprint-state SNF. Discharge to community runs about 52.0% in the footprint (national 51.5%) and potentially preventable readmission about 10.6% (national 10.7%). The tail is what matters for transport: 286 footprint SNFs sit in the worst NATIONAL decile for rehospitalization - each is a structural bounce-back node that generates SNF-to-hospital transport demand rather than discharging patients home. Panel C names the worst of them by CCN and city so they can be joined to the corridor and hub-spoke maps. What is NOT measured: actual transport counts - a readmission implies a transport leg but the dataset carries quality rates, not trips; and managed-care or non-Medicare stays are outside the claims-based measure.



SNF return-leg structure: who bounces patients back (ownership, scale, star rating)

X-A.4 measured the return-leg rates; this tab asks whether a SNF's ownership, scale and quality rating predict how much return-leg transport it generates. It joins the SNF QRP claims measures (snf\_grp, dataset fykj-qjee) to the CMS Nursing Home provider-info file (dataset 4pq5-n9py) on the CCN - a 1:1 join over all reporting SNFs - and cross-tabs the risk-standardized potentially-preventable readmission (PPR = bounce-back transport) and discharge-to-community (DTC) rates by ownership, certified-bed scale, and 5-star rating. Higher PPR / lower DTC = a stronger structural transport-demand node.  
DATA QUALITY: both inputs are national CMS facility files joined on the CMS Certification Number; the join covers 14,695 SNFs that appear in both. Rates are CLAIMS-BASED and RISK-STANDARDIZED (case-mix adjusted), so residual variation across ownership / scale / rating is facility signal, not case mix; transport demand is INFERRED (a readmission implies a SNF-to-hospital transport leg), not counted. Ownership buckets collapse the CMS sub-categories (For profit - LLC / Corporation / Individual / Partnership -> For profit, etc.). The worst-decile floor is the national 90th-percentile PPR.

| Panel A. Return-leg rates by ownership bucket (higher PPR = more bounce-back transport) |               |                     |                     |                |                   |                   |                                                      |
|-----------------------------------------------------------------------------------------|---------------|---------------------|---------------------|----------------|-------------------|-------------------|------------------------------------------------------|
| Ownership                                                                               | National SNFs | Natl median PPR (%) | Natl median DTC (%) | Footprint SNFs | Fp median PPR (%) | Fp median DTC (%) | Note                                                 |
| For profit                                                                              | 10,855        | 10.7                | 50.8                | 2,644          | 10.7              | 51.1              | for-profit tends to bounce back more than non-profit |
| Non profit                                                                              | 2,897         | 10.4                | 55.4                | 1,060          | 10.3              | 54.0              |                                                      |
| Government                                                                              | 943           | 10.6                | 50.5                | 293            | 10.5              | 53.4              |                                                      |

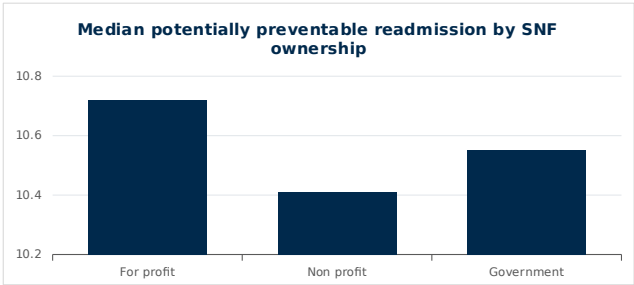
| Panel B. Return-leg rates by certified-bed scale |               |                     |                     |                |                   |                   |                      |
|--------------------------------------------------|---------------|---------------------|---------------------|----------------|-------------------|-------------------|----------------------|
| Bed scale                                        | National SNFs | Natl median PPR (%) | Natl median DTC (%) | Footprint SNFs | Fp median PPR (%) | Fp median DTC (%) | Note                 |
| 1. Under 60 beds                                 | 2,669         | 10.5                | 56.5                | 1,141          | 10.4              | 52.9              | scale vs bounce-back |
| 2. 60 to 119 beds                                | 6,642         | 10.7                | 51.7                | 2,078          | 10.6              | 52.1              |                      |
| 3. 120 to 179 beds                               | 4,041         | 10.8                | 50.8                | 641            | 10.7              | 51.0              |                      |
| 4. 180 or more beds                              | 1,335         | 10.6                | 48.1                | 137            | 10.7              | 50.2              |                      |

| Panel C. Return-leg rates by overall 5-star rating (the quality gradient) |               |                     |                     |                                            |
|---------------------------------------------------------------------------|---------------|---------------------|---------------------|--------------------------------------------|
| Overall rating                                                            | National SNFs | Natl median PPR (%) | Natl median DTC (%) | Note                                       |
| 1                                                                         | 2,873         | 10.7                | 46.4                | 1 star = worst care; note the PPR gradient |
| 2                                                                         | 3,025         | 10.8                | 49.9                |                                            |
| 3                                                                         | 2,844         | 10.7                | 52.0                |                                            |
| 4                                                                         | 2,783         | 10.7                | 53.1                |                                            |
| 5                                                                         | 3,045         | 10.4                | 57.6                |                                            |

| Panel D. For-profit vs non-profit gap (live formulas over Panel A national medians) |            |            |               |                                         |
|-------------------------------------------------------------------------------------|------------|------------|---------------|-----------------------------------------|
| Measure                                                                             | For profit | Non profit | Gap (FP - NP) | Read                                    |
| Median PPR - bounce-back (lower is better)                                          | 10.7       | 10.4       | 0.3           | positive = for-profit bounces back more |
| Median DTC - sent home (higher is better)                                           | 50.8       | 55.4       | (4.6)         | negative = for-profit sends fewer home  |

Read panel

What is measured: the same two risk-standardized SNF QRP claims measures as X-A.4, cut by facility structure. Ownership sorts the bounce-back: for-profit SNFs run a median PPR of about 10.72% against 10.41% for non-profit - the for-profit facilities return more patients to the hospital, so they generate proportionally more SNF-to-hospital transport. The 5-star rating is a clean gradient: 1-star SNFs sit near 10.74% PPR versus about 10.42% for 5-star, i.e. the worst-rated facilities are the heaviest bounce-back nodes. For an operator these are targeting screens: the return-leg transport demand concentrates in for-profit, lower-rated SNFs, and ownership concentration among them (chains) is a consolidation and partnership signal. What is NOT measured: actual transport counts (a readmission implies a leg but the file carries quality rates, not trips), and non-Medicare / managed-care stays outside the claims measure.



# Style standard: the CIM-presentation rules every tab must meet

These rules are the written standard behind the committed format gate (format\_gate.py); the gate checks them on every build and no revision ships unless it passes.

| The rules     |           |                                                                                                                                                                                                                                                                                                     |
|---------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Dimension | Rule                                                                                                                                                                                                                                                                                                |
| Title block   |           | Title in A1: Arial 15 bold, navy 00294C. The question line and the data-quality line in grey 9pt beneath it (rows 2-3).                                                                                                                                                                             |
| Panel headers |           | Panel/banner headers Arial 10 bold navy; table header rows white-on-navy fill.                                                                                                                                                                                                                      |
| Body          |           | Body Arial 9. Labels left-aligned, numbers right-aligned. Wrapped text with row heights sized so nothing clips.                                                                                                                                                                                     |
| Source line   |           | A source line under every panel naming the source IDs that power it.                                                                                                                                                                                                                                |
| Numbers       |           | Every numeric cell carries an explicit format: thousands separators on counts; dollars with \$ and no decimals above 1,000 (two decimals only for per-unit rates); percentages at one decimal; zeros as dashes; years as bare text; no raw floats, no unformatted integers, no scientific notation. |
| Layout        |           | Gridlines off; freeze panes below the title block; column widths sized to content so nothing truncates or shows #####; cursor at A1; zoom 100 on save.                                                                                                                                              |
| Tab colours   |           | Tab colours grouped by section (governance navy, demand teal, supply green, analysis slate, assembly amber, receiving/reference grey) matching the Index groupings; sheet order matches the Index.                                                                                                  |
| Charts        |           | Every chart passes the V9 style gate plus: title present, axis labels present, single value axis, category axis on the bottom (left for horizontal bars) with explicit delete=0, no smoothed lines, no default-grey Excel styling.                                                                  |
| The gate      |           | These rules are enforced by format_gate.py, committed beside the verification gates and run on every final build. No revision ships without the format gate passing and a render proof attached.                                                                                                    |

## Read panel

What this tab is: the presentation contract. The evidence tabs answer what the market looks like; this tab answers what a finished exhibit looks like. Every tab in this workbook is built to be screenshot-worthy at 100 percent zoom - a titled, sourced, cleanly formatted exhibit rather than a working file. The format gate turns that intent into a test: gridlines off, a section tab colour, freeze panes, a bold A1 title, and an explicit number format on every numeric cell, on all tabs, or the build fails.